9 CHAPTER

Advanced Biotechnology

RECOMBINANT DNA TECHNOLOGY

This field of Biotechnology encompases a number of methodologies that enables new combination of genetic material to be artificially construted in the laboratory. These "recombinant DNA" (rDNA) molecules are then introduced into host cells where they can be propagated and multiplied. *i.e.* to Create, Replicate & Analyze recombinant DNA molecules. First commerical application of this technology was in 1982 when the Food and Drug Administration permitted the marketing of Human Insulin produced by bacterial cells.

Steps involved in a typical rDNA experiment

(i) Isolation of the DNA fragments that are to be used or manipulated usually contains a gene sequence that needs to be cloned (commonly known as insert)

(ii) Generation of rDNA molecules by insertion of these DNA fragment(s) into a carrier DNA molecule termed vector

rDNA molecule = vector + insert

(iii) Transfer of these rDNA molecules into an appropriate host cell.

(iv) Selection of those host cells that carry the desired rDNA molecule and their replication to create a population of genetically identical cells or a 'clone'. Each cell in the clone contains multiple copies of the insert.

Gene Cloning

The basic steps in a gene cloning experiment are as follows (Figure).

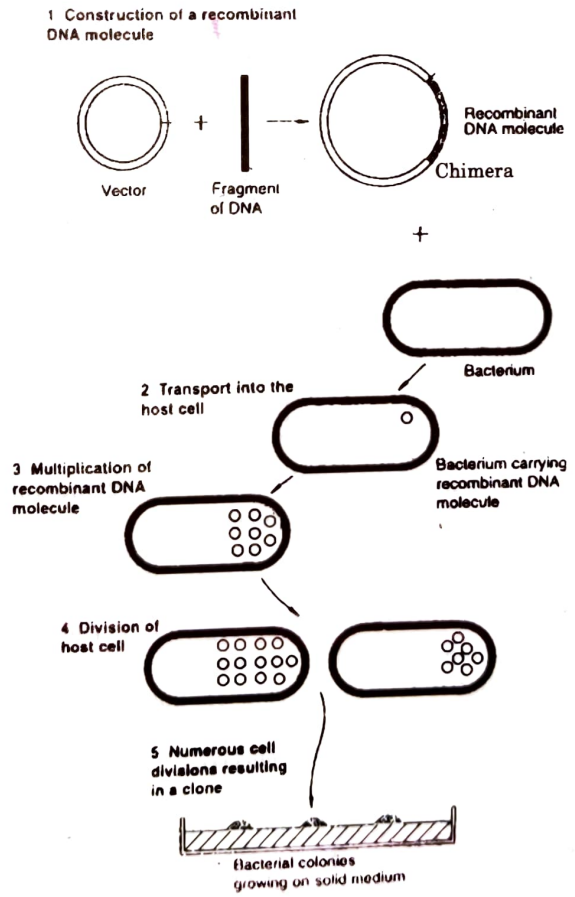


Figure. The basic steps in gene cloning.

(1) A fragment of DNA containing the gene to be cloned, is inserted into a circular DNA molecule called a **vector**, to produce a **chimera** or **recombinant DNA** molecule.

(2) The vector acts as vehicle that transports the gene into a host cell, which is usually a bacterium, although other types of living cell can be used.

(3) Within the host cell the vector multiplies, producing numerous identical copies not only of itself but also of the gene that it carries.

(4) When the host cell divides, copies of the recombinant DNA molecule are passed to the progeny and further vector replication takes place.

(5) After a large number of cell divisions, a colony, or **clone**, of identical host cells is produced. Each cell in the clone contains one or more copies of the recombinant DNA molucele; the gene carried by the recombinant molecule is now said to be cloned.

DNA & RNA Purification

Chromosomal DNA Purification

The chromosomes of eukaryotic cells are located within the nucleus of the cell. Each eukaryotic chromosome consists of a single DNA molecule wrapped around proteins called histones. All the DNA molecules that make up the chromosomes of a cell are referred to as the **genomic DNA** of the cell. In order to release the genomic DNA from a cell, the membranous envelopes (plasma membrane and nuclear membrane) that enclose and protect the genetic material must be dissolved and the histones and other chromosomal proteins digested. The membranes are dissolved using detergents that solubilize the phospholipids that make up these membranes. Very common detergents used for the extraction of genomic DNA are sodium dodecyl sulfate (SDS), Triton X-100® and Tween®. Once the membranes are dissolved, the resulting cell **lysate** (consisting of cytoplasmic and nuclear material, organelles and chromosomes) is treated with **proteases** (e.g., protease K) to digest the proteins away from the chromosomes. After the protease treatment, any DNA will be in aqueous solution. The cell lysate is treated with organic solvents such as a mixture of phenol and chloroform to remove proteins and other cell debris, leaving the genomic DNA in the aqueous interphase between the phenol and chloroform.

Collection of this partially purified DNA is the next and final step. When absolute (100%) ethanol is layered on top of the aqueous DNA solution an interphase develops between the two liquids due to their difference in densities. Genomic DNA in the aqueous phase will precipitate in this interphase as long, thin, white threads gathered into an opaque-looking mass. Genomic DNA can be **spooled** from the interphase with a glass rod. To do this, a glass rod is inserted into the interphase and twisted to wrap around it the strands of DNA. This spooled DNA can be dried, redissolved in an aqueous solution, and used in a variety of procedures.

Plasmid DNA and RNA Purification

The genome of a bacterial cell consists of a single, circular DNA molecule ("chromosome") Plasmids are small, self-replicating DNA molecules that may exist independently of the bacterial chromosome. Plasmids carry the genetic information to direct their own replication and a few genes, many of which impart some selective advantage to the cell. Examples of these genes include resistance to antimicrobial agents, resistance to heavy metals, and the production of antibiotics. Since many bacterial cells possess both the bacterial chromosome and several plasmids, the isolation of plasmid DNA involves a step in which the desired plasmid DNA is separated from the bacterial chromosome (genomic) DNA.

As with eukaryotic cells, bacterial cells are treated with detergents in order to disrupt the plasma membrane. Since bacteria possess a cell wall, treatment with the enzyme **lysozyme** is also required to remove this outer envelope. This treatment, along with that of detergents, releases both plasmid and chromosomal DNA from the cell. For the selective isolation of plasmid DNA from cells, sodium hydroxide (NaOH) is also added to the detergent solution to **denature** the DNA (i.e., cause the two strands to separate). The denaturation of bacterial DNA with NaOH and detergents is called **alkaline lysis**. Chromosomal DNA and many of the proteins and membrane components are removed by adding potassium (usually in the form of potassium acetate) to the mixture, followed by centrifugation. The selective precipitation of chromosomal DNA is based on the much larger size of the genomic DNA; the larger size makes it more vulnerable to precipitation by potassium acetate

than the small plasmid DNA molecule. After this step, plasmid DNA remains in solution along with cellular RNA. This solution is also known as a **cleared lysate**. The RNA can be eliminated by treatment with the enzyme **RNase**, which can then be removed with orgaric solvents (phenol-chloroform). Plasmid DNA is then precipitated with ethanol, dried, redissolved in an aqueous solution and used in a variety of applications.

Enzymes Used in Genetic Engineering

There are many enzymes which are used in genetic engineering as an important biological tool.

Table : Enzymes used in genetic engineering.

Enzyme	*Use*
Alkaline phosphatase	Removes PO_4 from 5′end of double stranded or single stranded DNA or RNA
DNA ligase	It joins sugar-phosphate backbone of dsDNA with 5′-PO_4 and a 3′-OH in an ATP-dependent reaction. The two ends of DNA must be blunt or complementary cohesive ones.
DNA Pol I	It synthesises DNA complementary to a DNA template in 5′→3′ direction beginning with a primer with a free 3′-OH. The Klenow fragment is a truncated version of DNA Pol I lacking 5′→3′ exonuclease activity
Exonuclease III	It cleaves from the ends of linear DNA and digests dsDNA from 3′end only
Mung bean nuclease	It digests ssDNA or RNA but leaves intact double helical region
Nuclease S1	It behaves as mung bean nuclease and cleaves a strand opposite to a nick on the complementary strand
Polynucleotide kinase	It adds to 5′-OH end of dsDNA or ssDNA or RNA in an ATP-dependent reaction. DNA becomes radiolabelled if [γ^{32}P]-ATP is used.
Restriction enzymes	These cut both strands of dsDNA within a symmetrical recognition site resulting in blunt or sticky ends (5′- or 3′ - ovelhangs)
Reverse transcriptase	It is a RNA-dependent DNA polymerase which synthesises DNA complementary to an RNA template in 5′→3′ direction by using deoxynucleotide triphosphates (dNTPs) beginning with a primer with a free 3′-OH
RNaseA	It is nuclease which digests RNA, but not DNA.
RNaseH	It is also a nuclease which digests the RNA or RNA-DNA heteroduplex
T3 T7, SP6 RNA polymerases	These are specific RNA polymerases encoded by the bacteriophage T3, T7 and SP6, respectively. These recognise only promoters from its own phage DNA and can be used to transcribe DNA downstream of each promoter
Taq polymerase	It is a DNA polymerase isolated from *Thermus aquaticus* which operates at 72°C and is stable above 90°C. It is used in PCR
Terminal transferase	It adds several nucleotides to 3′-ends of linear ssDNA/dsDNA or RNA molecule

1. Exonucleases

These enzymes act upon genome and digest the base pairs on 5′ or 3′ ends of a single stranded DNA or at single strand nicks or gaps in double stranded DNA. On the basis of mode of action the exonuclease are grouped into different types:

λ Exonuclease :

This enzyme removes nucleotides from 5′ end of double stranded DNA. Consequently an improved substrate for terminal transferase is formed. Therefore, this enzyme is used for 5′ end modification. The mode of action of this enzyme is given below :

$$\begin{matrix}5'\text{-G G A A T T}_{OH}\text{-}3' \\ 3'\text{-C C T T A Ap-}3'\end{matrix} \xrightarrow{\lambda\ \text{exonuclease}} \begin{matrix}5'\text{-GAATT}_{OH}\text{-}3' \\ 3'\text{-Cp-}5'\end{matrix} + \text{Np-}5'$$

(Np = nucleotide phosphate)

Exonuclease III :

This enzyme removes nucleotides from 3'-OH end of the double stranded DNA fragment. Consequently an improved substrate is formed which is used in gene manipulation. This enzyme is used for 3'end modification of DNA fragment as below :

$$\begin{matrix}5'\text{-G G A A T T}_{OH}\text{-}3' \\ 3'\text{-C C T T A A}_{P}\text{-}5'\end{matrix} \xrightarrow{\text{Exonuclease III}} \begin{matrix}5'\text{G}_{OH}\text{-}3' \\ 3'\text{-CCTTAA}_{P}\text{-}5'\end{matrix} \quad +\text{N}_{P}\text{-}5'$$

2. Endonucleases

They act upon genetic material and cleave the double stranded DNA at any point except the ends, but their action involves only one strand of the duplex.

3. Restriction Endonucleases

The restriction enzymes are called as *'molecular scissors'*. These act as foundation of recombinant DNA technology. These enzymes are present in bacteria and provide a type of defense mechanism called the *"restriction-modification system'*. Molecular basis of these systems was elucidated first by Werner Arber in 1965.

There are three different types of restriction endonucleases: Type I, Type II and Type III. Each enzyme differs slightly by different mode of action.

(a) Type I Restriction Enzyme : The B-K restriction and modification enzyme systems of *E. coli* have been termed as Type I endonucleases such as *EcoB* and *EcoK*. They are complex nucleases which function simultaneously as an endcnuclease and a methylase. They move along the DNA in a reaction and require Mg^{++}, S-adenosly methionine and ATP as co-factor. They are single multifunctional enzymes consisting of three different sub-units: *(i)* restrictive sub-unit, *(ii)* modification sub-unit, and *(iii)* specificity sub-unit. The third sub-unit specifies the recognition sites. The enzymes show specificity for recognition site but not for cleavage site *i.e.* they recognise specific sites within the DNA but do not cleave at these sites. Hence, heterogeneous population of DNA fragments is produced.

Recognition site of Type I enzyme is 15 bp long and cleavage site is about 1000 bp away from the 5'-end of the recognition sequence TCA which is present within 15 bp long recognition site. The adenine position in the restriction site may be methylated. Such DNA molecules are resistant to Type I endonuclease whose recognition sequence is methylated at A residues on both the strands. However, when the recognition sequence is unmethylated, only then DNA is cleaved as below :

EcoK 5'-A A A *C N N N N G T C G C-3'
3'-T T T G N' N' N' N' C A *G C G-5'

Here A shows methylation at adenine position, N is unspecified bases and N' is complementary bases to N. The unmodified DNA molecules is restricted i.e. cleaved by restriction endonucleases.

(b) Type II Restriction Enzymes: These restriction enzymes were first isolated by Hamilton Smith. They are simple consisting of single polypeptide chain and require no ATP for degradation of DNA. They have separate activities for cleaving (endonuclease) and methylation (methylase) of DNA molecules. They are most stable enzymes that require Mg^{++} as co-factor. Daniel Nathans first used these enzymes for mapping and analysing genes and genomes. The first (I) Type II restriction enzyme isolated and characterised was from *Escherichia coli* strain RY. Hence, it was named as *Eco*RI. For the discovery of restriction enzymes, W. Arber, H. Smith and D. Nathans were awarded with Nobel Prize in 1978.

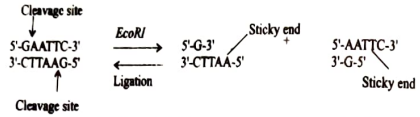
Out of above three types of restriction endonucleases, Type II are used in gene manipulation.Type I enzymes recognise a specific sequence of DNA molecule but cut elsewhere, and Type II make cuts only within the restriction sites and produce two single strand breaks, one break in each strand. Type II enzymes are the most important ones. Recognition sites are the **palindromes** or palimdromic sequences where most of Type II restriction enzymes bind and cut a DNA molecule. Example: *Eco*RI restriction enzyme isolated from *E. coli* cleaves DNA at the hexanucleotide 5' GAATTC3', whereas *Eco*RII isolated from the same bacterium recognises- PO_4 end of pentanucleotide 5'CCTGG3'.

Table : Source of type II restriction enzymes, cleavage sites and productions of cleavage

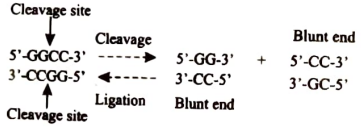
Microorganisms	*Restriction enzymes*	*Cleavage sites*	*Cleavage products*	
Bacillus amyloliquefaciens H	*Bam*HI	5'-GGATCC-3' 3'-CCTAGG-5'	5'-G 3'-CCTAG	GATCC-3' G-5'
B. globigii	*Bgl*II	5'-AGATCT-3' 3'-TCTAGA-5'	5'-A 3'-TCTAG	GATCT-3' A-5'
Escherchia coli RY13	*Eco*RI	5'-GAATTC-3' 3'-CTTAAG-5'	5'-G 3'-CTTAA	AATTC-3' G-5'
E. coli $R_2$45	*Eco*RII	5'-GGATCC-3' 3'-CCTAGG-5'	5'-G-3' 3'-CCTAG-5'	5'-GATCC-3' 3'-G-5'
Haemophilus influerzae Rd	*Hind*II	5'-GTPyPuAC-3' 3'-CAPuPyTG-5'	5'-GTPy-3' 3'-CAPu-5'	5'-PuAC-3' 3'-PyTG-5'
	*Hind*III	5'-AAGCTT-3' 3'-TTCGAA-5'	5'-A 3'-TTCGA	AGCTT-3' A-5'
H. parainfluenzae	*Hpa*I	5'-GTTAAC 3' 3'-CAATTG-5'	5'-GTT 3'-CAA	AAC-3' TTG-5'
	*Hpa*II	5'-CCGG-3' 3'-GGCC-5'	5',-C-3' 3'-GGC-5'	5'-CGG-3' 3'-C-5'
Klebsiella preumoniae OK 8	*Kpn*I	5'-GGTACC-5' 3'-CCATGG-3'	5'-GGTAC 3'-C	C-3' CATGG-5'
Nocardia otitidis caviarus	*Not*I	5'-GCGGCCGC-3' 3'-CGCCGGCG-5'	5'-GC-3' 3' CGCCGG-5'	5'-GGCCGC-3 3'-CG-5'
Providencia stuartii	*Pst*I	5'-CTGCAG-3' 3'-GACGTC-5'	5'-CTGCA-3' 3'-G-5'	5'G-3' 3'-ACGTC-5'
Streptomyces albus G	*Sal*I	5'-GTCGAC-3' 3'-CAGCTG-5'	5'-G 3'-CAGCT	TCGAC-3' G-5'
Staphylococus aureus 3AI	*Sau*3AI	5'-GATC-3' 3'-CTAG-5'	5'- 3'-CTAG	GATC-3 5'

* Arrows indicate the recognition sites.

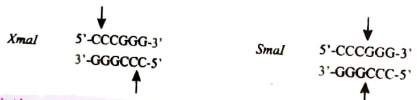
In addition, a very useful feature of many restriction nucleases is that they make **staggered cuts.** For example, *Eco*RI enzyme binds to a region of having specific palindromic sequence. The length of this region is 6 base pairs *i.e* hexanucleotide palindrome. It cuts between G and A residues of each strand and produces two single tranded complementary cut ends which are asymmetrical having 5' overhangs of 4 nucleotides. These ends are called **sticky ends** or *cohesive ends.* Because nucleotide bases of this region can pair and stick the DNA fragments again as given below:



On the other hand some other Type II restriction enzymes cleave both strands of DNA at the same base pairs but in the centre of recognition sequence, and results in DNA fragments with **blunt ends** *or flush ends.* For example *HaeIII* (isolated from *Haemophilus aegypticus,* the order of enzyme III), flour nucleotide long palindromic sequence and cuts symmetrically the both DNA strands and forms blunt ends.



There are some restriction enzymes which are **isozymes** e.g. *Hind*III and *Hsu*I. They are isolated from two different sources but possess similar recognition and cleavage sites. Both *Hind*III and *Hsu*I possess 5'AAGCTT3' recognition sequence. In contrast, some restriction enzymes in spite of having identical recognition sites, differ in cleavage sites. For example, *Xma*I and *Sma*I have identical recognition site but different cleavage sites as below :



There are some restriction enzymes whose recognition sites differ. Even then they produce identical sticky ends. For example, recognition sequence of *Bgl*II is 5'AGATCT3', that of *Bam*HI is 5'GGATCC3' and *Sau* is 5'GATC3'. But all the three enzymes cut recognition sequences and produce 5'GATC3' sticky ends. Since, fragments generated by each enzyme have identical sticky ends, they can join complementary sticky ends of one another.

(c) Type III Restriction Enzymes : The Type III enzyme is made up of two sub-units, one specifies for site recognition and modification and the other for cleavage. In a reaction, it moves along the DNA and requires ATP as source of energy and Mg^{++} as co-factor. ATPase activity is lacking in these enzymes. Some examples of Type III enzyme are *Hpa*I, *Mbo*II, *Fok*I, etc. They have symmetrical recognition sites and cleave DNA at specific non-palindromic sequences. For cleaving double stranded DNA two sites in opposite orientation must be present. One strand of double stranded DNA is cleaved about 25-27 bp away from the recognition site which is located in its immediate vicinity. Recognition sequence of some of Type III enzymes are given below:

*Mbo*II
5'-G A A G A $(N)_8$-3'
3'-C T T C T $(N)_7$-5'

*Fok*I
5'-G G A T G C $(N)_9$-3'
3'-C C T A C G $(N)_{13}$-5'

*Hgl*I
5'-G A C G C $(N)_5$-3'
3'-C T G C G $(N)_{10}$ -5'

4. S1 Nuclease

It degrades the single stranded DNA or single strand protrusion of double stranded DNA with cohesive ends. As a result of action of S1 nuclease cohesive ends are converted into blunt ends.

Therefore, S1 nuclease is used to remove the incompatible ends so that overlapping ends may be developed for annealing of two fragments of DNA molecules. The mode of action of S1 nuclease is given-below:

5'-AG-3' / 3'-TCTTAA-5' —S1 nuclease→ 5'-AG-3' / 3'-TC-5'

5. DNA Ligases

Mertz and Davis (1972) for the first time demonstrated that cohesive termini of cleaved DNA molecules could be covalently sealed with *E, coli* DNA ligase and were able to produce recombinant DNA molecules. DNA ligase seals single strand nicks in DNA which has 5'→3' -OH (hydroxyl) termini. There are two enzymes which are extensively used for covalently joining restriction fragments : the ligase from *E. coli* and that encoded by T4 phage. The main source of DNA ligase is T4 phage, hence, the enzyme is known as T4 DNA ligase.

6. Alkaline Phosphatase

When plasmid vector, for joining a foreign DNA fragment, is treated with restriction enzyme, the major difficulty arises at the same time. Because the cohesive ends of broken plasmids, instead of joining with foreign DNA join the cohesive end of the same DNA molecules and get recircularized. The restricted plasmid (*i.e.* plasmid treated with restriction enzymes) is, treated with an enzyme, alkaline phosphatase, that digests the terminal 5' phosphoryl group. The restriction fragments of the foreign DNA to be cloned are not treated with alkaline phosphatase. Therefore, the 5' end of foreign DNA fragment can covalently join to 3' end of the plasmid. The hybrid or recombinant DNA obtained has a nick with 3' and 5' hydroxy ends. Ligase will only join 3' and 5' ends of recombinant DNA together if the 5' end is phosphorylated. Thus, alkaline phosphatase and ligase prevent recircularization of the vector and increase the frequency of production of recombinant DNA molecules. The nicks between two 3' ends of DNA fragment and vector DNA are repaired inside the bacterial host cells during the transformation.

The mechanism of action of alkaline phosphatase is as below:

5'-GG_{OH}-3' / 3'-$CCAATTA_P$-5' —Alkaline phosphatase→ 5'-GG-3' / 3'-$CCAATTA_{OH}$-5'

7. Reverse Transcriptase

In addtion to these enzymes, reverse transcriptase is used to synthesize the copy DNA or complementary DNA (cDNA) by using mRNA as a template. Reverse transcriptase is very useful in the synthesis of cDNA and construction of cDNA clone bank.

8. DNA Polymerase

This enzyme polymerizes the DNA synthesis on DNA template (or cDNA template) and also catalyses a 5'→3' and 3'→5' exonucleolytic degradation of DNA (Kornberg, 1974). The DNA polymerase, investigated by A. Kornberg and coworkers in *E. coli* in 1956 is now known as DNA polymerase I (DNA pol I). The other two enzymes are DNA polymerase II (DNA pol II) and DNA polymerase III (DNA pol III). These have almost similar catalytic activity. DNA pol I (mol wt 109,000) has a single polypeptide chain of about 1,000 amino acid residues. The addition of mononucleotide to the free -OH end of a DNA chain is catalysed by this enzyme. Also it catalyses the other two reaction *i.e.*. 3'→5' exonuclease activity (hydrolysing single nucleotide residues from 5' - terminus) and 5→3' exonuclease activity (hydrolysing single nucleotide residues from 5'- terminus). Function of DNA pol II (mol wt 120,000) is little understood. However, it catalyses 3'→5' exonuclease activity. DNA pol III (mol wt about 140,000) is about several time more active than the other two. It is a dimer of DNA pol III. It requires an auxillary protein DNA copolymerase III and after combination, yields a DNA pol III - copol III complex. Where there is preformed DNA template it produces a parallel strand in the presence of ATP.

DNA Pol I is susceptible to protease action and can be degraded into fragments. The larger fragments possess polymerase and 3'→5' exonuclease activity but lack 5'→3' exonuclease activity.

This portion of DNA Pol I is called **Klenow polymerase**. Instead of intact Pol I, this enzyme is used for polymerase activity as below:

5'-G-3' — Klenow polymerase (dGTP, dTTP, dATP) → 5'-GGAATT-3'
3'-CCTTAA-5' — 3'-CCTTAA-5'

9. T4 Polynucleotide Kinase

The T4 polynucleotide kinase catalyses the transfer of $\gamma^{32}PO_4$ of ATP to 5' terminus of a dephosphorylated DNA (i.e. DNA lacking -PO_4 residue) or RNA as below:

$5'\text{-}GG_{OH}\text{-}3'$ — Polynucleotide kinase (γ^{32}P-ATP) → $5'GG_{OH}\text{-}3'$
$3'\text{-}CCTTAA_{OH}\text{-}5'$ — $3'\text{-}CCTTAA_{P^{32}}\text{-}5'$

This enzyme shows both phosphorylation as well as phosphatase activities; therefore, two types of reaction can be used accordingly.

10. Terminal Transferase

This enzyme is also known as deoxynucleotidyl transferase. It has been isolated and purified from calf thymus. It has a property to add 'oligodeoxynucleotide tail' to 3'-OH end of double stranded DNA fragments. Thus it extends homopolymer tails and this phenomenon is called homoplymer tailing.

Linkers and Adaptors

Treatment of DNA with Type II reatriction enzymes results in formation of fragment having sticky ends or blunt ends. Such DNA fragments cannot be ligated with cloning vectors. Therefore, artificially cleavage sites at blunt end can be added as linker or adaptor molecules. Chemically synthesised double stranded oligonucleotides are called **linkers.** They are added to double stranded DNA molecules that act as recognition site for restriction enzyme. For example the following oligonucleotide possess recognition sequence for *Eco*RI (asterisk nucleotides). It can be ligated to blunt end of any double stranded DNA and cut by *Eco*RI to generate sticky ends to DNA fragments.

5'-C C G A* A* T* T* C* G G-3'
3'-G G C T T A A G C C-5'

On the other hand chemically synthesised DNA molecules containing pre-formed cohesive ends are called adaptors. When recognition site is present within the DNA molecules to be coloned, adaptors are used to develop sticky ends. Adaptor is used in the form of 5'-OH so that self polymerisation could not be done. Thereafter, phosphorylation of foreign DNA ligated adaptor is done at 5' end. They joined to restriction enzyme treated fragments;

Cloning Vectors for Prokaryotes

Cloning Vectors Based on *E.coli* Plasmids

The simplest cloning vectors, and the ones in most widespread use in gene cloning, are those based on small bacterial plasmids.

The nomenclature of plasmid cloning vectors

The name 'pBR322' conforms with the standard rules for vector nomenclature:

- 'p' indicates that this is indeed a plasmid.
- 'BR' identifies the laboratory in which the vector was originally constructed (BR stands for Bolivar and Rodriguez, the two researchers who developed pBR322).
- '322' distinguishes this plasmid from others developed in the same laboratory (there are alos plasmids called pBR325, pBR327, pBR328, etc).

The useful properties of pBR322

The first useful feature of pBR322 is its size. pBR322 is 4363bp, which means that not only can the vector itself be purified with ease. But so can recombinant DNA molecules constructed with it.

Figure : A map of pBR322 showing the positions of the ampicillin resistance (amp^R) and tetracycline resistance (tet^R) genes, the origin of replication (ori) and some of the most important restriction sites.

The second feature of pBR322 is that, it carries two sets of antibiotic resistance genes. Either ampicillin or tetracycline resistance can be used as a selectable marker for cells containing the plasmid, and each marker gene includes unique restriction site's that can be used in cloning experiments. Insertion of new DNA into pBR322 that has been restricted with *Pst*I, *Pvu*I or *Sca*I inactivates the *amp^R* gene, and insertion using any one of eight restriction endonucleases (notably *Bam*HI and *Hind*III) inactivates tetracycline resistance. This great variety of restriction sites that can be used for insertional inactivation means that pBR322 can be used to clone DNA fragments with any of several kinds of sticky end.

A third advantage of pBR322 is that it has a reasonably high copy number. Generally there are about 15 molecules present in a transformed *E.coli* cell, but this number can be increased, upto 1000-3000, by plasmid amplification in the presence of a protein synthesis inhibitor such as chloramphenicol. An *E.coli* culture therefore provides a good yield of recombinant pBR322 molecules.

pBR327 - a higher copy number plasmid

pBR327 was constructed by removing a 1089 bp segment from pBR322. This delection left the *amp^R* and *tet^R* genes intact, but changed the replicative and conjugative abilities of the resulting plasmid. As a result pBR327 differes from pBR322 in two important ways.

(1) pBR327 has a higher copy number than pBR322, being present at about 30-45 molecules per E.coli cell. pBR327, with its high copy number, is therefore a better choice that pBR322 for this kind of work.

(2) The deletion also destroys the conjugative ability of pBR322, making pBR327 a non-conjugative plasmid that cannot direct its own transfer to other *E.coli* cells. This is important for **biological containment,** averting the possibility of a recombinant pBR327 molecule escaping from the test tube and colonizing bacteria in the gut of a careless molecular biologist.

pUC8- a Lac selection plasmid

pUC8 is derived from pBR322, although only the replication origin and the *amp^R* gene remain. The nucleotide sequence of the *amp^R* gene has been changed so that it no longer contains the unique restriction sites; all these cloning sites are now clustered into a short segment of the *lacZ'* gene carried by pUC8.

pUC8 has three important advantages that have led to it becoming one of the most popular ***E.coli*** cloning vectors. The first of these is fortuitous: the manipulations involved in construction of pUC8 were accompanied by a chance mutation, within the origin of replication, that results in the plasmid having a copy number of 500-700 even befor amplification. This has a significant effect on the yield of cloned DNA obtainable from E.coli cells transformed with recombinant pUC8 plasmids.

The second advantage is that identification of recombinant cells can be achieved by a single step process, by plating onto agar medium containing ampicillin plus X-gal.

The third advantage of pUC8 lies with the clustering of the restriction sites, which allows a DNA fragment with two different sticky ends (say *Eco*RI at one end and *Bam*HI at the other) to be cloned without resorting to additional manipulations such as linker attachment.

Cloning vectors based on M13 bacteriophage

The normal M13 genome is 6.4 kb in length, but most of this is taken up by ten closely packed genes each essential for the repiication of the phage. There is only a single. 507 nucleotide intergenic sequence into which new DNA could be inserted without disrupting one of these genes, and in fact this region includes the replication origin which must itself remain intact. Clearly there is only limited scope for modifying the M13 genome.

Nevertheless, it will be remembered that the great attraction of M13 is the opportunity it offers of obtaining single-stranded versions of cloned DNA. This feature has acted as a stimulus for the development of M13 cloning vectors.

Development of the cloning M13mp2

The first step in construction of an M13 cloning vector was to introduce the *lacZ'* gene into the intergenic sequence. This gave rise to M13mpl. Which forms blue plaques on X-gal agar. M13mpl does not possess any unique restriction sites in the *lacZ'* gene. It does, however, contain the hexanucleotide GGATTC near the start of the gene. A single nucleotide change would make this GAATTC, which is an EcoRI site. This alteration was carried out using *in vitro* mutagenesis resulting in M13mp2. M13mp2 has a slightly altered *lacZ'* gene (the sixth codon now sepecifies asparagine instead of aspartic acid), but the β-galactosidase enzyme produced by cells infected with M13mp2 is still perfectly functional.

M13mp2 is the simplest M13 cloning vector. DNA fragments with *EcoRI* sticky ends can be inserted into the cloning site, and recombinants are distinguished as clear plaques on X-gal agar.

M13mp7-symmetrical cloning sites

The next step in the development of M13 vectors was to introduce additional restriction sites into the *lacZ'* gene. This was achieved by synthesizing in the test tube a short oligonucleotide, called a **polylinker**, that consists of a series of restriction sites and has *EcoRI* sticky ends. This polylinker was inserted into the *EcoRI* site of M13mp2, to give M13mp7 a more complex vector with four possible cloning sites (*EcoRI, BamHI, SalI* and *PstI*). The polylinker is designed so that it does not totally disrupt the *lacZ'* gene; a reading frame is maintained throughout the polylinker, and a functional, though altered, β-galactosidase enzyme is still produced.

More complex M13 vectors- M13mp8

The most sophisticated M13 vectors have more complex polylinkers inserted into the *lacZ'* gene. An example is M13mp8 which is the counterpart of the plasmid pUC8. As with the plasmid vector, one advantage of M13mp8 is its ability to take DNA fragments with two different sticky ends.

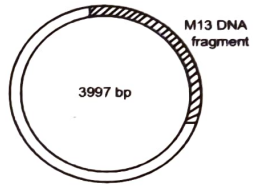
A second feature is provided by the sister vector M13mp9 which has the same polylinker but in the reverse orientation. A DNA fragment cloned into M13mp8, if excised by double restriction, and then inserted into M13mp9, will now itself be in the reverse orientation.

Hybrid plasmid-M13 vectors : Phagemids

Although M13 vectors are very useful for the production of single-stranded versions of cloned genes they do suffer from one disadvantage. There is a limit to the size of DNA fragment that can be cloned with an M13 vector, with 1500bp generally being looked on as the maximum capacity, though fragments up to 3kb have occasionally been cloned. To get around this problem a number of novel vectors **('Phagemids')** have been developed by combining a part of the M13 genome with plasmid DNA.

An example is provided by pEMBL8, which was made be transferring into pUC8 a 1300bp fragment of the M13 genome. This piece of M13 DNA contains the signal sequence recognized by the enzymes that convert the normal double-stranded M13 molecule into single-stranded DNA secretion of new phage particles. This signal sequence is still functional even though detached from the rest of the M13 genome, so pEMBL8 molecules are also converted into single-stranded DNA and secreted as defective phage particles. All that is necessary is that the *E.coli* cells used as hosts for a pEMBL8 cloning experiment are subsequently infected with normal M13 to

act as a **helper phage** providing the necessary replicative enzymes and phage coat proteins pEMBL8 being derived from pUC8, has the polylinker cloning sites within the *lacZ'* gene, so recombinant plaques can be identified in the standard way on agar containing X-gal. With pEMBL8, single-stranded versions of cloned DNA fragments up to 10kb in length can be obtained, greatly extending the range of the M13 cloning system.



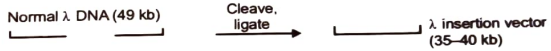
(a)

Figure. pEMBL8 a hybrid plasmid-M13 vector that can be converted into single-stranded DNA.

Cloning Vectors Based on λ Bacteriophage

Insertion and replacement vectors

The first two classes of vector to be produced were λ **insertion** and λ **replacement** (or substitution) vectors.



(a) Construction of a λ insertion vector

(b) λ gt10 ***(c) λZAPII***

Figure. λ insertion vectors. P = polylinker in the lacZ' gene of λZAPII, containing unique restriction sites for SacI, NotI, XbaI, SpeI, EcoRI and XhoI.

Insertion vectors

With an insertion vector a large segment of the non-essential region has been deleted, and the two arms ligated together. An insertion vector possesses at least one unique restriction site into which new DNA can be inserted. The size of the DNA fragment that an individual vector can carry depends of course on the extent to which the non-essential region has been deleted. Two popular insertion vectors are:

- **λgt10** which can carry up to 8kb of new DNA, inserted into a unique *Eco*RI site located in the cI gene. Insertional inactivation of this gene means that recombinants are distinguished as clear rather than turbid plaques.
- **λZAPII** with which insertion of up to 10kb DNA into any of six restriction sites within a plylinker inactivates the *lacZ'* gene carried by the vector. Recombinants give clear rather than blue plaques on X-gal agar.

Replacement vectors

A λ replacement vector has two recognition sites for the restriction endonuclease used for cloning. These sites flank a segment of DNA that is replaced by the DNA to be cloned. Often the replaceable fragment (or 'stuffer fragment' in cloning jargon) carries additional restriction sites that can be used to cut it up into small pieces of DNA than insertion vectors can handle. Recombinant selection is often on the basis of size, with non-recombinant vectors being too small to be packaged into λ phage heads.

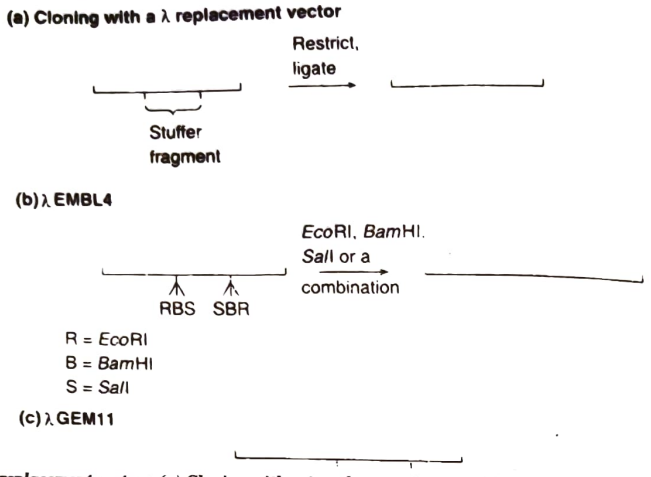


Figure. λ replacement vectors. (a) Cloning with a λ replacement vector. (b) Cloning with λEMBL4. (c) The structure of λGEM11, showing the order of restriction sites in the two polylinkers.

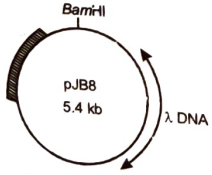
Two popular replacement vectors are :

- **λEMBL4** can carry up to 20kb of inserted DNA by replacing a segment flanked by pairs of *Eco*RI. *Bam* HI and *Sal*I sites (any of these three restriction endonucleases can be used to remove the stuffer fragment, so DNA fragments with a variety of sticky ends can be cloned). Recombinant selection with λEMBL4 can be on the basis of size, or can utilize the Spi phenotype.
- **λGEM11 and λGEM12,** each of which has a capacity of 23kb (near the theoretical maximum) with the stuffer fragment flanked by polylinkers that contain seven different restriction sites. The polylinkers are slightly different in the two vectors, but in both cases the outermost sites are for the restriction enzyme *Sfi*I which recognizes the sequence GGCCNNNNNGGCC. This sequence is very rare and unlikely to be present in the DNA fragment that has been cloned. Restriction of the recombinant vector with *Sfi*I can therefore be used to cut out the cloned fragment, with a high likelihood that the fragment will be recovered intact. As with λEMBL4, size or the Spi phenotype is used to select recombinants.

Cosmid

The final and most scphisticated type of λ-based vector is the **cosmid.** Cosmids are hybrids between a phage DNA molecule and a bacterial plasmid, and their design centres on the fact that the enzymes that package the λ DNA molecule into the phage protein coat need only the *cos* sites in order to function. The *in-vitro* packaging reaction works not only with λ genomes, but also with any molecule that carries *cos* sites separated by 37-52kb of DNA.

A cosmid is basically a plasmid that carries of *cos* site. It also needs a selectable marker, such as the ampicillin resistance gene, and a plasmid origin of replication, as cosmid lack all the λ genes and so do not produce plaques. Instead colonies are formed on selective media, just as with a plasmid vector.



(a) A typical cosmid

Figure. A typical cosmid

λ and other high capacity vectors

One solution is to develop new cloning vectors able to handle large DNA inserts. During the last few years progress in this area has centred on **bacterial artificial chromosomes (BACs),** which are novel vectors based on the F plasmid. The F plasmid is relatively large and vectors derived from it have a higher capacity than normal plasmid vectors. BACs can handle DNA inserts up to 300 kb in size reducing the size of the human genomic library to just 30000 clones. Other high capacity vectors have been constructed from bacteriophage PI, which has the advantage over λ of being able to squeeze 110kb of DNA into its capsid structure. Cosmid-type vectors based on PI have been designed and used to clone DNA fragments ranging in size from 75 to 100kb. Vectors that combine the features of PI vectors and BACs called **PI-derived artificial chromosomes (PACs),** also have a capacity of up to 300kb.

Cloning Vectors for Eukaryotes

Vectors for yeast and other fungi

The yeast *Saccharomyces cerevisiae* is one of the most important organisms in Biotechnology. The 2μm plasmid is in fact an excellent basis for a cloning vector. It is 6kb in size, which is ideal for a vector, and exists in the yeast cell at a copy number of between 70 and 200.

REP1 and REP2 are involved in replication of the plasmid and FLP codes for a protein that can convert the A form of the plasmid to the B form, in which the gene order has been rearranged by intramolecular recombination. The function of D is not exactly known.

Vectors based on the 2mm plasmid-yeast episomal plasmids

Vectors derived from the 2μm plasmid are called **yeast episomal plasmids (YEps).** Some YEps contain the entire 2μm plasmid. Other include just the 2μm origin of replication. An example of the latter type is YEp13.First it is a **shuttle vector.** As well as the 2μm origin of replication and the selectable LEU2 gene, YEp13 also includes the entire pBR322 sequence, and can therefore replicate and be selected for in both yeast and E.*coli.*

Other types of yeast cloning vector

In addition to YEps, there are several other types of cloning vector for use with *S.cerevisiae*. Two important ones are as follows.

(1) Yeast integrative plasmids (YIps) are basically bacterial plasmids carrying a yeast gene. An example is YIp5, which is pBR322 with an inserted *URA3* gene. The gene codes for orotidine-5′- phosphate decarboxylase (an enzyme that catalyses one of the steps in the biosynthesis pathway for pyrimidine nucleotides) and is used as a selectable marker in exactly the same way as *LEU2*. A YIp cannot replicate as a plasmid as it does not contain any parts of the 2μm plasmid, and instead depends for its survival on integration into yeast chromosomal DNA. Integration occurs just as described for a YEp.

(2) Yeast replicative plasmids (YRps) are able to multiply as independent plasmids because they carry a chromosomal DNA sequence that includes an origin of replication. Replication origins are known to be located very close to several yeast genes, including one or two which can be used as selectable markers. YRp7 is an example of a replicative plasmid. It is made up of pBR322 plus the yeast gene TRPI. This gene, which is involved in tryptophan biosynthesis, is located adjacent to a chromosomal origin of replication. The yeast DNA fragment present in YRp7 contains both TRPI and the origin.

Yeast Artifical chromosome

The final type of yeast cloning vector to consider is the **yeast artificial chromosome (YAC),** a totally new approach to gene cloning. The development of YACs has been a spin-off from fundamental research into the structure of eukaryotic chromosomes work that has identified the key components of a chromosome as being.

(1) The centromere, which is required for the chromosome to be distributed correctly to daugher cells during cell division.

(2) Two telomeres, the structures at the ends of a chromosome, which are needed in order for the ends to be replicated correctly and which also prevent the chromosome from being nibbled away by exonucleases.

(3) The origins of replication which are the positions along the chromosome at which DNA replication initiates similar to the origin of replication of a plasmid.

Once chromosome structure had been defined in this way the possibility arose that the individual components might be isolated by recombinant DNA techniques and then joined together again in the test tube creating an artificial chromosome. As the DNA molecules present in natural yeast chromosomes are several hundred kilobases in length it might be possible with an artificial chromosome to clone large pieces of DNA.

The Structure and use of a YAC vector

Several YAC vectors have been developed but each one is constructed along the same lines, with pYAC3 being a typical example. At first glance pYAC3 does not look much like an artificial chromosome, but on closer examination its unique features become apparent. pYAC3 is essentially a pBR322 plasmid into which a number of yeast genes have been inserted. Two of these genes, *URA3* and *TRPI* have been encountered already as the selectable markers for Ylp5 and YRp7 respectively. As in YRp7 the DNA fragment that carries *TRPI* also contains an origin of replication, but in pYAC3 this fragment is extended even further to include the sequence called CEN4, which is the DNA from the centromere region of chromosome 4. The *TRPI*-origin-*CEN4* fragment therefore contains two of the three components of the artificial chromosome.

(a) pYAC3

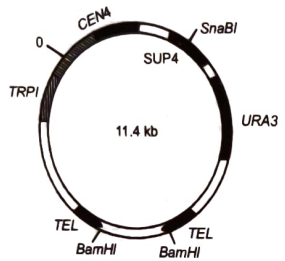


Figure. A YAC vector.

The third component, the telomeres, is provided by the two sequences called *TEL*. These are not themselves complete telomere sequences, but once inside the yeast nucleus they act as seeding sequences onto which telomeres will be built . This just leaves one other part of pYAC3 that has not been mentioned: SUP4 which is the selectable marker into which new DNA is inserted during the cloning experiment.

The cloning strategy with pYAC3 is as follows. The vector is first restricted with a combination of *Bam*HI and *Sna*BI,cutting the molecule into three fragments. The *Bam*HI fragment is removed, leaving two arms, each bounded by one *TEL* sequence and one *Sna*BI site. The DNA to be cloned which must have blunt ends (*Sna*BI is a blunt end cutter, recognizing the sequence TACGTA), is ligated between the two arms, producing the artificial chromosome.

Cloning Vectors for higher plants

Three types of vector system have been used with varying degrees of success with higher plants :

(1) Vectors based on naturally occurring plasmids of *Agrobacterium*.

(2) Direct gene transfer using various types of plasmid DNA.

(3) Vectors based on plant viruses.

Using the Ti plasmid to introduce new genes into a plant cell

It was realized very quickly that the Ti plasmid could be used to transport new genes into plant cells. All that would be necessary would be to insert the new genes into the T-DNA and then the bacterium could do the hard work of integrating them into the plant chromosomal DNA. Novel strategies have to be developed for inserting new DNA into the plasmid. Two are in general use:

(1) The binary vector strategy is based on the observation that the T-DNA does not need to be physically attached to the rest of the Ti plasmid. A two plasmid system, with the T-DNA on a relatively small molecule and the rest of the plasmid in normal form, is just as effective at transforming plant cells. In fact some strains of *A.tumefaciens*, and related Agrobacteria, have natural binary plasmid systems. The T-DNA plasmid is small enough to have a unique restriction site and to be manipulated using standard techniques.

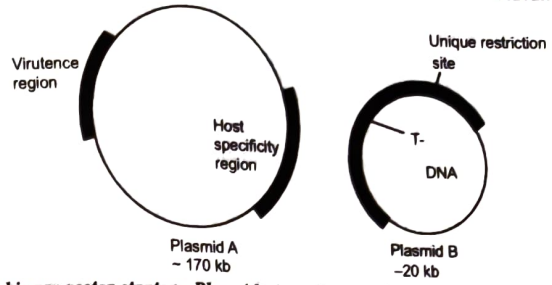


Figure. The binary vector strategy. Plasmids A and B complement each other when present together in the same A. tumefaciens cell.The T-DNA carried by plasmid B is transferred to the plant chromosomal DNA by proteins coded by genes carried by plasmid A.

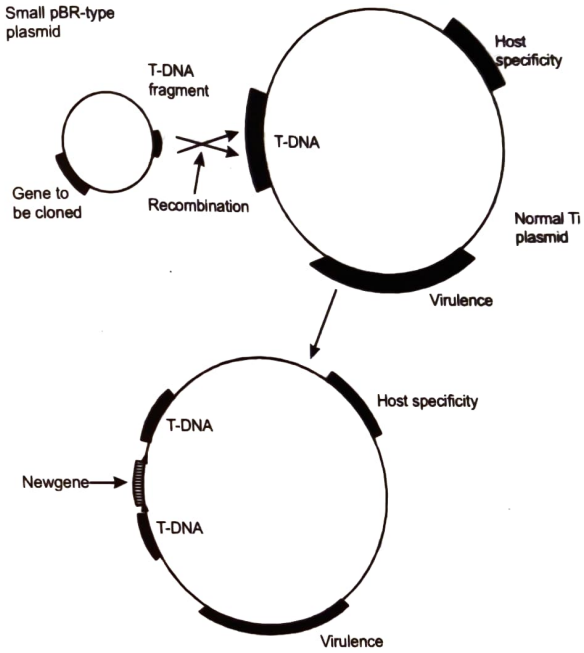


Figure. The cointegration strategy.

(2) The cointegration strategy uses an entirely new plasmid, based on pBR322 or a similar *E.coli* vector, but carrying a small portion of the T-DNA. The homology between the new molecule and the Ti plasmid means that if both are present in the same *A.tumefaciens* cell, recombination can integrate the pBR plasmid into the T-DNA region. The gene to be cloned is therefore inserted into a unique restriction site on the small pBR plasmid, introduced into *A.tumefaciens* cells carrying a Ti plasmid and the natural recombination process left to integrate the new gene into the T-DNA. Infection of the plant leads to insertion of the new gene, along with the rest of the T-DNA, into the plant chromosomes.

Cloning Vectors for animals

Cloning vectors for insects

The fruit fly, *Drosophila melanogaster*, has been and still is one of the most important model organisms used by biologists.

p elements as cloning vectors for Drosophila

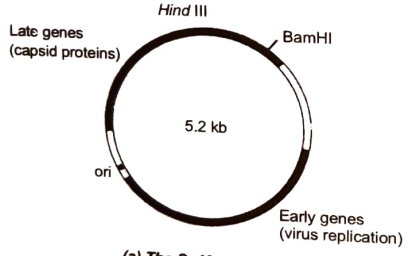
Cloning in *Drosophila* makes use of a **transposon** called the **P element.**

Transposons are common in all types of organisms. They are short pieces of DNA (usually less than 10kb in length) that can move from one position to another in the chromosomes of a cell. P elements, which are one of several types of transposon in *Drosophila*, are 2.9kb in length and contain three genes flanked by short inverted repeat sequences at either end of the element. The genes code for transposase, the enzyme that carries out the transposition process, and the inverted repeats form the recognition sequences that enable the enzyme to identify the two ends of the inserted transposon.

Insertion of the new DNA into this P element results in disruption of its transposase gene, so this element is inactive. The second P element carried by the plasmid is therefore one that has an intact version of the transposase gene. Ideally this second element should not itself be transferred to the *Drosophila* chromosomes, so it has its 'wings clipped': its inverted repeats are removed so that the transposase does not recognize it as being a real P element. Once the gene to be cloned has been inserted into the vector, the plasmid DNA is microinjected into fruit fly embryos. The transposase from the wings-clipped P element directs transfer of the engineered P element into one of the fruit fly chromosomes. If this happens within a germline nucleus then the adult fly that develops from the embryo will carry copies of the cloned gene in all its cells. P element cloning was first developed in the 1980s and has made a number of important contributions to *Drosophila* genetics.

Cloning vectors for mammals

The first cloning experiment involving mammalian cells was carried out in 1979 with a vector based on simian virus 40(SV40). This virus is capable of infecting several mammalian species, following a lytic cycle in some hosts and a lysogenic cycle in others. The genome is 5.2kb in size and contains two sets of genes, the 'early' genes, expressed early in the infection cycle and coding for proteins involved in viral DNA replication, and the late genes, coding for viral capsid proteins. SV40 suffers from the same problem as λ and the plant caulimoviruses, in that packaging constraints limit the amount of new DNA that can be inserted into the genome. Cloning with SV40 therefore involves replacing one or more of the existing genes with the DNA to be cloned.



(a) The Sv40 genome

Figure. SV40 and an example of its use as a cloning vector

Selection of Transformed Cells

Identification of cells containing the vector molecule requires the presence of suitable marker gene on the vector molecule whose expression provides a means of identifying cells containing it. Two popularly used marker gene systems are based on:

Antibiotic resistance genes

A host cell strain is chosen that is sensitive to a particular antibiotic, often ampicillin, tetracycline or chloramphenicol. The corresponding vector has been engineered to contain a gene which confers resistance to the antibiotic. After transformation, cells are plated on agar containing the antibiotic to rescue cells transformed by the vector.

β-galactosidase gene complementation

The *E.coli* enzyme β-galactosidase is a large polypeptide that is the product of the *lac*Z gene. The active form of the enzyme is a homo-tetramer. Certain mutations in the *lac*Z gene prevent subunit association of the resultant peptide (ω peptide) and the monomers lack enzyme activity. In such mutants, subunit assembly and enzyme activity can be restored by the presence of a small (50- or so amino-acid) amino-terminal fragment of the *lac*Z product (the α polypeptide). Such *lac*Z mutants are said to be subject to α **complementation.**

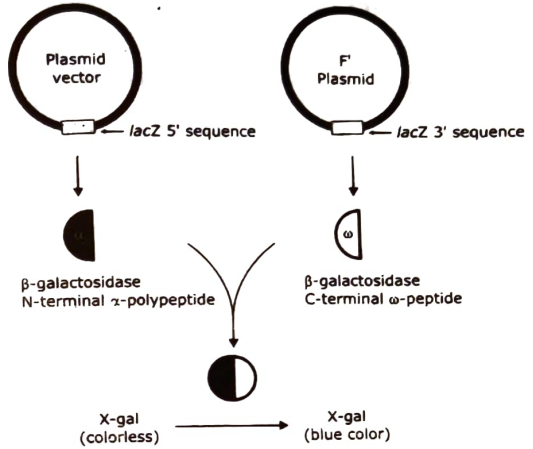


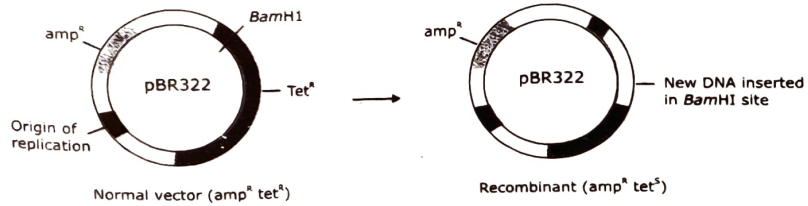
Figure. β-galactosidase activity can be used as an indicator of gene function. Plasmids that express the N-terminal α-polypeptide of β-galactosidase can be used in E.coli strains expressing the C-terminal ω peptide of β-galactosidase on an F′ plasmid.

In β-galactosidase gene complementation method, the host cell is a mutant which contains a fragment of the β-galactosidase gene and thus cannot make any functional β-galactosidase. The vector is engineered to contain a different fragment of the β-galactosidase gene. After transformation by the vector, *functional complementation* occurs resulting in active β-galactosidase which can be assayed by acting on a colorless substance, *Xgal* (5-bromo, 4-chloro, 3-indolyl β-d-galactopyranoside), to make a blue product (Blue-white screening).

Recombinant screening

A selective medium enables transformants to be distinguished from non-transformants. The next problem is to determine which of the transformed colonies comprise cells that contain recombinant DNA molecules, and which contain self-ligated vector molecules. With most cloning vectors insertion of a DNA fragment into the plasmid destroys the integrity of one of the genes present on the molecule. Recombinants can therefore be identified because the characteristic coded by the inactivated gene is no longer displayed by the host cells called **insertional inactivation.**

Most commonly recombinant selection is carried out by insertional inactivation of antibiotic resistance gene. In this case insertion of new DNA fragments (insert) occurs at the site within the gene that confers resistance towards a particular antibiotic.



Insertional inactivation does not always involve antibiotic resistance genes. For example in pUC8, gene LacZ', which codes for part of enzyme β-galactosidase is used for insertional inactivation. Recombinant pUC8 involves insertional inactivation of the lac Z' gene, can be identified because of their inability to synthesize β-galactosidase. β-galactosidase, coded by lacZ' gene, causes the breakdown of lactose to glucose plus galactose. lacZ', a modified lacZ gene, codes for the α peptide portion of β-galactosidase.

Introduction of DNA into Host Cells

Introducing Genes into Prokaryotes

There are no regular sexual processes in prokaryotes that ensure imports of alleles into a cell from other members of the species. The prokaryote relies on a few parasexual channels for such acquisitions. They include the gene transfer methods referred to as natural parasexual modes of gene transfer, cell transformation and tronsfection with phage vector.

1. **Natural Parasexual modes of gene transfer in prokaryotes.** (1) Transformation by naked DNA, (2) transduction by phage carried DNA, (3) conjugational transfer via a tube connecting donor and recipient cells introduces exogenous alleles via homologous recombination, and (4) the fourth mode, transposition, employs an entirely different method for joining totally unrelated DNA sequences. Gene cloning by and large uses the first two and the last modes for transforming cells with foreign DNA.
2. **Cell Transformation with Plasmids**

   Although bacteria are known to be transformed by exogenous DNA in nature, the efficiency of transformation in the natural habitat is very low. For improving the efficiency of uptake of free phase DNA by *E. coli* cells, that involves treating the cells in succession by chilled $CaCl_2$ and heat shock.

   The cell transformation protocol consists of treating $CaCl_2$-treated (0-5°C for 0.5 to 24 hours) *E. coli* cells with heat shock (42°C). The cells thus made competent are mixed with the DNA to be introduced into the cells. The Ca^{++} appears to make the DNA adhere to the cell surface and the heat treatment allows the membrane to be temporarily premeable to the exogenous DNA. The mechanisms underlying these steps are not clear.
3. **Transfection with Phage Vectors**

   The phage vector-carried gene/DNA of interest may be introduced into an *E. coli* cell either by transforming with the naked vector DNA, or via infectious particles carrying the gene to be introduced. The latter particles are constructed around replicated recombinant DNA copies with the help of a λ-packaging mix that has been described earlier.

   Transfection by packaged particles has a slightly higher efficiency than even the most efficient protocol using transformation protocols.

Introducing Genes in Eukaryotes

1. **The Recombinant Viral Technique**

   The recombinant viral technique involves the transfer of an exogenous gene into an animal cell via a recombinant DNA made with a viral vector. So far, SV40, adenovirus and papilloma viruses have been utilized.
2. **DNA Mediated Gene Transfer Methods**

   There are a few techniques by which naked DNA, which is not necessarily spliced to vectors, may be introduced directly into mammalian cells.

   (*i*) **The Calcium Phosphate Method**

   The calcium ion-facilitated method involves the formation of calcium phosphate, in a solution containing donor DNA, by the action of calcium chloride in a phosphate saline buffer. The DNA becomes complexed with the freshly formed calcium phosphate. The recipient cells are mixed with the DNA phosphate-containing solution and incubated for to days at 37°C. The complexed particles enter the cells through endocytosis or phagocytosis (it is presumed); some of the DNA become integrated into the chromosomal DNA in the cells (Fig.). The 48-hour incubation allows the stably integrated gene to express itself.

(*ii*) **Gene transfer by somatic cell fusion**

PEG (polyethylene glycol) is used as a fusing agent. It destabilizes cell membranes. Cells with destabilized membranes, when in contact, become enclosed within one membrane. These cell hybrids at first have separate nuclei (heterokaryons). Later, the nuclei also fuse; the hybrid cell is now known as a synkaryon. Somatic cell fusion is achieved when the partners are both eukaryotes, one a eukaryote and the other a prokaryote, as also between organelles and cells.

(*iii*) **The DEAE-dextran method**

DEAE (diethylaminoethyl)-dextran-treated viruses have been shown to increase the efficiency of infection of cultured cells of several mammalian species. Based on this fact, a technique has been evolved that transfects mammalian cells with donor DNA.

3. **Microcell fusion Technique**

   Microcell fusion method. Treatment with colecemid (a mitotic spindle poison) and cytochalasin causes the nucleus to become disintegrated and each fragment (one or more chromosomes) becomes enclosed in its own membrane. These micronuclei are isolated by centrifugation with added cytochalasin B. They are now called microcells. These microcells may be utilized to transfer single chromosomes by fusing to a somatic cell. It helps if the microcell carries a selectable marker. Microcell fusion has been useful for specific mammalian gene mapping projects.
4. **Metaphase Chromosome Transfer**

   Genes that are arranged in the native chromosome may be transferred into a mammalian cell in the form of whole chromosomes. As the latter are condensed the most at the metaphase stage of mitosis, it is easier to isolate chromosomes at this stage.
5. **Via Liposomes**

   Liposomes are spheres with phospholipid membranous walls that may be filled with a drug or any other substance of choice. The latter are entrapped during the process of making the liposomes.

   Liposomes with entrapped metaphase chromosomes, or vectors with donor DNA, have been fused with target cells. Wong (1980) introduced the Amp^R gene (on pBR322) into several cultured animal cell strains using this method.
6. **Microinjection Technique**

   DNA may be injected directly into cells (nuclei, if necessary) through glass pipettes that are pulled to produce very fine bores (= 0.1 – 0.5) at the tip.
7. **Electroporation**

   Exogenous DNA can be introduces into a cell by subjecting the cell to electrical pulses. This technique is especially suitable for introducing DNA into plant cells for both transient gene expression and long-term integrative studies.

Polymerase Chain Reaction

The polymerase chain reaction (PCR) provides a simple and ingeneous method for exponentially amplification of specific DNA sequences by *in vitro* DNA synthesis. This technique was developed by Kary Mullis in 1985. This technique has made it possible to synthesize large quantities of a DNA fragment without its cloning.

The PCR required the following :

(*i*) **DNA Template :** Any source that contains one or more target DNA molecules to be amplified can be taken as template.

(*ii*) **Primers :** A pair of oligonucleotides of about 180-30 nucleotides with similar G+C contents act as primers. They direct DNA synthesis towards one another. The primers are designed to anneal on opposite strands of target sequence so that they will be extended towards each other by addition of nucleotides to their 3′ ends.

(*iii*) **Enzyme :** The most common enzyme used in PCR is a thermostable enzyme called *Taq* polymerase. It is isolated from a thermostable bacterium called *Thermus aquaticus*. It survives at 95°C for 1-2 minutes and has a half life for more than 2 hours at this temperature. The other thermostable polymerases can also be used in PCR.

1. Working Mechanism of PCR

The action of PCR involves several cycles. However, there are three steps in one amplification cycles e.g *denaturation (melting), annealing* and *polymerisation (extension)*

(*a*) **Denaturation (Melting of Target DNA):** The target DNA containing sequence (between 100 and 5,000 bases) to be amplified is heat denatured (95°C for 15 second) to separate its complementary strands (step 1). This process is called melting of target DNA. After separation each strand acts as template for DNA synthesis.

(*b*) **Primer Annealing :** The second step is the annealing of two oligonucleotide primers to the denatured DNA strands. Since nucleotide sequence of each of oligonucleotide primer is complementary to 3′ end of single stranded template, the primers anneal (hybridises) the each template. Primers are added in excess and the temperature lowered to about 68°C for 60 seconds. Consequently the primers form hydrogen bonds i.e. anneal to the DNA on both sides of the DNA sequence (step 2). The annealing temperature varies, but the too low temperature favours mispairing. The annealing temperature (in °C) can be calculated using the formula: T = 2 (AT) + 4 (G + C).

(*c*) **Primer Extension (Polymerization):** Finally, nucleoside triphosphate (dATP, dGTP, dCTP, dTTP) and a thermostable DNA polymerase are added to the reaction mixture. The DNA polymerase accelerates the polymerization process of primers and, therefore, extends the primers (at 68°C) resulting in synthesis of copies of target DNA sequence (step 3). Only those DNA polymerases which are thermostable *i.e.* function at the high temperature are employed in PCR technique. For this purpose the two popular enzymes, *Taq* polymerase (of a thermophilic bacterium, *Thermus aquaticus)* and the *vent* polymerase (from *Thermococcus litoralis)* are used in PCR technology. These enzymes exhibit relative stability at DNA-melting replenishment after each cycle of synthesis. Also it reduces the cost of PCR and allows automated thermal cycling.

The concentration of Mg^{++} is maintained between 1 and 4 mM. Thus in the first step the target DNA is copied from the primer sites for various distances until the start of second cycle. However, after completion of step 3 (of one cycle) the targeted sequences on both strands are copied and four strands are produced. Now, the three step cycle (first cycle) is repeated which yields 8 copies from four strands. Always after *n* number of cycles, 2^n molecules of DNA are generated using single stranded DNA as template.

PCR efficiency is expressed following the formula as given below :

PCR product yield = Input target amount × $(1+\%\ \text{efficiency})^n$; where n is the number of cycles.

You can calculate that by using an amplification efficiency value of 70%, about 26 cycles are required to produce 1 g of PCR product from 1 pg of a target sequence (10^6 amplification [1 g PCR product = (1 pg target) × $(1 + 0.7)^{26}$].

Variations of PCR

PCR is highly versatile technique; it has been modified in a variety of ways to suit specific situations and aplications. Some important variations of PCR are briefly described in the follwing sections.

Inverse PCR

PCR can be used to amplify the sequence flanking (located on either side of) a segment the border sequences of which are known. This is done by using as primers oligonucleotides complementary to the 5′-ends of the segment in question. This orients the free 3′–OH of the primers outward of the sequence; as a result, the newly synthesized chain grows from the borders of the concerned segment. This is called ***inverse PCR.***

Anchored PCR

When sequence of only one end of the desired segment or gene is known, the primer complementary to the 3′ -end of this stand is used to produce several copies of only one strand of the desired segement. Now a poly-G (or any other homopolymer) tail is added to the 3′-ends (the end for which primer is not available) of the single-strand DNA copies produced by PCR. This allows the use of complementary homopolymer, poly-C, to be used as primer for copying the DNA single-strand generated by PCR and gives rise to the complete DNA duplex that can be amplified normally (we now have primers for both the ends of the newly produced DNA duplex). This approach has been termed as ***anchored*** PCR and the poly-G tail added to the 3′-ends of the single strand copies is called ***anchor***.

Asymmetric PCR

This variation of PCR generates single-strand copies of one strand of template DNA and use one primer at a very low concentration as compared to that of the other

Thermal Cycle Sequencing PCR

In this type of PCR, only one primer is used in combination with dideoxynucleotides; the primer is either radio- or fluorescence-labelled. The product of PCR reaction is subjected to PAGE for determining the base sequence of the template DNA

Nested PCR

In case of ***nested PCR,*** the target sequence is amplified using a pair of PCR primers, and a portion of the amplification product is reamplified using another pair of PCR primers complementary to the regions located immediately beyond the 3′ -ends of the first pair of primers.

Touchdown PCR

This variation of PCR has been devised to increase the specificity of PCR without lowring the efficiency. The PCR is initiated at very high annealing temperatures, which allow only perfectly matched primer-template DNA hybrids to form and support amplification. The annealing temperature is dropped in a stepwise fashion with each cycle of PCR. The logic behind this modification is as follows. The first few cycles will amplify only the target sequence due to the high annealing temperature. These products will be the major template for the subsequent cycles; therefore, annealing temperature becomes less critical during the later cycles. Thus the higher temperature during the initial cycles increases specificity, while lower annealing temperatures during later cycles increase the efficiency without lowering the specificity.

Hotstart PCR

In case of ***hotstart PCR,*** a critical component like Mg^{2+} or *Taq* polymerase, is left out from the reaction mixture. Once the reaction mixture is heated to the denaturation temperature, this reagent is added to the PCR tubes by opening their tops. The logic for hotstart PCR is as follows. Most DNA samples used as template will have some single-stranded DNA. After the reaction mixture is prepared, primers can pair to the single-stranded regions in a nonspecific manner. Since *Taq* polymerase shows a degree of activity at room temperature, it could extend such primers before the temperature is raised to denaturation, causing amplification of spurious sequences. Hotstart PCR was devised to prevent such nonspecific amplifications.

ARMS PCR

The ***amplification refractive mutation system (ARMS) PCR*** was devised to achieve sequence-specific amplification. *Taq* polymerase can initiate DNA synthesis only when there is perfect match at the 3′-end of the primer. Therefore, two alleles can be differentially amplified by designing an appropriate pair of primers: one primer is complementary to a region of the gene that is identical in both the alleles, while the other primer is specific for a region that differs in the two alleles. The 3′-nucleotide of the second primer is that nucleotide, which consistently differs between the two alleles. Such a pair of primers will amplify one allele, but will fail to amplify the other allele. An example of this variant of PCR is the minisatellite variant repeat PCR (MVR-PCR;).

Reverse Transcription-mediated PCR (RT - PCR)

Amplification of double stranded DNA is not limited only to the genomic DNA. It can also be applied to double stranded cDNA also which is synthesised from mRNA using the enzyme reverse transcriptase. Based on this principle RT-PCR was developed. RT-PCR includes a single application combining the process of cDNA synthesis (by reverse transcription) and PCR amplification. Using this principle one can isolate very rare mRNA transcripts from cell samples through RT-PCR. The thermostable enzyme *rTth* uses RNA templates from cDNA synthesis and thus allows single enzyme RT-PCR viral reverse transcriptase (RTase) from avian murine virus (AMV RTase) or mammalian leukemia virus (MLV RTase) can also be used for cDNA synthesis. Both the enzymes i.e. RTase and *rTth* DNA polymerase are used alternatively in the RT-PCR.

Random Amplified Polymorphic DNA (RAPD) marker

It is a PCR-based molecular marker (RAPD analysis) technique. In this method the single short oligonucleotide primers are arbitrarily selected. These primers amplify a set of discrete regions of DNA which are randomly distributed throughout the genome. On the basis of this principle closely related three techniques were developed at the same time such as RAPD (randomly analysed polymorphic DNA), DAF (DNA amplified fingerprinting) and AP-PCR (arbitrarily primed PCR). Therefore, these techniques are collectively called 'multiple arbitrary amplification profiling' (MAAP). All these methods amplify DNA by using single random primers using the same principle with some differences.

Like PCR, RAPD amplification can be carried out by using genomic cDNA (from desired individuals or species) and a single short oligonucleotide primer (about 10 bases). The portion of DNA which is flanked by 10 bp primer in suitable orientation is amplified to get PCR product. The genomic DNA of two different individuals show different RAPD and represents DNA polymorphism. Hence, these differences can be exploited as genetic marker.

Arbitrary Primed PCR (AP-PCR)

It is a type of RAPD where single primers of 10-50 bases are used to amplify genomic DNA in PCR. In the first two cycles, annealing in under non-stringent conditions. AP-PCR differs from RAPD in some respects: (*i*) in AP-PCR amplification occurs in three parts, with its own stringency and concentration of constituents, (*ii*) higher concentrations of primers are used in the first cycles of AP-PCR. Primers of different length are arbitrarily selected e.g. M13 universal sequence primer, (*iii*) products of AP-PCR are analysed on acrylamide gels and detected by autoradiography, whereas these of RAPD are analysed on agarose gels, stained with ethidium bromide and visualised under UV light.

DNA Amplification Fingerprinting (DAF)

Single arbitrary primer (about 5 bases long) is used to amplify DNA by using PCR. In this technique the parameters are carefully optimised. But for early and easy determination of amplified products, automation and fluorescent tagging of primers are extremely amenable. DAF differs from RAPD as (*i*) it requires higher concentration of shorter primers (5-8 nucleotides), (*ii*) two temperature cycles are used in contrast to three cycles in RAPD, (*iii*) it produces a very complex banding pattern, (*iv*) DAF products are analysed on acrylamide gels and detected by using silver staining.

Nucleic Acid Hybridization

The phenomenon of complementary base pairing between two DNA strands, and between one DNA and one RNA strand having similar base sequences has been exploited to develop a very large number of sensitive techniques;

In situ hybridization has been used to establish the location of sat-DNA in chromosomes. For this purpose, radioactive copies of sat-DNA or its complementary RNA (using sat-DNA as templete) are prepared. Chromosomes in squash preparations are specially pretreated to expose and denature their DNA without affecting their structural integrity. These chromosomes are then loaded with the radioactive single-stranded sat-DNA or its complementary RNA. After an appropriate interval, the squash preparations are washed to remove the nonhybridized radioactive DNA (or RNA) probe and the location of radioactivity in the chromosomes is determined through the radioautographic technique. The ***radioautographic technique*** consists of covering the squash preparations with a special type of photofilm or photographic emulsion. These slides are then stored in dark for a suitable duration of time after which the film is developed. When such a squash preparation is now viewed under light microscope the areas having radioactivity would show silver grains on the photographic film. Using this technique it was shown that sat-DNA is mainly located in the centromeric and telomeric regions of chromosomes.

In situ or cytological hybridization is also used to locate specific genes in chromosomes, especially in giant chromosomes. For this purpose, a radioactive clone representing a gene, most often labelled cDNA copies of the mRNA produced by the gene, is used as probe. This approach has enabled. The location of specific sequences in particular bands of salivary gland chromosomes of Drosophilla Colony hybridization is also a form of *in situ* hybridization.

Labeling of Nucleic Acids

The techniques of labeling of nucleic acids are used chiefly to produce radioactively or otherwise labelled probes (RNA or DNA), which are essential in Southern hybridization, Northern blotting, colony hybridization, *in situ* hybridization etc. The various techniques for labeling nucleic acids are as follows : (1) nick translation, (2) primer extension, (3) methods based on RNA polymerase, (4) end labeling method, and (5) direct labeling methods.

Nick Translation

This technique is quite flexible with respect to probe size, specific activity and concentration, it is particularly suited for the production of large quantities of probe for use in multiple hybridization reactions and/or where a high probe concentration is required.

1. The DNA preparation to be labelled is treated with DNAase I for a very short time; this induces nicks in the DNA molecules, the positions of the nicks being random. A ***'nick'*** is the point in one strand of a double-stranded DNA molecule where the phosphodiester bond has been broken; thus each nick has a free 3′—OH group.
2. The nicked DNA sample is subjected to *E. coli* DNA polymerase I, which successively adds new nucleotides to the free 3′—OH group. As the new segment of DNA is synthesized in this way, the existing copy of the strand in the duplex is progressively displaced and digested away by the 5′ —> 3′ exonuclease activity of the enzyme.

The process is called 'nick translation' because there is a movement (translation) of the 'nick' along the DNA duplex due to activities of *E. coli* DNA polymerase I. Nick translation can be used with a variety of labels to generate probes suitable for most hybridization applications after which the film is developed.

When such a squash preparation is now viewed under-light microscope, the areas having radioactivity would show silver grains on the photographic film. Using this technique it was shown that sat-DNA is mainly located in the centromeric and telomeric regions of chromosomes.

Blotting

Southern Hybridization

The name of this technique is derived from the name of its inventor, E.M. Southern.

In Southern hybridization, a sample of DNA containing fragments of different sizes is subjected to electrophoresis using either polyacrylamide or agarose gel. The DNA sample may either be subjected to mechanical shearing or restriction endonuclease digestion in order to generate the fragments. Agarose gel is useful in separating DNA fragments of few hundred to 20 kb in size, while polyacrylamide is preferred for smaller fragments. Very large DNA fragments of upto 1000-2000 kb are separated in agarose gel with pulsed electrical fields, or field inversion.

The gel provides a complex network of polymeric molecules through which DNA fragment migrate, depending on their sizes, under an electric field since DNA molecules are negatively changed Smaller molecules of DNA migrate relatively faster than do the larger ones. Marker DNA fragment of known sizes are run in a separate lane; this permits an accurate determination of size of an unknown DNA molecule by interpolation. The gels are stained with intercalating dye ethidium bromide, which gives visible fluorescence or illumination of the gel with UV light; as little as 0.05 g of DNA in one band can be detected by using this dye.This approach is useful when few DNA fragments with considerable lengths are to be separated and studied. This approach also separates the closed circular (supercoiled), nicked (relaxed) and linear configurations of a single DNA molecule.

The Southern blotting technique is extremely sensitive. It can be used to map the restriction sites around a single copy gene sequence in any genome (even of man). It is used for DNA fingerprinting, preparation of RFLP maps, detection and identification of the transferred genes in transgenic individuals etc.

Northern Hybridization

In this technique, RNAs are separated by gel electroporesis, the RNA bands are transferred onto a suitable membrane, e.g., diazobenzyloxymethyl (DBM) paper or nylon membranes, and immobilized the bands are hybridized with radioactive single-stranded DNA probes, and the bands showing hybridization are detected by autoradiography.

Northern hybridization is useful in the identification and separation of the RNA that is complementary to a specific DNA probe; this is a sensitive test for the transcription of a DNA sequence that is used as probe.

Western Blotting

In Western blotting, proteins are electrophoresed in polyacrylamide gel, transferred onto a nitrocellulose or nylon membrane (to which they bind strongly), and the protein bands are detected by their specific interaction with antibodies, lectins or some other compounds. The various steps of this technique are briefly described below.

1. Proteins are separated by polyacrylamide gel electrophoresis.
2. The protein bands are transferred onto a nitrocellulose or nylon membrane.

3. The specific protein bands are identified in a variety of ways, (*i*) Antibodies are the most commonly used as probes for detecting specific antigens. (*ii*) Lectins are used as probes for the identification of glycoproteins. Often the identification process is based on a ***'sandwich' reaction***. In such an approach, a species-specific second antibody or protein A of *Staphylococcus aureus* (protein A binds to certain subclasses of IgG antibodies) or streptavidin (it binds to biotinylated antibodies) is used to bind to the antibodies bound to the protein bands. These second molecules may be labelled with radioactive, enzyme or fluorescent tags; a single preparation of these labelled molecules can be employed as a general detector for various probes.

Dot Blot Technique

This technique is used to detect the presence of a given sequence of DNA/RNA in the nonfractionated DNA; sample DNA's from several tissues/individuals can be tested in a single test-run. Dot blots are useful in detecting the presence of the sequence being transferred in a number of suspected transgenic individuals, the presence of the specific mRNA in several such individuals or in different tissues of a single individual. The generalized procedure of dot blot is briefly described below.

1. The sample DNAs (or RNAs) from different individuals or tissues are transferred onto a nitro cellulose in form of dots; several samples are dot-blotted onto a single filter. A sample DNA or RNA is the total DNA or RNA extracted from one individual
2. The DNA is first denatured and then the filter is baked at 80°C to fix the DNA firmly onto the filter. The filter is then pretreated appropriately to prevent nonspecific binding of the probe to the filter (see, Southern hybridization).
3. The filter is then treated with the appropriate radioactive single-stranded DNA probe under conditions favouring hybridization. The filter is then washed repeatedly to remove the free probe.
4. The dots having the appropriate DNA or RNA sequence will hybridize with the radioactive probe. These dots are detected by autoradiography; the intensity of a dot in the autoradiograph corresponds fairly well with the extent to which DNA or RNA is represented in the concerned clone.

DNA Liberary

A. Genomic Library

A ***genomic library*** is a collection of plasmid clones or phage lysates containing recombinant DNA molecules so that the sum total of DNA inserts in this collection, ideally, represents the entire genome of the concerned organism.

For preparation of a genomic library, the total genomic DNA of an organism is extracted. The DNA is broken into fragments of appropriate size either by mechanical shearing (this generates blunt-ended fragments), or sonication, or by using a suitable restriction endonuclease for partial digestion of the DNA; complete digestion is avoided since it generates fragments that are too heterogeneous in size. For partial digestion, restriction enzymes having 4-base (tetrameric) recognition sequences are employed in preference to those having 6-base (hexameric) target sites.

The partial digests of genomic DNA are subjected to agarose gel electrophoresis or sucrose gradient centrifugation for separation from the mixture of fragments of appropriate size. These fragments are then inserted into a suitable vector for cloning. This constitutes the ***shotgun approach*** to gene cloning.

The minimum size (number of clones or bacterial clolonies) of a genomic library depends on the following two factors :

1. Complexity of the genome and
2. Insert or fragment length used for cloning

B. cDNA Library

A ***cDNA library*** is a population of bacterial transformants or phage lysates in which each mRNA isolated from an organism or a tissue is represented as its cDNA insertion in a plasmid or a phage vector. The frequency of a specific cDNA in such a library would ordinarily depend on the frequency of concerned mRNA in the tissue/organism.

Preparation of cDNA

cDNA is the copy or complementary DNA produced by using mRNA (usually) as a template. In fact, any RNA molecule can be used to produce cDNA. DNA copy of an RNA molecule is produced by the enzyme ***reverse transcriptase*** (*RNA-dependent DNA polymerase* : discovered by **Temin** and **Baltimore** in 1970) generally obtained from avain myeloblastosis viruse (AMV). This enzyme performs similar reactions as DNA polymerase, and has an absolute requirement for a primer with a free 3′-OH.

Method of Clone Identification

Colony Hybridization

This technique is used to identify those bacterial colonies in a plate, which contain a specific DNA sequence. These bacterial colonies are obtained from bacterial cells into which this sequence was introduced through genetic engineering, and the given sequence is represented by the probe used in the hybridization experiment. The procedure for colony hybridization is briefly described below.

1. The bacterial cells subjected to transformation (*i.e.,* introduction of the vector containing a given sequence into bacterial cells) are plated on to a suitable agar plate; this is the ***master plate.***
2. The colonies of the master plate are replica-plated on to a nitrocellulose filter membrane placed on agar medium. For ***replica plating,*** a block of wood or cork, suitable for the master plate is covered with velvet cloth. This block is sterilized and then lowered into the master plate till the velvet touches all the colonies; the block is withdrawn and gently lowered onto the nitrocellulose filter so that the bacterial cells sticking on to the velvet are transferred on to the filter. The master plate is retained intact for later use. A reference point is marked both on the master plate and on the replica plate to facilitate later comparisons.
3. After colonies appear on the filter, the filter is removed from agar plate and treated with alkali to lyse the bacterial cells. This also dinatures the DNA of these cells.
4. The filter is treated with proteinase K to digest and remove the proteins; the denaturated DNA remains bound to the filter.
5. The filter is now baked at 80°C to fix the DNA onto the filter; this yields DNA-print of the bacterial colonies in the same relative positions as those of the colonies themselves in the master plate.

Expression Vector

An expression vector contains regulatory elements allowing the expression of any foreign DNA it carries. A foreign gene present on expression vector can be efficiently transcribed and translated by the host cell. The simplest expression vectors, *transcription vectors,* allow transcription but not translation of cloned foreign DNA, Typical *protein expression vector* allow both the transcription and translation of cloned DNA, and thus facilitate the production of recombinant protein. Such vectors are equipped with transcriptional regulatory sequences and sequences that control RNA processing and protein synthesis.

For transcription, a *promoter site* and a *terminator site* are necessary. Transcription of the desired gene begins at the promoter site and ends at the terminator site. A *ribosome binding site* upstream from the start codon is also present in many of the expression vectors. This site is required for the efficient initiation of in bacteria.

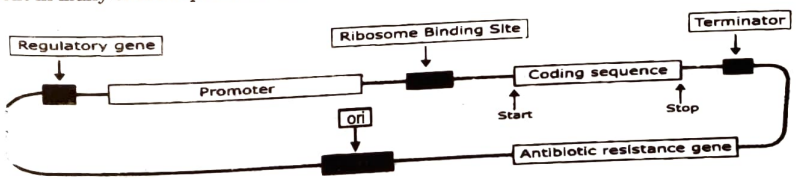


Figure. Components of a typical prokaryotic expression vector.

Promoter is the most critical component of an expression vector since it controls the very first stage of gene expression and also regulates the rate of transcription. An expression vector should carry a *strong* promoter so that highest possible rate of gene expression could be achieved. Regulation of promoter is important factor to be considered during construction of an expression vector. Two important ways of regulating a promoter in *E.coli* are :

Induction : Where transcription of a gene is switched on by addition of a chemical.

Repression : Where gene transcription is switched off upon addition of a regulatory chemical.

Most frequently used promoter for an E.coli expression vector :

The *lac* promoter : It regulates transcription of *lacZ* gene coding for β-galactosidase. It can be induced by isopropylthiogalactoside (IPTG).

The *trp* promoter : It regulates transcription of a cluster of genes involved in tryptophan biosynthesis. It is repressed by tryptophan and easily induced by 3-β-indoleacrylic acid.

The *tac* promoter : It is a hybrid of *trp* and *lac* promoter but is stronger than either of them. It is induced by IPTG.

The λP_L promoter : It is a very strong promoter that regulates transcription of λDNA molecule in infected bacterium. It is repressed by product of λcI gene called λ repressor.

Reporter gene

Reporter gene is a gene producing a protein that can be detected and quantified using a simple assay. They can be used to visualize and measure the efficiency of gene expression, and to investigate intracellular protein localization.

The rate at which a structural gene is expressed depends on its upstream control sequences. Consequently, the rate of expression of a gene can be monitored by replacing its protein-encoding portion with or fusing it in frame to a gene expressing a protein whose presence can be easily determined. Many of the studies to investigate the function of promoters, enhancers and the factors which bind to them have made use of reporter genes. Several reporter genes are used. Most commonly used examples of reporter are :

- *lacZ* gene of *E. coli :*

  The *lacZ* gene acts as reporter gene in the presence of X-gal because its level of expression is readily quantitated by the intensity of the blue color that is generated.

- *Chloramphenicol acetyl transferase* (CAT) encoding gene of *E.coli :*

  The CAT gene encodes the enzyme *chloramphenicol acetyltransferase,* which transfers acetyl groups from acetyl CoA onto the antibiotic chloramphenicol.

- *Luciferase* encoding gene of firefly, *Photinus pyralis :*

  Luciferase catalyzes the oxidation of luciferin with the emission of yellow-green light which can be detected easily even at low levels.

- *Green fluorescent protein* (GFP) encoding gene of bioluminescent jellyfish *Aequorea victoria :*

  GFP (discovered by Shimomura) is an autofluorescent protein (238 amino acid residues) of bioluminescent jellyfish *Aequorea victoria.* In GFP, eleven beta-strands make up the beta-barrel and an alpha-helix runs through the center. The chromophore is located in the middle of the beta-barrel. Amino acid residues 65-67 (Ser-Tyr-Gly) form the fluorescent chromophore, p-hydroxybenzylideneimidazolinone.The chromophore of GFP is responsible for its fluorescence. It fluoresces with a peak wavelength of 508 nm (green light) when irradiated by UV or blue light (optimally 400 nm).

  GFP serve as a unique reporter and used as a fusion tag for monitoring protein localization. GFP is used as a tag in a fusion protein where it is coupled to the protein whose expression is to be tracked. In such cases, the principal aim is to investigate the subcellular localization of the protein under investigation. Genetic engineering can be used to produce vectors containing a GFP coding sequence into which a coding sequence for an uncharacterized protein, X, can be cloned. The resulting GFP-X fusion construct can be transfected into suitable target cells and expression of the GFP-X fusion protein can be monitored to track the subcellular location of the protein.

Table : List of the common reporter genes

Reporter	*Origin*
Chloramphenicol acetyl transferase (CAT)	*E. coli*
β-galactosidase	*E. coli*
β-glucuronidase (GUS)	*E. coli*
Luciferase	Firefly
Green fluorescent protein (GFP)	Jellyfish *Aequorea victoria*
β-lactamase	*E. coli*

Expression system

The heterologous proteins production involves suitable *expression system.* Prokaryotic and eukaryotic systems are the two general categories of expression systems. There is no universal expression system for heterologous proteins production. All expression systems have some advantages as well as some disadvantages that should be considered in selecting which one to use.

Prokaryotic expression systems

E. coli is by far the most widely employed host, provided post-translational modifications of the product are not essential. It combines high growth rates along with the ability to express high levels of heterologous proteins. Strains used for recombinant production have been genetically manipulated so that they are generally regarded as safe for large-scale fermentation. Purification has been greatly simplified by producing recombinant fusion proteins which can be affinity-purified, e.g. glutathione-S-transferase and maltose-binding fusion proteins.

Eukaryotic expression systems

In prokaryotic expression systems, most protein products of cloned eukaryotic genes become insoluble aggregates called *inclusion bodies* and are very difficult to recover as functional proteins. Another problem is that prokaryotes do not carry out the same kinds of post-translational modification such as glycosylation, phosphorylation as eukaryotes do. This can affect a protein's activity or stability, or at least its response to antibodies. A more serious problem is that the interior of a bacterial cell is not as conducive to proper folding of eukaryotic proteins as the interior of a eukaryotic cell. Frequently, the result is improperly folded, inactive products of cloned genes. These problems can be mitigated by using eukaryotic expression system. Eukaryotic systems for the expression of protein include: yeast, mammalian cells and baculovirus cells (insect).

Advantages of eukaryotic protein expression systems include very high levels of expression. The proteins are easy to purify using special tags which are included into the vectors including His, Myc and other tags. The disadvantages of eukaryotic protein expression systems include the fact that eukaryotic cells do grow slower than prokaryotic cells.

Fusion protein

Proteins may be expressed by expression vectors as native polypeptide or fusion proteins, the latter often to facilitate protein purification or analysis. *Fusion proteins* (also called hybrid protein, chimeric protein) are product of two or more coding sequences from different genes that have been cloned together and that, after translation, form a single polypeptide sequence. Fusion protein protects the cloned gene product from attack by host cell proteases. In a number of cases cloned gene proteins have been found to be resistant to degradation when they are part of a fusion protein, whereas, when they are expressed as separate intact proteins, they are susceptible to degradation by proteolytic enzymes (proteolysis).

Fusion proteins are constructed at the DNA level by ligating together a portion of the coding regions of two or more genes. In its simplest form, a fusion vector system entails the insertion of a target gene or gene segment into the coding region of a cloned host gene. Knowledge of the nucleotide sequences of the various coding segments that are joined at the DNA level is necessary to ensure that ligation produces the correct reading frame.

Protein-protein interaction

Study of interaction between different proteins is very useful as it yields information about the tentative function of a protein through its interaction with well characterized protein with known role or function. **Phage display** and **yeast two hybrid system** are two well known and most useful methods for studying protein-protein interactions.

Phage display method was first described by **George Smith** is 1985. It involves display of a heterologous target protein under study on phage particles. Phage display is a form of expression cloning of foreign genes using phage. Genetic engineering techniques are used to insert foreign DNA fragments into a suitable phage coat protein gene. Initially phage display involved the use of filamentous phages such as fd, fl, M13, where the foreign gene was incorporated into a gene specifying a minor coat protein. The recombinant gene can then be expressed as a fusion protein which is incorporated into the virion and displayed on the surface of the phage which, however, retains infectivity. If an antibody is available for a specific protein, phage displaying the protein can be selected by preferential binding to the antibody.

There are two main methods in which phage display can be used to study protein interaction. In one method test protein is displayed and its interaction is sought with a series of purified proteins or protein fragments of known function. In second method a phage display library is prepared. Phage display library made up of many recombinant phage, each displaying a different protein. These libraries can be prepared by cloning a mixture of cDNAs from a particular tissue by cloning genomic DNA fragments. The library consists of phage displaying a range of different proteins and is used to identify those that interact with a test protein.

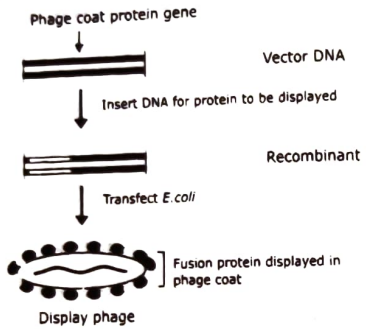


Figure. DNA coding for test protein is cloned into phage vectors in order to express hybrid protein (test protein fusion to the coat protein) on the phage surface. Phage particles produced by these transformed bacteria therefore display the test protein in their coats.

Yeast two hybrid assay detects protein interactions in a comparatively more complex way. It is based on the fact that eukaryotic gene expression is promoted by transcription factors called *activators* that contain two distinct domains. One is *DNA binding domain* that binds to regulatory sequence and another is *activation domain* that interacts with RNA polymerase enzyme and stimulates expression of associated gene. These two domains can be physically separated from each other in which case protein loses its activity but is reactivated when the two domains are brough together.

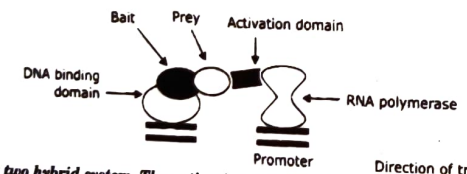


Figure : Yeast two hybrid system. The activation domain of the yeast transcription factor is fused with a polypeptide called prey. Prey interacts with another polypeptide, bait, which in turn is fused with the DNA-binding domain of the same transcription factor. Successful transcription of an essential gene is possible only when the bait and the prey interact, bringing together the DNA-binding and the activation domains of the transcription factor.

Yeast two hybrid system makes use of a *Saccharomyces cerevisiae* strain which lacks activator for a particular reporter gene. Gene for one of the interacting protein called *bait* is fused to the gene coding for DNA binding domain of the activator and specifies synthesis of fusion protein made up of DNA-binding domain and test protein. The recombinant yeast strain is co-transformed with a second construct carrying gene, for the second interacting protein called *prey*, from a c-DNA expression library and gene for activation domain. Reporter gene is expressed only in transformed cells in which *bait* and *prey* genes code for proteins that interact with each other. This given a *bait* gene its interacting pair the *prey* can be cloned using a yeast two hybrid system.

Microarrays

An ***array*** is an orderly arrangement of data. The strategy of ***microarrays*** is the exact opposite of that of dot blot assay. In the ***case of microarrays,*** a series of probes are immobilized on a glass slide as microdots, which are then hybridized with a mixture of test DNA sequence that are labelled with a fluorochrome. Microarrays are used to determine by a single assay the identity of genes expressed in a specific tissue of an organism at a given time.

Microarrays were first used in case of yeast.

Types of Microarrays

At present, microarrays are basically of two types :

1. DNA microarrays (being widely used), and
2. Antibody microarrays.

DNA microarrays are of two types (*i*) spotted microarrays and (*ii*) oligonucleotide microarrays. In case of ***spotted DNA microarrays,*** DNA sequences representing different genes of an organism are spotted onto the slides; these microarrays are used to determine genes being transcribed in the cells concerned, *i.e.,* their transcriptome. ***Transcriptome*** is the total mRNA content (= composition) of a cell, and reflects the overall pattern of gene expression in that cell The DNA microarrays can be prepared by using DNA sequences derived in one the following ways.

1. Sequences obtained from individual cloned genes.
2. A systematized collection of cDNAs, often derived from collections of expressed sequence tags (ESTs). ***ESTs*** are short sequences obtained from the ends of cDNAs, *i.e.,* partial gene sequences, and are useful in identifying the locations of genes or, now, in construction of microarrays. Thus each EST represents a different gene, although its sequence may not enable a direct identification of the gene.

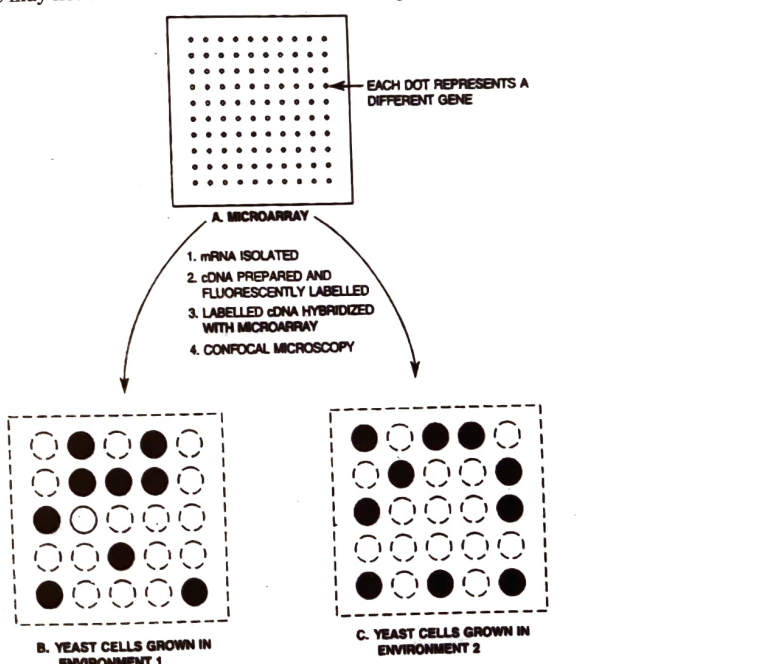


Fig. Use of DNA microarrays in yeast. A. A DNA microarray (only 10 x 10 spots are shown). B and C. Microarrays as seen by confocal microscopy; the dark spots represent fluorescence (= hybridization with labelled cDNA) and hence the transcribed genes. A comparison of B and C reveals the identity of genes that are differentially expressed under these two environments. If cDNAs from the two yeast populations were labelled with different fluorophores, both of them can be hybridized with the same microarray, and differences in the colours of the spots will reveal the pattern of expression.

3. ESTs or cDNAs from a related species.

The ***antibodies microarrays***, on the other hand, consit of spots of different antibodies and are used in measure the abundance of thousands of different proteins in the samples; these are in developmental stages.

DNA Chips (Oligonucleoitde Microarrays)

DNA chips are thin wafers of silicon glass carrying many different oligonucleotides synthesized at a very high density (300,000 to over 1 million oligonucleotides/cm^2) directly onto the wafer.

Applications of DNA Microarrays

1. Studies of the gene expression pattern of an organism as affected by the stage of development and/or environment.
2. Identification of common regulatory elements shared by coregulated genes.
3. DNA chips can be prepared for the detection of SNPs (single nucleotide polymorphisms); these are called ***SNP chips***.
4. DNA chips can be used for the detection of genetic diseases, *e.g.*, for detection of mutant alleles causing cystic fibrosis, mutant alleles of gene *BRCA1* (gene involved in breast cancer), etc.

Site-Directed Mutagenesis

Introduction of predetermined changes at specific sites (also predetermined) of a DNA molecule is known as ***site-directed mutagenesis***. Several different techniques allow site-directed mutagenesis all these techniques are based on recombinant DNA technology. Some of these techniques are briefly outlined below.

A. Oligonucleotide-Directed Mutagenesis

An oligonucleotide (15-20-mer) comlementary to the region in which mutation is to be introuduced serves as the primer. This primer has the desired change in base sequence idealty this change should be located in the centre of the primer. The primer is mixed with the single-stranded version of the concerned gene that is integrated in a phage M13 vector; obviously this strand must be complementary to the primer. At sufficiently low temperature and high salt concentration, the primer will pair with the template DNA inspite of the mismatch resulting from the change in base sequence of the primer. The primer is extened by DNA polymerase to produce the complementary strand of the whole recombinant DNA molecule and the nick is sealed with DNA ligase. This generates the double-standed version of the M13 vector plus the gene; the newly synthesized strand contains the desired mutation.

The recombinant DNA is now introduced into *E. coli*. Replication will produce the following two kinds of recombinant DNAs :

(1) The one having the wild type gene and

(2) The other having the mutant gene. The recombinant clones of *E. coli* having the mutant gene are identified using the labelled primer as probe; at carefully selected temperature and salt concentration, this probe will hybridize with the mutant gene but not with the wild type gene. The recombinant DNA is recovered from such clones, and the mutant gene is isolated and expressed to generate the mutant protein. This technique also allows deletion or insertion of predetermined bases (ordinarily, complete codons are added or deleted). In such a case, the primer (usually, 30-mer) contains the desired deletion or addition (in the place of base substitution) located, ideally, at its centre.

B. Casette Mutagenesis

This technique can be used when the mutation is to be introduced in a region that is flanked by unique restriction sites located close to this region. The target region is excised by using the concerned restriction enzyme(s), and a synthetic oligonucleotide is inserted in its place. The base squence of this oligonucleotide is defined by the changes (= base substitutions, deletions, additions) that were planned in the base sequence of the excised region. A mixture of oligonucleotides, each containing, a different change in the base sequence, may be used to produced variety of changes in this region of the gene. The reconstructed gene(s) is/are expressed to recover the mutant protein(s) for further analyses.

Genome Mapping

Genome means a single set of chromosomes, refers ordinarily to nuclear chromosomes and is denoted by x. A genome map may be defined as a detailed schematic description of the structural and functional organization of the complete genome of an organism.

Genetic Maps

A ***genetic map*** is a schematic representation of the various genetic markers in the specific order in which they are located in a chromosome as well as the relative distances between these markers. Three different strategies have been used to construct genetic maps, which are usually referred to as linkage maps, cytogenetic or cytological maps and physical maps.

1. Linkage Maps

A ***linkage map*** depicts the order of genetic markers and the relative distances between them as measured in terms of recombination frequencies between the markers. A ***map unit*** or ***centrimorgan*** (cM) is that distance, which allows 1% recombination between the concerned genes.

2. Cytogenetic Maps

A ***cytogenetic map*** depicts the locations of various genes in a chromosome relative to specific microscopically visible landmarks in the chromosome. In most cases, each chromosome has a characteristic banding pattern, which may be either naturally present, *e.g.*, in polytene chromosomes of *Drosophila*, or more commonly generated by specific staining protocols, *e.g.*, in case of human chromosomes; the genes are mapped cytologically relative to these band locations.

Fluorescence *in situ* hybridization (FISH). *In situ* hybridization (ISH) is widely used to map the cytological locations of genes and other DNA sequence within large eukaryotic chromosomes. In this approach, the DNA of concerned gene/DNA sequence is used as a probe. The probes are usually fluorescently labelled so that this form of *in situ* hybridization is called ***fluorescent in situ hybridization (FISH)***. A ***fluorescent molecule*** absorbs light at a particular wavelength and then emits light at a longer wavelength.

3. Physical Maps

In a ***physical map,*** genes are depicted in the same order as they occur in chromosome, and the distances between them are shown as number of base pairs separating the genes. For example, genes *sc* and *w* of *Drosophila* are separated by ~ 1.5×10^6 bp. *The ultimate of physical mapping is the complete sequencing of a genome.*

Genetic Markers

Any trait of an organism that can be identified with confidence and relative ease, and can be followed in a mapping population is called ***genetic marker***. Genetic markers can be detected with the naked eye (naked eye polymorphism), as differences in electrophoretic mobility of specific proteins, or as differences in specific DNA sequence (detectable as RFLP or differential amplification by PCR).

Naked eye Polymorphisms (NEP)

Those genetic markers whose inheritance can be followed with the naked eye constitute ***nake-eye polymorphisms***. The traits included in this group are plant height, disease response, photoperiod sensitivity, shape or colour of flowers, fruits or seeds and many other visually scorable traits.

Sequence-Tagged Sites (STSs)

A ***sequence-tagged site (STS)*** is any site on the genome that is unambiguously defined in terms of flanking primers that are used for PCR amplification of this site. Thus when a pair of PCR primers amplifies only a single sequence within a haploid genome, the amplified region of the genome is called a ***sequence tagged site***. STSs can be created in the following ways.

1. Sequencing the two ends of a PCR amplification product (***amplicon*** - the DNA sequence amplified using a set of primers) beyond the ends of the RAPD primer makes it possible to create improved primer sequences that are far more reproducible than are RAPDs.
2. The ends of an RFLP DNA fragment may be sequenced and the information can be used to create primers that allow PCR amplification of this RFLP locus, and thus convert this RFLP into an STS locus.

3. STSs are often created by determining the unique sequences flanking mini- and microsatellite sites and using this information to design primers for PCR amplification of these sites.
4. Sequences of 200-400 bp long DNA segments obtained randomly from the genome may be determined and primers may be designed to amplify these sites in the genome. A proportion of these sites may turn out to be useful STSs.

DNA Fingerprinting

The technique of DNA fingerprinting was developed by **Alec Jeffreys** in 1985. DNA fingerprinting or DNA profiling is generally used for the indentification of criminals from blood stains, semen etc. and for establishing parentage in case of dispute. The procedure for DNA fingerprinting is briefly described below.

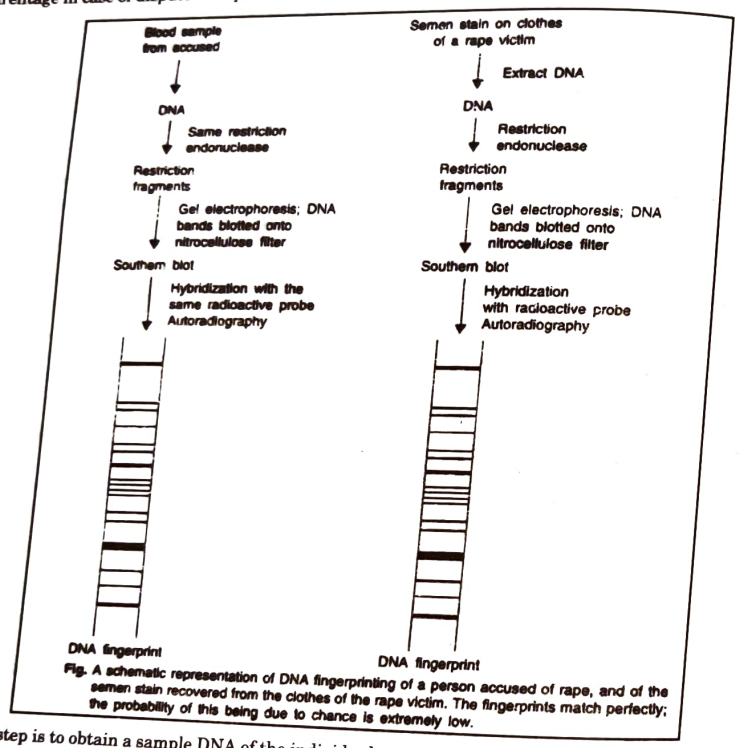


Fig. A schematic representation of DNA fingerprinting of a person accused of rape, and of the semen stain recovered from the clothes of the rape victim. The fingerprints match perfectly; the probability of this being due to chance is extremely low.

1. The first step is to obtain a sample DNA of the individual.
2. The DNA is digested with a suitable restriction endonuclease, and the digest is subjected to gel electrophoresis.
3. The DNA is denatured and then transferred from the gel on to a nitrocellulose filter membrane, where it is fixed by baking the filter at 80°C.
4. The DNA bands on the filter are hybridized with the appropriate radioactive probe (single-stranded DNA probe); the free probe is washed off and the bands to which the probe has hybridized are detected through autoradiograph.

The probes used for DNA fingerprinting are usually prepared from mini-satellite DNA.

The bands appearing in the DNA fingerprint of the suspected criminal(s) are compared with those obtained from the test DNA recovered from the evidence for crime there should be a perfect match in case the suspect had been involved.

PLANT BIOTECHNOLOGY

The beginning of plant tissue culture was made as early as 1898, when a German botanist, **G. Haberlandt** successfully cultured fully differentiated individual plant cells, isolated from different tissues in several plant species. During 1934-1939, due to discovery of the importance of auxins and B vitamins, the foundation of plant tissue culture was laid down by three scientists **(Gautheret, White and Nobecourt),** even though only small pieces of tissue and not individual differentiated cells could be grown in cultures.

Culture Media

MS media, which evolved due to a variety of modifications of the original MS medium **(Murashige and Skoog,** 1962). The pH of the medium is usually adjusted between 5.0 and 6.0. B_5 and White medium are also used as a media for plant cell suspeusion cultures.

The major constituents of a culture medium include (*i*) inorganic nutrients, (*ii*) organic nutrients, (*iii*) growth hormones and (*iv*) agar.

Cell Culture

(A) Isolation of single cells

Single cells can be isolated either from cultured tissues or from intact plant organs.

(B) Suspension cultures

Suspension cultures can be maintained in either of the following two forms:

(*i*) **Batch cultures,** are initiated as single cells in 100 - 250 ml flasks and are propagated by transferring regularly small aliquots of suspension to a fresh medium.

(*ii*) **Continuous cultures,** are maintained in a steady state for long period by draining out the used medium and adding fresh medium; in this process either the cells separated from drained medium are added back to suspension culture (these are called **'close continuous cultures')** or the addition of medium is accompanied by the harvest of an equal volume of suspension culture (these are called **'open continuous cultures').**

(C) Culture of single cells (Bergmann's cell plating technique)

In this technique, free cells are suspended in a liquid medium (if cell aggregates are there, it is filtered). Culture medium with agar (0.6 - 1%) is cooled and maintained at 35°C in a water bath. Equal volumes of liquid and agar media are mixed and rapidly spread in a Petri dish, so that cells are evenly distributed in a thin layer, after solidification. The Petri dishes are sealed with parafilm and examined with inverted microscope to mark single cells (marking is done on outer surface of the dish). The Plates are incubated in dark at 25°C and cell colonies developing from marked single cells, are used to obtain single cell cultures.

Applications of Cell Culture

(A) Mutant selection

An important use of cell cultures is in mutant selection in relation to crop improvement.

(B) Biotransformations

Plant cell cultures are also being utilized for producing valuable products including secondary metabolites through biotransformation, a technique also utilized with the help of microbes.

(C) Single cell proteins (SCP)

Single cell cultures are also being used for production of **single cell proteins** (SCP).

In plant tissue culture experiments, unless single cell cultures are absolutely necessary, more often we use **explants** (pieces of differentiated tissues) to initiate their growth in culture.

Cellular Totipotency

The phenomenon of the reversion of mature cells to the meristematic state leading to the formation of **callus** is called **'dedifferentiation'.** The component cells of callus have the ability to form a whole plant, a phenomenon described as **'redifferentiation'.** These two phenomenons of dedifferentiation and redifferentiation are inherent in the capacity described as **'cellular totipotency',** a property found only in plant cells and not in animal cells.

Types of Cultures

Embryo culture.

For embryo culture, embryos are excised from immature seeds, the immature seeds are surface sterilized and soaked in water for a few hours, before the embryos are excised. The excised embryos are directly transferred to a culture dish or culture tube containing synthetic nutrient medium.

The entire operation is carried out in the 'laminar flow cabinet' and the culture plates or culture tubes with excised embryos are transferred to a culture room maintained at a suitable temperature, photoperiod and humidity. The frequency of excised embryos that give rise to seedlings generally varies greatly and the medium may even have to be modified in difficult cases. Most extensive use of this embryo culture technique has been made for making interspecific and intergeneric crosses within the tribe Triticease of the grass family (Poaceae). The hybrids raised through embryo culture have been utilized for (*i*) phylogenetic studies and genome analysis, (*ii*) transfer of useful agronomic traits from wild genera and species to the cultivated crops and (*iii*) to raise synthetic crops like triticale by producing amphiploids from the hybrids.

Embryo culture has also been used for haploid production through distant hybridization followed by elimination of the chromosomes of one parent in the hybrid embryos cultured as above. A popular example includes hybridization of barley and wheat with *Hordeum bulbosum* leading to the production of haploid barley and haploid wheat respectively. Haploid wheat plants have also been successfully obtained through culture of hybrid embryos from wheat × maize crosses.

Ovule culture

For a variety of difficult interspecific/intergeneric crosses involving members of the families Malvaceae, Fabaceae, Cruciferae, Solanaceae, etc., ovules after fertilization have been successfully cultured to obtain mature embryos/seeds. Depending upon, when the embryo aborts, the ovules have to be excised any time from soon after fertilization to almost developed fruits, which may sometimes be lost due to premature abscission. However, ovule culture is mainly tried only in those cases, where embryo aborts very early, and embryo culture is not possible due to difficulty of its excision at a very early stage. In some cases, the medium may need to be supplemented with some fruit/vegetable juice to accelerate initial growth.

Ovary culture

In vitro rooting of the pedicel and a limited growth of ovaries, on a culture medium, were obtained rather early, although technique for growing excised ovaries on culture medium was developed later in several plant materials, including tobacco, tomato, beans, etc. Subsequently mature fruits containing seeds could be successfully obtained from ovaries excised from pollinated flowers (e.g. *Cucumis,* tomato).

Ovary culture is often used either for *in vitro* pollination and fertilization or for embryo rescue, when the embryo culture and ovule culture either fail, or are not feasible due to very small ovules. Interspecific hybrids, using ovary culture, have successfully been obtained in several genera including *Brassica (B. campestris* x *B. oleracea).* For interspecific or intergeneric crosses, ovaries are excised at the zygote stage or at the two celled proembryo stage and normal development is completed *in vitro*. The development of fruit may be promoted by application of chemicals like IAA, or by addition of coconut milk to the medium. Ovary culture, prior to fertilization, may also lead to the development of parthenogenetic haploids. This has been achieved successfully in materials like wheat, barely and other crops.

Ovary culture has also been used for a study of the understanding of physiology of fruit development. For this purpose, specific chemicals may be added and their effect observed by comparing the development with that in control. This is ordinarily not possible, when the treatment is given to the entire plant, because the chemical may undergo alterations, exercising a different effect on ovary growth and development.

Endosperm culture

Tissue culture methods are also used for culturing endosperm, which is unique *firstly,* in its function of supplying nutrition to the developing embryo and *secondly,* in being triploid in its chromosome constitution. Triploid plants are useful for production of seedless fruits (e.g. apple, banana, watermelon, etc.) and for production of trisomic for cytogenetic studies.

The technique of endosperm culture involves the following steps :

(*i*) the immature seeds are disected under aseptic conditions and endosperms along with embryos, are excised; sometimes mature seeds can also be used;

(*ii*) the excised endosperms are cultured on a suitable medium and embryos are removed after initial growth;

(*iii*) the initial callus phase is followed by 'embryogenesis' or 'shoot bud differentiation';

(*iv*) the shoots and roots may subsequently develop and complete triploid plants can be established for further use.

Nucellus culture

Nucellus from pollinated flowers can also be used for micro propagation, particularly in cases where adventive embryos develop from nucellar cells (e.g. *Citrus*). It can be grown on casein hydrolysate supplemented White's medium, giving a callus. The callus may then given rise to pseudobulbils, differentiating into embryoids, which can eventually develop into seedlings. Nucellus from unfertilized ovules may also develop embryoids, when grown under suitable culture conditions. These embryoids can be excised and grown on medium supplemented with GA3, and plantlets can be obtained.

The seedlings obtained from the nucellar tissue are of parental type and thus may be important for maintaining purity of horticultural stocks. In *Citrus,* these may be better than those obtained from cuttings due to following reasons: (*i*) nucellar seedlings have tap root, while those from cuttings have lateral roots; (*ii*) the vigour lost due to repeated propagation by cuttings is restored in nucellar seedlings, (*iii*) nucellar seedlings are disease free.

Protoplast Culture

The culture methods for isolated protoplasts are similar to those for single cells. For instance protoplasts can be cultured following **'Bergmann's cell plating technique'.** Both semi-solid medium and liquid medium can be used, although the liquid medium is preferred. Protoplasts can be suspended in a liquid medium in Erlenmeyer flasks without shaking and can be cultured in small quantities in **'hanging drops'** or in **'microchambers'. Multidrop array (MDA)** techniques may also be used. Within 2-4 days, protoplasts start developing cell walls, which can be detected by staining with 0.1% **calcofluor white** (CFW) fluorescent stain.

While the presence of a proper wall is essential for a regular division, it is not always a prerequisite for nuclear division; not all protoplasts with cell walls formed, embark upon division. The protoplasts, which are capable of dividing, undergo first division within 2-7 days and form multicellular colonies after 2-3 weeks. After another two weeks, these colonies can be treated as standard tissue cultures. From these colonies or tissues in culture, plants can be regenerated, although success in this effort has been achieved only in a few species.

Somatic Embryogenesis

A somatic embryo (SE) is an embryo derived from a somatic cell, other than zygote, and obtained usually on culture of the somatic cells *in vitro;* this process is known as ***somatic embryogenesis.*** In contrast, embryos developing from zygotes are called ***zygotic embryos*** or often simply ***embryos,*** while those derived from pollen are known as ***pollen embryos*** or ***androgenetic embryos.*** By 1978, somatic embryogenesis was reported from 80 species belonging to 33 families.

Developmental Pattern of SEs. SEs generally originate from single cells, which divide to form a group of meristematic cells. Usually, this multicellular group becomes isolated by breaking cytoplasmic connections with the other cells around it and subsequently by cutinization of the outer walls of this differentiating cell mass. The cells of this meristematic mass continue to divide to give rise to globular (round ball shaped), heart-shaped, torpedo and cotyledonary stages of SEs. In general, the essential features of SE development, especially after the globular stage, are comparable to those of zygotic embryos.

Somatic embryos are bipolar structures in that they have a radicle and a plumule. The radicular end is always oriented toward the centre of callus or cell mass, while the plumular end always sticks out from the cell mass. In contrast, a shoot bud is monopolar as it does not have a radicular end. In many SEs, radicle is suppressed so that they often do not produce roots; in such cases, roots have to be regenerated from the shoots produced by germinating SEs.

SEs often show abnormal developmental features, *e.g.*, there or more cotyledons, bell-shaped cotyledon, larger embryo size, etc., these problems are often overcome by the presence of ABA or a suitable concentration of mannitol. In some species, normal looking somatic embryos are formed, but they fail to germinate; at least some SEs do not germinate in most of the species.

The SEs regenerating from an explant or a callus are termed as ***primary somatic embryos.*** In many cases, SEs regenerate from the tissues of other SEs or the parts of germinating SEs; such SEs are called ***secondary somatic embryos.*** Ordinarily SEs originate from cells at the surface of callus or explant, *e.g.*, from epidermal cells of *Ranunculus* stem.

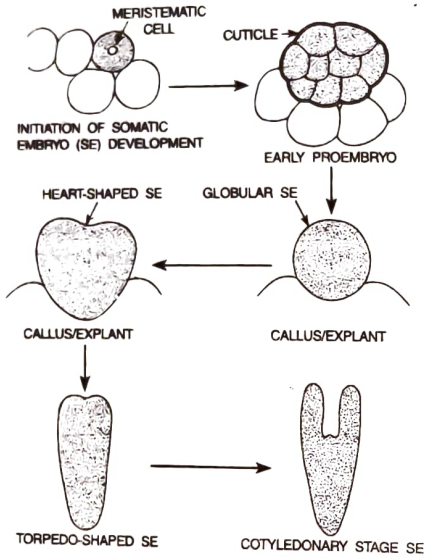


Fig. Development of a somatic embryo from a single superficial cell of the explant.

Somatic embryogenesis is influenced by several factors, *e.g.*, (1) GRs, (2) exposure to a stress, (3) arabinogalactan proteins (AGPs), (4) nitrogen source, (5) explant genotype and (6) other factors.

Organogenesis

Regeneration refers to the development of organised structures like roots, shoots, flower buds. Somatic embryos (SEs), etc. from cultured cells/tissues; the term ***organogenesis*** is also used to describe these events. Root regeneration occurs quite frequently, but it is useful only in case of shoots and embryo germination. Only shoot and SE regenerations give rise to complete plants, which is essential for applications of tissue culture technology in agriculture and horticulture. Often differentiation is used as synonym for regeneration. But differentiation describes the development of different cell types, *e.g.*, vascular elements, etc., ***(cytodifferentiation)*** as well. Therefore, it is more appropriate to use phrases like shoot differentiation, SE differentiation, etc. than using the term differentiation alone. Regeneration may occur either directly from the explant or may follow an intervening callus phase.

Somaclonal Variation and Gametoclonal Variation

The genetic variability present among cultured cells, plants derived from such cells or progeny of such plants is called ***somaclonal variation.*** Another term, ***gametoclonal variation,*** has been proposed for the variability present among pollen derived cells/plants, but the term is rarely used. Generally, the term somaclonal variation is used for genetic variability present among all kinds of cells/plants obtained from cells cultured *in vitro.* Plants regenerated from tissue and cell cultures show heritable variation for both qualitative and quantitative traits. Somaclonal variation has been described in sugarcane, potato, tomato etc. Some variants are obtained in homozygous condition in the plants regenerated from cells cultured *in vitro* ***(R_0 generation)***, but most variants are recovered in the selfed progeny of tissue culture-regenerated plants ***(R_1 generation).***

Somatic Hybridization and Cybridization

The technique of hybrid production through the fusion of somatic cells under *in vitro* conditions, bypassing sex altogether, is called *somatic hybridization*. First report on protoplast fusion between *Nicotiana glauca* and *N. longsdorfii*, was published by Carlson et al., 1972 Unlike sexual reproduction in which organelles genomes are generally contributed by the maternal parent, somatic hybridization combines cytoplasmic organelles from both the parents.

Protoplast fusion can be induced by chemical or electrical methods. Most commonly used chemical fusogens (agents that helps in fusion) are UV-inactivated sendai virus, polyethylene glycol, polyvenyl alcohol, artificial sea-water, and lysozyme. The most successful chemical fusion method is the addition of polyethylene glycol (PEG) in the presence of a high concentration of calcium ions and a pH between 8-10. Electrical fusion of protoplasts involves the application of short pulses of direct electrical current. Electro-fusion is more reproducible than chemical fusion and generally is less damaging to the protoplasts. The frequency of electrical fusion of protoplasts may be higher than chemical fusion.

Fusion of protoplast at the level of plasma membrane is nonspecific and there is no barrier to interspecific or intergeneric fusion of cells. However even if barriers preventing somatic hybridization between two remote species are bypassed by protoplast fusion, barriers may still exist at the genomic level, resulting in spontaneous elimination of chromosome during culture of the hybrid cells. Consequently some of the hybrid plants regenerated from the fusion products lack some or most of the chromosomes of one of the fusion partners. The somatic hybrids that have lost chromosomes of one of the parents and are, therefore, phenotypically closer to the other parent are called **asymmetric hybrids.** If somatic hybrid cells and the plants regenerated from them contain full nuclear genomes from both the fusion partner then it is termed **symmetric hybrids.** Symmetrical nuclear hybrids are generally rare. Most somatic hybrids are predominantly asymmetrical hybrids.

Cybrids, in contrast to conventional hybrids, possess a nuclear genome from only one parent but cytoplast genes from both parents. The process of protoplast fusion resulting in the development of *cybrids is* known as *cybridization*. In cybridization, heterozygosity of extrachromosomal material can be obtained.

Methods to produce cybrids

a. Fusion of normal protoplast with an enucleate protoplast;

b. Fusion between a normal protoplast and a protoplast containing nonviable nucleus;

c. Elimination of one of the nuclei after heterokaryon formation; or

d. Selective elimination of chromosomes after fusion.

Application of Cell and Tissue Culture

Clonal Propagation or Micropropagation

It has been demonstarated that a variety of plant species can be conveniently propagated through the techniques of cell, tissue or organ culture. This is popularly described as **clonal propagation** or **micropropagation**. The major benefits of this method include the following

(*i*) rapid multiplication of superior clones and maintenance of uniformity

(*ii*) multiplication of disease free plants and

(*iii*) multiplication of sexually derived sterile hybrids.

In most cases, clonal propagation is achieved by placing sterilized shoot tips or axillay buds onto a culture medium that is sufficient to induce formation of multiple buds. This method has already been used to propagate a large number of marketable ornamentals. Following stages are involved in the method of clonal propagation:

(*i*) Stage I involves establishement of tissue *in vitro*

(*ii*) Stage II involves multiplication of shoots (often media are not changed between stage I and stage II

(*iii*)Stage III concerns root formation and conditioning of propagules prior to transfer to the green house; this stage requires high intensity and alteration of media for promotion of root formation.

(*iv*) Stage IV involves growth in pots followed by field trials. The number of steps in microporpagation may sometimes be reduced to three or two.

Synthetic Seeds

Where somatic embryos are encapsulated in a suitable matrix (e.g. sodium alginate) along with substances like mycorrhizae, insecticids, fungicids and herbicides. Synthetic seeds have many advantages including the following

(*i*) they can be stored upto a year without loss of viability ;

(*ii*) they are easy to handle, and useful as units of delivery ;

(*iii*) they can be directly sown in the soil like natural seeds and do not need hardening in green house. The only limitation of synthetic seeds, is the high cost of their production.

(*iv*) callus which is usually virus free like meristems.

Virus -Free Plants

Seeds from virus-infected plants generally do not contain the virus. Therefore sexual progeny are usually virus-free, except for new infections. Many viruses are now known to be seed-transmissible. This list is increasing as more and more viruses are closely and extensively investigated. But in case of asexually reproducing crops, virus infections spread rapidly. This is because the vegetative propagules from virus-infected plants contain virus particles. But the shoot apical meristems and some young tissues surrounding them are often free form viruses or contain very low titre of the virus. Meristem culture, therefore, is often useful in recovering viru-free plants from virus-incected plants or clones. Many valuable clones of potato (*Solanum tubersum*), sugarcane (*Saccharum officinarum*) and other clonal crops have been lost due to viral infections.

Transgenic Plants

Transgenic plants are those plants which carry additional stably integrated and expressed foreign genes from transpecies. In higher plants cells or protoplasts can be cultured and used for regeneration of whole plants. Therefore, these protoplasts can be used for gene transfer followed by regeneration leading to the production of transgenic plants. Besides cultured cells and protoplasts, other meristem cells (immature embryos or organs), pollen or zygotes can also be used for gene transfer in plants. The enormous diversity of plant species and the availability of diverse genotypes in a species, made it necessary to develop a variety of techniques, suiting different situations. Cultured cells or protoplasts are often used for gene transfer.

Transformation Techniques

The uptake of foreign DNA or transgenes by plant cells is called ***transformation.*** A variety of techniques have been used to introduce transgenes into plant cells; these can be grouped into the following two broad categories : (1) *Agrobacterium*-mediated and (2) direct gene transfers.

1. *Agrobacterium*-Mediated Gene transfer

Gene transfer through *Agrobacterium* is achieved in the following two ways : (*i*) coculture with tissue explants and (*ii*) in planta transformation.

Coculture with Tissue Explants: The appropriated gene construct is inserted within the T-region of a disarmed Ti plasmid; either a cointegrate or a binary vector is used. The recombinant DNA is placed in *Agrobacterium*, which is then cocultured with the plant cells or tissues to be transformed for about 2 days.

During the leaf disc-*Agrobacterium* coculture, acetosyningone released by plant cells induces the *vir* genes, which together bring about the transfer of recombinant T-DNA into many of the plant cells. The T-DNA would become integrated into the plant genome, and the transgene would be expressed. As a result, the transformed plant cells would become resistant to kanamycin (due to the expression of *neo* gene). After 2 days, the leaf discs are transferred onto a regeneration medium containing appropriate concentrations of kanamycin and carbenicillin. Kanamycin allows only transformed plant cells to divide and regenerate shoots in about 3-4 weeks, while carbenicillin kills *Agrobacterium* cells. The shoots are separated, rooted and finally transferred into soil.

In Plant Transformation: Inhibition of *Arabidopsis* seeds in fresh cultures of *Agrobacterium* leads to stable integration of T-DNA in the *Arabidopsis* genome; the majority of about 14,000 transformants screened represent independent and unique events. It appears that *Agrobacterium* cells enter the seedlings during germination, are retained within the plants, and when flowers develop they transform either the zygotes or the cells that give rise to zygotes. Alternatively, *Arabidopsis* plants about to flower are immersed in a fresh culture of *Agrobacterium*, and partial vaccum is created to facilitate the entry of bacterial cells into the plants. The plants are grown, selfed, and the progeny so obtained are screened for the identification of transformants.

Agroinfection and Gene Transfer

Agroinfection is a phenomenon, in which a virus infects a host as a part of T-DNA of Ti plasmid carried by *Agrobacterium*. Viral DNA can be integrated into the T-DNA and can be delivered into plant cells with the normal *Agrobacterium* T-DNA transfer process. After infection viral DNA is released to form functional virus that replicates and spreads systemically. **Agroinfection** may also lead to the integration of viral DNA so that transgenic plant containing integrated viral DNA can be produced. In maize, agroinfection with maize streak virus has been demonstrated. This suggested that *Agrobacterium* based vector system can be used for genetic engineering in cereals, although ordinarily *Agrobacterium* does not infect moncotyledons. Thus agroinfection can lead to the production of transgenic plants, even though it has no better chances of yielding trangenic cereals than does *Agrobacterium* infection alone. However, agroinfection has great potential for studies in virus biology, because it can transfer deletion mutations or even single viral genes.

2. Direct Gene Transfer

Chemically stimulated DNA uptake by protoplasts

Direct DNA uptake by protoplasts can be stimulated by chemicals like polyethylene glycole (PEG). The technique is so efficient that virtually every protoplast system has proven transformable. PEG is also used to stimulate the uptake of liposomes and to improve the efficiency of electroporation.PEG at high concentration (15-25%) will precipitate ionic macromolecules such as DNA and stimulate their uptake by endocytosis without any gross damage to protoplasts. This is followed by cell wall formation and initiation of cell division. These cells can now be plated at low denisty on selection medium.

Microinjection and Macroinjection

When cells or protoplasts are used as targets in the technique of microinjection, glass micropipettes with 0.5-10μm diameter tip are used for transfer of macromolecules into the cytoplasm or the nucleus of a recipient cell or protoplast. The recipient cells are immobilized on a solid support (cover slip or slide,e tc) or artificially bound to a substrate or held by a pipette under suction (as done in animal sytems). Often a specially designed micromanipulator is empolyed for microinjectin the DNA. DNA **macroinjection** employing needles with diameters greater than cell diameter has alos been tried. In rye (*Secale cereale*), a marker gene was macroinjeced into the stem below the immature floral meristem, so as to reach the sporogenous tissue to successful production of transgenic plants.

Microprojectiles (biolistics or particle gun) for gene transfer

In recent years, it has been shown that DNA delivery to plant cells is also possible, when heavy **microparticles** (tungsten or gold) coated with the DNA of interest are accelerated to a very high initial velocity (1,400 ft per sec). These **microprojectiles**, normally 1-3 μm in diameter, are carried by a **'macroprojectile'** or the **'bullet'** and are accelerated into living plant cells (target cells can be pollen, cultured cells, cells in differentiated tissures and meristems) so that they can penetrate cell walls of intact tissue. The acceleration is achieved either by an explosive charge (**cordite explosion**) or by using shock waves initiated by a high-voltage electric discharge.

Electroporation- method for gene transfer

This method is based on the use of short electrical impulses fo high field strength. These impulses increase the permeability of protoplast membrane and facilitates entry of DNA molecules into the cells, if the DNA is in direct contact with the membrane. In view of this, for delivery of DNA to protoplasts, electroporation is one of the several routine techniques for efficient transformation.

Liposome mediated gene transfer

Liposomes are small lipid bags, in which large number of plasmids are enclosed. They can be induced to fuse with protoplasts using devices like PEG, and therefore have been used for gene transfer. The technique, offers following advantages

(*i*) Protection of DNA/RNA from nuclease digestion

(*ii*) Low cell toxicity

(*iii*) Stability and storage of nucleic acids due to encapsulation in liposomes

(*iv*) High degree fo reproducibility

Calcium Phosphate Precipitation Method

Foreign DNA can also be carried with the Ca^{++} ions, to be released inside the cell due to the precipitation of calcium in the form of calcuim phospate. In the past, this method was considered to be very important for gene transfer in plants.

Transformation using pollen or pollen tube

There has been a hope that DNA can be taken up by the germinating pollen and can either integrate into sperm nuclei or reach the zygote through the pollen tube pathway.

Transgenic Plant : Application

1. Herbicide Resistance

Development of transgenic plants resistanat to certain biodegradable herbicides was the first major achievement from genetic engineering in plants. In fact, this activity generated the basic tools and techniques for gene transfer in plants, and several genes that confer herbicide resistance serve as useful selectable markers. Transgenic plants resistant to several herbicides, *e.g.* Glyphosate, glufosinate bialaphos or phosphinothricin), sulfonylureas.

2. Insect Resistance

The efforts to produce insect resistant transgenic plants contain either a gene from the bacterium *B. thringiensis* or some other gene. *B.*thuringiensis strains are usually classified on the basis of serological tests, and so far over 30 different serotypes have been identified. However, the serotype has very little relevance to the insecticidal activity of a strain.

The Crystal (*Cry*) Proteins

The *cry* gene of *B. Thuringiensis* produces a protein, which forms crystalline inclusions in the bacterial spores. These crystal proteins are responsible for the insecticidal activities of the bacterial strains. The *cry* genes (or *Cry* proteins) have so far been grouped into 18 distinct groups, which either code for a 180 kDa or a 70kDa protein.

3. Virus Resistance

Several approaches have been used to engineer plants for virus resistance, which are as follows :

(1) coat protein gene

(2) cDNA of satellite RNA

(3) defective viral genome

(4) antisense RNA approach

(5) ribozyme-mediated protection.

All these strategies are based on genes derived from the pathogenic viruses themselves. The disease resistance generated by empolying pathogen genes is called **Pathogen-derived resistance (PDR).**

4. Resistance to Fungal and Bacterial Diseases

Ordinarily, hosts and pathogens show a *gene- for- gene relationship*. Thus for every gene for resistance possessed by the host, the pathogen has a corresponding gene for virulence. Disease development depends on the specific interaction between the products of genes governing host resistance and those conditioning pathogenecity or virulence. The *gene-for-gene relationship* may be classified into the following two general groups:

(1) incompatible reaction

(2) compatible reaction

5. Drought Resistance

A number of such genes have been identified, isolated, cloned and expressed in plants, which are potential sources of resistance to abiotic stresses. These genes include *Rab* (responsive to abscisic acid and *SalT* (induced in sesponse to salt stress) genes of rice; genes for enzymes involved in betaine synthesis *etc*.

6. Haploid Plants

Haploid plants can be produced in large numbers by anther and ovary culturs and from crosses with *Hordeum bulbosum* in case of barley and wheat. When *H.vulgare* or *T.aestivum* are crossed with *Hordeum bulbosum* the chromosome complement of *H. bulbosum* is gradually and selectively eliminated from the developing embryos. Most of the seedling obtained from such crosses are haploid, having only one set of chromosomes from the *H.vulgare* or *T.aestivum* parent. There is some seed set in *H.bulbosum* × *T.aestivum* crosses (13.7% seed set in crosses with diploid and 43.7% in crosses with tetraploid *H.bulbosum*), but caryopses tend to degenerate after 10 days following pollination in the case of *H. bulbosum* × *H. vulgare* crosses. Culture of 8-10 day old embryos enables the recovery of a large number of haploid seedlings in both the cases. In case of *H.bulbosum* × barley crosses, >90% of the plants are haploid.

7. Hairy Root Cultures

Hairy roots are induced by transforming plant cells with *Agrobacterium rhizongenes*. These roots can be grown indefinitely, and in large fermenters at growth rates comparable to that of suspension cultures. Hairy roots synthesize those secondary metabolities and at such levels that are characteristic of the roots of parent plants. A *rhizogenes* may be used to transform leaf discs, other organs or even protoplasts. Roots may arise directly from the site of transformation or they may differentiate form a callus that may develop at that site. The roots are excised and used to initiate root cultures; usually, a culture flask is inoculated with 3-4 roots of 2-3 cm length. Hairy root cultures are easily developed in most dicot plants.

Hairy root cultures

(*i*) are suitable for the production of all root-derived biochemicals,

(*ii*) are faster growing, higher producing and much easier to maintain than untransformed root cultures,

(*iii*) present no risk of instability common with cell cultures,

(*iv*) in some cases a portion of the product is released into the medium, e.g., upto 20% of the shikonin produced by *Lithospermum* hairy roots (in such cases, a two-phase system may be used to advantage

(*v*) provide opportunities for isolation of stable high producing hairy root lines by exploiting somaclonol variations.

8. Golden Rice

Vitamin A-deficiency caused blindness among children and may even lead to their death. Vit A-deficiency often occurs where rice is the staple food, since rice grain does not contain provitamin A, *i.e* β-carotene. Three transgenes providing phytoene synthase, phytoene desaturase, zeta-carotene desaturase and lycopene cyclase activities were transferred into rice by *Agrobacterium* mediated transformation. All the transgenes were introduced together in a single co-transformation experiment. The resulting transgenic rice, popularly called "***golden rice***" contains good quantities of β-carotene, which gives the grains a golden colour. In one transgenic line, the β-carotene content was as high as 85% of the total carotenoids present in the grain.

Iron- deficiency anaemia is the most common nutritional disorder in the world. Rice grain has the lowest iron content among crops; it also has phytate, which reduces iron absorption in human intestine by up to 98%. In addition, iron absorption form a vegetarian diet is rather poor. ***Phytate*** is used for phosphate storage in seeds, and it is used during seed germination. The high iron rice was produced expressing in rice three transgenes , viz., ferretin encoding gene, metallothionein gene and thermostable phytase encoding gene.

ANIMAL BIOTECHNOLOGY

Animal Cell Culture

Cells in animals exist in an organized tissue matrix which require for their controlled growth and differentiation. These cells from infact organisms may be isolated, maintained and grown *in vitro* in culture media aseptically containing a suitable mixture of nutrient and growth factors. This process is called animal cell culture.

Primary Culture

When cells are surgically removed from an organism and placed into a suitable culture environment they will attach, divide and grow. This is called a Primary Culture. Primary cultures are derived from intact or dissociated tissues or organ fragments. A culture is considered a primary culture until is subcultures after which it is termed a cell line.

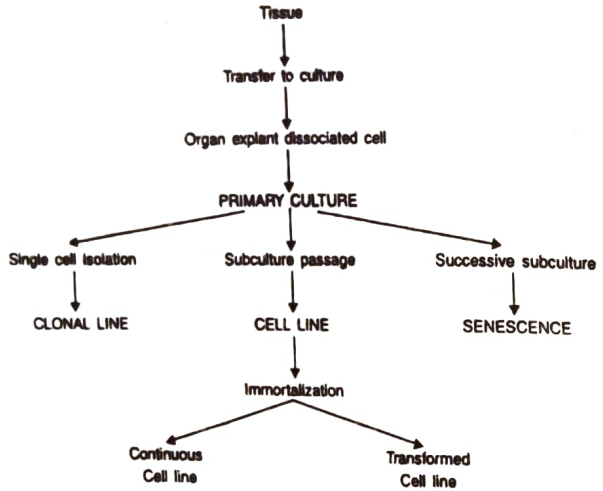


Fig. Diagrammatic Representation of Culturing of Animal Cells.

Sub-Culturing (Secondary cell culture or continuous cell culture)

The process of inoculating cultured cells into fresh culture vessels is termed as sub culturing. When the cells in the primary culture vessel have grown and filled up all of the available culture substrate, they must be subculture to give them room for continued growth. This is usually done by removing them as gently as possible from the substrate with enzymes. These are similar to the enzymes used in obtaining the primary culture and are used to break the protein bonds attaching the cells to the substrate. Some cell lines can be harvested by gently scraping the cells off the bottom of the culture vessel. Once released, the cell suspension can be treated with suitable cryoprotective agents, such as dimethyl sulfoxide (DMSO) or glycerol, carefully frozen and then stored at cryogenic temperatures (below 130°C) until they are needed.

When starting with a primary culture, the product of the first subculture is also called secondary culture. The term tertiary culture is rarely used. The culture becomes more stable by the third subculture.

Cell Lines

A cell line is a cell population derived from a primary culture at the first subculture. A cell line is grown and maintained in the favorable conditions (37°C and 5% CO_2) in a cell incubator. All cell lines originate from cell cultures with a limited lifetime, but occasionally some cells keep on multiplying because they are mutated. These cells can be cultured infinitely. Sometimes, it is possible to fuse normal cells with an immortal cell line.

Example : Monoclonal antibodies.

Thousands of cell lines have been developed from human and other animal tissues; many of them are finite, while others are continuous cell lines. A cell line is multiplied and divided into a seed stock and working stock.

Continuous Cell Lines

The phenomenon that generates a continuous cell lines is called *in vitro* transformation which may either occur spontaneously, or may be chemically or virally induced. The word transformation is used because continuous cell lines have marked similarities with malignant cells and are often tumerogenic. The features of continuous cell lines are as follows :

Smaller, more rounded, less adherent cells with a higher nucleus/cytoplasm ratio, Reduced dependence on serum, Increased cloning deficiency, Ability to grow upto higher cell densities, Many cell lines become malignant, Increased growth ratio, Aneuploid chromosome in cell.

Continuous cell lines themselves may have originated from normal or tumour tissues. The faster growth rate, low serum requirement, ability to grow in suspension, independence from cell density is the advantages of continuous cell line. Disadvantages include chromosome instability, loss of tissue specific markers and difference from the phenotypes of donor tissue.

Animal Culture Media

The culture media used for animal cell culture must be able to provide nutritional, hormonal and stromal factor. Successful grwoth and maintenance of animal cell *in vitro* requires culture conditions that mimic *in vivo* condition with respect to temperature, oxygen and carbon dioxide concentration, pH, osmolality, and nutrition. Growth of vertebrate cells in culture requires rich media containing essential amino acids, vitamins, and peptide or protein growth factors, frequently provided by serum. For this reason, early tissue culture media were based on biological fluids such as plasma, lymph and serum, and tissue extracts especially of embryonic origin. Most cultured vertebrate cells will grow only when attached to a negatively charged substratum that mimics the extra cellular matrix in animal tissues.

1. Primary cells, which are derived directly from animal tissue, have limited growth potential in culture and may give rise to a cell strain.
2. Transformed cells derived from animal tumors or arise spontaneously from primary rodent cells, grow indefinitely in culture. They usually have an unstable, aneuploid complement of chromosomes, including abnormal chromosomes. Transformed cells derived from a single parental cell are called cell lines.
3. Cultured cells can be induced to fuse into heterokaryons (hybrids) by treatment with certain viruses or polyethylene glycol. Heterokaryons between cells of different species tend to lose the chromosomes of one species as they divide.

Culture medium is one single most important factor for culturing cells and tissues. A complete cell culture medium should be composed of two distinct parts :

(*a*) A basal medium, that satisfies cellular requirements for nutrients, salts and pH control.

(*b*) A set of supplement, which satisfies other cellular requirement and permit growth of cells in the basal medium.

It provides :

1. The optimum conditions of factors like pH, osmotic pressure, etc.
2. The chemical constituents, which the cells are incapable of synthesizing.

Application of Animal Tissue Culture

Animal cells are good hosts for the expression of rDNA molecules and a number of commercial products are being developed. A variety of different areas of research in animal biotechnology depend largely on tissue culture. For production of interferson's, Hybrid antibodies and recombinant proteins, cell culture has become one of the major tools used in cell and molecular biology. Some of the important areas where cell culture is currently playing a major role are briefly described below :

Model Systems

Cell cultures provide a good model system of studying :

1. Basic cell biology and biochemistry.
2. The interactions between disease causing agents and cells.
3. The effects of drugs on cells.
4. The process and triggers for aging.
5. Nutritional studies.

Toxicity Testing

Cultured cells are widely used alone or in conjunction with animal tests to study the effects of new drugs, cosmetics and chemicals on survival and growth in a wide variety of cell types. Especially important are liver and kidney-derived cell cultures.

Cancer Research

Since both normal cells and cancer cells can be grown in culture, the basic differences between them can be closely studied. In addition, it is possible, by the use of chemicals, viruses and radiation, to convert normal cultured cells to cancer causing cells. Thus, the mechanisms that cause the change can be studied.

Virology

One of the earliest and major uses of cell culture is the replication of viruses in cell cultures (in place of animals) for use in vaccine production. Cell cultures are also widely used in the clinical detection isolation of viruses, as well as basic research into how they grow and infect organisms.

Cell Based Manufacturing

While cultured cells can be used to produce many important products, three areas are generating the most interest.

1. The first is the large-scale production of viruses for use in vaccine production. These include vaccines for polio, rabies, chicken pox, hepatitis B and measles.
2. Second is the large-scale production of cells that have been genetically engineered to produce proteins having medicinal or commercial value. These include monoclonal antibodies, insulin, hormones , etc.
3. Third is the use of cells as replacement tissue and organ. Artificial skin for use in treating burns and ulcers is the first commercially available product. However, testing is underway on artificial organs such as pancreas, liver and kidney. A potential supply of replacement cells and tissues may come out of work currently being done with both embryonic and adult stem cells. These are cells that have the potential to differentiate into a variety of different cell types.

Genetic Counseling

Amniocentesis, a diagnostic technique that enables doctors to remove and culture fetal cells from pregnant women, has given doctors an important tool for the early diagnosis of fetal disorders. These cells can then be examined for abnormalities in their chromosomes and genes using karyotyping, chromosome painting and other molecular techniques.

Genetic Engineering

The ability to transfect or reprogram cultured cells with new genetic material (DNA and genes) has provided a major tool to molecular biologists wishing to study the cellular effects of the expression of these genes (new proteins). These techniques can also be used to produce these new proteins in large quantity in cultured cells for further study. Insect cells are widely used as miniature cells factories to express substantial quantities of proteins that they manufacture after being infected with genetically engineered baculoviruses.

Drug Screening and Development

Cell based assays have become increasingly important for the pharmaceutical industry, not just for cytotoxicity testing but also for high throughput screening of compounds that may have potential use as drugs.

Karyological Studies

Karyotyping is the most important application in medicine. Tissue culture technique helps in the early diagnosis of some congenital abnormalities.

Example : Klinefelter's syndrome, Turner's syndrome, Down's syndrome, etc.

Identification and Study of Hereditary Metabolic Disorders

The nature of may genetic diseases can be determined only by demonstrating the metabolic abnormality commonly the absence of an enzyme in tissue. Disorders such as absence of ctalase in acatalasia can be demonstrated in primary fibroblast cultures. Some human genetic disorders which can be studied in this way are glycogen storage diseases; Hunter's Syndrome, Xeroderma pigmentosa, etc.

Somatic Cell Genetics

Tissue culture technique have been very useful at understanding the :

1. Behavior of heterokaryons.
2. Behavior of hybrid cells derived from heterokaryons.
3. Interaction between different genes.
4. Mapping loci on different chromosomes.

Monoclonal Antibodies

The generation of monoclonal antibodies is indispensable in many areas of biotechnology and it is completely dependent on tissue culture techniques (Hybridoma Technology).

Recombinant Proteins

Important recombinant protein like lymphokines, interferons, and human growth hormone has been produced from cultured animal cells. In 1986, human interferon gamma produced from lymphoblastoid cells in culture was licensed as a therapeutic agent.

Reconstitution and Replacement of Damaged Tissue and Cells

Cultured cells can be themselves act as product and can be used in the reconstruction of damaged tissue or as a replacement of non-functional cells and tissue.

Example : Tissue grafting autograft.

Gene Targeting

Tissue cultured cells can be suitable replace for faulty genes. Advances have been made in introducing foreign genes by isertional mutagenesis into cultured skin and blood vessel cells.

Amniocentesis, Infertility and Embryo Transplantation

Culturing cells collected from the amnion during early pregnancy may identify genetic abnormalities of foetus. The techniques of animal tissue culture are also directly relevant to *in vitro* fertilization and embryo transplantation that are employed to overcome some problems of infertility.

Cytotoxicity Testing

Now-a-days, evaluations of the effects of carcinogens, toxins and drugs on specific cell types are done using tissue culture cells. It would be more suitable both in terms of cost and ethical reasons.

Transfection of Animal Cell Lines

Exogenous DNA can be introduced in bacteria, fungi, plants and animals. Introduction of exogenous DNA into targeted plants and animals and their stable expression gives rise to transgenic plants and animals. The transferred gene is TRANSGENE and whole process is referred to as 'TRANSGENESIS'. The uptake of genes by cells in animals is called 'TRANSFECTION'.

All though techniques of DNA transfer differ for different organisms; there are three basic approaches of doing it :

1. The natural methods
2. Artificial methods
3. Novel methods.

Natural Methods

Bacterial Transformation

Transformation involves the uptake of naked DNA molecules from one bacterium to another bacterium. Transformation consists of numerous steps which can be divided into two main categories.

(*i*) Entry of exogenous DNA into a recipient cell.

(*ii*) Recombination of the donor DNA with the homologous region in the recipient chromosomes.

Conjugation

Conjugation is a parasexual process in which the genetic information of one bacterium is transferred to and combined with that of another bacterium. It is a process during which DNA from a donor or male cell is transferred to a recipient or female cell through a sex pilus of "conjugation tube". The transfer of genetic information is thus a one-way transfer during conjugation.

Transposition

Transposition involves the integration of a copy of a defined segment of DNA from another locus, either in the same or in a different molecule of DNA. Such movable elements have terminal repeat regions to mediate the transposition event. Different mobile elements are Insertion Elements (IS), Transposons (Tn) and certain macrotransposons like bacteriophage genome.

Phage Transduction

Phage transduction occurs when bacterial genes are carried from a donor cell to a recipient cell by a bacteriophage. The process phage transduction begins when a prophage enters the lytic cycle and progeny viruses subsequently infect and lyse the bacterium. During infection, the bacterial DNA is degraded into small fragments and the viral DNA is replicated.

Agrobacterium like Vector Mediated Transfer

Agrobacterium tumefaciens is most impotant tool for engineering the plants. This bacterium is the causative agent of dicots and not in monocots. The foreign genes can be introduced into T-DNA region to transform plants via DNA. After infection viral DNA is released to form functional virus that replicates and spreads systematically. Agroinfection may also lead to integration of viral DNA so that transgenic plant containing integrated viral DNA can be produced.

Retroviral Transduction

Retro viruses have a high efficiency of gene transfer, are able to incorporate DNA fragments than plasmids, and infect host cells spontaneously (Ausbuel et. Al. 1995). The introduced gene becomes permanent, and the process of insertion is achieved by normal cellular processes, is not harmful to the host cell, and does not cause any other genetic alterations.

Transformation with Plasmids

The cell transformation with plasmid includes the treatment of *E.coli* cells with $CaCl_2$ at 4°C along with heat shock treatment. The cells become competent by this are now mixed with the DNA to be introduced into the cells. The calcium makes DNA to adhere to the cell surface whereas heat shock treatment allows the membrane to be open to the exogenous DNA temporarily.

Artificial Methods

A. Physical Methods

Microinjection

It can be applied to both plant and animal cells. But, it is more common in animal cells.

Intra nuclear microinjection is a direct physical approach that overcomes many of the biological and other obstacles. The microinjection technique uses a fine capillary needle to deliver DNA precisely into cells in such a way that the cells are not killed but survive the treatment and can develop further into clones. The DNA may be into cells or protoplasts with use of very fine needles with 0.5 to 10 micrometer in diameter. T.P. Lin in 1966 described the technique of microinjection of mouse eggs for the first time. Microinject DNA may become stably integrated into chromosome of host cell through recombination and may be inherited in a Mendelian manner. 500-100 cells can be microinjected per hour by skilled person having proper practice.

Macroinjection

The macroinjection technique involves the use of injection needles with diameters far wider than cell diameters for delivering DNA into wound sites within tissues. This technique can be used for those cereals which show inability to regenerate and develop into whole plants from cultures.

Biolistic Transformation (Gene Gun Method)

Biolistic transformation avoids the need of protoplasts and can be used for any plant cells, leaves, root sections, embryos, seeds and pollens. The method involves bombardment of particles carrying DNA of interest into target cells using high velocity transfer mechanism. This method was first introduced by Sanford and coworkers in 1987.

Protoplast Fusion

This is a general method for introducing macromolecules into cells and is therefore not restricted to DNA. To obtain efficient transfection, care must be taken to ensure no lysis of sphaeroplasts take place, and that the optimal protoplast cell ratio is achieved. The protoplast mixture is treated with 28-50% PEG for 15-30 minutes followed by gradual washing of the protoplast to remove PEG; protoplast fusion occurs during washing. The washing. The washing medium may be alkaline and contain a high Ca^{2+} ion concentration. This treatment is more effective than either treatment alone. PEG is negatively charged and may bind to cation like Ca^{2+} which, in turn, may bind to the cationic molecules of the plasma membrane. During the washing process, PEG molecules cationic molecules of the plasma membrane. During the washing process, PEG molecules may pull out the plasma components bound to them. This would disturb plasmalemma organization and may lead to the fusion of protoplasts located close to each other.

One obvious advantage to this technique is that the cloned DNA to be transfected need not be purified from *E. coli*.

B. Chemical Methods

Calcium Phosphate Mediated DNA transfer

This technique was first introduced by Graham and Vander in 1973. It is the most common method of DNA transfection. The process of transfectin involves mixture of isolated DNA with solution of calcium phosphate and potassium phosphate under condition which allow fine precipitation of calcium phosphate to be formed. Cells are then incubated with precipitated DNA in tissue culture dish. A fraction of cells will take up the calcium phosphate DNA precipitate by endocytosis.

Use of Polyethylene Glycol (PEG)

Direct DNA uptake by protoplast can be stimulated by using hydrophilic, long chain polycations such as PEG, poly L lysine, poly L ornithine with or without metal ions such as zinc, sodium, magnesium etc. This technique is efficient and can be applied virtually to every protoplast system and for transfer of even small quantities of DNA. Chemical used here is PEG which alters surface charge of the plasma membrane of protoplat. It is an extremely hydrophilic molecules and it removes 'free water' when placed in solution. Free water indicates such molecules which interact with charged molecules soluble water.

Use of DEAE-Dextran

DEAE-Dextran (dimethylaminoethyl-dextran) is water soluble and polycationic has a multiple positive charge.

It is added to the transfection solution containing the DNA.

Use of Liposome

Liposomes are the artificial phospholipids vesicle. Animal cells, protoplasts of yeasts, plants and bacteria are susceptible to transformation by liposomes. The use of liposome for transformation is called lipofection. Liposomes, as defined by Bongham, in 1965, are special type of lamellar phase in which water is self contained and the lipid molecules are disposed in bimolecular layers attached by their non polar interfaces. Liposomes possess many properties akin to those biological membranes, they are easy to manipulate, their lipid contents can be varied at will and many substances may be trapped inside the interlammellar spaces. They are found to be ideal carrier systems to deliver foreign DNA molecules into a desired cell. Papandjopopulos and coworkers demonstrated this technique in 1988. The use of liposome for delivery of gene was first described for delivering polio virus RNA. Later liposomes were used in combination with polyethylene glycol for delivery of plasmid DNA into tobacco protoplast.

C. Electrical Methods

Electroporation

Electroporation is relatively new, simple and rapid technique which can be applied to artificially introduce gens into wide variety of cells. This technique was devised by Zimmermann (1983). It was found that when subjected to electrical shock the cells take up exogenous DNA from the suspending solutions through the hole which are temporarily produced. A proportion of these cells become stably transformed.

Electrofusion

The more selective and less drastic approach is Electrofusion technique which utilizes low voltage electric current pulses to align the protoplasts in a single row like a pearl-chain. The aligned protoplasts are pushed, with a micromanipulator, at a gentle pace through the narrow gap between the two electrodes. When two protoplasts that are to be fused are appropriately oriented opposite the electrodes, a short pulse of high voltage is released which induces the protoplasts to fuse. The high voltage creates transient disturbances in the organization of plasma lemma, which leads to the fusion of neighbouring protoplasts. The entire operation is carred out manually in specially designed equipment, called electroporator, under a microscope. This fusion technique is more desirable than the others for a number of important reasons.

Novel Methods

Use of Laser Rays

A new method of introducing gene materials into cells is laser microbeam irradiation. DNA can be incorporated into whole plant cells and has been successfully applied to tobacco. DNA enters the cell through a self healing hole made with a laser microbeam.

Sonication

The latest method developed uses ultrasound to facilitate the uptake of nuclei acids into plant and animal cells. By this technique, plant protoplasts and cells in suspension and intact pieces of plant tissue can be readily transformed.

Transgenic Animals

The term '***transgenic***' in case of animals was first given by **Gorden** and **Ruddle** (1983) and is now used to describe animals into which a new gene(s) has been inserted. New gene transfer techniques and introduction of cloned gene into fertilized eggs, successful implantation of modified eggs into receptive female to get progeny carrying cloned gene to get transgenic animals.

Production of Transgenic Animals

The ES cells are allowed to get transfected by vector containing desirable genes. In transfected cells targeting of gene to specific site by homologous recombination occur. Thereafter, the transfected cells are identified and isolated from bulk. They are multiplied and introduced into blastocyst through microinjection. The blastocyst is transferred into uterus of a surrogate mother for further developmental stages. The animal, which is born, are checked for the presence of transgene. The transgenic animal is crossed with a normal one to study the inheritance of introduced foreign gene. The flow chart of the same is as follows

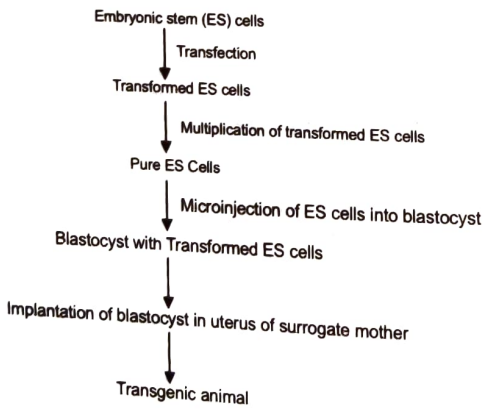


Fig. Transgenic Animal Production.

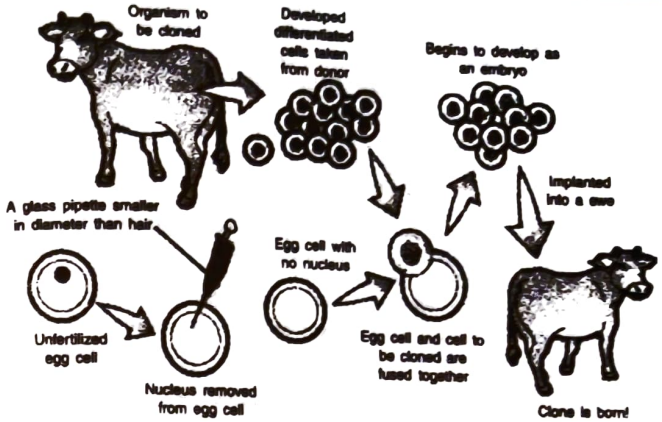


Fig. Steps Involved in Transgenic Animal Production

Knockout Mice

A ***knockout mouse*** has one of its genes disrupted by the technique of targetted gene transfer. The name '*knockout*' signifies the fact that a specific gene in the genomes of these animals has been made nonfunctional due to gene disruption, i.e., has been 'knocked out'. Production of knockout mice employs the embryonic stem cell technology. Thus knockout mice are transgenic mice, which lack the function of a known specific gene that was disrupted by the targetted gene transfer. The gene that is disrupted by this procedure is strictly under experimental control. because it is the same gene that is introduced into the embryonic cell lines during transfection.

Knockout mice are invaluable in various kinds of basic and applied research. For example, these mice permit the study of the role of specific individual genes in development, in immune response etc., and provide animal models for evaluation of various therapeutic regimens. For example, a variety of knockout mice are being used in immunological research, including mice that lack particular cytokines of MHC molecules. 'Oncomouse' is a special type of knockout mouse, that is predisposed to develop cancer, these mice are extremely useful in various studies of cancer.

The findings from many studies with knockout mice were surprising in that many seemingly important genes in development were in fact not necessary for the production of viable individuals. Examples of such genes are *myo*D (concerned with muscle differentiation and development), retinoic acid-γ-receptor, etc. These findings have prompted the suggestion that genetic redundancy is common in development. Further, when a gene is inactivated, e.g., in the case of knockout mice, there is a compensatory upregulation of some other member of the concerned gene family. For example, mgf-5 is upregulated in mice lacking *myo*D function.

The targetted gene transfer can be used to replace the normal endogenous gene with a specific mutant form of the same gene or the whole gene may be replaced by a DNA sequence of choice. This technology is described as ***"knock-in"-technology*** since it replaces one gene function by another one.

This strategy can be exploited to seek answers to various developmental questions. For example in one experiment, the coding sequence of CD4 gene was replaced by *E.* coli β-galactosidase gene while the CD4 promoter was left intact and functional. The pattern of β-galactosidase gene, expression was followed in the developing lymphocytes, and the data so gathered were useful in tracing CD4/CD8 lineage commitment in developing T cells. It may be pointed out that CD4 gene is expressed primarily in T_H cells, and that β-galactosidase expression can easily be followed by blue colour production in the presence of X-gal by cells expressing this enzyme.

The cre / lox System

The deletion of gene functions by gene targeting leads to a loss of function from every cell of the organism. In contrast, certain strategies allow the specific deletion of a gene of interest in precisely the tissue of choice. The technologies rely on the use of site-specific recombinase genes isolated from bacteria or yeast. The most commonly used recombinase is Cre, isolated from *E. Coli* bacteriophage pi-Cre recognizes a specific 34-bp site in DNA known as *lox*P and, upon recognition, catalyzes a site-specific recombination between two *lox*p sites. Therefore, a DNA sequences that is flanked by *lox*p sites is deleted as a result of this recombination event. In other words, animals that ubiquitously express Cre recombinase will delete all *lox*p flanked sequences. The real innovation of

this technique is that expression of the Cre recombinase gene can be controlled by the use of a tissue-specific promoter. This allows tissue-specific expression of the recombinase and thus tissue-specific deletion of the DNA sequence flanked by *lox*P. For example, one could express Cre in B cells using the immunoglobulin promoter, and this would result in the targeted of *lox*-P-flanked DNA sequence only B cells.

This technology is particularly useful when the targeted deletion of a particular gene is lethal. This is true in many circumstances. For example, the DNA polymerase β gene is required for embryonic development. In experiments designed to test the *Cre*/*lox* system, scientists flanked the mouse DNA polymerase β gene with *loxp* and mated these mice with mice carrying a Cre transgene under the control of a T-cell promoter. The result of this mating are offspring that express the Cre recombinase specifically in T cells. Using such mice, the scientists were able to examine the effects of deleting the enzyme DNA polymerase β specifically in T cells. The effects of the deletion of this gene could not be examined in a conventional gene-targeting experiment, because deletion of DNA polymerase β throughout the animal would be lethal. However, with the *Cre*/*lox* system, it is now possible to examine the effects of the deletion of this gene in a specific tissue of the immune system.

The *Cre*/*lox* system also can be used to turn on gene expression in a particular tissue. Just as the lack of a particular gene may be lethal during embryonic development, the expression of a gene can also be toxic. In order to examine tissue-specific expression of such a gene, it is possible to insert a translational stop sequence flanked by *lox*/P into an exon at the beginning of the gene. Using a tissue-specific promoter driving Cre expression, the stop sequence may be deleted in the tissue of choice and the expression of the potentially toxic gene examined in this tissue. This technology has been used in the terminator technology designed to protect transgenes from being used in breeding programmes without proper authorization. These modifications of gene-targeting have been very useful in determining the effects of particualr genes in cells and tissue of the immune system.

Animal Cloning

When complete animals are obtained from somatic cells of an animal, it is called ***animal cloning***. But early in 1997, British scientists announced successful cloning of sheep by fusing an udder cell of an adult sheep with an eunucleated fertilized egg. The egg was then transplanted into the uterus of a surrogate mother where it developed like a normal zygote into a normal lamb, which grew into a normal adult sheep, called 'Dolly'.

Cells from mammary gland of an adult sheep (6 yr-old ewe) were first cultured *in vitro*. The cultured cells were arrested, induced to enter the *Go* phase (quiescent stage) by reducing the concentration of serum in the medium from 10% to 0.5% for 5 days. The *Go* cells were then fused *in vitro* with ennucleated ova of appropriate stage. Oocytes were recovered between 28 and 33 hr after injection of gonadotropin releasing hormone and enucleated as soon as possible. Cell fusion was induced by electrical pulse, which also activated the oocyctes. The fusion products were cultured *in vitro* in ligated oviducts of sheep upto morula or blastula stage, before their transfer into the uteri of surrogate mothers. The rate of success in obtaining normal embryo development was rather low; a total of 277 oocytes were fused with cultured mammary gland cells; of these 29 reached morula/blastula stage, and were transferred into surrogate mothers leading to 13 pregnancies, but only one live birth.

It has been suggested that inducing the differentiated nucleus of donor cells to enter G_0 phase causes such changes in chromatin structure that facilitate reprogramming of gene expression. Therefore, if nuclei from G_0 cells were transferred into enucleated oocytes, the resulting diploid cell may be expected to develop lime a normal zygote into a normal animal. But there is considerable concern about the normalcy of health of cloned animals. Some cloned animals have been much larger than normal at birth, and many of such animals have become abnormally obese after birth. In addition, animals obtained by cloning showed a variety of developmental defects like heart defects, lung abnormalities and malfunctioning of immune system. Obviously, the cloning techniques need considerable refinement to ensure a much higher success rate and, more importantly, a defect-free normal development and after-birth life of animals.

Cloning is, in many situations, highly desirable since this allows

(1) indefinite multiplication of an elite desirable genotype without the risk of segregation and recombination during meiosis, which must precede sexual reproduction. Obviously,

(2) the technique holds a great promise in genetic research, especially in understanding ageing and curing genetic diseases, *e.g.*, to study the possible persistence and impact of epigenetic changes, such as, imprinting and telomere shortening, etc. In addition,

(3) this technique should make it feasible to target transgenes in livestock by nuclear transfer from transgenic cell populations developed *in vitro* enucleated oocytes to recover nonchimaeric transgenic animals. Obviously this approach would obviate the generations of breeding necessary for the recovery of transgenic animals from the chimaeric ones that are generated by embryonic stem cell transfer.

Following the above report, the first cloned mouse, 'Cumulina', was produced by injecting diploid.

Gene Therapy

There are many diseases which can be cured by using specific medicine synthesised biochemically. Now-a-days techniques have been developed to produce recombinant therapeutic biochemicals, for example insulin, interferon, somatotropin, somatostatin, endorphin, human blood clotting factor VIII:C.

Genes are the ultimate molecular switches that control various cellular process. The abnormal gene expression can manifest in the form of specific genetic disorders. Until the last decade, delivering genes into humans to correct diseases has been accepted as scientifically viable and recognised as an independent discipline and christened '*gene therapy*'.

The ultimate goal of gene therapy is the gene replacement therapy. Gene replacement therapy permits physiological regulation of the transgenes and elimination of the possibility of insertional activation of other cellular genes which occur at the time of random integration of the foreign gene. At present the current strategy for gene therapy largely centres around gene augmentation therapy, where the foreign gene replaces the defective or missing gene.

Overall, there are two strategies for gene transfer

(*i*) the *in vivo* approach which involves introduction of genes directly into the target organs of an individual (it is done in patients therefore, also called *patient therapy*),

(*ii*) *ex vivo* approach where cells are isolated for gene transfer *in vitro* followed by transplantation of genetically modified cells back into the patients.

(*a*) **Types of Gene Therapy :** All the gene therapies that can be done in humans and classified into the following four types :

(*i*) **Somatic gene therapy :** The genetic defects are corrected in somatic cells of the body. It was initially formulated for the treatment of monogenetic defects, but now holds promises for a wide range of disorders such as cancer, neurological disorders, heart diseases and infectious diseases.

Table- Genetic disorders and acquired diseases amenable to gene therapy

Disease	*Therapeutic agent*	*Strategy*	*Vector*	*Target cell / tissue*
Genetic Disorders				
Cystic fibrosis	CFTR	*in vivo*	Adenovirus	Nasal epithelium
Familial hyper-cholesterolaemia	LDL	*in vivo*	Cationic lipid	Nasal epithelium
SCID	ADA	*Ex vivo*	Retrovirus	T cells
Haemophilia	Factor VIII/IX	*In vivo*	Retrovirus	Hepatocytes, skin, muscles
DMD	Dystrophin	*In vivo*	Retrovirus	Skeletal muscles
		Ex vivo	Retrovirus	Myoblasts
Acquired disorders				
Alzheimer's disease	NGF	*Ex vivo*	Retrovirus	Tumour cells
AIDS	HIV antigen	*Ex vivo*	Retrovirus	T cells
	Rev M10	*Ex vivo*	Retrovirus	Hepatocyte
	Cytokine	*Ex vivo*	Retrovirus	Hematopoietic stem cells
Cancer	Interleukins,	*Ex vivo*	Retrovirus	Tumour cells
	HSV-TK	*Ex vivo*	Retrovirus	Tumour cells
	HLA-B4	*In vivo*	Catonic lipid	Tumour cells
	Tumour suppressor gene	*In vivo*	Catonic lipid	Tumour cells
Cardiovascular disease	tPA	*In vivo*	Adenovirus	Tumour cells
Parkinson's disease	TH	*Ex vivo*	Retrovirus	Fibroblast

CFTR, cystic fibrosis transmembrane regulator; SCID, severe combined immunodeficiency syndrome; DMD, Duchenne muscular systrophy, ADA, adenosine deaminase; HSV-TK, herpes simplex virus thymidine virus, NGF, nerve growth factor TH, tyrosine hydroxylase, tPA, tissue plasminogen activator.

(ii) Germ-line gene therapy : The functional genes are introduced into the germ cells for correction of genetic defects in the offspring. This therapy is being carried out in laboratory and farm animals. However, it has not been attempted in humans due to technical ethical problems. One of its types is the *embryo therapy* where embryos are diagnosed for genetic defects. If any such disease is present the patients are advised for embryo therapy or abortion. In young embryo a functional gene is transferred through microinjection technique.

(iii) Enhancement genetic engineering : This type of gene transfer is done for the improvement of a specific trait in animals; for example introduction of growth hormone gene to increase height. It is being carried out in laboratory and farm animals.

(iv) Eugenic genetic engineering : Novel genes can be introduced in humans to alter or improve complex traits such as intelligence and personality. This type of therapy is not being attempted in humans because is is far beyond our technical capabilities, and ethical problems.

In 1990, for the first time, **Michaele Blease** and **W. French Andresco** of National Institute of Health, Bethesda, U.S.A. attempted gene therapy on a human patient. A four year girl was suffering from 'severve combined immunodeficiency' (SCID). This disease is caused by a faulty gene which expresses the enzymes adenosine deaminase (ADA). Deficiency of ADA results in the production of a chemical which selectively destroys the T- and β-cells of the immune systems. Finally, the patients die. The scientists introduced a healthy ADA gene into the body of the girl who protected her immune system from damage. This successful trial has given the signal for the dawn of a new era in the field of medical sciences.

(b) Methods of Gene Transfer : A variety of gene transfer strategies have been developed during the last decade for the treatment of human diseases which can be grouped into the two major categories: the viral and non-viral methods.

(i) Viral approaches : After 1980, much work has been done on retroviruses as gene transfer vectors, more specifically on murine-leukemia virus (MLV) for gene therapy. Efforts are being made to develop HIV-based vectors so that even non-dividing cells can be injected. The steps of developing a replication-defective recombinant retroviral vectors are :

(*i*) the replacement of viral structural genes *e.g. gag pol* and *env* by the therapeutic foreign genes of interest,

(*ii*) transfection of this vector into packaging cell line (*i.e.* producer cells) that provide the viral structural proteins *in trans* so that the recombinant retroviral genome is packed and replication defective retroviruses are generated,

(*iii*) transfection of host cells by such viruses, and reverse transcription of recombinant retroviral RNA and random integration into the host genome. In the absence of viral genes, the foreign gene (therapeutic in nature) is transcribed from the viral LTRs, (the long terminal repeats) and desired protein is synthesised (Fig.). The retroviral vectors are used in *ex vivo* gene transfer experiment, although it has been shown that they can infect a regenerating liver when administered intravenously into hepatectomized animal.

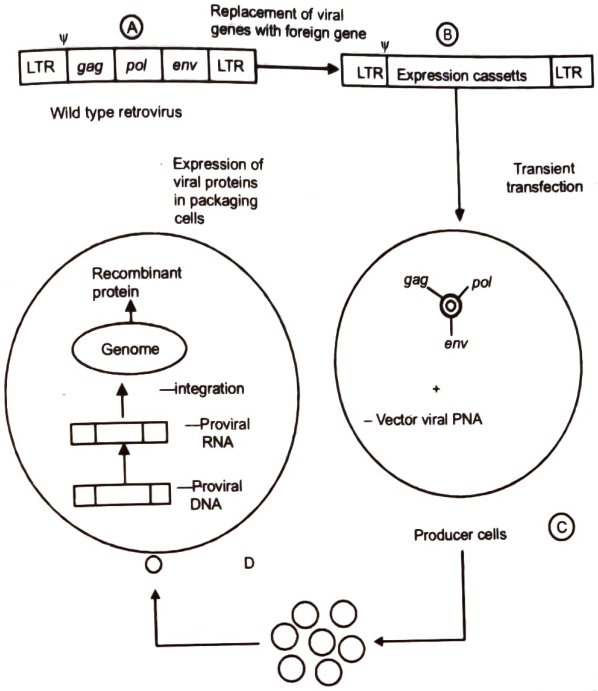


Fig. Retroviral gene transfer A. Construciton of a retroviral vector; B. Productin of recombinant retrovirus; C. Transfection of target cells; D. Synthesis of recombinant protein.

(ii) Non-viral approaches : Most of approaches in gene therapy has been focused on development of viral vectors rather than clinical efficacy and biosafety. Furthermore, techniques of virus-based gene therapy is very costly and complex. Therefore, this led to the development of non-viral gene therapy. Majority of non-viral methods follow the *in vivo* approaches. Therefore a gene can be used successfully as a drug. In general, the non-viral methods for gene transfer can be grouped into two methods, physical and chemical methods.

Physical methods : The various forms of physical methods are given below :

1. **Microinjection :** It is a very tedious method. However, it is used in oocytes, eggs, and embryos. **Jaffrey S. Chamberlain** have cured mice that inherited a neuro-muscular disease which is similar to muscular dystrophy of humans The mutant mice lacking 427 K gene in brain and muscle calls had a weak diaphragm which gradually degenerates like that in dystrophy affected humans. Chamberlain and coworkers microinjected the DNA containing 427 K gene into the zygote of mutant mice. The transplanted genes worked properly and produced necessary protein and thus, prevented the diaphragm.
2. **Direct DNA injection:** Direct injection of DNA into skeletal muscle has created a lot of excitement and led to the possibility of using gene as vaccines. Direct gene transfer was attempted since early 1960s but these studies were not done seriouly due to inefficient gene transfer and low level of expression. Due to low level of expresstion theraputic benefits for the treatment of genetic disorder could not be derived.
3. **Gene gun:** It was developed originally for transformation of plant cells, but in animal system it is also being used for delivery of genes into tissues such as liver. skin, mammary glands and muscle. Direct gene delivery is now being used to map promoter elements that regulate gene expression *in vivo*, in transfection of herpes simplex thymidine kinase genes into melanomas to render than sensitive to gancyclovir and as a vaccination strategy.

4. **Magic bullets – cells as drug delivery system:** Magic bullet is a drug delivery system through a cell loaded with an appropriate treatment molecule which upon injection into a patients migrate through the circulatory system directly to the diseased site. Still, it is only a dream to the patients suffering from genetic diseases. In future it may be possible to cure such diseases such as cystic fibrosis, when a cell can reach to the lung tissues and deliver an enzymes to breakdown the thick getatinous mucus, and help in breath and food digestion.

Chemical Methoths :

1. **Using detergent mixture:** Certain charged chemical compounds e.g. calcium phosphate, dextran or lipids are mixed with functional cDNA of desired function. The mixture is introduced near the vicinity of recipient cells. Thereafter, it is spread to the interior of the body organ. The chemical mixed with cDNA disturbs the cell membranes, widens pore size and allows the cDNA to pass in the cell. In this method the number of cells allowing the entry of cDNA is very small.

2. **Receptor-meditated gene delivery:** It is an attractive strategy. It takes advantage of normal physiological pathway. The receptors are exclusively present on hepatocytes. They bind to certain glycoproteins lacking the terminal asialic acid. The concept of delivering genes into specific tissue was examined extensively with the liver asialoglycoprotein receptors. An asialoglycoprotein (orosomucoid) is coupled with poly-L-lysine. This conjugate is then condensed with plasmid DNA via electrostatic interactions. This soluble complex i.e. asialoglycoproteinpoly-L-lysin-DNA complex (ASGP-PL-DNA) is injected intravenouly/intraperitoneally, which gets entered into hepatocytes via the asialoglycoprotein receptors (ASGPR). DNA is delivered into the nucleus. The transgene expression occurs, therefore, recombination proteins is detected for several weeks (Wu and Wu, 1988).

3. **Embryo therapy through IVF-technology:** In 1993, adopting IVF-technology. **Dr. A. Handyside** got success in producing a genetically engineered female baby, whose parents transmitted the genetic disease (cystic fibrosis) in earlier four babies, who died later on. Even after confirmation through prenatal diagnosis that the babies would be sufferer or carrier of this disease they decided to give birth to babies, which failed. This disease clogs the lungs of sufferer and makes them them unable to digest the food properly.

Dr. Handyside admitted seven couples. Their eggs. were collected and fertilized with their respective husbands sperm in test tubes. After 3 days incubation *in vitro*, when embryo reached 8 cell stage, a single cell was removed for analysis. The cell was magnified one million times by using a magnifying Thermal Cycler. The individual defective gene which causes cystic fibrosis was identified and screened out. Thereafter, the embryo was implanted in the womb of mother. Consequently, a female baby (**C. O' Brien**) was born who is neither sufferer nor carrier of this diseases, but a normal one.

(c) Success of Gene Therapy : The success of gene therapy depends on gene delivery mechanism as well as on the choice of target tissue different conditions leading to success of gene therapy:

(i) Cell types capable of dividing *in vitro* (e.g myeloblasts, hepatocytes, keratinocytes, endothelial cells, etc.) are amenable for *in vitro* and *in vivo* gene therapy, both the in *vivo* methods are preferred for cell types such as neuronal cells.

(ii) The function of gene products also govern the selection of tissues, for example in case of haemophilia a gene can be delivered in any tissue provided the gene product is released into blood stream.

(iii) Another attractive strategy for the treatment of several disorders is antisense gene transfer, for example in β-thalassemia, α-globin chains are accumulated in RBCs that result in their premature decay. Such types of destruction can be prevented by infection of K562 erythro-leukemia cells with AAV expressing human α-globin gene in antisense orientation.

(d) Potential of a Gene Delivering System : The potential of a gene delivery system is first confirmed in cultured cells of experimental animals by using reporter genes. On the basis of the results so obtained clinical trials are conducted in experimental animals to measure the levels and time of expression of therapeutic genes which were introduced. Thereafter, clinical efficacy of procedures of gene therapy in humans is evaluated. But before conducting gene therapy in humans, permission is taken from specific regulatory agencies.

EXERCISE – I

MCQ TYPE QUESTIONS

1. Which restriction enzyme has no methylase activities?
(a) Type I RE (b) Type II RE
(c) Type III RE (d) None of these

2. Which type of vector is used in cloning of very long segment of DNA?
(a) Plasmid vector (b) BAC vector
(c) Bacteriophage (d) None of these

3. Other name of DNA profiling is
(a) DNA foot printing (b) DNA finger printing
(c) DNA chips (d) All of these

4. Techniques used for study of gene Expression
(a) DNA microarrays (b) DNA hybridization
(c) Both (d) None of these

5. Cloning vectors with the transcription and translation signals, called
(a) Plasmid vector (b) Expression vector
(c) pBR 322 (d) pUC

6. Yeast artificial chromosomes have
(a) Origin of replication (b) Two telomere
(c) Both (d) None of these

7. *Agrobacterium tumefaciens* infect plant and produce tumor that is called
(a) Club (b) Apple scab
(c) Crown gall (d) Leaf curl

8. Ti plasmid found in
(a) *Agrobacterium tumeficians*
(b) *Agrobacterium Jaminifera*
(c) *Agrobacterium tumifacien*
(d) All of these

9. Which statement is wrong?
(a) Ti plasmid has T DNA flanking with two 25 base pair repeats
(b) T DNA encodes enzymes that convert plant metabolites to two classes of compounds that benefit bacterium
(c) Both (a) and (b) above
(d) None of the above

10. Luciferase gene code for
(a) Enzyme (b) Luciferin
(c) Antibiotics (d) All of these

11. For delivering DNA into cell, which is not applicable?
(a) Micro injection (b) Viral vector
(c) Liposome (d) Electro fusion

12. A small nucleotide sequence used to detect the presence of complementary sequences in nucleic acid sample called
(a) Primer (b) Oligonucleotide
(c) Probe (d) All of these

13. Nick translation done by
(a) DNA polymerase II
(b) DNA ligase
(c) DNA polymerase I
(d) Polynucleotide kinase

14. Hybridomas are employed for
(a) Production of somatic cells
(b) Killing cancer cells
(c) Synthesis of antibiotics
(d) Synthesis of monoclonal antibodies

15. DNA probe used in finger printing are
(a) Highly sensitive electron microscope
(b) UV beams
(c) DNA segment having radioactive isotope
(d) X- ray scanners

16. In DNA finger printing, the sequences which are matched are?
(a) Multisatellite DNA
(b) Moderately repetitive sequences
(c) Microsatellite DNA
(d) Satellite DNA

17. Nuclease enzyme which begins its attack from free end of polynucleotide is
(a) Endonuclease (b) Exonuclease
(c) Polymerase (d) Kinase

18. The technique of invertion of describing gene into DNA of plasmid vector is
(a) Gene splicing (b) Gene dressing
(c) Gene cloning (d) Gene drafting

19. These vectors can be grown in one host and, then moved into another without any extra manipulation called
(a) ARS vector (b) YAC vector
(c) Shuttle vector (d) pUC vector

20. Restriction endonuclease cleave DNA molecule at specific nucleotide sequence called
(a) Transnucleotide site (b) Restriction site
(c) Internal site (d) Hexanucleotide site

21. Extra chromosomal piece is known as
(a) Cosmid (b) Episome
(c) Plasmid (d) Bacteriophage

22. The technique of transfer of DNA molecules separated by gel electrophoresis to the nitrocellulose or nylon membrane is called
(a) Northern blot (b) Southern blot
(c) Western blot (d) None of these

23. DNA is visualized using UV transillumination after staining of
(a) Giemsa stain (b) Cosmic blue
(c) India ink (d) Ethidium bromide

24. In polymerase chain reaction which of the following is required essentially
(a) DNA ligase (b) DNA primer
(c) DNA polymerase (d) None of these

25. The initiation codon for translation in prokaryotes is
(a) UGA (b) AUG
(c) GAU (d) UUA

26. The uptake of plasmid DNA into bacterial cell is facilitated in the presence of
(a) Calcium chloride (b) Magnesium chloride
(c) Potassium chloride (d) None of these

27. The double stranded DNA molecule loses its viscosity upon
(a) Denaturation (b) Filtration
(c) Sedimentation (d) Concentration

28. Functional unit of DNA that specifies of one polypeptides is
(a) Codon (b) Cistron
(c) Muton (d) Recon

29. Regulated unit of genetic engineering is
(a) Operator gene (b) Promoter gene
(c) Regulator gene (d) Operon

30. The Eco RI enzyme is obtained from
(a) Virus (b) Salmonella
(c) Escherichia coli (d) Penicillium

31. Which of the following would not be used in preparing recombinant DNA?
(a) Plasmids (b) Phages
(c) Restriction enzymes (d) DNA polymerase III

32. The process which can be used to bring two genetically different cells together for common desired characters?
(a) Hybridization (b) Protoplast fusion
(c) PCR (d) Sexual reproduction

33. The Biolistic PDS- 1000 is an
(a) Antibiotics drug
(b) Gene therapy operation technique
(c) Instrument used to transfer DNA to cell
(d) Important cell fusion technique

34. Taq polymerase is used in PCR be of
(a) Law thermal stability
(b) High thermal stability
(c) High fidality
(d) High speed

35. DNA polymerase II used in PCR is concerned ith
(a) Initiation of DNA synthesis
(b) Termination of DNA synthesis
(c) Proof reading of DNA synthesized by taq polymerase
(d) Inhibition of taq polymerase

36. Poly acryl amide gel electrophoresis is used for DNA sequencing because, it
(a) Can separate different DNAs
(b) Can resolve DNA molecules differing in length
(c) Can break DNA into small fragments
(d) Can separate purine and pyrimidine

37. Purines and pyrimidines are cleaved by
(a) Dimethyl sulphate
(b) Hydrazine
(c) Piperazine
(d) Dimethyl sulphate and Hydrazine respectively

38. A vector should have which of the following properties
i. Small size *ii.* Multiple ori
iii. MCS *iv.* Low replication speed
(a) *i, ii, iii* (b) *i, ii, iv*
(c) *ii, iii, iv* (d) *i, iii, iv*

39. The enzyme used in foot printing technique
(a) Permease (b) DNA polymerase
(c) DNA ligase (d) RNA polymerase

40. pBR 322 is a
(a) Plasmid (b) Cosmid
(c) Bacteriophage (d) Phage

41. The cluster of unique restriction sites of cloning vectors are
(a) Polylinker (b) Marker
(c) Palindrome (d) All of these

42. Which of the following serves to alter the sequence so that the restriction enzymes will not damage the strand?
(a) Polymerase (b) Modification enzyme
(c) Endonuclease (d) Mutation

43. Cos stands for
(a) Cosmid (b) Complementary strand
(c) Cohesive site (d) Cloning site

44. Which of the following is/ are the restriction enzyme which produces the DNA fragment with blunt ends?
i. Sma I
ii. Eco RI
iii. Hind III
(a) I, II (b) II, III
(c) I, III (d) All

45. Reverse transcriptase which helps in the synthesis of DNA from RNA is encoded by
(a) Retrovirus (b) Polyomavirus
(c) Adenovirus (d) Herpes simplex virus

46. The catabolic plasmid is
(a) Ti plasmid (b) Cosmid
(c) YAC (d) Lambda 10

47. In the Ti plasmid, which gene causes the nick in DNA?
(a) Arc (b) Ocs
(c) Vir (d) Nos

48. Which of the following is an insertion bacteriophage vector?
(a) Lambda gt 10 (b) EMBL 3
(c) Charons (d) None of these

49. YAC is less stable than BAC, because of
(a) Chimeric effects
(b) Cloning of larger fragments
(c) Low frequency of crossing over
(d) All of these

50. An 8 base pair sequence recognized by rare cutter restriction enzyme is
(a) HincII (b) Not I
(c) Hpa II (d) Xho I

51. After the construction of recombinant lambda or cosmid library, the total DNA transferred into *E.coli* using a technique called
(a) Transformation (b) *In vitro* packaging
(c) Transduction (d) Conjugation

52. The element of yeast which contains the origin of replication is a/an
(a) Autonomously replicating sequence
(b) Self replicative
(c) Semi- conservative replication
(d) Bi- directional

53. Permanent transfection is also called as
(a) Insertional transfection
(b) Transient transfection
(c) Replicating vector
(d) Integrative trnasfection

54. The reason for integration of DNA at random places in the genome is
(a) Homologous recombination
(b) Non homologous recombination
(c) Crossing over
(d) Mutation

55. The sequencing methods used for determining the order of the nucleotide in a molecule of DNA is
(a) DNA labeling (b) DNA sequencing
(c) Gene therapy (d) Gene transfer

56. In the chain termination methods, the termination of process occurs because of
(a) Deoxy nucleotides (b) DNA polymerase
(c) DNA primer (d) Dideoxy nucleotides

57. Transposons are also called
(a) Jumping genes (b) Clock genes
(c) Multiple genes (d) Entrez genes

58. The number of distinct types of transposons is
(a) 2 (b) 1
(c) 4 (d) 3

59. Evidence for Transposons in the genome of an organism is
(a) Variegated phenotypes
(b) Variegated genotypes
(c) Marcus Rhoades
(d) Lerner

60. The phenomenon of the movement of genetic locus on the chromosome location is known as
(a) Transformation (b) Transposition
(c) Mutation (d) None of these

61. The concept of replacement of the mutant alleles with functional ones is
(a) Gene therapy (b) Transformation
(c) Gene cloning (d) Transposition

62. The DOP in DOP- PCR labeling indicates
(a) Degenerate oligonucleotide priming- PCR
(b) Degenerate oligonucleotide polymerase- PCR
(c) Degenerate optimized priming- PCR
(d) Degenerate optimize polymerase- PCR

63. Insertion of therapeutic genes into vegetative cells of the patient is known as
(a) Germ line gene therapy
(b) Somatic gene cloning
(c) Somatic gene therapy
(d) Germ line gene cloning

64. The type of gene transfer in which the organism receives the genetic material from its ancestor is
(a) Horizontal gene transfer
(b) Lateral gene transfer
(c) Vertical gene transfer
(d) None of these

65. The genetic technique which is used by the gene targeting is
(a) Crossing over
(b) Gene transfer
(c) Homologous recombination
(d) DNA labeling

66. Random insertion of genes is involved in
(a) Gene transfer (b) Gene trapping
(c) Gene targeting (d) All of these

67. The transplantation which includes transfer of living cells from one species to another is
(a) Allotransplantation
(b) Xenotransplantation
(c) Autograft
(d) Isograft

68. Bacteria that are capable of being transformed, whether naturally or artificially is known as
(a) Competent (b) Coefficient
(c) Compatable cell (d) Capable cell

69. The efficiency with which a competent culture can take up exogenous DNA is
(a) Transformation co efficiency
(b) Transformation efficiency
(c) Transformation compatibility
(d) Transformation competence

70. A genetic technique in which an organism is engineered to carry genes that have been made inoperative is
(a) Gene knockout (b) Gene targeting
(c) Gene transferring (d) Gene trapping

71. PCR is an indispensable adjunct to forensic DNA typing- commonly called
(a) Southern blotting (b) Northern blotting
(c) DNA fingerprinting (d) ELISA

72. In southern blotting, the DNA fragments are electrophoreses on an agarose gel to separate them by
(a) Molecular weight
(b) Size
(c) Ion concentration
(d) Agarose concentration

73. Western blotting technique is used to detect
(a) Proteins
(b) DNA
(c) Carbohydrate epitopes
(d) RNA

74. RAPD markers use DNA fragments from
(a) Southern blotting (b) PCR
(c) Genomic library (d) cDNA

75. RFLP loci is measured in
(a) Centi Morgan (b) Centimeter
(c) Picometre (d) Angstroms

76. Site directed mutagenesis is also known as
(a) Mononucleotide directed mutagenesis
(b) Polynucleotide directed mutagenesis
(c) Dinucleotide directed mutagenesis
(d) Oligonucleotide directed mutagenesis

77. The principle involved in southern blotting for the conversion of DNA fragments on agarose gel to nitrocellulose membrane is
(a) Cooling (b) Heating
(c) Adsorption (d) Absorption

78. In southern blotting, the process of using radio labeled probe for observing band position on x-ray film is
(a) Radioimmunoassay (b) X- ray diffraction
(c) Autoradiography (d) X- ray crystallography

79. In PCR, annealing occurs due to the change in the temperature from
(a) High temperature to low temperature
(b) High pressure to low pressure
(c) Low pressure to low temperature
(d) Low temperature to high temperature

80. RFLP is also known as
(a) DNA polymorphism (b) DNA profiling
(c) DNA sequencing (d) DNA hybridization

81. RFLP are detected as differential movement of a band on their gels. Each band is regarded as
(a) RFLP point (b) RFLP locus
(c) RFLP junction (d) RFLP origin

82. RAPD are produced by using
(a) PCR equipment (b) Thermo cycler
(c) Both *i* and *ii* (d) None of these

83. DNA profiling uses repetitive sequences that are highly variable, called
(a) RAPD (b) VNTR
(c) RFLP (d) SSR

84. Nylon membrane is used in northern blotting because
(a) It is positively charge
(b) It is negatively charge
(c) It is neutral
(d) None of these

85. Genomic library is constructed using
(a) Complete genome
(b) Partial DNA sequence
(c) Expressed DNA sequence
(d) Conserved sequence

86. Restriction endonuclease recognizes a specific Palindromic nucleotide sequence in the DNA. This Palindromic sequence is
(a) Direct repeat (b) Inverted repeat
(c) No repeat (d) (a) and (b)

87. "Biolistics" or "gene gun" is used to
(a) Kill a harmful gene
(b) Introduce the alien DNA into host cells
(c) To burst the nuclear membrane
(d) To make hole in nuclear membrane

88. DNA ligase is used to join DNA between
(a) 3'- OH and 5' P
(b) 5'- OH and 3' P
(c) Hydrogen bond between purine and pyrimidine
(d) None of the above

89. Two bacteria used in the genetic engineering are:
(a) Rhizobium and Nitrosomonas
(b) Rhizobium and Klebsiella
(c) Azotobacter and Rhizobium
(d) Agrobacterium and Escherichia

90. Type I restriction endonuclease enzyme is
(a) Eco PI (b) Eco RI
(c) Eco K (d) Pst

91. SI Nuclease cleave the
(a) ds DNA (b) ds RNA
(c) ss DNA (d) ss RNA

92. The enzyme DNase I acts on the
(a) ss DNA (b) ds DNA
(c) RNA (d) (a) and (b)

93. Complementary DNA (cDNA) is synthesized from
(a) DNA (b) mRNA
(c) Protein (d) Genetic engineering

94. Which of the following statement about the shuttle vector is false?
(a) It contains two origin for replication
(b) It carry two selectable marker one for each host
(c) It can live only in one host cell
(d) Its host belongs to two divergent groups such as bacteria and yeast, *E.coli* and human beings.

95. Phage lambda contains
(a) ds DNA (b) ds RNA
(c) ss DNA (d) ss RNA

96. In PCR the number of copies of DNA increases
(a) Logarithmically (b) Exponentially
(c) Arithmetically (d) Geometrically

97. Which of the following is not a scorable marker?
(a) Lux (luciferase) gene
(b) Gus (B- galacturinidase) gene
(c) Chloramphenical acetyl transferase (CAT)
(d) Thymidine kinase

98. DNA after isolation from the cells is ultimately precipitated in
(a) Chloroform (b) Ethanol
(c) Methanol (d) Acetaldehyde

99. In which of the following method of DNA transfer vector is not required?
(a) Particle or gene gun method
(b) Agro- infection
(c) Micro- injection
(d) (a) and (c)

100. Which of the following is not a reporter gene?
(a) Kan' (b) Nos
(c) Gag (d) Tet'

101. Beta-galactosidase enzyme produces which of the following compound to yield blue color?
(a) 5-Bromo-4-chloro-3-indolyl-b-D- galactosidase
(b) X- gal
(c) Galactose
(d) (a) and (b)

102. What is the function of linker in genetic engineering?
(a) Create cohesive end on blunt end
(b) Chemically synthesize DNA double strand
(c) To link Histone with DNA
(d) To bind ribosome to mRNA

103. Which of the following is first transgenic cow which produce human protein enriched milk
(a) Dolly (b) Rosie
(c) Romie (d) Romy

104. In the study of DNA fingerprinting which of the following is technique is used
(a) Southern blot (b) Northern blot
(c) Western blot (d) All of the above

105. Human preproinsulin contains following chains
(a) Alpha-chain and beta-chain
(b) Alpha-chain, beta chain and c-chain
(c) Alpha-chain, beta chain, c-chain and signal peptides
(d) None of the above

106. The first gene therapy was given in 1990 to a 4- year old girl. What is the disease from which she was suffering?
(a) Cystic fibrosis
(b) Huntington's Chorea
(c) Adenosine deaminase (ADA)
(d) Hunter's syndrome

107. A person suffering from ADA deficiency can be treated by gene therapy. The type of gene therapy employed for the treatment of patient is
(a) Gene replacement therapy
(b) Whole animal transformation
(c) Germline gene therapy
(d) (a) and (b)

108. A transgenic organism contains the genes from
(a) Related organism
(b) Unrelated organism
(c) Within a species
(d) None of the above

109. Which of the following human protein is used for treatment of "emphysema"?
(a) Trypsinogen (b) Chymotrypsinogen
(c) Chymotrypsin (d) Alpha- I- antitrypsin

110. Emphysema is a
(a) Reproductive disease
(b) Nutritional disorder
(c) Respiratory disorder
(d) Cardiovascular disease

111. The crystal (*Cry*) protein, which has insecticidel property is synthesized by
(a) *Pseudomonas putida*
(b) *Bacillus thuringiensis*
(c) *Streptomyces griseus*
(d) *Escherichia coli*

112. The crystal proteins inactive in bacteria are called protoxin. This protoxin isconverted into active toxin in the gut of insect due to
(a) Alkaline environment
(b) Acidic environment
(c) Neutral environment
(d) Heat

113. RNA i is referred to as
(a) Interference RNA (b) Infectious RNA
(c) Inserted RNA (d) Inoculated RNA

114. Proinsulin is matured into insulin by removal of
(a) A- chain of insulin
(b) B- chain of insulin
(c) C- chain of insulin
(d) Both B- chain and C- chain of insulin

115. Which of the following fungi is used for biological control of plant disease?
(a) *Trichoderma viridae*
(b) *Penicillium notatum*
(c) *Aspergillus flavus*
(d) *Saccharomyces cerevisiae*

116. The ribozymes is a
(a) Ribosome degrading enzyme
(b) RNA molecules having catalytic action
(c) Special type of ribosome
(d) Ribosome found in bacteria

117. Which of the following enzymes does not need a template for DNApolymerization but insists on a primer?
(a) DNA polymerase
(b) Terminal nucleotidyl transferase
(c) Reverse transcriptase
(d) Nucleases

118. The substances used by the proliferating infecting *Agrobacteria* cells as sources of carbon and nitrogen are called
(a) Tumors (b) Transposons
(c) Opines (d) Cosmids

119. Which of the following was found to possess the genes for functions like oncogenesis, opine synthesis, opine breakdown, differentiation of the tumor into root, shoot or root and shoot on a hormone less synthetic medium?
(a) Ti- plasmids
(b) Ti- opine- genes
(c) Ri- plasmids
(d) T- DNAs

120. Which of the following methods makes it possible to insert a gene at specific sites in the T- region of a Ti- plasmid that has all its normal properties of transfer and T- DNA insertion and maintenance of the state of integration?
(a) Somatic cell hybridization
(b) Recombinant virus mediated gene transfer
(c) Site- specific mutagenesis
(d) Microinjection

121. The technique devised to produce vast quantities of strong and healthy plantlets by rapid vegetative multiplication under controlled conditions is known as
(a) Micro propagation (b) Micro induction
(c) Microinjection (d) Suspension

122. Why are DNA polymerases from thermophilic organisms used in the polymerase chain reaction?
(a) Because they are required to keep the two strands separated
(b) Because they cannot add new nucleotides at low temperatures
(c) Because the primers will not attach to a complementary sequence unless the polymerases warm the reaction tube
(d) Because the priming and extension steps must be carried out at high temperatures to prevent the single strands from reannealing

123. Transgenic micro- organisms have been used to improve or benefit all but which of the following?
(a) Meat yield (b) Medical diagnosis
(c) Crop improvement (d) Pest reduction

124. In what type of gene therapy are genes inserted into an egg, sperm or early embryo?
(a) *Ex vivo* therapy (b) Germline therapy
(c) *In vivo* therapy (d) Antisense therapy

125. What type of genome map is the most ideal for understanding the nature of gene, what they code for, and their functions?
(a) Linkage maps (b) Physical maps
(c) Sequence maps (d) Fingerprint maps

126. All methods of DNA fingerprinting depend on some variation of what strategy?
(a) RFLP (b) GMOs
(c) Gene therapy (d) Microarray analysis

127. Microarray analysis has allowed scientists to view what phenomenon?
(a) The genome sequence in a cell
(b) The cDNA of a cell
(c) The RFLPs of a cell
(d) The expression of specific genes in a cell

128. In a sequence genome, candidate protein-encoding genes are identified by searching for
(a) STRs (b) RFLPs
(c) ESTs (d) ORFs

129. DNA markers are
(a) DNA polymorphisms that are useful for genetic mapping
(b) Unique introns spliced out of genes during intron processing
(c) Extremely rare occurrences in the non coding regions of the genome because mutations are exceedingly rare events
(d) Both A and B are correct

130. An STR is
(a) A chloroplast enzyme
(b) Repetitive sequence used in DNA fingerprinting
(c) Short oligonucleotide probe
(d) Identified by mRNA

131. An SNP is
(a) An enzyme found in chloroplasts used in PCR
(b) A repetitive sequence used in DNA fingerprinting
(c) A short oligonucleotide probe found in a microarray
(d) A single base substitution, addition, or deletion in the sequence of a gene

132. RFLP analysis is a technique that
(a) Uses hybridization to detect specific DNA restriction fragments in genomic DNA
(b) Is used to determine whether a gene is transcribed in specific cells
(c) Measures the transfer frequency of genes during conjugation
(d) Is used to detect genetic variation at the protein level

133. The best choice to measure the rate of RNA synthesis in a culture of human cellswould be to measure the rate of incorporation of radioactive
(a) Adenine (b) Cytidine
(c) Guanosine (d) Uridine

134. One strand of a double stranded DNA fragment contains the following number of bases: G = 20, A = 10, T =15, C = 10. What numbers of each base are found in thedoubles stranded fragment?
(a) G = 30, A = 25, T = 25, C = 30
(b) G = 30, A = 30, T = 25, C = 25
(c) G = 35, A = 20, T = 35, C = 20
(d) G = 35, A = 25, T = 25, C = 35

135. A male with hemophilia type A is found to have a point mutation which changes alysine residue to a glutamate residue in factor VIII. The point mutationresponsible for hemophilia in the patient is located in which of the following structures?
(a) Enhancer (b) Exon
(c) Intron (d) Poly A tail

136. By using "reference sequence" developed in the human genome project,individual differences are now being detected using
(a) NPs (b) PSNs
(c) SNPs (d) "snaps"

137. In the discovery of introns, a DNA molecule called __ was formed that had the same nucleotide sequence as the gene that produced the mRNA
(a) mDNA (b) rDNA
(c) sDNA (d) cDNA

138. All fragments cut by most restriction endonucleases have
(a) Complementary double- stranded ends
(b) Supplementary single- stranded ends
(c) Double- stranded "sticky" ends
(d) Complementary single- stranded ends

139. Humans are not affected by glyphosphate in plant crops because
(a) While it is toxic to plants it is a normal metabolite of humans
(b) Humans do not produce aromatic amino acids
(c) It is used only on those crops that convert it to a harmless substance
(d) Humans lack the enzymes to interact with it

140. Plant cells receiving Ti plasmids with a TMV viral gene encoding the viral coat protein become immune to the TMV virus because
(a) The introduction of the viral gene triggered an immune response in the plant's own immune system
(b) The gene causes the cell walls to become impermeable to the virus
(c) TMV does not infect cells already infected with TMV
(d) In plants, the gene encodes an antiviral substance2

141. Wilt- proof flowers are being engineered by making them insensitive to
(a) Methylene (b) Propylene
(c) Butylenes (d) Ethylene

142. The ingestion of BST in milk is no threat to humans because
(a) It is destroyed in the stomach
(b) It cannot be digested and will pass through as waste
(c) They do not have the hormones to utilize it
(d) It is destroyed in the pasteurization process

143. Engineering plants that are resistant to glyphosate was an important advancement because
(a) Glyphosate promotes frost damage
(b) Glyphosate encourages the production of fruit that is lower protein
(c) Glyphosate is the active ingredient in herbicide
(d) Glyphosate prevents the transfer of genes into the plants

144. Match the following :

List-I	List-II
i) Adenovirus	a. ss DNA
ii) Parvovirus	b. ds DNA
iii) Retrovirus	c. RNA virus
iv) Reoviridae	d. ds RNA

(a) *i-c, ii-a, iii-d, iv-b* (b) *i-c, ii-d, iii-b, iv-a*
(c) *i-b, ii-a, iii-c, iv-d* (d) *i-b, ii-d, iii-a, iv-c*

145. A scientist has a processed mRNA transcript for a gene which he/she wants toclone into a bacterial vector. What must he/she do as a first step in this process?
(a) Use PCR to create a cDNA molecule
(b) Generate primers to the processed mRNA
(c) Sequence the mRNA transcript
(d) Digest the mRNA and cloning vector with the same restriction endonuclease

146. Following insertion of a cloning vector into a cloning host, why are the host cells plated on selective medium, which usually contains an antibiotic?
(a) Because all the host cells are naturally resistant to the antibiotic
(b) Because all the host cells are naturally sensitive to the antibiotic
(c) Because the physical process of transformation causes the host cells to become resistant to the antibiotic
(d) Because only those host cells that have been transformed by the vector will grow in the presence of the antibiotic

147. Which is not a benefit of using mammalian cells as cloning and expression hosts?
(a) They may prevent allergic reactions to animal products
(b) They allow large scale production of enzymes, hormones, and antibodies
(c) Scientists do not have to use human cadavers as sources for hormones
(d) Mammalian clones replicate faster than prokaryotic clones and will thus express a protein product much quicker

148. A vector that is obtained as a linear molecule containing single T overhanging bases at each end and also contains a selectable marker gene and an origin ofreplication would likely be called a(n)
(a) Shuttle vector (b) Transcribable vector
(c) PCR cloning vector (d) Expression vector

149. Gene cloning can involve inserting genes into
(a) Bacteria (b) Protein
(c) Humans (d) Machines

150. Whole animal cloning puts a nucleus from a cell of one animal into
(a) The chromosome of a bacterium
(b) The plasmid of a bacterium
(c) An egg cell from another animal
(d) A sperm cell from another animal

151. Positional cloning refers to:
(a) Using a selection procedure to clone a cDNA
(b) Cloning a portion of a gene using PCR
(c) Isolating a gene by PCR using primers from another species
(d) Mapping a gene to a chromosomal region and then identifying and cloning a genomic copy of the gene from the region

152. Plasmids and phages. Most high copy number plasmid vector used for cloning in Escherichia coli are based on which natural replicon?
(a) Col E1 (b) F
(c) R6K (d) SV 1

153. What is the difference between a PAC and a BAC?
(a) One is a plasmid- based vector and the other is derived from a yeast chromosome
(b) One is derived from bacteriophage P1 and one is derived from bacteriophage B1
(c) One is derived from the E. coli F- plasmid and one is derived from bacteriophage P1.
(d) None of these

154. Cloning strategies. In higher eukaryotes, which of the following are differencesbetween genomic and cDNA clones?
(a) Only genomic DNA contains genes
(b) Only genomic DNA contains introns
(c) Only cDNA contains a coding region
(d) None of these

155. Which of the following vectors does not contain an origin of replication from a yeast chromosome?
(a) yCp (b) YAC
(c) Yep (d) YRp

156. The principle behind yeast two- hybrid system is :
(a) The detection of protein- protein interactions by assembling a functional transcription factor from two fusion proteins
(b) The detection of protein- protein interactions in a pair if hybrid yeast strains
(c) The detection of protein- protein interactions by studying the hybridization of two cDNA sequences
(d) None of these

157. Which of the following viral vectors has the largest capacity for foreign DNA?
(a) Adenovirus (b) Retrovirus
(c) Baculovirus (d) None of these

158. Gene transfer to plants, T- DNA is :
(a) DNA of plasmid origin which is transferred to the Agrobacterium chromosome
(b) DNA from the chromosome of Agrobacterium species which is transferred to the plant genome
(c) DNA of plasmid origin which is transferred to the plant genome
(d) None of these

159. Which of the following enzymes is used to synthesize a cDNA library, but not to synthesize a genomic library?
(a) Restriction enzymes
(b) DNA ligase
(c) Reverse transcriptase
(d) SI nuclease

160. Endonuclease is an enzyme that hydrolyses:
(a) A nucleotide at 3' ends of oligonucleotides
(b) A phosphodiester bond located in the interior of a polynucleotide
(c) A nucleotide from either termini of an oligonucleotide
(d) A peptide bond located in the interior of polypeptide

161. In Cohen and Boyer's transfer of toad genes to a bacterial cell, restrictionendonucleases were used to:
(a) Isolate sections of cloned bacterial plasmids
(b) Isolate sections of toad DNA
(c) Cleave bacterial plasmids
(d) Both b and c are correct

162. A probe is used in which stage of the gene transfer process?
(a) Cleaving DNA (b) Recombining DNA
(c) cloning (d) screening

163. The cofactor required by the type I restriction Enzymes for the methylation of DNA
(a) Methane
(b) S- adenosyl methionine
(c) Methionine
(d) Glutamine

164. The short single stranded oligonucleotide sequences which can anneal with single stranded DNA strands during PCR are called
(a) RNA's (b) cDNAs
(c) Annealers (d) Primers

165. The enzyme that can change the covalently closed circular DNA into opencircular DNA is
(a) Conformational enzymes
(b) Super coil removers
(c) Topoisomerases
(d) Telomers

166. Determining relative positions of different genes on the DNA molecule iscalled ____
(a) Gene mapping (b) Gene localization
(c) Gene sequencing (d) Gene marking

167. The set of recombinant clones that contain all the DNA present in an individual organism is referred to as
(a) DNA collection (b) Genomic library
(c) cDNA library (d) Gene library

168. The sequences which are required by the bacterial expression vectors for the gene product to get expressed
i. Shine- Dalgarno sequences
ii. Transcription termination sequences
iii. Sequence to promote post- translational modification of protein product
iv. Oligonucleotide sequences with poly A tailing
v. Adaptor sequences with transcription terminations
(a) *i* & *iv* (b) *i, ii, v*
(c) *ii, iii, iv* (d) *i, ii, iii*

169. The single stranded RNA that is complementary to a transcribed mRNA sequence of a cell is
(a) Invert of RNA (b) Reversible RNA
(c) Antisense RNA (d) Transcribable RNA

170. Arrange the Yeast derived vectors (Yep; Yip; YRp) based upon theirtransformation efficiency
(a) YEp >YRp >YIp (b) YEp >YIp >YRp
(c) YIp >YRp >YEp (d) YIp >YEp >YRp

171. Following restriction digestion, self annealing of the fragments having sticky ends can be prevented by treating with
(a) Hydrolase
(b) Topoisomerase
(c) Alkaline phosphatase
(d) Polynucleotide kinase

172. The method of localizing specific DNA or RNA sequences present in a portion or section of a tissue or even cells, which uses a labeled complementary DNA orRNA strands as probes
(a) Gene mapping (b) RT PCR
(c) Blotting method (d) *In situ* Hybridization

173. The cofactor required by E. coli DNA ligase is
(a) NAD^+ (b) NADH
(c) ATP (d) ADP

174. In RAPD, the characteristics of primers include
i. Arbitrary primers having random sequences
ii. Primer length from 8 to 12 nucleotides
iii. Lengthy Primers of 13 to 18 nucleotides to reduce number of fragments
iv. Primers are complementary for some known gene are used
v. Primers bind to random sites on the DNA strands
(a) *i, iii, iv* (b) *ii, iv*
(c) *i, iii, v* (d) *i*, ii, *v*

175. The recognition site chosen by RFLP restriction enzymes for genetic disease analysis
(a) coding regions (b) VNTR segments
(c) Telomeres (d) Centromere regions

176. The water soluble vitamin that is incorporated into dUTP as a non- radioactive label for the DNA probe is
(a) Vit C (b) Vitamin B6
(c) Vitamin B7 (d) Ascorbic acid

177. The biotin binding protein, Avidin is a
(a) Polymeric
(b) Dimer
(c) Tetramer
(d) Monomer

178. The approach where larger, 150,000 base pairs go together to create chromosomes is known as
(a) "hierarchical shotgun" approach
(b) Hydrolysis
(c) Methylation of bases
(d) Methylation

179. If DNA ligase was not used in the creation of a recombinant plasmid:
(a) Base-pairing would occur but the sugar phosphate backbone would not be connected.
(b) Links between adenine and thymine would not occur.
(c) Links between guanine and cytosine would not occur.
(d) Hydrogen bonds between complementary bases could not form.

180. Taq polymerase is similar in activity to:
(a) DNA polymerase I
(b) DNA polymerase II
(c) DNA polymerase III
(d) RNA polymerase

181. Psi sequence in retrovirus is responsible for
(a) Replication (b) Transcription
(c) Packaging (d) None of these

182. A heterologous protein for its expression in the milk of a transgenic animal should be under the control of the promoter of the gene coding for
(a) Beta- globin (b) Beta- Lactoglobulin
(c) Preproinsulin (d) Lac Z

183. Which of the following has been produced commercially from mammalian culture?
(a) Plasminogen activator
(b) Antibacterial antibody
(c) Insulin
(d) Rennin

184. Recombinant DNA human health care product developed for treating haemophiliaces is
(a) Interferon
(b) Lymphokines
(c) Coagulation Factor IX
(d) Phytohaemaglutanin

185. Aminopterine is used during the production of hybridoma cells because it
(a) Blocks the de novo pathway
(b) Prevents the growth of B cells
(c) Prevents the growth of myeloma cells
(d) Blocks the synthesis of Ig by B cells

186. In animal cell cultures, the addition of serum to media is essential for providing?
(a) Amino acid for protein synthesis
(b) Nucleotide for DNA synthesis
(c) Growth factors
(d) All of these

187. What is primary purpose of neomycin in creating mice with knock- outs in gene X?
(a) Neomycin selects for the survival of embryonic stem cells (ES) that have incorporated the mutant gene X anywhere in the genome
(b) Neomycin selects for the survival of ES cells that have incorporated the mutant gene in the place of the wild- type gene
(c) Neomycin prevents candida infection during ES cell culture that does not have gene X
(d) Neomycin makes the gene X- knock out mice resistant to candida infection

188. Cell line TCL 598 is used in
(a) Collagenase
(b) Pepsin
(c) Plasminogen activator
(d) All of these

189. In cellular Biochemical's beta and gamma interferon's cell lines used are
(a) Chinese hamster ovary
(b) Human tumor
(c) Mouse fibroblast
(d) Human leucocytes

190. Cell line isolated from normal, non cancerous human embryonic tissue
(a) MRC- 5 (b) TCL 598
(c) Human tumor (d) None of these

191. The transgene construct is prepared in a buffer solution and injected into
(a) Male nuclei of fertilized egg
(b) Female nuclei of fertilized egg
(c) Male pronuclei of fertilized egg
(d) Female pronuclei of fertilized egg

192. Bacterial genes cys E and cys M are used in
(a) Gene farming
(b) Increase body growth
(c) Improved wool production
(d) Disease diagnosis

193. Hybridoma cells are
(a) Non anchorage dependent and can't be grown in suspension culture
(b) Non anchorage dependent and can be grown in suspension culture
(c) Anchorage dependent and can be grown in suspension culture
(d) Anchorage dependent and can't be grown in suspension culture

194. Which of the following mice are used for immunization in the Hybridoma technology
(a) Swiss mice (b) Balb/c mice
(c) Out bred mice (d) Indigenous mice

195. Hayflick limit cells undergo the process
(a) Apoptosis (b) Transition
(c) Senescence (d) Duplication

196. Which of the following statement/s are true about cell lines?
i. Continuous Cell Lines shows the property of ploidy
ii. Finite cell line shows the property of contact inhibition
iii. Continuous cell line shows the property of contact inhibition
(a) *i, ii, iii* (b) *i, ii*
(c) *i* (d) *iii*

197. Consider the following statements

I. The myeloma cell is actually a double mutant

II. Myeloma cell is capable of producing thymidine kinase

III. Myeloma cell is incapable of producing antibodies

Which of the above statements are true about myeloma cell?

(a) I, II (b) I, III

(c) II, III (d) I, II, III

198. Cell- cell adhesion molecules (CAMS) are

(a) Calcium dependent

(b) Sodium independent

(c) Calcium independent

(d) Potassium independent

199. The conditions which favor the cell proliferation is

(a) Low cell density (b) Absence of EGF

(c) High cell density (d) High calcium levels

200. Plating density is the ratio of

(a) Number of cells per volume of culture medium

(b) Number of cells per volume of biomass

(c) Number of cells per volume of reactor

(d) Number of cells per volume of air

201. Disrupting the normal relationship of animal cells with their neighboring cell wall result in

(a) Tissues

(b) Cell fragments

(c) Debris

(d) Cell suspension cultures

202. The time between successive generation of cells is

(a) Delay time (b) Generation time

(c) Dead time (d) Pause

203. The terms ATCC, ECACC and NIGMS are the abbreviated forms for

(a) Animal cell culture collection repositories

(b) Isolated plant cells

(c) Animal cell lines

(d) Animal cell products

204. The act of transfer of a portion of culture and inoculating into fresh medium is termed

(a) Filtration (b) Addition

(c) Passage (d) Transfection

205. In the term He La cell lines, the immortal cell lines, He La stands for

(a) The tissue of origin

(b) The patient name and type of cancer

(c) Type of cancer

(d) Patient name

206. Cells in culture require nutrient optimization for

(a) Promoting anchorage to substrate

(b) Maximize nutritional uptake

(c) Obtain maximum cell density

(d) All of these

207. Introduction of an exogenous gene into a living organism is called

(a) Transgene (b) Transgenesis

(c) Transposon (d) Transposition

208. During the process of fertilization, after the sperm enters the ovum, but before they fuse, the nucleus of a sperm or an egg cell is called

(a) Micronucleus

(b) Bipolar nucleus

(c) Pro nucleus

(d) Pro fertilization nucleus

209. Carboxymethylcellulose, BSA (100 ug/ml) or Pluronic F-68 (0.1%) are added to the animal cell culture media for

(a) Increase osmolality (b) Decrease osmolality

(c) Decrease Viscosity (d) Increase Viscosity

210. Match the following

Cell line	Organism
I. YAR	(P) Hamster
II. YAC- 1	(Q) African Green Monkey
III. Vero cells	(R) Mouse
IV. CHO	(S) Human

(a) I- S; II- R; III- Q; IV- P

(b) I- R; II- S; III- Q; IV-P

(c) I- Q; II- R; III- S; IV- P

(d) I- S; II- R; III- P; IV- Q

211. During the preparation of cell culture media the ingredients are to be prepared in WFI which is an abbreviated form for

(a) Water filter incubation

(b) Whatman filtered injection

(c) Water for Injection

(d) Water for Introduction

212. Supplementation of Fibronectin is essential for

(a) Transformed cell line

(b) Anchorage dependent cell line

(c) Immortal cell line

(d) He La cell line

213. Disabling a gene to understand its function is the principle involved in

(a) Gene knockout studies

(b) Gene elimination studies

(c) Gene gun method

(d) Vaccination

214. Loss of the phenotypic characteristics typical of the tissue from which the cell has been isolated is referred to as

(a) Differentiation (b) Dedifferentiation

(c) Redifferentiation (d) Cell transfer

215. Which method is the most effective method used to produce transgenic animals

(a) Fusion method

(b) Biolistic method

(c) CaMV mediated method

(d) Retrovirus method

216. The cells which transfers the desired gene into progeny of the transfected animals are

(a) Somatic cells (b) Germ cells

(c) Bone marrow cells (d) Adult stem cells

217. Which of the substrates offer maximum surface area for the cultivation of anchorage dependent cells?

(a) Hollow fiber perfusion systems

(b) Macro carriers

(c) Petriplates

(d) Nunc cells

218. Cell culture medium is supplemented with serum extracted from

(a) Bovine (b) Equine

(c) Both (a) and (b) (d) Human

219. For the detoxification of free radicals, cell culture is supplemented with the element

(a) Fe (b) Ca

(c) Se (d) S

220. A monolayer culture in which all the cells are in contact with other cells around periphery and there is no substrate left uncovered is known as

(a) Complete culture (b) Confluent

(c) Stacked culture (d) Explants

221. The percentage of anchorage dependent cells inoculums that attaches to the substrate within a stated period of time is referred to as

(a) Saturation density (b) Attachment

(c) Seeding efficiency (d) Split ratio

222. The type of engineering involved in changing genes in eggs, sperms, or very early embryos

(a) Germline engineering

(b) Gamete engineering

(c) Cell line engineering

(d) Macrospore engineering

223. The gene therapy is opened by Monochromosomal Transfer Techniques with the help of

(a) Yeast artificial Chromosome

(b) Fungal artificial Chromosome

(c) Mammalian artificial Chromosome

(d) Bacterial artificial Chromosome

224. The method is used to disrupt endogenous loci, resulting in targeted null alleles is

(a) Allelic elimination (b) Gene knockout

(c) Gene elimination (d) Allelic nullification

225. Name the enzyme that is expressed by the cells which are being continuous cultivated for several generations

(a) Lipase (b) Telomerse

(c) Polymerase (d) Endonuclease

226. Supplementation of $NaCO_3$ is done for one of the following purposes

(a) To maintain pH and supply sodium ions

(b) To supply Sodium ions only

(c) Maintain pH alone

(d) None of these

227. Identify the product which does not require mammalian cells to be cultured

i. Erythropoietin

ii. t- PA

iii. Monoclonal antibodies

iv. Blood clotting factors

v. Atropine

vi. Pheromones

(a) (*iii*) (b) (*i*) & (*ii*)

(c) (*v*) & (*vi*) (d) (*iii*), (*v*) & (*vi*)

228. Transgenic cattle milk containing human proteins, which may help in the treatment of

(a) Rheumatoid arthritis

(b) Huntington's disease

(c) Emphysema

(d) Polio

229. A transgenic cow exists ___ that produces a substance to help to grow human

(a) RBC (b) WBC

(c) Insulin (d) Blood platelets

230. The business of micro propagation is due to

(a) Division (b) Totipotency

(c) Uptake of nutrients (d) Attachment

231. The hydrocarbon secreted by plants which is used in varnishes is

(a) Cellulose (b) Tannin

(c) Resin (d) Rubber

232. *In vitro* fusion of isolated protoplasts to form a hybrid cell is commonly referred to

(a) Cell hybridization

(b) Plant cell hybridization

(c) Somatic hybridization

(d) Cybridization

233. The borrowed substrates of the primary metabolic pathways that have a defensive role in plants are referred to as
(a) Plantibodies (b) Resins
(c) Static agents (d) Secondary metabolites

234. After subculture, the growth factor which inhibits the root development is
(a) Auxins (b) Cytokinins
(c) Gibberlins (d) Inositol

235. The isoprenoids extensively used for their aromatic qualities are
(a) Alkaloids (b) Flavonoids
(c) Carotenoids (d) Terpenoids

236. Match the following

Essential elements	Examples
I. Macroelement	P. Sucrose
II. Microelement	Q. Nicotinic acid
III. Organic supplement	R. KI
IV. Carbon source	S. KNO3

(a) I-S; II-R; III-Q; IV-P
(b) I-R; II-S; III-Q; IV-P
(c) I-S; II-P; III-Q; IV-R
(d) I-S; II-R; III-P; IV-Q

237. *Ex situ* conservation of rare and endemic or threatened plants by tissue culture techniques is also called
(a) Excavation (b) Conservation
(c) Plant regeneration (d) Tissue regeneration

238. Which of the following is a vector dependent method of gene transfer?
(a) Microinjection
(b) *Agrobacterium* mediated transfer
(c) Biolistic method
(d) Electroporation method

239. Diploidisation of haploid plants in culture can be achieved by treatment with
(a) Sucrose (b) $ZnSO_4$
(c) Colchicine (d) Gibberlic acid

240. Haploid plant production from microspores of anthers is referred to as
(a) Antherogenesis (b) Androgenesis
(c) Gametogenesis (d) Gamete cultures

241. Artificial culture of haploid plants is done by
(a) Culturing from anther tissues
(b) Culturing from microspores
(c) Both A and B
(d) Fusion of male and female gamete

242. The other name for organic culture is
(a) Intensive farming
(b) Industrial agriculture
(c) Permaculture
(d) All of these

243. *Agrobacterium rhizogenes* mediated transformation efficiency was assessed under the influence of
(a) Sonication (b) Calcium treatment
(c) Acetosyringone (d) All of these

244. Match the components with their appropriate functions

COMPONENTS
I. Ultrasonic frequency
II. Calcium
III. Macerating enzymes
IV. Acetosyringone

FUNCTION
A. protoplast isolation
B. activates vir gene in Agrobacterium
C. signaling pathways
D. cell wall disruption

(a) I-D;II-C;III-A;IV-B (b) I-D;II-C;III-B;IV-A
(c) I-C;II-D;III-A;IVB (d) I-D;II-A;III-C;IV-B

245. The selecting agent used to detect high- producing cell line is
(a) Pimelic acid (b) Alkaloids
(c) Gibberellic acid (d) Anthocyanins

246. Elicitors produced by microorganisms are called
(a) Endogenous elicitors
(b) Abiotic elicitors
(c) Exogenous elicitors
(d) None of these

247.

GROWTH	REGULATORS FUNCTION
I. Batasins	A. Potent growth inhibitor
II. Brassins	B. Saturated primary alcohol
III. Triacontanol	C. Steroid growth promoters
IV. Xanthoxin	D. Causes dormancy in bulbs.

(a) I-D;II-C;III-B;IV-A (b) I-D;II-C;III-A;IV-B
(c) I-C;II-D;III-B;IV-A (d) I-D;II-A;III-B;IV-C

248. Abiotic elicitors include
(a) UV light (b) Peptides
(c) Oligogalacturans (d) Chitosan

249. Biodiesel produced from hemp is known as
(a) Hempoline (b) Oakum
(c) Retting (d) Manila hemp

250. Resistance to herbicides glyphosate in transgenic plants is brought about by
(a) Over production of RNA polymerase
(b) Underproduction of EPSP synthase
(c) Overproduction of EPSP synthase
(d) Overproduction of Aspartate amino transferase

251. What role do opines play in crown gall diseases?
(a) Source of carbon, nitrogen and energy for the Agrobacterium
(b) Transfer of T-DNA to plant cell
(c) Attachment of Agrobacterium to plant
(d) Induction of the expression of vir genes

252. Synthetic seed is produced by encapsulating somatic embryo with
(a) Sodium alginate
(b) Sodium nitrate
(c) Sodium acetate
(d) Sodium sulphate

253. Match the following

Group I	Group II
A. Meristem culture	1. Virus elimination
B. Suspension culture	2. Homozygosity
C. Protoplast culture	3. Packed cell volume
D. Anther culture	4. Embryo rescue
	5. Macerozyme
	6. Liposome

Codes:

	A	B	C	D
(a)	2	1	4	6
(b)	1	3	5	2
(c)	3	2	5	4
(d)	5	6	2	1

254. Match the following

Group I	GroupII
A. Biolistic	1. Gene pulser
B. Agrobacterium	2. PDS 1000 He
C. Electroporation	3. micromanipulator
D. Microinjection	4. Silicon carbide
	5. Vir protein
	6. Rolc

Codes:

	A	B	C	D
(a)	2	5	1	3
(b)	2	1	6	4
(c)	5	6	2	4
(d)	1	5	4	2

255. The most widely used chemical for protoplast fusion, as fusogen, is
(a) Manitol
(b) Polyethylene glycol (PEG)
(c) Sorbitol
(d) Glycerol

256. Commonly used reporter gene in plant expression vector is
(a) Ti. gene of Agrobacterium tumifaciens
(b) GUS gene
(c) Beta- Lactamase gene
(d) Alpha- anylase gene

257. 'Golden rice' is a promising transgenic crop. When released for cultivation, it will help in
(a) Herbicide tolerance
(b) Producing a petrol like fuel form rice
(c) Alleviation of vitamin A deficiency
(d) Pest resistance

258. White's medium is used for?
(a) Culture of shoot (b) Root culture
(c) Protoplast culture (d) Anther culture

259. Which of the following has been found a suitable medium for anther culture?
(a) Agar- Agar (b) M.S. medium
(c) L.S. medium (d) Potato extract medium

260. Variations observed during tissue culture of some plants are known as
(a) Clonal variations
(b) Somatic variations
(c) Somaclonal variations
(d) Tissue culture variations

261. Plants developed in vitro culture from pollen grain are :
(a) Androgenic haploids
(b) Pollen plants
(c) Male plants
(d) Sterile plants

262. Cryoprotectant used by arctic frog is
(a) Polyols
(b) Glycerol
(c) Both (a) & (b)
(d) Glucose

263. Cryoprotectants operate by
(a) Increasing the solute concentration
(b) Increasing the permeability of cells
(c) Decreasing the cell permeability
(d) Increasing the temperature

264. Which of these sentences are true regarding the identifying organisms?

I. By using RFLP technology, DNA fingerprinting cannot be generated

II. The fingerprint can be used to determine family relationship

III. Hybrid line can be identified by minor differences in their DNA sequences

(a) I, II (b) II, III

(c) I, III (d) I, II, III

265. Which of these sentences are true regarding the cloning of animal cells?

I. The developmental process in animals doesn't usually permit cloning as easily as in plants.

II. Worms are capable of regenerating a whole organism from a small piece

III. Vertebrates are capable of regenerating a whole organism from a small piece

(a) I, II (b) II, III

(c) I, III (d) I, II, III

266. Match the following diseases with their medicines used for treatment

P. Insulin

Q. Erythropoietin

R. Tissue plasminogen activator

I. dissolves clot after heart attack

II. Diabetes

III. Anemia associated with dialysis for kidney disease

(a) P-II; Q-III; R-I

(b) P-III; Q-I; R-II

(c) P-I; Q-II; R-III

(d) P-II; Q-I; R-III

267. Which of these sentences are true regarding the efficiency of animal cell division?

I. The rate of cell division of adult cells is much slower than that of the cells of the early frog embryo.

II. Gene expression is not regulated by the loss of DNA but by the turning off of specific genes.

III. The role of a particular cell type depends on the combination of regulatory proteins it expresses.

(a) I, II

(b) II, III

(c) I, III

(d) I, II, III

268. Match the following organisms with their respective cell stages used for successful nuclear transfer

Organism	Cell stage
P. Pig	I. 8-celled stage
Q. Rabbit	II. 120-cell blastocyst stage
R. Cow	III. 32 to 64 cell early embryos
S. Mice	IV. 4-celled balstomere nuclei

(a) P-IV; Q-III; R-II; S-I (b) P-III; Q-I; R-IV; S-II

(c) P-II; Q-I; R-IV; S-III (d) P-II; Q-III; R-;IV; S-I

269. Match the following item at which embryonic gene activation occurs between species

P. Mice I. 8-16 cell stage

Q. Humans II. 2-cell stage

R. Sheep III. 4-8 cell stage

(a) P-II; Q-III; R-I (b) P-III; Q-I; R-II

(c) P-I; Q-II; R-III (d) P-II; Q-I; R-III

270. Which of these sentences are true regarding the reproductive choice?

I. Somatic cell nuclear transfer cloning could be the only technique for individuals to create a genetically related child

II. When individuals have alternative means of procreating, cloning might be chosen

III. Somatic cell nuclear transfer is merely a 'manufacturing of humans' than of reproduction

(a) I, II (b) II, III

(c) I, III (d) I, II, III

271. Match the following animal with their proteins

P. Antithrombin III I. Pig

Q. Lysostaphin II. Sheep

R. Alpha-1-antitrypsin III. Cow

S. Factor IX IV. Goat

(a) P-IV; Q-;III; R-II; S-I (b) P-III; Q-I; ;R-IV; S-II

(c) P-II; Q-I; R-IV; S-III (d) P-II; Q-III; R-IV; S-I

272. Which of the methods can be used for gene transfer into animal cells?

P. Coprecipitation method using Calcium Phosphate.

Q. Coprecipitation using cesium chloride.

R. Lipofection

S. Electroporation

T. Retroviral method

U. Baculoviral method

V. Agrobacterium.

(a) P; R; T; U; V (b) P; R; S; T; U

(c) P; R; S; T; U; V (d) P; R; S; T; U; V

273. To maintain the membrane potential of animal cell in culture, growth medium requires supplementation of

(a) K^+; Ca^{2+} and Cl^- (b) K^+; Mg^{2+} and Br^-

(c) K^+; Na^+ and Cl^- (d) K^+; Ca^{2+} and Cl^-

274. Match the types of acyl-CoA dehydrogenase with their oxidations

P. Type-I I. C4-C14 fatty acids

Q. Type-II II. C4 and C6 acyl-CoA substrates

R. Type-III III. C12-C18 fatty acids

(a) P-III; Q-I; R-II (b) P-II; Q-I; R-III

(c) P-II; Q-III; R-I (d) P-III; Q-II; R-I

275. Which of these sentences are true regarding the continuous culture?

I. Cell lines of a finite life are usually diploid

II. Cell lines senesce after approximately thirty cycles of division

III. Continuous cultures are comprised of a single cell type

(a) I, II (b) II, III

(c) I, III (d) I, II, III

276. Match the following cell lines with their classification

P. Endothelial I. MRC-5

Q. Epithelial II. SH-SY5Y

R. Neuronal III. HeLa

S. Fibroblasts IV. BAE-1

(a) P-IV; Q-III; R-II; S-I

(b) P-III; Q-IV; R-II; S-I

(c) P-II; Q-III; R-IV; S-I

(d) P-IV; Q-III; R-I; S-II

277. Match the tissue of origin with their morphology

P. Human lung I. Neuroblast

Q. Human cervix II. Endothelial

R. Bovine aorta III. Epithelial

S. Human neuroblastoma IV. Fibroblasts

(a) P-IV; Q-III; R-II; S-I (b) P-III; Q-IV; R-II; S-I

(c) P-II; Q-III; R-IV; S-I (d) P-IV; Q-III; R-I; S-II

278. Which of these sentences are true regarding the cell growth cycle?

I. Cell death is not due to the reduction in nutrient supplements

II. Cellular proliferation slows down due to the cell population becoming confluent

III. Each cell line will show same cell proliferation kinetics during the log phase

(a) I, III (b) II, III

(c) I, II (d) I, II and III

279. Which of these sentences are true regarding the hybridoma technology?

I. ike myeloma cells, hybridoma cells are mortal

II. Each hybridoma produces the monoclonal antibody encoded by its B-lymphocyte partner

III. Primary cultures of normal B lymphocytes do not grow indefinitely

(a) I, II (b) II, III

(c) I, III (d) I, II, III

280. Match the following

Type of transplantation

I. Autograft

II. Isograft

III. Xenograft

IV. Allograft

Transport between

P. Genetically non-identical members

Q. One species to another species

R. Between twins

S. Same person

(a) I-S; II-R; III-Q; IV-P (b) I-R; II-S; III-Q; IV-P

(c) I-Q; II-R; III-P; IV-S (d) I-S; II-R; III-P; IV-Q

281. Which of these sentences are true regarding shear stress?

I. There is no effect of the impeller position was observed for h/HL less than 0.35

II. In blood flow shear stress is parallel to blood flow

III. In blood flow tensile stress, is perpendicular to the vessel wall

(a) I, II (b) III

(c) I, II, III (d) II, III

282. Which of these sentences are true regarding cultivation of organism under the effect of shear stress?

I. Cultivation conditions should be different during the cell growth stage and production stage.

II. Agitation speed and aeration rate were varied during the exponential growth phase and stationary phase.

III. By decreasing the agitation speed during the stationary phase, product formations get increases.

(a) I, II

(b) III

(c) I, II, III

(d) II, III

283. Arrange essential steps of antibody engineering in order

1. To develop antibody design
2. Introduce into vector into myeloma cell line
3. To mass culture the transfected clones of myeloma cell lines
4. To harvest the recombinant antibodies produced

(a) 1, 2, 3, 4 (b) 2, 3, 4, 1
(c) 1, 3, 4, 2 (d) 2, 3, 4, 1

284. Arrange the following steps according to the process involved in invivo multiplication of hybridoma clones

1. Separation of antibodies
2. Collection of ascitic fluid
3. Injection of hybridoma cells into peritoneal cavity of isogenic animals

(a) 3, 2, 1 (b) 1, 2, 3
(c) 2, 3, 1 (d) 1, 3, 2

285. The monoclonal antibody is known as

(a) OKT2 (b) OKT3
(c) OKT (d) OKT4

286. Which of these sentences are true regarding Hybridoma Technology?

I. Monoclonal antibodies cannot distinguish subsets of B cells and T cells

II. Unfused B cells die as they have a short life span

III. The antibodies in a well are produced by the same B cell, they will be directed towards the same epitope, and are thus monoclonal antibodies.

(a) I (b) II, III
(c) III (d) I, II

287. Which of these sentences are true regarding interaction of aminoacid with energy metabolism?

I. Intracellular concentration of aminoacid are much higher than the k_m for t-RNA

II. Intracellular non essential animoacid concentration is affected by glutamine concentration in medium.

III. The intracellular concentration of essential aminoacids are in the same order of magnitude of the extracellular concentration.

(a) I, II
(b) II, III
(c) I, II, III
(d) I, III

288. Which of these sentences are true regarding nutrients and their relation to transport consumption?

I. The transport of a compound across a membrane is same as its metabolic rate

II. Decreasing the concentration of nutrient species increases its uptake

III. The transporters is shared by many nutrients

(a) I, II (b) II, III
(c) I, II, III (d) I, III

289. When human fibroblasts were cultured on a surface coated with collagen, cell attachment and subsequent cell growth were

(a) Enhanced (b) Inhibited
(c) Remains same (d) No effect

290. Anoikis is a

(a) Detachment induced cell death
(b) Suspension induced apoptosis
(c) Both (a) & (b)
(d) Phagocytosis

291. Which of these sentences are true regarding growth kinetics?

I. IN the late stationary phase (2nd stationary phase) more oxygen is consumed whereas the demand for oxygen in the 1st stationary phase is nearly zero

II. Optimum pH for product formation may be different from that for growth

III. Hybridomas are known to produce antibodies at a higher rate at pH 6.2 than at 7.2

(a) I, II (b) II, III
(c) I, II, III (d) I, III

292. Heritability is formulated by

(a) $h2 = \frac{1}{3} VA/VP$

(b) $h2 = \frac{1}{2} VA/VP$

(c) $h2 = \frac{1}{4} VA/VP$

(d) $h2 = VA/VP$

293. In conventional breeding strategies, applications of molecular markers include

I. Parentage determination
II. Determination of twin zygosity
III. Genetic distance estimation
IV. Gene mapping

(a) I, II, III (b) II, III, IV
(c) II, IV (d) I, III, IV

294. I. In arbitrary primed PCR (AP-PCR), shorter primer used is universal M13 primer

II. In DNA amplification fingerprinting (DAF) analysis, longer primers are used

III. Semi-arbitrary primers designed on the basis of RE sites

Which of the above sentences are true regarding primers used in markers?

(a) I (b) II
(c) III (d) I, II

295. Match the following

Genetic markers	*Uses*
I. Morphological	P. Genetic variation at the DNA level
II. Chromosomal	Q. Age-dependent
III. Biochemical markers	R. Numerical variations
IV. Molecular markers	S. Pigmentation

(a) I-S; II-R; III-P; IV-Q (b) I-R; II-S; III-Q; IV-P
(c) I-S; II-R; III-Q; IV-P (d) I-Q; II-R; III-P; IV-S

296. I. Parentage testing using molecular markers yields much higher exclusion probability than the testing with blood groups.

II. DNA fingerprinting with oligoprobes OAT18 and ONS1 has been successfully used for determining the parentage of IVF buffalo calf

III. Molecular markers also serve as an useful tool for verification of the semen used for artificial insemination

Which of the above sentences are true regarding parentage determination?

(a) I (b) II
(c) I, II, III (d) III

297. I. Serum and milk proteins have been used extensively for the estimation of genetic divergence of different livestock species

II. A great amount of genetic variations at protein loci remain undetected

III. The PCR-based RAPD fingerprinting assays cannot be used for detection of genetic variations in cattle and sheep

Which of the above sentences are true regarding the genetic estimations?

(a) I (b) II
(c) I, II (d) III

298. I. Carboxymethyl benzylamide sulfonate dextrans are especially studied for heparin like properties

II. Plant derived macromolecules do not enhance adherence

III. Different dextran derivatives are capable of increasing or decreasing the adherence of cells.

Which of the above sentences are true?

(a) I
(b) II
(c) I, II
(d) I, II, III

299. I. Ca2+ and Mg2+ are cofactors for many attachment proteins

II. Ca2+ concentration in media is often reduced to increase adherence of cells cultured in suspension mode

III. Ca2+ concentration in media is often increased to culture anchorage dependent cells on microcarrier beads

Which of the above sentences are true regarding Co-factors adhesion?

(a) I (b) II
(c) II, III (d) I, III

300. Match the following

I. Monoclonal antibodies	P. Toxicological testing
II. Insect virus	Q. Interferons
III. Immunoregulators	R. Bioinsecticides
IV. Whole cells	S. Diagnostic tools

(a) I-S; II-R; III-Q; IV-P
(b) I-R; II-S; III-Q; IV-P
(c) I-S; II-R; III-P; IV-Q
(d) I-S; II-P; III-Q; IV-R

301. I. Cross-linked dextran has been found to be the best material for making microcarriers

II. Collagen cannot be used for producing microcarriers

III. Microcarriers absorption of proteins like IgG or albumin present in the serum was low

Which of the above sentence are true regarding microcarriers?

(a) I (b) II
(c) I, III (d) I, II

302. I. Microcarriers with a layer of surface charge on the periphery of the bead tends to show better attachment of cells as compared with others

II. Macrocarriers can be used when cells are extremely shear sensitive

III. The choice of the microcarrier doesn't depend on the morphology of the cells

Which of the above sentences are true regarding the choice of microcarriers?

(a) I
(b) II
(c) I, II
(d) I, III

303. Match the following

Junction	*Transmembrane linker protein*
1. Adherence (cell-cell)	P. Integrin
2. Adherence (cell-matrix)	Q. Cadherin
3. Desmosome	R. Actin filaments
4. Hemidesmosome	S. E-cadherin

(a) 1-S; 2-R; 3-Q; 4-P (b) 1-S; 2-R; 3-P; 4-Q
(c) 1-R; 2-S; 3-Q; 4-P (d) 1-Q; 2-R; 3-S; 4-P

304. I. In cell culture, the risk of contamination is related to the number of handling steps
II. Microcarrier culture increases the number of handling steps
III. There is a much-reduced risk of contamination when producing a large quantity of cells in a single microcarrier culture than in several hundred roller bottles
Which of the above sentences are true regarding the risk of contamination in microcarrier culture?
(a) I, II (b) III
(c) I, III (d) II

305. I. Microcarrier technology is one method for cell retention
II. The very large culture surface area to volume ratio offered by the microcarrier system provides high cell yields
III. Microcarrier systems allow excellent control of culture pH
Which of the above sentences are true regarding microcarrier culture?
(a) I, II (b) III
(c) II (d) I, II, III

306. I. Gelatin macrocarriers are dissolvable
II. For solid microcarriers, cell can be detached more readily than cytodex
III. Cells tend to extend membrane into the pores on microcarriers, making detachment easier
Which the above sentences are true regarding the microcarriers?
(a) I, II (b) II, III
(c) III (d) I, III

307. I. The adhesion of cells to microcarriers differs to the adhesion to tissue culture surface
II. The frequency of cell-microcarrier collision is stable
III. The probability of collisions between cell and microcarriers are crucial
Which of the above sentences are true regarding the cell attachment to the microcarrier?
(a) I, II (b) I
(c) III (d) I, III

308. I. Stem cells produce progeny with a restricted mitotic potential.
II. In human beings about 100,000 haematopoietic stem cells produce on billion RBC.
III. Stem cells retain the capacity to self renew
Which of the above sentences are true regarding the stem cells?
(a) I, II (b) II, III
(c) I, III (d) I, II, III

309. I. Organ culture lack vascular system
II. Due to the inadequate supply of the nutrients and oxygen, some degree of necrosis at the central part of the organ may occur.
III. The organ cultures can also be grown on top of a stainless steel grid.
Which of the above sentences are true regarding the organ culture?
(a) I, II (b) II, III
(c) I, III (d) I, II, III

310. Rearrange the steps in an order regarding the procedure involved in the organ culture
P. Incubation in a CO_2 incubator
Q. The tissue is placed on a gas medium interface support
R. The size of the tissue is reduced to less than 1mm in thickness.
S. The organ tissue is collected after the dissection
(a) P, Q, R, S (b) S, Q, R, P
(c) P, R, Q, S (d) S, R, Q, P

311. Match the following

P. YM	I. Gram positive bacteria
Q. Blood agar	II. Gram positive bacteria
R. MacConkey agar	III. Streptococcus
S. Mannitol salt agar	IV. Bacterial growth

(a) P-IV; Q-III; R-II; S-I (b) P-III; Q-IV; R-II; S-I
(c) P-IV; Q-I; R-II; S-III (d) P-IV; Q-III; R-I; S-II

312. I. In prokaryotes, importers mediate the uptake of nutrients into the cell
II. Eukaryotes do not possess any importers
III. The third subgroup of ABC proteins do not function as transporters
Which of the above sentences are true regarding ABC transporters?
(a) I, II, III
(b) I, II
(c) I, III
(d) II, III

313. Which of these sentences are true regarding the subculture of cells?
I. Cell Concentration, which should not exceed 1×10^6 cells/mL. for most suspension-growing cells.
II. pH, which is linked to cell concentration, and declines as the cell concentration rises.
III. Transformed cells should also be subcultured on reaching confluence
(a) I, II (b) II, III
(c) I, III (d) I, II, III

314. Which of these sentences are true regarding the properties acquired as a result of infection with pathogenic agents?
I. When manipulating cell cultures, the presence of adventitious contaminating agents constitutes the main hazard to humans.
II. The biological risk of infected cell cultures will depend on the biological risk of the infecting pathogen(s).
III. M. pneumaniae account for more than 96% of mycoplasma contaminants in cell cultures.
(a) I, II (b) II, III
(c) I, III (d) I, II, III

315. Match the following restriction endonucleases with the organism in which they naturally occur.

List-I	List-II
P. ECORI	1. E-Coli R 245
Q. Hind II	2. Bacillus globigii
R. EcBII	3. Escherichia Coli RyB
S. BgCII	4. Haemophilus influenzae

Code :

	P	Q	R	S
(a)	1	2	3	4
(b)	2	1	3	4
(c)	4	3	2	1
(d)	3	4	1	2

316. Match the following restriction endonucleases, with their corresponding examples.

List-I	List-II
P. Type I	1. EcoP1
Q. Type II	2. EcoK
R. Type II	3. HindII

Code :

	P	Q	R
(a)	1	2	3
(b)	3	3	2
(c)	2	3	4
(d)	3	1	2

317. During modification of cut ends, removal of the protruding ends by digestion with S_1 nuclease is done, this enzyme digests
(a) Only 3′ ends (b) Both 3′ and 5′ ends
(c) Only 5′ end (d) Either 3′ or 5′ ends

318. Match the following restriction endonucleases, with recognition sequences.

A	B
P. BamHI	1. G/AA TTC C TT AA/G
Q. EcoRI	2. G/G AT CC CCTA G/G
R. HindIII	3. G/G CC CC G/G
S. HindII7ujm	4. A/A GCTT TTCG A/A

Code :

	P	Q	R	S
(a)	1	2	3	4
(b)	2	1	4	3
(c)	3	4	1	2
(d)	3	2	4	1

319. In L γ–DNA technology. Reverse transcriptase is used for
(a) For removing 5′ – phosphate from the DNA ends
(b) For addition of phosphate group to an end having a free 5′ - OH
(c) To make the protruding ends double – stranded by extending the shorter stand
(d) Producing CDNA copies of m – RNA for creation of C.DNA libraries

320. Which of the following enzyme is not used as a tool in gene cloning experiments ?
(a) DNA ligase
(b) Alkaline phosphatase
(c) Reverse transcriptase
(d) Topoisomerase

321. Which of the following enzyme is termed as molecular glue ?
(a) Reverse transcriptase
(b) DNA polymerase
(c) Alkaline phosphatase
(d) DNA ligase

322. What is linker ?
(a) Short single stranded DNA molecule used for convert blunt end to stiky end
(b) Short RNA molecule used for converting blunt end to stiky end
(c) Short double stranded DNA molecule to convert blunt end to sticky ends
(d) All of these

323. The sequence recognized by restriction enzymes is
(a) Repetitive sequences
(b) Palindromic sequences
(c) Variable sequences
(d) Highly variable sequences

324. Matcn the following enzymes of Group A with thus activity in group B

A	B
A. DNA poll III	1. Cutting of DNA at specific
B. Endo nucleases	2. Separation of two DNA strands
C. Helicases	3. Joining two strands of DNA
D. DNA ligase	4. Addition of nucleotides at 3′

Code :

	A	B	C	D
(a)	4	1	2	3
(b)	1	2	3	4
(c)	1	3	4	2
(d)	2	3	4	1

325. Match the following antibiotic resistance genes found in R plasmids with the corresponding protein produced by them.

A	B
P. Ampicillin	1. Aminoglycoside phosphotransferase
Q. Kanamycin	2. β – lactamase
R. Neomycin	3. Kanamycin acetyl transferase
S. Streptomycin	4. Streptomycin phosphor transferase

Code :

	P	Q	R	S
(a)	2	3	1	4
(b)	2	4	3	1
(c)	4	3	1	2
(d)	3	1	2	4

326. Plasmids which have genes for enzymes that enables the bacterium to metabolize unusual substrates such as toluene, xylene and Salicylic acid are called
(a) Dissimilation plasmids
(b) Degradative plasmids
(c) Both (a) and (b)
(d) Assimilation plasmids

327. For the expression of the desired gene, the vector should contain
(a) A promoter
(b) A termination sequence
(c) A shine-Dalgarno sequence
(d) All of these

328. Which of the following is the advantage of Cosmid?
(a) The cosmid transfers a somewhat large foreign gene into bacterial cells. Cosmid pick up relatively larger DNA fragments than the plasmid DNA's do.
(b) The cosmids are used for the establishment of gene libraries of lower and higher organisms.
(c) Cosmids are used in the study of some non-sense sequences found in the genome of the organisms.
(d) All of these

329. Match the following organisms in Group A with their tissue specific proteins in Group B

A	B
P. Chicks	1. Immunoglobulin
Q. Mouse	2. Ovalbumin
R. Cereals	3. Amylase
S. Barley	4. Storage protein

Code :

	P	Q	R	S
(a)	1	2	3	4
(b)	4	3	2	1
(c)	2	1	4	3
(d)	2	3	1	4

330. Match the following

Enzyme	Effective For
I. Tdt	(P) DNA, RNA
II. T4 PNK	(Q) RNA
III. Periodate	(R) ssDNA, RNA
IV. EDC	(S) ssDNA

(a) I-S; II-R; III-Q; IV-P (b) I-S; II-P; III-Q; IV-R
(c) I-R; II-S; III-Q; IV-P (d) I-Q; II-R; III-S; IV-P

331. The nucleotide of hypoxanthine is
(a) AMP (b) GMP
(c) IMP (d) ATP

332. I. Pyrosequencing uses luciferase to generate light for detection of the individual nucleotides added to the nascent DNA
II. DNA nanoball sequencing is a type of high throughput sequencing technology used to determine the entire genomic sequence of an organism
III. In dye-terminator sequencing, each of the four dideoxynucleotide chain terminator is labelled with fluorescent dyes, each of which emit light at same wavelengths.
Which of the above sentence are true regarding DNA sequencing ?
(a) I, II (b) III
(c) II (d) I, III

333. The basic principle of Real time PCR
(a) Ct (b) Baseline
(c) Threshold (d) ΔRn

334. Southern blotting method is used for the separation of
(a) RNA (b) DNA
(c) Both (a) and (b) (d) Proteins

335. What is DNA finger printing ?
(a) It is a DNA isolation technique
(b) It is a hybridization technique used to identify
(c) It is the labeling procedure
(d) None of these

336. Northern blotting technique was devised by
(a) James alwine (b) David kemp
(c) George stark (d) All of these

337. The DNA binding prcteins are identified for
(a) S. blotting (b) Western blotting
(c) Northern blotting (d) South western blotting

338. DNA finger prints are valuable ?
(a) Solving disputed problem of parentage, identity of criminals, rapists.
(b) To determine whether the child belongs to the couple (or) not
(c) In forensic science
(d) All of these

339. I. Negatively charged amino acids contributing to a law isoelectric point.
II. Positively charged amino acids contributing to a high isoelectric point.
III. In 2-D SDS-PAGE, polypeptides are separated according to isoelectric point according to their molecular weight
Which of the above sentences are true regarding the 2D SDS-PAGE ?
(a) I (b) II
(c) III (d) I, II, III

340. In order to make the proteins accessible to antibody detection, they are moved from within the gel onto a membrane made of
(a) Nitrocellulose membrane
(b) PVDF
(c) Both (a) and (b)
(d) None of these

341. The protein transfer on a suitable membrane with protein-binding capacity is highly dependent on the
(a) Biochemical properties of the target protein.
(b) Physical properties of the target protein.
(c) Molecular weight of the target protein.
(d) Charge of the target protein.

342. I. Genetic fingerprints are closely related between parents and child
II. The DNA fingerprinting process was developed in 1985 by the British scientist Alec Jeffreys
III. RFLP profiling takes months to complete
Which of the above sentences are true regarding the DNA fingerprinting ?
(a) I, II
(b) III
(c) I, III
(d) II

343. What is a knock out mice ?
(a) A mice in which specific genes have been replaced, by their disrupted counter parts through a process of homologues recombination.
(b) A mice in which gene are inserted in addition to the disrupted counter parts by direct gene transfer method
(c) A mice in which certain gene are deleted
(d) Both (b) and (c)

344. What is Hybridization ?
(a) The pairing of complementary RNA and DNA strands to give an RNA – DNA hybrid.
(b) Pairing of complementary RNA and DNA strands to give an RNA – DNA hybrid.
(c) Both (a) and (b)
(d) It is the pairing of two complementary RNA strands to give RNA molecule

345. Hargobind khoranas work related to
(a) Synthesis of simple DNA
(b) Genetic code
(c) Synthesis of RNA from branch
(d) None of these

346. What is In situ hybridization ?
(a) It is a technique used to identify a virus carrying recombinant gene
(b) It is a technique used to identify a bacteria containing recombinant gene
(c) It is a technique used to identify as animal all carrying recombinant gene
(d) All of these

347. In situ hybridization is also called as
(a) Simple hybridization
(b) Colony hybridization
(c) Both (a) and (b)
(d) None of these

348. SCID syndrome indicate
(a) Severe combined immune deficiency
(b) Severe congenital immuno disorder
(c) Severe common immuno disorder
(d) None of these

349. SCID syndrome is caused due the deficiency of
(a) Adenosine deaminase (ADA)
(b) Thymidine kinase (TK)
(c) di hydrofolate reductase (DH FR)
(d) Xanthine – guanine phosphori bosyl transferse

350. I. RAPD markers are recessive
II. PCR is an enzymatic reaction
III. Mismatches between the primer and the template may result in the total absence of PCR product
Which of the above sentences are true regarding RAPD ?
(a) I, II (b) II, III
(c) III (d) I, III

351. I. RFLPs cannot be used determine the disease status of an individual.
II. RFLP loci measured in centimorgans.
III. RFLPs determine if a person were homozygous, heterozygous to disease
Which of the above statements are true regarding RFLP ?
(a) II, III (b) II
(c) III (d) I, III

352. I. PCR site directed mutagenesis generates a nicked, curcular DNA which can undergo repair by endogenous bacterial machinery.
II. A clear disadvantage of PCR as a DNA cloning method has been the size range of the DNA sequences that can be cloned.
III. The amount of PCR product obtained in a single reaction is also much more limited than the amount that can be obtained using cell-based cloning where scale-up of the volumes of cell cultures is possible.
Which of the above sentences are true regarding site directed mutagenesis ?
(a) I, II, III (b) III
(c) I, III (d) II

353. The process of the injection of DNA's directly into the cells using a micropipette is called.
(a) Micro projectile (b) Micro suction
(c) Macro projectile (d) Micro injection

354. Cloning an entire genome in the form of library of random genomic clones is often called as
(a) Gene gun experiment
(b) Random gene expression
(c) Shot gun experiment
(d) All of these

355. Direct DNA uptake by protoplast can be stimulated by chemicals like
(a) Agarose (b) REG
(c) PEG (d) Both (b) and (c)

356. Micro projectiles for gene transfer is also called as
(a) Micro gun method (b) Ballistics
(c) Particle gun (d) Both (b) and (c)

357. The process of introduction of a foreign DNA into a bacterial cell through a virus is called
(a) Transfection (b) Transduction
(c) Transformation (d) Transcription

358. The trade name of PEG
(a) Carbowax (b) Poly ethane glycerol
(c) Poly ethylene oxide (d) Poly ethylene Glycol

359. Which of these below method is a rapid method to introduce more number of recombinants in short time ?
(a) Micro injection (b) Electroporation
(c) Shot Gun (d) Transduction

360. Electroporation is a dynamic phenomenon that depends on
(a) Local transmembrane voltage
(b) Membrane pumps
(c) Molecular weight
(d) Ion channels

361. Solutions with high salt concentrations might cause an electrical discharge
(a) Arcing (b) Electrolytic effect
(c) Impedance (d) Transfection

362. I. Many strains of A. tumefaciens do not contain a pTi
II. A. tumefaciens have flagella
III. A. tumefaciens produces autoinducer
Which of the above sentences are true regarding agrobacterium ?
(a) I (b) II
(c) III (d) I, II, III

363. The process of infeding the divot plants with tumefaciens for gene transfer is called
(a) Bacterial infection (b) Agro transformation
(c) Agro infection (d) None of these

364. A. tumefaciens causes
(a) Root Gall
(b) Mosaic disease
(c) Crown Gall
(d) Tikka disease

365. I. Homologous recombination via the SDSA pathway occurs in cells that divide through meiosis
II. The double Holliday junctions are then converted into recombination products by nicking endonucleases
III. A displacement loop (D-loop) is formed during strand invasion between the invading 3' overhang strand and the homologous chromosome
Which of the above sentences are true regarding homologous recombination ?
(a) I, II (b) I, III
(c) I, II, III (d) III

366. What is Gene therapy ?
(a) It is the process of elimination of defective gene, from the patient
(b) It is the process of in which the patient is treated with gene product
(c) In this approach, normal and functional gene copies of the defective gene are introduced into the patient
(d) None of these

367. The ability of bacterial cells to uptake (intake) DNAs from the medium is said to be
(a) Compitancy
(b) Competence
(c) Transformancy
(d) None of these

368. Match the following

Disease	Genetic effect
I. Hemophilia A	(P) Absence of clotting factor IX
II. Cystic fibrosis	(Q) Defective muscle protein
III. Muscular dystrophy	(R) Defective chloride channel protein
IV. Hemophilia B	(S) Absence of clotting factor VIII

(a) I-S; II-R; III-Q; IV-P
(b) I-R; II-S; III-Q; IV-P
(c) I-Q; II-R; III-S; IV-P
(d) I-S; II-R; III-P; IV-Q

369. All are right in case of single cell cloning except
(a) Simplest procedure
(b) Each cell population is screened for altered phenotype
(c) Difficulties with the isolation and culture of single cells that limit its applicability
(d) For easier than cell aggregate cloning and more successful

370. In a cell aggregate cloning variation may be
(a) Spontaneous (b) Genetic
(c) Chemical (d) None of these

371. Which media is used for obtaining highly productive cell cultures ?
(a) Solid media (b) Liquid media
(c) Semi-solid media (d) (a) and (b)

372. From Lavandula vera, we obtain :
(a) Nicotine (b) Anthocyanin
(c) Biotin (d) All of these

373. Who showed the enzymatic permeabilization in Morinda citrifolia culture ?
(a) Yamada
(b) Sujitha
(c) Dornberg and Knorr
(d) All of these

374. Colchicine is a
(a) Secondary metabolite
(b) Spindle fiber inhibitor
(c) Spindle fiber activator
(d) Both (b) and (c)

375. Role of colchicines in cell division of suspension culture is
(a) Arrests the cells at metaphase stage of the cell division
(b) Arrests the cells at telophase stage of the cell division
(c) Arrests the cells at anaphase stage of the cell division
(d) Inhibits the cell division at mitosis-telephase

376. The use of TdR and HU in synchronization of suspension culture cells is
(a) Accumulation of cells at G_1/s interface stage
(b) Accumulation of cells at G_1 interphase stage
(c) Accumulation of cells at S interphase stage
(d) Accumulation of cells at G_1-G_2 interphase stage

377. The method employed to bring the synchronization is
(a) Cold treatment
(b) Starvation method
(c) Use of inhibitors
(d) All of these

378. The cell cycle can be divided into
(a) Inter-phase and mitosis
(b) Mitosis and meiosis
(c) Inter-phase and meiosis
(d) Inter-phase only

379. Plating efficiency is calculated as :

(a) $PE = \frac{\text{No. of colonies/plate at the end of the experiment}}{\text{No. of cellular units initially/plate}} \times 100$

(b) $PE = \frac{\text{No. of colonies}}{\text{plate}} \times 100$

(c) $PE = \frac{\text{No. of cells/plate}}{\text{No. of colonies initially/plate}} \times 100$

(d) $PE = \frac{\text{No. of cells}}{\text{No. of colonies}} \times \frac{\text{No. of cellular units}}{\text{plate}} \times 100$

380. For plant tissue cultures, the sugar used most frequently is

(a) Sucrose (b) Glucose
(c) Raffinose (d) None of these

381. The yield of secondary product is effected by

(a) Nature of sugar
(b) Concentration of sugar
(c) Both (a) & (b)
(d) Temperature of sugar

382. Nicotine production of tobacco cell cultures is best achieved by

(a) One-stage approach
(b) Two-stage approach
(c) Three-stage approach
(d) Seven-stage approach

383. Match the following

	Compound		Activity
P	Pyrethrins	1.	Antineoplastic
Q	Baccharin	2.	Antimalarial
R	Vincristine	3.	Caffeine
S	Crocin	4.	Insecticidal

(a) P-2; Q-3; R-4; S-1 (b) P-2; Q-3; R-1; S-4
(c) P-4; Q-3; R-2; S-1 (d) P-3; Q-2; R-1; S-4

384. Most successes in development of transgenic plants in monocots have been to

(a) Vector systems (b) Vector less system
(c) Both (a) and (b) (d) None of these

385. Genes conferring insect resistance to plants have been obtained from

(a) Virus (b) Plants
(c) Microorganisms (d) All of these

386. Resistance genes of animal origin

(a) Serine inhibitors
(b) Serine proteinase inhibitors
(c) Phenylalanine inhibitors
(d) Both (b) and (c)

387. Both the full length & truncated forms of Bt δ-endotoxins conferring resistance to

(a) Tobacco pests (b) Tomato pests
(c) Potato pests (d) All of these

388. The function of α – Amylase inhibitor in tobacco is

(a) Blocks the larval feeding in the foregut
(b) Synthesis the larval feeding by inhibits the α – amylase synthesis
(c) Blocks the larval feeding in the midgut
(d) All of these

389. Which of the transgenic plant (maize) gene is transferred to control insect european corn borer?

(a) Bt cry II (b) Bt cry I
(c) Corn cystine (d) All of these

390. Satellite RNA depends on

(a) Virus for it's replication
(b) Virus for it's transmission
(c) Both (a) & (b)
(d) Virus for it's translation

391. 2′ → 5′ oligo-adenylate synthase is induced by

(a) Interleukins (b) Immunoglobulins
(c) Interferons (d) None of these

392. Which proteins accumulate the significant levels in infected plant tissues ?

(a) Anti-microbial proteins
(b) Pathogenesis related proteins
(c) Satellite-RNA proteins
(d) None of these

393. Name the gene when present with chitinase showed fungal resistance in tobacco & tomato?

(a) Osmotin (b) Lectins
(c) Glucanase (d) Proteinase

394. Cecropin gene casette when transferred to tobacco showed resistance against which pathogen

(a) Pyricularia oxyzea (b) Rhizoctonia solani
(c) P. Syringae (d) Bacterial pathogen

395. Cause of phaseolotoxin which is produced by pseudomonas phaseolicola is

(a) Chlorotic halos (b) Florotic halos
(c) Atroseptia (d) All of these

396. Phytoalexins are

(a) Low mol. Wt substances
(b) Secondary metabolites
(c) Which contribute to the resistance to an plants to diseases.
(d) All of these

397. HML gene from maize encode NADPH dependent HC – toxin reductase that inactivates

(a) Bacterial HC toxin (b) Viral HC toxin
(c) Fungal HC toxin (d) All of these

398. Which transgenic plant is resistant to xanthomonas expression AvrXa21 ?

(a) Tomato (b) Tobacco
(c) Rice (d) Maize

399. Following abiotic stresses is/are

(a) Drought, low temperature
(b) Drought, salinity
(c) Salinity, & alkalinity
(d) All of these

400. The limiting enzyme in the synthesis of proline is

(a) P5CS (b) BADH
(c) PCCS (d) P3CS

401. The use of herbicides in modern agriculture

(a) To control weeds
(b) To control pathogens
(c) To enhance abiotic stress
(d) None of these

402. Glufosinate herbicide is inhibited by

(a) Glutamine synthase (b) ALS
(c) EPSPS (d) All of these

403. Which gene is isolated from Achromobacter bacterial strain ?

(a) bar (b) 2v0A
(c) gox (d) bxn

404. The approach which improved storage of tomato that is suitable for food processing can be developed by

(a) Antisense RNA technology
(b) Using gene for ACC deaminase
(c) Both (a) & (b)
(d) Antisense DNA technology

405. Which gene is used antisense RNA technology in tomato ?

(a) Pal gene (b) Cal gene
(c) Cos gene (d) bal gene

406. Which of the following plays a major role in regulating fruit ripening & flower senescence process ?

(a) Cytokins (b) Ethylene
(c) Acetylene (d) Ethylchloride

407. What are the major flower pigments in higher plants ?

(a) Auxins (b) Cytokinins
(c) Chloroplasts (d) Anthocyanins

408. How the transgenic violet carnations have been successfully produced

(a) Introduction of SAM hydrolase
(b) Introduction of the petunia F3′5′-H gene & it's high expression in the petals
(c) Introduction of the petunia F3′5′-H gene & it's low expression in the petals
(d) All of these

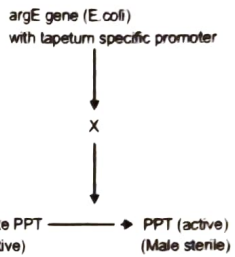
409. Male sterility in plants is inherited by

(a) Nuclear (b) Cytoplasmic
(c) Both (a) & (b) (d) Mitochondria

410. Transgenics with male sterlity and fertility restoration are produced in

(a) Maize (b) Rice
(c) Brassica napus (d) Tomato

411.



X represents ?

(a) Ornithine synthase (b) Ornithine acetylase
(c) Ornithine lyase (d) Ornithine deacetylase

412. The patent number of terminator gene technology is

(a) No. 5,723,767 (b) No. 5,723,777
(c) No. 5,723,765 (d) No. 2,123,824

413. When does the repressor protein becomes inactive ?

(a) When it binds to a specific chemical
(b) When it binds to specific repressor
(c) When it binds to inducer
(d) None of these

414. Plant species used for production of medium chain fatty acids is

(a) Tobacco (b) Arabidopsis
(c) Oilseed rape (d) Tomato

415. Nicotiana benthamiana is used for the production of

(a) Antibodies
(b) Antigens
(c) Alpha-trichasantin
(d) Antiotensin converting enzyme inhibitor

416. In the transgenic plants, which carbohydrate level is increased with a concomitant reduction in starch?

(a) Fructose (b) Mannitol
(c) Glucose (d) Sucrose

417. Application of α-amylase is

(a) Subunit vaccine
(b) Liquefaction of starch
(c) Animal feed
(d) Bleaching & pulping of paper

418. Match the following

Gene		Transgenic over - expressing	
(1)	γ-pyrroline - 5 - carboxylate synthetase	(A)	LEA
(2)	Barely lea gene (HVAI)	(B)	Glycine betaine
(3)	Cod a from Arthrobacter globiformis encoding choline oxidase (Rice)	(C)	Promote glycosidic flux through NADH oxidation
(4)	VHb gene from vitroscilla sterconassia	(D)	Proline

(a) 1 – B, 2 – C, 3 – D, 4 – A (b) 1 – A, 2 – D, 3 – B, 4 – C

(c) 1 – D, 2 – C, 3 – B, 4 – A (d) 1 – C, 2 – B, 3 – D, 4 – C

NUMERICAL TYPE QUESTIONS

1. The Ti plasmid is lost when the Agro bacterium is grown at
2. The range of DNA fragment which can be cloned in YAC is
3. The number of dideoxy nucleotides in the chain termination method is
4. The length of the labeled DNA fragment in FISH (fluorescent *in situ* hybridization) should be of
5. If you have started a protocol with 10 double stranded molecules of DNA, then how many double stranded molecules of product would you have after 5 cycles of PCR?
6. Calculate the plating efficiency of a cell line for which if the inoculated cell count is 200 cell/ ml and colonies formed are 40
7. The temperature and humidity requirements for animal cell in culture are
8. In polyacrylamide gel electrophoresis the larger ribosomal subunit used as marker is
9. In $cacl_2$ treatment, the DNA is taken up by the cells when they are subjected to heat shock for 2 min at
10. How much percentage of the global area is covered by herbicide tolerant transgenic crops under transgenic in 2007 ?

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2016

1. Which one of the following is **NOT** a therapeutic agent based on nucleic acid for the treatment of genetic disorders?
   (a) Antisense oligonucleotide
   (b) Ribozyme
   (c) Aptamer
   (d) Avidin
2. Which one of the following is **NOT** used for the measurement of cell viability in animal cell culture?
   (a) Trypan blue dye exclusion
   (b) Tetrazolium (MTT) assay
   (c) LDH activity in the culture medium
   (d) Coulter counter
3. A biological process is involved in the _____ treatment of industrial effluent.
   (a) primary (b) secondary
   (c) tertiary (d) quaternary
4. Select the **CORRECT** combination of genetic components that are essential for the transfer of TDNA segment from *Agrobacterium tumefaciens* to plant cells.
   (a) Border repeat sequences and oncogenes
   (b) Border repeat sequences and *vir* genes
   (c) Opine biosynthetic genes and *vir* genes
   (d) Opine biosynthetic genes and oncogenes
5. Match the secondary metabolites **(Column-I)** with the corresponding plant species **(Column-II)**.

Column-I	Column-II
P. Morphine	1. Datura stramonium
Q. Pyrethrins	2. Catharanthus roseus
R. Scopolamine	3. Papaver somniferum
S. Vincristine	4. Tagetes erecta

Codes:

	P	Q	R	S
(a)	4	3	1	2
(b)	3	4	1	2
(c)	2	3	4	1
(d)	4	1	2	3

6. A variety of genetic elements are used in the transgenic plant research. Match the genetic elements (Column-I) with their corresponding source (Column-II).

Column-I

P. Ubiquitin1 promoter
Q. Nos transcriptional terminator
R. bar selection marker gene
S. gus reporter gene

Column-II

1. Agrobacterium tumefaciens
2. Streptomyces hygroscopicus
3. Escherichia coli
4. Zea mays

Codes:

	P	Q	R	S
(a)	2	1	3	4
(b)	2	3	4	1
(c)	3	4	1	2
(d)	4	1	2	3

7. In animal cell culture, a CO_2 enriched atmosphere in the incubator chamber is used to maintain the culture pH between 6.9 and 7.4. Which one of the following statements is **CORRECT**?
   (a) Higher the bicarbonate concentration in the medium, higher should be the requirement of gaseous CO_2
   (b) Lower the bicarbonate concentration in the medium, higher should be the requirement of gaseous CO_2
   (c) Higher the bicarbonate concentration in the medium, lower should be the requirement of gaseous CO_2
   (d) CO_2 requirement is independent of bicarbonate concentration in the medium

2015

8. Which one of the following features is **NOT** required in a prokaryotic expression vector ?
   (a) *oriC* (b) Selection marker
   (c) CMV promoter (d) Ribosome binding site
9. In DNA sequencing reactions using the chain termination method, the ratio of ddNTPs to dNTPs should be
   (a) 0 (b) < 1
   (c) 1 (d) > 1
10. Which one of the following is **NOT** a product of denitrification in *Pseudomonas*?
    (a) N_2 (b) N_2O
    (c) NO_2^- (d) NH_4^+
11. Which one of the following is a second generation genetically engineered crop ?
    (a) Bt brinjal
    (b) Roundup soyabean
    (c) Golden rice
    (d) Bt rice
12. Choose the appropriate pair of primers to amplify the following DNA fragment by the polymerase chain reaction (PCR).

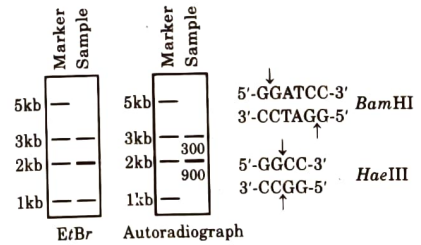
    5′–GACCTGTGG————ATACGGGAT–3′

    3′–CTGGACACC————TATGCCCTA–5′

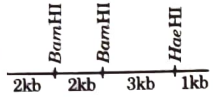
    Primers

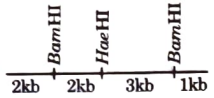
    P. 5′–GACCTGTGG–3′
    Q. 5′–CCACAGGTC–3′
    R. 5′–TAGGGCATA–3′
    S. 5′–ATCCCGTAT–3′

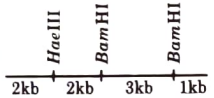
    (a) P and R (b) P and S
    (c) Q and R (d) Q and S
13. A linear double stranded DNA of length 8 kbp has three restriction sites. Each of these can either be a *Bam*Hi or a *Hae*III site. The DNA was digested completely with both enzymes. The products were purified and subjected to an end-filling reaction using the Klenow fragment and [α^{-32}P]-dCTP. The products of the end-filling reaction were purified, resolved by electrophoresis, stained with ethidium bromide (EtBr) and then subjected to autoradiography. The corresponding images are shown below.

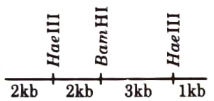


The numbers below each band in the sample lane in the autoradiograph represent their mean signal intensity in arbitrary units. Which one of the following options is the correct restriction map of the DNA ?

(a)


(b)


(c)


(d)


2014

14. The statistical frequency of the occurrence of a particular restriction enzyme cleavage site that is 6 bases long can be estimated to be

(a) once every 24 bases

(b) once every 256 bases

(c) once every 1024 bases

(d) once every 4096 bases

15. The growth medium for mammalian cells contains serum. One of the major functions of serum is to stimulate cell growth and attachment. However, it must be filter sterilized to

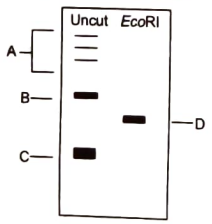
(a) remove large proteins

(b) remove collagen only

(c) remove mycoplasma and microorganisms

(d) remove foaming agents

16. The plasmid DNA was subjected to restriction digestion using the enzyme *Eco*RI and analysed on an agarose gel. Assuming digestion has worked (the enzyme was active), match the identity of the DNA bands shown in the image in **Group I** with their identity in **Group II**.



Group–I	Group–II
P. Bands labeled as A	1. Nicked
Q. Band labeled as B	2. Supercoiled
R. Band labeled as C	3. Concatemers
S. Band labeled as D	4. Linear

	P	Q	R	S
(a)	3	1	2	4
(b)	1	4	3	2
(c)	4	3	1	2
(d)	4	1	2	3

17. Match the following plant sources with their secondary metabolites and medical uses.

Source plant	Secondary metabolites
P. Belladona	1. Menthol
Q. Foxglove	2. Atropine
R. Pacific yew	3. Digitalin
S. Eucalyptus	4. Taxol

Medical use

A. Cancer treatment

B. Heart disease

C. Eye examination

D. Cough

	P	Q	R	S
(a)	2-C	3-B	4-A	1-D
(b)	3-C	2-A	1-D	4-B
(c)	2-C	4-B	1-A	3-D
(d)	1-B	4-C	2-D	3-A

18. The pungency of mustard seeds is primarily due to secondary metabolites such as isothiocyanate and nitrile. The pungency is usually felt only when the seeds are crushed. This is because of

(a) the coat of the intact seeds blocks the pungent volatiles from being released

(b) the pungent chemicals are stored as inactive conjugates and compartmentalized from the enzymes that convert them into active chemicals

(c) the pungent chemicals are formed only after the reaction with atmospheric oxygen

(d) the pungent chemicals are formed only after the reaction with atmospheric carbondioxide

19. The length of the minimum unique stretch of DNA sequence that can be found only once in a 3 billion base pairs long genome is

(a) 14 (b) 15

(c) 16 (d) 18

2013

20. Human genome sequencing project involved the construction of genomic library in

(a) bacterial artificial chromosome

(b) pBR322

(c) bacteriophage

(d) pcDNA3.1

21. In nature, the horizontal gene transfer across bacteria is mediated by

(a) gene cloning followed by transformation

(b) conjugation and transformation

(c) conjugation only

(d) transformation only

22. In nature, *Agrobacterium tumefaciens* mediated infection of plant cells leads to

P. crown gall disease in plants

Q. hairy root disease in plants

R. transfer of T-DNA into the plant chromosome

S. transfer of Ri-plasmid into the plant cell

(a) S only

(b) P and R only

(c) Q and S only

(d) Q only

23. Match the herbicides in **Group I**with the target enzymesin **Group II.**

Group I	Group II
P. Glyphosate	1. Nitrilase
Q. Bromoxynil	2. Acetolactatesyn thetase
R. Sulphonylureas	3. Dehalogenase
S. Dalapon	4. 5-Enolpyruvylshikimate 3-phosphate synthase

(a) P-4, Q-1, R-2, S-3

(b) P-2, Q-1, R-4, S-3

(c) P-4, Q-3, R-2, S-1

(d) P-3, Q-2, R-4, S-1

24. A complete restriction digestion of a circular plasmid (5000bp) was carried out with *Hin*dIII, *Bam*HI and *Eco*RIindividually. Restriction digestion yielded following fragments.

Plasmid + *Hin*dIII'→ 1200bp and3800bp

Plasmid + *Bam*HI'→ 5000bp

Plasmid + *Eco*RI'→ 2500bp

The number of sites for *Eco*RI, *Bam*HI-and*Hin*dIIIpresent on this plasmid are

(a) *Eco*RI-2, *Bam*HI-1, *Hin*dIII-2

(b) *Eco*RI-1, *Bam*HI-1, *Hin*dIII-2

(c) *Eco*RI-3, *Bam*HI-2, *Hin*dIII-1

(d) *Eco*RI-2, *Bam*HI-2, *Hin*dIII-1

Statement for Linked Answer Q. 25 and 26:

A DNA fragment of 5000bp needs to be isolated from *E.coli* (genome size 4×10^3kb) genomic library.

25. The minimum number of independent recombinant clones required to represent this fragment ingenomic library are

(a) 16×10^2 (b) 12×10^2

(c) 8×10^2 (d) 1.25×10^2

26. The number of clones to represent this fragment in genomic library with a probability of 95% are

(a) 5.9×10^3 (b) 4.5×10^3

(c) 3.6×10^3 (d) 2.4×10^3

2012

27. The basis for blue-white screening with pUC vectors is

(a) intraallelic complementation

(b) intergenic complementation

(c) intragenic suppression

(d) extragenic suppression

28. Dimethyl sulfoxide (DMSO) is used as a cryopreservant for mammalian cell cultures because

(a) it is an organic solvent

(b) it easily penetrates cells

(c) it protects cells by preventing crystallization of water

(d) it is aiso utilized as a nutrient

29. Protein-DNA interactions in vivo can be studied by

(a) gel shift assay

(b) southern hybridization

(c) chromatin immunoprecipitation assay

(d) fluorescence in situ hybridization assay

30. Which one of the following is NOT a protoplast fusion inducing agent?

(a) Inactivated Sendai virus

(b) Ca^{2+} at alkaline pH

(c) Polyethylene glycol

(d) Colchicine

31. A single base pair of DNA weighs 1.1×10^{-21} grains. How many picomoles of a plasmid vector of length 2750 bp are contained in 1 ug of purified DNA?

(a) 0.30

(b) 0.55

(c) 0.25

(d) 0 91

32. Determine the correctness or otherwise of the following **Assertion** (A) and **Reason** (R).

Assertion: The production of secondary metabolites in plant cell cultures is enhanced by the addition of elicitors.

Reason: Elicitors induce the expression of enzymes responsibie for the biosynthesis of secondary metabolites.

(*a*) Both (A) and (R) are true but (R) is not the correct reason for (A)

(*b*) Both (A) and (R) are true and (R) is the correct reason for (A)

(*c*) (A) is true but (R) is false

(*d*) (A) is false but (R) is true

33. Determine the correctness or otherwise of the following **Assertion** (A) and **Reason** (R)

Assertion: In direct somatic embry-ogenesis, embryos are developed without going through callus formation.

Reason: This is possible due to the presence of pre-embryonically determined cells.

(*a*) Both (A) and (R) are true but (R) is not the correct reason for (A)

(*b*) (A) is false but (R) is true

(*c*) (A) is true but (R) is false

(*d*) Both (A) and (R) are true and (R) is the correct reason for (A)

2011

34. Embryonic stem cells are derived from

(*a*) fertilized embryo

(*b*) unfertilized embryo

(*c*) sperm

(*d*) kidney

35. Yeast artificial chromosomes (YAC's) are used for cloning

(*a*) large segments of DNA

(*b*) mRNA

(*c*) bacterial DNA

(*d*) yeast DNA

36. The product commercially produced by animal cell culture is

(*a*) insulin

(*b*) tissue plasminogen activator

(*c*) interferon

(*d*) hepatitis B vaccine

37. Hydrated synthetic seeds which are produced by ion exchange reaction involve mixing the somatic embryos in a solution of

(*a*) sodium alginate and dropping it in a solution of calcium nitrate

(*b*) calcium alginate and dropping it in a solution of sodium nitrate

(*c*) calcium alginate and dropping it in a solution of ammonium nitrate

(*d*) mannitol and dropping it in a solution of sodium nitrate

38. Shoot organogenesis by tissue culture results into

(*a*) a bipolar structure that has no vascular connection with the explant

(*b*) a monopolar structure that has a strong connection with the pre-existing vascular tissue of the explant

(*c*) a monopolar structure that has no vascular connection with the explant

(*d*) a bipolar structure that has a strong connection with the pre-existing vascular tissue of the explant

39. 'Hairy roots' induced in *vitro* by the infection of *Agrobacterium rhizogenes*, are characterized by

P. a high degree of lateral branching

Q. genetic instability of culture

R. an absence of geotropism

S. poor biomass production

(*a*) P and R only (*b*) P and Q only

(*c*) Q and R only (*d*) R and S only

40. Restriction endonucleases which recognize and cut same recognition sequences are known as

(*a*) isoschizomers

(*b*) isozymes

(*c*) isoaccepting endonucleases

(*d*) abzymes

41. The study of evolutionary relationships is known as

(*a*) genomics (*b*) proteomics

(*c*) phylogenetics (*d*) genetics

42. Protein-protein interactions are studied by

P. DNA foot printing

Q. Yeast two hybrid system

R. Ligase chain reaction

S. Mass spectrometry

(*a*) P and S only (*b*) Q and S only

(*c*) P and R only (*d*) Q and R only

43. Identify the CORRECT statements

P. 5′ and 3′ ends of the transcripts can be mapped by utilizing polymerase chain reaction

Q. S_1 nuclease can cleave the DNA strand of a DNA-RNA hybrid

R. T_4 polynucleotide kinase is used for labeling 3′ end of DNA

S. Baculovirus (*Autographa californica*) can be used as an insect expression vector

(*a*) P and Q only (*b*) R and S only

(*c*) P and S only (*d*) Q and R only

44. HAT (hypoxanthine, aminopterin and thymidine) is used for selecting the hybridomas based on the following

I. Only hybridoma will grow since it inherited the HGPRT genes from B-cells and can synthesize DNA from hypoxanthine.

II. Myeloma cells will not grow in cultures since *de novo* synthesis is blocked by aminopterin and due to the lack of HGPRT enzyme.

(*a*) only I is true

(*b*) only II is true

(*c*) both I and II are true

(*d*) I is true and II is false

Statement for Linked Answer Questions 45 and 46:

A 200 µl of polymerase chain reaction has 100 template DNA molecules and the reaction was performed for 10 cycles.

45. How many molecules of amplicons will be generated?

(*a*) 1.024×10^4 (*b*) 1.024×10^5

(*c*) 2.048×10^4 (*d*) 2.048×10^5

46. How many molecules of amplicons will be present in 0.1 µl of reaction?

(*a*) 102.4 (*b*) 1024

(*c*) 51.2 (*d*) 512

2010

47. The bacteria known to be naturally competent for transformation of DNA is

(*a*) Escherichia coli

(*b*) Bacillus subtilis

(*c*) Mycobacterium tuberculosis

(*d*) Yersinia pestis

48. Antibiotic resistance marker that CANNOT be used in a cloning vector in Gram negative bacteria is

(*a*) Streptomycin (*b*) Ampicillin

(*c*) Vancomycin (*d*) Kanamycin

49. An example for template independent DNA polymerase is

(*a*) DNA Polymerase I

(*b*) RNA polymerase

(*c*) Terminal deoxynucleotidyl transferase

(*d*) DNA polymerase III

50. Somatic cell gene transfer is used for

P. transgenic animal production

Q. transgenic diploid cell production

R. *in-vitro* fertilization

S. classical breeding of farm animals

(*a*) P, R and S (*b*) P, Q and R

(*c*) P and R (*d*) P only

51. Expressed Sequence Tag is defined as

(*a*) a partial sequence of a codon randomly selected from DNA library

(*b*) the characteristic gene expressed in the cell

(*c*) the protein coding DNA sequence of a gene

(*d*) uncharacterized fragment for DNA presence in the cell

52. Determine the correctness or otherwise of the following **Assertion (*a*)** and the **Reason** (*r*)

Assertion : Somatic embryogenesis in plants is a two step process comprising of embryo initiation followed by embryo production.

Reason : Embryo initiation is independent of the presence of 2, 4 dichlorop-henoxyacetic acid whereas embryo production requires a high concentration of 2, 4-dichloroph-enoxyacetic acid.

(*a*) both (*a*) and (*r*) are true and (*r*) is the correct reason for (*a*)

(*b*) both (*a*) and (*r*) are true and (*r*) is not the correct reason for (*a*)

(*c*) (*a*) is true but (*r*) is false

(*d*) (*a*) is false but (*r*) is true

53. Match Group I with Group II

Group I	Group II
P. Fibronectin	1. Uptake of amino acids and glucose
Q. Insulin	2. Trypsin inhibitor
R. α-Macroglobulin	3. Binds iron
S. Transferrin	4. Cell attachment to substratum

(*a*) P-2, Q-1, R-4, S-3

(*b*) P-3, Q-2, R-1, S-4

(*c*) P-4, Q-2, R-1, S-3

(*d*) P-4, Q-1, R-2, S-3

54. Match the promoters listed in Group I with the tissues listed in Group II

Group I	Group II
P. α–Amylase	1. Endosperm
Q. Glutenin	2. Tuber
R. Phaseollin	3. Aleurone
S. Patatin	4. Cotyledon

(a) P-3, Q-1, R-4, S-2 (b) P-3, Q-4, R-1, S-2
(c) P-4, Q-2, R-1, S-3 (d) P-1, Q-3, R-2, S-4

55. Consider the following statements,
I. T4 DNA ligase can catalyze blunt end ligation more efficiently than *E.coli* DNA ligase
II. The ligation efficiency of T4 DNA ligase can be increased with PEG and ficoll.
(a) only I is true (b) both I and II are true
(c) only II is true (d) I is true and II is false

56. Match the items in Group I with Group II

Group I (*Vectors*)	Group II (*Maximum DNA packaging*)
P. λ phage	1. 35–45 kb
Q. Bacterial Artificial Chromosomes (BACs)	2. 100–300 kb
R. P1 derived Artificial Chromosomes (PACs)	3. ≤ 300 kb
S. λ cosmid	4. 5–25 kb

(a) P-3, Q-4, R-1, S-2 (b) P-1, Q-3, R-2, S-4
(c) P-4, Q-3, R-2, S-1 (d) P-1, Q-2, R-3, S-4

57. A mutant G_α protein with increased GTPase activity would
(a) not bind to GTP
(b) not bind to GDP
(c) show increased signaling
(d) show decreased signaling

58. A cell has five molecules of a rare mRNA. Each cell contains 4×10^5 mRNA molecules. How many clones one will need to screen to have 99% probability of finding at least one recombinant cDNA of the rare mRNA, after making cDNA library from such cell?
(a) 4.50×10^5 (b) 3.50×10^5
(c) 4.20×10^5 (d) 4.05×10^5

59. Match Group I with Group II

Group I	Group II
P. Real Time-PCR	1. Biochips
Q. 2-D Electrophoresis	2. Syber Green
R. Affinity chromatography	3. Antibody linked sephrose beads
S. Micoroarray	4. Ampholytes

(a) P-1, Q-2, R-4, S-3 (b) P-2, Q-3, R-4, S-1
(c) P-2, Q-4, R-3, S-1 (d) P-3, Q-2, R-1, S-4

2009

60. To produce plants that are homozygous for all traits, the best choice is
(a) Protoplast culture
(b) Cell suspension culture
(c) Anther and pollen culture
(d) Apical merisem culture

61. Restriction endonucleases from two different organisms that recognize the same DNA sequence for cleavage are called
(a) Isoschizomers (b) Isozymes
(c) Concatamers (d) Palindromes

62. Baculovirus expression system is used to express heterologous genes in
(a) Mammals (b) Plants
(c) Insects (d) Yeasts

63. Virus resistant transgenic plants can be developed by the expression of
(a) Cowpea trypsin inhibitor
(b) Crystalline toxin protein
(c) Defective movement protein
(d) Snowdrop lectin

64. Which of the following statements are true about glyphosate tolerant transgenic plants?
P. Transgenic plants detoxify glyphosate.
Q. Transgenic plants produce an altered enzyme that is not affected by glyphosate.
R. Transgenic plants sequester glyphosate in vacuoles.
S. Transgenic plants overcome the inhibition of aromatic amino acid biosynthesis.
(a) P, Q (b) R,S
(c) Q,S (d) P, R

65. Match the items in Group 1 with their functions in Group 2:

Group 1	Group 2
P. *rol* genes	1. Food and energy source
Q. Opines	2. Tumor formation
R. Virulence genes	3. Hairy root induction
S. *Aux* and *cyt* genes	4. T-DNA transfer and integration

(a) P-4, Q-3, R -2, S-1 (b) P-3, Q-2, R-4, S-1
(c) P-1, Q-3, R-4, S-2 (d) P-3, Q-1, R-4, S-2

66. Which of the following statements hold true for pluripotent stem cells (PSCs) under in *vitro* conditions?
P. PSCs can be maintained in an undifferentiated state
Q. PSCs exhibit abnormal and unstable karyotypes
R. PSCs can differentiate into a wide variety of cell types
S. PSCs cannot be passaged continuously
(a) P, Q (b) P, R
(c) Q, R (d) Q, S

67. Match the items in Group 1 with correct options in Group 2 :

Group 1	Group 2
P. DNA footprinting	1. Protein-protein interaction
Q. Yeast two-hybrid system	2. VNTR
R. DNA fingerprinting	3. DNA binding protein
S. SAGE	4. Transcriptome analysis

(a) P-1, Q-2, R-4, S-3 (b) P-3, Q-1, R-2, S-4
(c) P-3, Q-4, R-1, S-2 (d) P-4, Q-2, R-1, S-3

68. Match the products in Group 1 with their possible applications in Group 2:

Group 1	Group 2
P. Erythropoietin	1. Blood clot
Q. Anti-fibrin 99	2. Binding and transport of iron
R. Collagenase	3. Anaemia
S. Transferrin	4. Animal cell separation

(a) P-3, Q-1, R-4, S-2 (b) P-3, Q-4, R-1, S-2
(c) P-2, Q-3, R-1, S-4 (d) P-2, Q-1, R-4, S-3

69. A RNA polymerization assay was performed using ^{3}H UTP as the labelled nucleotide with a specific activity of 500 μCi/ μmol (1 μCi = 2.2×10^6 counts per min). After 10 min incubation, the trichloroacetic acid-insoluble radioactivity was found to be 692521 counts per min as determined in a liquid scintillation counter working at 60% efficiency for ^{3}H. The amount of UTP incorporated into the RNA will be
(a) 15 nmol (b) 105 nmol
(c) 150 nmol (d) 50 nmol

2008

70. Parthenogenetic embryos in plants are those which are formed by
(a) unfertilized eggs (b) fertilized eggs
(c) sporophytic cells (d) male gametophyte

71. Multiplication of genetically identical copies of a cultivar by asexual reproduction is known as
(a) aclonal propagation
(b) vegetative propagation
(c) polyclonal propagation
(d) clonal propagation

72. Identify the correct statement for the 'HAT medium'
(P) Includes drug aminopterin to block major pathway for synthesis of deoxyri-bonucleotides
(Q) Hypoxanthine is precursor for thymidine
(R) Includes drug aminopterin to block major pathway for synthesis of polypeptides
(S) Cells can grow in presence of aminopterin only if they have enzymes thymidine kinase and hypoxanthine-guanine phosphoribosyl transferase
(a) P, Q (b) P, S
(c) R, S (d) Q, S

73. Determine the correctness or otherwise of following **Assertion [a] and Reason [r]**
Assertion: The enzymatic degradation of cell wall to obtain single cell called protoplast has helped immensely in developing somatic cell genetics in plants
Reason: In plants or animals, fusion to two cells must occur through the plasma membrane
(a) Both [a] and [r] are true and [r] is the correct reason for [a]
(b) Both [a] and [r] are true but [r] is not the correct reason for [a]
(c) [a] is true but [r] is false
(d) [a] is false but [r] is true

74. Which of the following statements are true with respect to a special complex called 'dicer'?
(P) It consists of deoxyribonuclease and DNA fragments
(Q) It consists of ribonuclease and RNA fragments
(R) It is involved in gene silencing
(S) It triggers apoptosis
(a) P, R (b) Q, R
(c) P, S (d) Q, S

75. Match the items in groupl with the terms given in group 2

Group 1	Group 2
P. *Lactobacillus and Bifidobacteria*	1. Prebiotics
Q. Polychlorobenzenes (PCBs)	2. Probiotics
R. Fructo-oligosaccharides	3. Antibiotics
S. β-Lactams	4. Xenobiotics

	P	Q	R	S
(a)	2	4	1	3
(b)	3	4	1	2
(c)	4	1	2	3
(d)	1	3	4	2

76. A polymerase chain reaction was performed beginning with 400 template DNA molecules in a 100 l reaction. After 20 cycles of polymerase chain reaction, how many molecules of the amplified product will be present in 0.1 l of reaction ?

(a) 2.19×10^4 (b) 4.19×10^4

(c) 2.19×10^5 (d) 4.19×10^5

2007

77. Which of the following reagents is used for harvesting anchorage-dependent animal cells from culture vessels ?

(a) Trypsin/Collagensase

(b) Trypsin/Collagen

(c) Collagen/Fibronectin

(d) DMSO

78. Protein binding regions of DNA are identified by one of the following techniques

(a) finger printing (b) foot printing

(c) southern blotting (d) western blotting

79. Plant secondary metabolites

(a) help to increase the growth rate of plant

(b) help in plant reproduction processes

(c) provide defense mechanisms against microbial attack

(d) make the plant susceptible to unfavorable conditions

80. Oils rich in PUFA are NOT desirable for bio-diesel production because

(a) they form epoxides in presence of oxygen

(b) they do not form epoxides in presence of oxygen

(c) they have high ignition temperature

(d) they solidify at low temperature

81. Gynogenesis is a process of development of haploid plants

(a) from a fertilized cell of female gametophyte

(b) from an unfertilized cell of female gametophyte

(c) from isolated pollen grains

(d) by selective elimination of chromosomes following distant hybridization

82. Match the following marker genes in group 1 with suitable selecting agent in group 2

Group 1	Group 2
P. npt II	1. Glyphosate
Q. aroA	2. Phosphinothricin
R. hpt	3. Kanamycin
S. bar	4. Hygromycin B

(a) P-1, Q-2, R-4, S-3

(b) P-3, Q-2, R-4, S-1

(c) P-2, Q-3, R-4, S-1

(d) P-3, Q-1, R-4, S-2

83. Meristems escape virus invasion because

(a) vascular system is absent in the meristem

(b) of low metabolic activity in the meristem

(c) the 'virus inactivating system' has low activity in the meristem

(d) of low endogenous auxin level

2006

84. A thermostable DNA polymerase that can carry out both reverse transcription reaction and polymerizations has been isolated from

(a) Thermococcus litoralis

(b) Thermus aquaticus

(c) Thermotoga maritima

(d) Thermus thermophilus

85. When present in tissue culture medium, gibberellin

(a) helps to break dormancy of buds and bulbs

(b) promotes dormancy development in buds and bulbs

(c) is regarded as plant growth inhibitor

(d) prevents normal recognition of auxin molecule

86. To promote attachment and spreading of anchorage-dependent animal cells, the surface of the culture vessel needs to be coated with

(a) trypsin

(b) collagen

(c) pronase

(d) polyglycol

87. For amplificatio of GC rich sequences by polymerase chain reacstion, identify the reagent that binds and stabilizes AT sequences and destabilizes GC regions.

(a) Tetramethyl ammonium chloride

(b) Betaine

(c) 7-deaza-2'-deoxyguanosine triphosphate

(d) Sodium dodecyl sulphate

88. Which of the following statements is INCORRECT about immobilized plant cell cultures?

(a) It is possible to use high cell densities

(b) Cells remain active for long periods

(c) Cell products or inhibitors can be removed easily

(d) It provides low shear resistance to cells

89. All the cells that participate in immune responses originate from a population of

(a) neutrophils (b) stem cells

(c) macrophages (d) lymphocytes

90. Identify the natural plant growth regulators from the following list.

(P) Zeatin

(Q) Benzylaminopurine (BAP)

(R) Indole-3-acetic acid (IAA)

(S) 2, 4-Dichlorophenoxyacetic acid

(a) P, Q (b) Q, S

(c) P, R (d) R, S

91. What are the experimental steps needed for screening an expression library for a clone encoding a protein X that has been isolated and purified ?

(P) *m*-RNA isolation

(Q) Antibody preparation

(R) Cloning into an appropriate vector

(S) Western blotting on transferred plaques

(a) P, S (b) Q, S

(c) Q, R (d) R, S

92. Majority of the cereals are highly recalcitrant to *Agrobacterium*-mediated transformation, and so direct transfor-mation methods have been developed to transform such plants. Which of the following direct transformation methods is applicable to intact plant tissues?

(a) Calcium chloride and PEG-mediated transformation

(b) Liposome-mediated transformation

(c) Electroporation

(d) Transformation using microprojectiles

93. Match items in group 1 with correct options from those given in group 2

Group 1

P. Tissue plasminogen activator

Q. Gamman interferon

R. Podophyllotoxin

S. Polyhydroxyalkanoate

Group 2

1. Immunomodulator
2. Biodegradable plastic
3. Clot dissolution
4. Anti-tumor agent

(a) P–4, Q–3, R–1, S–2

(b) P–1, Q–3, R–4, S–2

(c) P–3, Q–1, R–2, S–4

(d) P–3, Q–1, R–4, S–2

94. Match items in group 1 with correct options from those given in group 2

Group 1

P. Amperometric biosensor

Q. Evanescent wave biosensor

R. Calorimetric biosensor

S. Potentiometric biosensor

Group 2

1. Light beam
2. Flux of redox electrons
3. Field effect transistors
4. Exothermic reaction

(a) P-3, Q-4, R-2, S-1

(b) P-2, Q-1, R-4, S-3

(c) P-3, Q-2, R-4, S-1

(d) P-2, Q-4, R-3, S-1

95. Which of the following would result in somaclonal variation in micropropagated plants?

(P) Propagation by axillary branching in the absence of plant growth regulators

(Q) Cell suspension maintained for five years before induction of somatic embryogenesis

(R) Cullus induction using 20μM 2, 4-Dichlorophenoxyacetic acid followed by shoot organogenesis

(S) Shoot organogenesis from an explant in the absence of an intermediate callus phase

(a) P, Q (b) Q, R

(c) P, S (d) Q, S

Common Data for Questions 96, 97 :

Lignocellulosic biomass was subjected to microbial composting. The microbial consortium produced an extracellular enzyme xylanase, which was a glycoprotein having a molecular weight of 68 kDa and a positive charge. An aqueous extract of the enzyme could be easily prepared from the compost.

96. What techniques would you recommend for confirming the molecular weight of the purified enzyme?

(P) Isoelectric focusing

(Q) SDS-PAGE

(R) Native PAGE

(S) Gel filtration

(a) P, Q (b) Q, S

(c) R, S (d) P, S

97. If Con A sepharose column was used for the purification of enzyme, then separation would be based on

(*a*) molecular exclusion

(*b*) affinity binding

(*c*) ion exchange

(*d*) hydrophobic interaction

Statement for Linked Answer Questions 98 and 99 :

DNA content of Caenorhabditis elegans was analysed and found to contain 1.0 × 10^8 bp.

98. How many standard λ-phage vectors carrying 20kb DNA fragments or YACs carrying 250 kb DNA fragments are theoretically required to constitute a complete C. *elegans* genomic library?

(*a*) 500 λ-phage vectors or 40 yeast clones

(*b*) 400 λ-phage vectors or 5000 yeast clones

(*c*) 5000 λ-phage vectors or 400 yeast clones

(*d*) 5 × 10^4 λ-phage vectors or 4000 yeast clones

99. How many λ-phage vectors/yeast clones should be prepared in order to ensure that every sequence is included in the library?

(*a*) 25 × 10^3 λ-phage vectors/2000 yeast clones

(*b*) 20 × 10^3 λ-phage vectors/1600 yeast clones

(*c*) 5 × 104 λ-phage vectors/4000 yeast clones

(*d*) 10 × 104 λ-phage/10000 yeast clones

NUMERICAL TYPE QUESTIONS

2015

1. Assuming random distribution of nucleotides, the average number of fragments generated upon digestion of a circular DNA of size 4.3 × 10^5 bp with *Alu*I(5′-AG↓CT-3′) is _____ × 10^3.

2. Plasmid DNA (0.5 μg) containing an ampicillin resistance marker was added to 200 μl of competent cells. The transformed competent cells were diluted 10,000 times, out of which 50 μl was plated on agar plates containing ampicillin. A total of 35 colonies were obtained. The transformation efficiency is _____ × 10^6 cfu. μg^{-1}.

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (*b*)	**2.** (*b*)	**3.** (*b*)	**4.** (*a*)	**5.** (*b*)	**6.** (*c*)	**7.** (*c*)	**8.** (*c*)	**9.** (*d*)	**10.** (*a*)
11. (*d*)	**12.** (*c*)	**13.** (*c*)	**14.** (*d*)	**15.** (*c*)	**16.** (*a*)	**17.** (*b*)	**18.** (*d*)	**19.** (*c*)	**20.** (*b*)
21. (*c*)	**22.** (*a*)	**23.** (*d*)	**24.** (*c*)	**25.** (*b*)	**26.** (*a*)	**27.** (*a*)	**28.** (*b*)	**29.** (*d*)	**30.** (*c*)
31. (*b*)	**32.** (*b*)	**33.** (*c*)	**34.** (*b*)	**35.** (*c*)	**36.** (*b*)	**37.** (*d*)	**38.** (*d*)	**39.** (*d*)	**40.** (*a*)
41. (*a*)	**42.** (*b*)	**43.** (*c*)	**44.** (*d*)	**45.** (*a*)	**46.** (*a*)	**47.** (*c*)	**48.** (*a*)	**49.** (*a*)	**50.** (*b*)
51. (*b*)	**52.** (*a*)	**53.** (*a*)	**54.** (*b*)	**55.** (*b*)	**56.** (*d*)	**57.** (*a*)	**58.** (*d*)	**59.** (*a*)	**60.** (*b*)
61. (*a*)	**62.** (*a*)	**63.** (*c*)	**64.** (*c*)	**65.** (*c*)	**66.** (*b*)	**67.** (*b*)	**68.** (*a*)	**69.** (*b*)	**70.** (*a*)
71. (*c*)	**72.** (*b*)	**73.** (*a*)	**74.** (*b*)	**75.** (*a*)	**76.** (*d*)	**77.** (*d*)	**78.** (*c*)	**79.** (*a*)	**80.** (*b*)
81. (*b*)	**82.** (*c*)	**83.** (*b*)	**84.** (*a*)	**85.** (*a*)	**86.** (*d*)	**87.** (*b*)	**88.** (*a*)	**89.** (*d*)	**90.** (*c*)
91. (*c*)	**92.** (*d*)	**93.** (*b*)	**94.** (*c*)	**95.** (*a*)	**96.** (*b*)	**97.** (*d*)	**98.** (*b*)	**99.** (*d*)	**100.** (*c*)
101. (*d*)	**102.** (*a*)	**103.** (*b*)	**104.** (*a*)	**105.** (*b*)	**106.** (*c*)	**107.** (*a*)	**108.** (*b*)	**109.** (*d*)	**110.** (*c*)
111. (*b*)	**112.** (*b*)	**113.** (*a*)	**114.** (*c*)	**115.** (*a*)	**116.** (*b*)	**117.** (*b*)	**118.** (*c*)	**119.** (*d*)	**120.** (*c*)
121. (*a*)	**122.** (*d*)	**123.** (*a*)	**124.** (*b*)	**125.** (*c*)	**126.** (*a*)	**127.** (*d*)	**128.** (*d*)	**129.** (*a*)	**130.** (*b*)
131. (*d*)	**132.** (*a*)	**133.** (*d*)	**134.** (*a*)	**135.** (*b*)	**136.** (*c*)	**137.** (*d*)	**138.** (*d*)	**139.** (*b*)	**140.** (*c*)
141. (*d*)	**142.** (*a*)	**143.** (*c*)	**144.** (*c*)	**145.** (*a*)	**146.** (*d*)	**147.** (*d*)	**148.** (*c*)	**149.** (*a*)	**150.** (*c*)
151. (*d*)	**152.** (*a*)	**153.** (*c*)	**154.** (*b*)	**155.** (*c*)	**156.** (*a*)	**157.** (*c*)	**158.** (*c*)	**159.** (*c*)	**160.** (*b*)
161. (*d*)	**162.** (*d*)	**163.** (*b*)	**164.** (*d*)	**165.** (*c*)	**166.** (*a*)	**167.** (*b*)	**168.** (*d*)	**169.** (*c*)	**170.** (*a*)
171. (*c*)	**172.** (*d*)	**173.** (*a*)	**174.** (*d*)	**175.** (*b*)	**176.** (*c*)	**177.** (*c*)	**178.** (*a*)	**179.** (*a*)	**180.** (*a*)
181. (*c*)	**182.** (*b*)	**183.** (*a*)	**184.** (*c*)	**185.** (*a*)	**186.** (*c*)	**187.** (*b*)	**188.** (*c*)	**189.** (*a*)	**190.** (*a*)

191. (*c*)	**192.** (*c*)	**193.** (*b*)	**194.** (*b*)	**195.** (*c*)	**196.** (*b*)	**197.** (*b*)	**198.** (*c*)	**199.** (*a*)	**200.** (*a*)
201. (*d*)	**202.** (*b*)	**203.** (*a*)	**204.** (*c*)	**205.** (*d*)	**206.** (*d*)	**207.** (*b*)	**208.** (*c*)	**209.** (*d*)	**210.** (*a*)
211. (*c*)	**212.** (*b*)	**213.** (*a*)	**214.** (*b*)	**215.** (*d*)	**216.** (*b*)	**217.** (*a*)	**218.** (*c*)	**219.** (*c*)	**220.** (*b*)
221. (*c*)	**222.** (*a*)	**223.** (*c*)	**224.** (*b*)	**225.** (*c*)	**226.** (*a*)	**227.** (*c*)	**228.** (*c*)	**229.** (*a*)	**230.** (*b*)
231. (*c*)	**232.** (*c*)	**233.** (*d*)	**234.** (*b*)	**235.** (*d*)	**236.** (*a*)	**237.** (*c*)	**238.** (*b*)	**239.** (*c*)	**240.** (*b*)
241. (*b*)	**242.** (*c*)	**243.** (*d*)	**244.** (*a*)	**245.** (*a*)	**246.** (*c*)	**247.** (*a*)	**248.** (*a*)	**249.** (*a*)	**250.** (*b*)
251. (*a*)	**252.** (*a*)	**253.** (*b*)	**254.** (*a*)	**255.** (*b*)	**256.** (*b*)	**257.** (*c*)	**258.** (*b*)	**259.** (*d*)	**260.** (*c*)
261. (*a*)	**262.** (*d*)	**263.** (*a*)	**264.** (*b*)	**265.** (*a*)	**266.** (*a*)	**267.** (*d*)	**268.** (*a*)	**269.** (*a*)	**270.** (*d*)
271. (*a*)	**272.** (*b*)	**273.** (*c*)	**274.** (*a*)	**275.** (*d*)	**276.** (*a*)	**277.** (*a*)	**278.** (*c*)	**279.** (*b*)	**280.** (*a*)
281. (*c*)	**282.** (*c*)	**283.** (*a*)	**284.** (*a*)	**285.** (*b*)	**286.** (*b*)	**287.** (*c*)	**288.** (*d*)	**289.** (*a*)	**290.** (*c*)
291. (*c*)	**292.** (*a*)	**293.** (*a*)	**294.** (*c*)	**295.** (*c*)	**296.** (*c*)	**297.** (*c*)	**298.** (*d*)	**299.** (*d*)	**300.** (*a*)
301. (*c*)	**302.** (*c*)	**303.** (*a*)	**304.** (*c*)	**305.** (*d*)	**306.** (*a*)	**307.** (*d*)	**308.** (*d*)	**309.** (*d*)	**310.** (*d*)
311. (*a*)	**312.** (*a*)	**313.** (*d*)	**314.** (*d*)	**315.** (*d*)	**316.** (*c*)	**317.** (*b*)	**318.** (*b*)	**319.** (*d*)	**320.** (*d*)
321. (*d*)	**322.** (*c*)	**323.** (*b*)	**324.** (*a*)	**325.** (*a*)	**326.** (*c*)	**327.** (*d*)	**328.** (*d*)	**329.** (*c*)	**330.** (*a*)
331. (*c*)	**332.** (*a*)	**333.** (*a*)	**334.** (*b*)	**335.** (*b*)	**336.** (*d*)	**337.** (*d*)	**338.** (*d*)	**339.** (*c*)	**340.** (*c*)
341. (*a*)	**342.** (*a*)	**343.** (*a*)	**344.** (*a*)	**345.** (*b*)	**346.** (*b*)	**347.** (*c*)	**348.** (*a*)	**349.** (*a*)	**350.** (*b*)
351. (*a*)	**352.** (*a*)	**353.** (*d*)	**354.** (*c*)	**355.** (*c*)	**356.** (*d*)	**357.** (*b*)	**358.** (*a*)	**359.** (*b*)	**360.** (*a*)
361. (*a*)	**362.** (*d*)	**363.** (*c*)	**364.** (*c*)	**365.** (*b*)	**366.** (*c*)	**367.** (*b*)	**368.** (*a*)	**369.** (*d*)	**370.** (*a*)
371. (*a*)	**372.** (*c*)	**373.** (*c*)	**374.** (*b*)	**375.** (*a*)	**376.** (*a*)	**377.** (*d*)	**378.** (*a*)	**379.** (*a*)	**380.** (*a*)
381. (*c*)	**382.** (*b*)	**383.** (*a*)	**384.** (*b*)	**385.** (*c*)	**386.** (*b*)	**387.** (*d*)	**388.** (*c*)	**389.** (*a*)	**390.** (*c*)
391. (*c*)	**392.** (*b*)	**393.** (*c*)	**394.** (*c*)	**395.** (*a*)	**396.** (*d*)	**397.** (*c*)	**398.** (*c*)	**399.** (*d*)	**400.** (*a*)
401. (*a*)	**402.** (*a*)	**403.** (*c*)	**404.** (*c*)	**405.** (*b*)	**406.** (*b*)	**407.** (*d*)	**408.** (*b*)	**409.** (*c*)	**410.** (*c*)
411. (*d*)	**412.** (*c*)	**413.** (*a*)	**414.** (*c*)	**415.** (*c*)	**416.** (*d*)	**417.** (*b*)	**418.** (*a*)		

Numerical Type Questions

1. 28°C	**2.** 100- 3000 Kb	**3.** 1	**4.** 200- 500 bp	**5.** 320	**6.** 20%
7. 37°C and 95%	**8.** 28 S	**9.** 42°C (C)	**10.** 63%		

EXERCISE – II

MCQ Type questions

1. (*d*)	**2.** (*d*)	**3.** (*b*)	**4.** (*b*)	**5.** (*b*)	**6.** (*d*)	**7.** (*a*)	**8.** (*c*)	**9.** (*b*)	**10.** (*d*)
11. (*c*)	**12.** (*b*)	**13.** (*a*)	**14.** (*d*)	**15.** (*c*)	**16.** (*a*)	**17.** (*a*)	**18.** (*b*)	**19.** (*c*)	**20.** (*a*)
21. (*b*)	**22.** (*b*)	**23.** (*a*)	**24.** (*a*)	**25.** (*c*)	**26.** (*d*)	**27.** (*a*)	**28.** (*c*)	**29.** (*c*)	**30.** (*d*)
31. (*b*)	**32.** (*d*)	**33.** (*d*)	**34.** (*a*)	**35.** (*a*)	**36.** (*b*)	**37.** (*a*)	**38.** (*b*)	**39.** (*a*)	**40.** (*a*)
41. (*c*)	**42.** (*b*)	**43.** (*c*)	**44.** (*c*)	**45.** (*b*)	**46.** (*c*)	**47.** (*a*)	**48.** (*c*)	**49.** (*c*)	**50.** (*b*)
51. (*a*)	**52.** (*c*)	**53.** (*d*)	**54.** (*a*)	**55.** (*b*)	**56.** (*c*)	**57.** (*d*)	**58.** (*b*)	**59.** (*c*)	**60.** (*c*)
61. (*a*)	**62.** (*c*)	**63.** (*c*)	**64.** (*c*)	**65.** (*d*)	**66.** (*b*)	**67.** (*b*)	**68.** (*b*)	**69.** (*)	**70.** (*a*)
71. (*d*)	**72.** (*b*)	**73.** (*b*)	**74.** (*b*)	**75.** (*a*)	**76.** (*d*)	**77.** (*a*)	**78.** (*b*)	**79.** (*c*)	**80.** (*a*)
81. (*b*)	**82.** (*d*)	**83.** (*a*)	**84.** (*d*)	**85.** (*a*)	**86.** (*b*)	**87.** (*b*)	**88.** (*b*)	**89.** (*d*)	**90.** (*c*)
91. (*b*)	**92.** (*b*)	**93.** (*d*)	**94.** (*b*)	**95.** (*b*)	**96.** (*b*)	**97.** (*b*)	**98.** (*c*)	**99.** (*c*)	

Numerical Type Questions

1. (1.68 to 1.7) **2.** (2.8)

MCQ TYPE QUESTIONS

1. Type II RE does not have the methylase activity. Type II restriction endonucleases combine's endonuclease and methyltransferase activities in a single polypeptide. These are used in recombinant DNA technology because they can be used in vitro to recognize and cleave within specific DNA sequence typically consisting of 4 to 8 nucleotides.
2. BAC vector is used in cloning of very long segment of DNA. It can accommodate up to 300 - 350 kb of foreign DNA and is also being used in genome sequencing projects.
3. Other name of DNA profiling is DNA finger printing. It is a method of distinguishing one organism from another (e.g. two people) by using a pattern of DNA segments separated by gel electrophoresis.
4. The technique used for the study of gene expression is called DNA microarray. This technology promises to monitor the whole genome on a single chip so that researchers can have a better picture of the interactions among thousands of genes simultaneously.
5. The cloning vectors which contain the signals for transcription and translation are called expression vectors. These signals are necessary for initiation and termination of transcription and translation.
6. YAC or Yeast Artificial Chromosomes have both the origin of replication and the two telomeric sequences.
7. Agro bacterium tumefaciens infect plant and produce tumor that is called Crown gall disease. It causes crown gall disease of a wide range of dicotyledonous (broad-leaved) plants, especially members of the rose family such as apple, pear, peach, cherry, almond, raspberry and roses. A separate strain, termed biovar 3, causes crown gall of grapevine.
8. Ti plasmid is named for a plasmid found in bacteria called Agro bacterium tumefaciens. Ti stands for tumor inducing. It causes "plant cancer" or what are known as galls.
9. All the above statements are true. Ti plasmid has T DNA flanking with two 25 base pair repeats and a T DNA encodes for an enzymes that convert plant metabolites to two classes of compounds that benefits the bacterium.
10. Luciferase gene codes for an enzyme. Luciferase is the enzyme that catalyzes the reaction of luciferin to generate light.
11. Electro fusion is a method which is not applicable for delivering DNA into the cells. Instead electro fusion is the process of combining two cell types, thus resulting in mingling of cytoplasmic contents including organelles and nuclei. While the other methods like microinjection viral vectors and liposome mediated transfer can be used to deliver a DNA into the cell.
12. A small nucleotide sequence used to detect the presence of complementary sequences in nucleic acid sample is called a probe. A probe binds very specifically with its complementary nucleic acid sequence.
13. Nick translation activity is only found in DNA polymerase I. Nick translation is also known as 5′ -> 3′ exo nuclease activity. In nick translation, a mis paired DNA is removed i.e. DNA repair is done.
14. Hybridomas are used for the synthesis of monoclonal antibodies by using a hybridoma technology. Hybridomas are hybrid cell lines. A monoclonal antibody is an antibody which is directed towards a single epitope.
15. The probes used in DNA finger printing are DNA segments having a radioactive isotope. These radioactive probes are used to tag certain genes or loci of a chromosome to visualize them. Adding radioactive or colored probes to the nylon sheet produces a pattern. Each probe typically sticks in only one or two specific places on the nylon sheet.
16. In DNA finger printing, the sequences which are matched are Multisatellite DNA. Satellite DNA consists of very large arrays of tandemly repeating, non-coding DNA. Satellite DNA is the main component of functional centromeres, and forms the main structural constituent of heterochromatin.
17. The nuclease enzyme which begins its attack from free end of polynucleotide is the exonuclease enzyme. It cleaves the nucleotides from the ends. They are of two types :

    5′ → 3′ exonuclease: it cleaves from the 5′ end towards the 3'end.

    3′ → 5′ exonuclease: it cleaves from the 3′ end towards the 5'end.
18. The technique of invertion of describing gene into DNA of plasmid vector is called gene drafting.
19. Shuttle vectors are those vectors which can be grown in one host and, then moved into another without any extra manipulation. These vectors can exist in both the eukaryotic cell and E.coli.
20. Restriction endonuclease selectively recognizes a specific DNA sequence and cleaves any DNA containing that sequence. That specific nucleotide sequence is called restriction site. These are those specific sites at which a restriction enzyme will cleave DNA.
21. Plasmids are extra chromosomal, self- replicating, circular and double- stranded DNA molecules that are found naturally in many bacteria and also in some yeast.
22. The technique of transfer of DNA molecules separated by gel electrophoresis to the nitrocellulose or nylon membrane is called northern blot. Northern blotting is used for the identification of RNA by using a formaldehyde gel.
23. DNA is visualized using UV transillumination after staining of ethidium bromide. Ethidium bromide is the most commonly used dye for DNA and RNA detection in gels. Ethidium bromide is a DNA intercalator, inserting itself between the base pairs in the double helix.
24. The most essential requirement in polymerase chain reaction is DNA polymerase which is stable at high temperature (> 90°C) to carry out synthesis of new DNA.
25. The initiation codon for translation in prokaryotes is AUG. AUG is the start codon and 10 bases from it, an AGG AGG sequence is present known as Shine Dalgarno sequence.
26. The uptake of plasmid DNA into bacterial cell is facilitated in the presence of cold calcium chloride. The bacterial cells become markedly competent to take up external DNA when these cells are briefly suspended in cold calcium chloride.
27. The double stranded DNA molecule loses its viscosity upon denaturation. DNA denaturation, is also called DNA melting and is the process by which double-stranded deoxyribonucleic acid unwinds and separates into single-stranded strands through the breaking of hydrogen bonding between the bases
28. Functional unit of DNA that specifies of one polypeptide is called a cistron. In other words, a cistron is a section of DNA that contains the genetic code for a single polypeptide and functions as a hereditary unit.
29. Regulated unit of genetic engineering is Operon. An operon is a sequence of genetic material that functions in a coordinated manner, consisting of an operator, a promoter, and one or more structural genes that are transcribed together. Operons were first found in prokaryotes.
30. Eco RI is isolated from bacterium Escherichia coli strain RY 13.

31. A plasmid is used as a vector in preparing recombinant DNA. The rDNA molecules are inserted into these vectors. Restriction enzymes are required to selectively recognize a specific DNA sequence and degrade any DNA containing that sequence. It restricts the propagation of any foreign DNA in a bacterium. While DNA polymerase is required to replicate the DNA. Therefore only Phage which is a foreign DNA is not used in preparing a recombinant DNA.
32. The process which can be used to bring two genetically different cells together for common desired characters is known as protoplast fusion. It is a technique by which two protoplasts are fused to form hybrid cells that can grow into mature hybrid organisms. It is usually performed on plants.
33. The biolistic PDS- 1000 is an instrument which is used to transfer DNA to cell. It is a He device. It is used to transform living organisms with foreign DNA. DNA is delivered to cells in association with microscopic metal particles, called micro carriers that are propelled at high velocity towards target tissues. The micro carriers are accelerated on a plastic cylinder, called a macro carrier, which is driven by a shock wave of helium gas.
34. The polymerase used in PCR is known as Taq polymerase. It is a thermostable polymerase which can withstand high temperatures of > 90°C to carry out synthesis of new DNA. Taq DNA polymerase synthesizes the DNA region between the primers, by using dNTPs and Mg^{2+} ions.
35. DNA polymerase II used in PCR is mainly concerned with the proof reading of DNA synthesized by taq polymerase. DNA polymerase II has the 3′ ′ 5′ exonuclease activity or proofreading activity. When an incorrect base pair is recognized, DNA polymerase reverses its direction by one base pair of DNA and excises the mismatched base. Following base excision, the polymerase can re-insert the correct base and replication can continue.
36. Poly acryl amide gel electrophoresis is used for DNA sequencing because, it can resolve the DNA molecules which are differing in their length. Traditional DNA sequencing techniques such as Maxam-Gilbert or Sanger methods used poly acryl amide gels to separate DNA fragments differing by a single base-pair in length so the sequence could be read.
37. Purines and pyrimidines can be cleaved by Dimethyl sulphate and Hydrazine. Purines react with dimethyl sulfate and pyrimidines react with hydrazine in such a way so as to break the glycoside bond between the ribose sugar and the base displacing the base. Piperidine then catalyzes the phosphodiester bond cleavage where the base has been displaced.

38. The properties of a good vector include that it should be relatively small in size since large molecules tend to break down during purification and are difficult to manipulate. It should have Multiple Cloning Site (MCS) or a polylinker which provides flexibility in the choice of restriction enzymes that can be used for cloning. Also the speed of replication of a vector should below.

39. The enzyme used in foot printing technique is RNA polymerase. In the foot printing technique the RNA polymerase-promoter complex is partially digested by DNase I. The DNA is then labeled on one strand. The fragments are then broken using the Maxam-Gilbert sequencing reagents.

40. pBR 322 is a plasmid vector. It contains two different antibiotic resistance genes and recognition sites for several restriction enzymes.

41. The cluster of unique restriction sites of cloning vectors are known as Polylinker or Multiple Cloning Sites (MCS). It is a short segment of DNA consisting of around 20 restriction sites. They provide flexibility in the choice of restriction enzyme that can be used for cloning.

42. A modification enzyme adds a methyl group to one or two bases usually within the sequence recognized by the restriction enzyme. Once a base in a DNA sequence is modified by the addition of a methyl group, the restriction enzymes fail to recognize and cleave that DNA. Thus a modification enzyme alters the DNA sequence so that restriction enzymes will not damage the strands.

43. Cos stands for cohesive sites. The extreme ends of the lambda DNA are known as Cos sites, each is single stranded and 12 nucleotides long. Cos sequences are ~200 base pairs long and essential for packaging. They contain a cosN site where DNA is nicked at each strand, 12bp apart, by terminase.

44. All the restriction enzymes given above (Sma I, Eco RI and Hind III) produce the DNA fragments with blunt ends. These restriction enzymes cleave both the strands of DNA at exactly the same nucleotide position, typically in the center of the recognition sequence, which results into a blunt or flush end.

45. Reverse transcriptase which helps in the synthesis of DNA from RNA is encoded by retrovirus. A retrovirus is an RNA virus that replicates in a host cell. It uses its own reverse transcriptase enzyme to produce DNA from its RNA genome.

46. The catabolic plasmid is the Ti plasmid. A Ti or tumour inducing plasmid is a circular plasmid that is a part of the genetic equipment that Agrobacterium tumefaciens and Agrobacterium rhizogenes use to transduce its genetic material to plants. The Ti plasmid is lost when Agrobacterium is grown above 28°C. These are catabolic plasmids classified into different types based on the type of opine produced by their genes.

47. The Vir region on the Ti/Ri plasmid is a collection of genes whose collective function is to excise the T-DNA region of the plasmid and promote its transfer and integration into the plant genome. Vir D1 + D2 have endonuclease activity, and make single-stranded cuts within the left and right borders. VirE acts as a ssDNA binding protein, protecting the single strand T-DNA region during the transport phase of the process. Once in the plant cell, the complementary strand of the T-DNA is synthesized.

48. Lambda gt 10 is an insertion bacteriophage vector. Lambda gt10 is widely used for cloning small inserts up to 7.6kb. The vector is a linear double stranded DNA that has a unique Eco RI site located near the carboxyl terminus of the lac Z gene in the phage repressor gene cI. Inserts cloned into the Eco RI site disrupt the lac Z gene, allowing for blue/white recombinant selection.

49. YAC is less stable than BAC because of the chimeric effects. Chimeric effect can be defined as the artifacts where the sequence of the cloned DNA actually corresponds not to a single genomic region but to multiple regions. Chimerism may be due to either co-ligation of multiple genomic segments into a single YAC, or recombination of two or more YACs transformed in the same host Yeast cell.

50. An 8 base pair sequence recognized by rare cutter restriction enzyme is Not I. It recognizes the 8 base pair target sequence: 5'-GCGGCCGC-3'. Not I is used to generate large genomic fragments and to map DNA methylation status.

51. After the construction of recombinant lambda or cosmid library, the total DNA transferred into *E.coli* is done by using a technique called in vitro packaging. It is a process in which the assembly of head and tail proteins of bacteriophages with recombinant bacteriophage DNA is done to generate infectious phage particles.

52. The element of yeast which contains the origin of replication is known as an autonomously replicating sequence. It is a short DNA sequence of a few hundred base pairs which is required to support the autonomous replication of the chromosomes. It contains four regions (A, B1, B2, and B3), named in order of their effect on plasmid stability; when these regions are mutated, replication does not initiate.

53. Permanent or stable transfection is also known as insertional transfection. In permanent/ insertional transfection, the transfected DNA integrates into the chromosomes or exists as an episome. The cells that have integrated, or episomal foreign DNA, can be distinguished by selectable markers located on the plasmids.

54. Integration of DNA at random places in the genome is a result of non homologous recombination. Non-homologous recombination is a random insertion. This is seen with transposons and VDJ recombination (changing gene sequences to facilitate antibody production in the mammalian immune system), and T-DNA insertions.

55. The sequencing method used for determining the order of the nucleotide in a molecule of DNA is the DNA sequencing. It is the process of determining the precise order of nucleotides within a DNA molecule. It includes any method or technology that is used to determine the order of the four bases-adenine, guanine, cytosine, and thymine-in a strand of DNA.

56. In the chain termination methods, the termination of process occurs because of the incorporation of Dideoxy nucleotides. These molecules terminate DNA chain elongation because after being incorporated they cannot form a phosphodiester bond with the next deoxynucleotide (formation of phosphodiester bond in DNA requires a hydroxyl group at 3'- position) leading to chain termination.

57. Transposons are also called as Jumping genes. These are DNA sequences that move from one location on the genome to another. These elements were first identified more than 50 years ago by geneticist Barbara McClintock of Cold Spring Harbor Laboratory in New York. They are found in almost all organisms (both prokaryotes and eukaryotes) and typically in large numbers. For example, TEs make up approximately 50% of the human genome and up to 90% of the maize genome.

58. There are 3 distinct types of transposons known as :–
    - Class I (retrotransposons): They copy themselves in two stages, first from DNA to RNA by transcription, then from RNA back to DNA by reverse transcription.
    - Class II (DNA transposons): The cut-and-paste transposition mechanism of class II TEs does not involve an RNA intermediate. The transpositions are catalyzed by various types of transposase enzymes.
    - Class III (MITEs): MITE stands for Miniature Inverted- repeats Transposable Elements. These are characterized by their short lengths, generally about 400 to 600 base pairs, and by a stretch of about 15 base pairs that occurs at each end of each element in an inverted fashion.

59. The genetic studies of Zea Maize by Barbara McClintock and also by Marcus Rhoades revealed that there are jumping elements in the genome. They have been detected through the abnormalities they produce in the activities of the genes near their new position to which they move. If a transposon has integrated into a gene involved in corn pigmentation, the resulting corn reveals a mosaic phenotype. Transposon-induced mutations are therefore said to be unstable and the resulting variegated phenotypes can be taken as evidence for the presence of transposons within the genome of this organism.

60. The phenomenon of the movement of genetic locus on the chromosome location is known as transposition. Or in other words, it can be defined as the ability of genes to change position on chromosomes, a process in which a transposable element is removed from one site and inserted into a second site in the DNA.

61. The concept of replacement of the mutant alleles with functional ones is called gene therapy. This technique uses genes to treat or prevent the disease.

    A typical gene therapy technique includes :
    - Replacing a mutated gene that causes disease with a healthy copy of the gene.
    - Inactivating, or "knocking out," a mutated gene that is functioning improperly.
    - Introducing a new gene into the body to help fight a disease.

62. DOP in DOP- PCR stands for Degenerate Oligonucleotide Priming PCR. It employs oligonucleotides of partially degenerate sequence and has been developed for genome mapping studies. This degeneracy, together with a PCR protocol utilizes a low initial annealing temperature and ensures priming from multiple evenly dispersed sites within a given genome. It is a species independent method.

63. Insertion of therapeutic genes into vegetative cells of the patient is known as Somatic gene therapy. It refers to the specific gene therapy that involves insertion of new genetic material into body cells and tissues other than the reproductive ones. As a result of this therapy, the new cells introduced into the body are not passed on to the next generation because they are not part of the sperm cells or egg cells but only of the body cells.

64. The type of gene transfer in which the organism receives the genetic material from its ancestor is known as vertical gene transfer. Vertical gene transfer uses reproduction as a means of gene transfer through generations, whereas horizontal gene transfer uses non-reproductive methods of gene transfer. In vertical gene transfer, transfer of DNA occurs during normal cell division. It produces new cells, and ultimately the offsprings of subsequent generations.

65. The genetic technique which is used by the gene targeting is the homologous recombination. Homologous recombination is a type of genetic recombination in which nucleotide sequences are exchanged between two similar or identical molecules of DNA. Gene targeting through homologous recombination (HR) is the most accurate and versatile mechanism for correction of mutations in genomic DNA in situ.

66. Gene trapping is an approach that randomly disrupts genes throughout the genome by inserting a DNA element (usually containing a reporter). These DNA elements are sometimes fused in frame of the endogenous gene so that the reporter will be expressed in a similar pattern as the endogenous gene. The disruption usually mutates the endogenous gene.

67. Xenotransplantation is any procedure that involves the transplantation, implantation or infusion into a human recipient of either live cells, tissues, or organs from a nonhuman animal source, or human body fluids, cells, tissues or organs that have had ex vivo contact with live nonhuman animal cells, tissues or organs.

68. Competent bacteria are bacteria adapted for easier receiving of heterogeneous DNA. Such bacteria thus enable fast and highly efficient transformation of recombinant plasmids in molecular cloning or protein expression. Many bacteria including E. coli do not naturally transform, however, simple chemical treatments make them competent for transformation.

69. Transformation efficiency is the efficiency by which cells can take up extracellular DNA and express the genes encoded by it. It is based on competence of the cells. It can be calculated as the number of transformed cells (transformants) generated by 1 g of supercoiled plasmid DNA in a transformation reaction.

70. A genetic technique in which an organism is engineered to carry genes that have been made inoperative is known as gene knockout. It allows inactivating specific genes within an organism and determining the effect this has on the functioning of the organism. Genes are knocked out by changing the region of the gene that codes for the protein. Genes can be knocked out in microorganisms as well as organisms of greater complexity, including flies, mosquitoes, and worms. However, the mouse is perhaps the most important species in which researchers can now use knockout technology.

71. PCR is of immense value in generating abundant amount of DNA for analysis in the DNA fingerprinting technique used in the forensic science to link a suspect's DNA to the DNA recovered at the crime scene.

72. In southern blotting, the DNA fragments obtained after digesting the genomic DNA with restriction endonuclease are resolved according to size by agarose gel electrophoresis.

73. Western blotting is a powerful and important procedure for the immune detection of proteins post-electrophoresis, particularly proteins that are of low abundance. It is also known as protein blotting or immune blotting.

74. RAPD markers are decamer (10 nucleotide length) DNA fragments from PCR amplification of random segments of genomic DNA with single primer of arbitrary nucleotide sequence and which are able to differentiate between genetically distinct individuals, although not necessarily in a reproducible way. It is used to analyse the genetic diversity of an individual by using random primers.

75. RFLPs can be used determine the disease status of an individual and the distance between its loci is measured in Centi Morgan.

76. Site-directed mutagenesis is also called site-specific mutagenesis or oligonucleotide-directed mutagenesis. It is a molecular biology technique often used in biomolecular engineering in which a mutation is created at a defined site in a DNA molecule. In general, this form of mutagenesis requires that the wild type gene sequence be known.

77. The DNA fragments on agarose gel are transferred to the nitrocellulose membrane by either capillary action or by vaccum blotting and electro blotting techniques which use the principle of absorption for transfer.

78. In Southern blotting Autoradiography is used to locate the positions of bands complementary to the probe. The sample is analyzed with an autoradiograph, or by the use of an X-ray film to reveal whether or not the sample contains the target DNA sequence.

79. In PCR, annealing occurs due to the change in the temperature from high temperature to low temperature. In annealing step, the two oligonucleotide primers anneal or hybridize to each of the single- stranded template DNA since the sequence of the primers is complementary to the 3' end of the template DNA. This step is carried out at a lower temperature (around 45- 55°C) depending on the length and sequence of the primers.

80. DNA profiling is the process that creates a virtually unique bar-code like pattern that can be used to identify an individual and to show who they are closely related to. RFLP (Restriction Fragment Length Polymorphism) is a DNA profiling method. RFLP DNA profiling is the process of separating an individual's unique, polymorphic fragments from the ones which are already present in everyone.

81. RFLP are detected as differential movement of a band on their gels and each band is regarded as RFLP locus. RFLP locus is definable only by combination of a specific restriction enzyme with a specific DNA probe.

82. RAPD stands for random amplification of polymorphic DNA. RAPD is produced by using both the PCR equipment and Thermo cycler. It is a type of PCR reaction, but the segments of DNA that are amplified are random.

83. DNA profiling is the collection, processing and analysis of VNTRs -- unique sequences on the loci (area on a chromosome). VNTR stands for Variable Number Tandem Repeats. These are tandem repeats, or pairs of nucleotides, which vary in number. Most DNA sequences in different people look too similar to tell apart. After processing, however, VNTRs result in bands that are unique enough to be used for identification. VNTRs are a kind of molecular marker that contains sequences of DNA that have end-to-end repeats of different short DNA sequences.

84. Nylon membrane is used in northern blotting because it is a positively charged membrane and because of high affinity of negatively charged nucleic acid towards the membrane, it is not used. The sensitivity of northern blots is increased with the use of nylon positively-charged membranes.

85. A genomic library is a collection of clones representing the complete genome of an organism. This means that the entire genome of an organism is represented as a set of DNA fragments inserted or cloned into a vector molecule that can be propagated in a suitable host. To prepare a genomic library, the total genomic DNA is isolated from the tissue.

86. The palindromic sequence which is recognized by restriction endonuclease enzyme is both a direct as well as inverted repeat. An inverted repeat (or IR) is a sequence of nucleotides that is the reversed complement of another sequence further downstream. When no nucleotides intervene between the sequence and its downstream complement, it is called a palindrome having a direct repeat.

E.g.: 5'GAATTC3'
3'CTTAAG5'

87. Gene gun/ particle gun method is used to introduce foreign/alien DNA into the host cells. Microscopic particles of gold or tungsten are coated with the DNA of interest and bombarded onto cells. This technique is known as biolistics.

88. DNA ligase enzyme requires a phosphate group at the 5' carbon of one nucleotide and a hydroxyl group at the 3' carbon of the adjacent nucleotide to form the phosphodiester bond between these two nucleotides.

89. The most commonly used bacteria in genetic engineering are Agrobacterium or A.tumifaciens and Escherichia or *E.coli*. Bacteria like *E.coli* have very short generation times. Combined with the high copy number of plasmid they become small factories that duplicate the plasmid with pretty good fidelity for other uses.

A. tumefaciens is capable of transferring a particular DNA segment Transfer (T)-DNA of the tumour-inducing (Ti) plasmid into the nucleus of infected cells where it is subsequently integrated into the host genome. A.tumefaciens is widely used in agricultural and plant transformation researches as mediated transformation of plants.

90. Eco K is a type I restriction endonuclease. It acts only on double-stranded DNA that contains a unique recognition sequence. AACN6GTGC is the recognition sequence for EcoK (N = any nucleotide).

91. SI nuclease is a single-strand-specific endonuclease that degrades DNA and RNA to nucleoside 5'-monophosphates. It has an acid pH optimum (4.0-4.5) and requires Zn2+ or Co2+ for maximal activity.

92. DNase I, (RNase-free) is an endonuclease that nonspecifically cleaves DNA to release di, tri- and oligonucleotide products with 5´-phosphorylated and 3´-hydroxylated ends. DNase I act on single- and double-stranded DNA, chromatin and RNA: DNA hybrids.

93. cDNA is a DNA copy synthesized from messenger RNA or mRNA. The enzyme used is reverse transcriptase an RNA-dependent DNA polymerase isolated from a retrovirus.

94. Shuttle vectors are capable of living in more than one host cell. They function in eukaryotic cell as well as in *E.coli*, as it has two origins of replication and two marker genes. Therefore the statement "it can live in only one host cell" is false.

95. Lambda phage is a bacterial virus which has a double stranded and linear DNA genome of 48,514 bp, in which the 12 bases at each end are unpaired but complementary.

96. As both strands are copied during PCR, there is an exponential increase of the number of copies of the gene. Suppose there is only one copy of the wanted gene before the cycling starts, after one cycle, there will be 2 copies, after two cycles, there will be 4 copies, and three cycles will result in 8 copies and so on.

97. Thymidine kinase (TK) is not a scorable marker while it is a ubiquitous enzyme that catalyzes the ATP-dependent phosphorylation of thymidine.

98. After isolation DNA is precipitated in ice- cold ethanol. DNA precipitates in ethanol after adding salt because ethanol creates an electrostatic environment that allows the Na+ to bind to the PO3- groups on the sugar-phosphate backbone of DNA strands. Water has too high of a dielectric constant to allow a significant amount of the Na+ to bond.

99. Microinjection and particle gun are vectorless or direct gene transfer methods. In microinjection the DNA is directly injected in to the host cell by using a fine tipped glass needle or micropipette. In particle gun method, foreign DNA is coated onto the surface of minute gold or tungsten particles and bombarded onto the host cell by using a particle gun.

100. The gag gene provides the basic physical infrastructure of the HIV virus. The gag (group-specific antigen) gene codes for the Gag polyprotein, which is processed during maturation to MA (matrix protein, p17).

101. Beta-galactosidase converts other substrates such as X-Gal or 5-bromo-4-chloro-3-indolyl-[beta]-D-galactosidase into a colored product. X-Gal is a colorless modified galactose sugar, however, when it is metabolized by beta-galactosidase the products are a bright blue.

102. The function of the linker is to create cohesive ends on blunt ends. They modify the DNA by the addition of new restriction site. They do not have any preformed ends. They are double stranded and the size of a linker is minimum 10 bases.

103. In 1997, the first transgenic cow, Rosie, produced human protein-enriched milk at 2.4 grams per litre. This transgenic milk is a more nutritionally balanced product than natural bovine milk and could be given to babies or the elderly with special nutritional or digestive needs. Rosie's milk contains the human gene alpha-lactalbumin.

104. DNA fingerprinting, also termed DNA profile analysis is based on the use of the "Southern" hybridization technique to analyze polymorphic regions of human DNA.

105. Once food enters the body, it immediately is detected and the insulin mRNA is translated as a single chain precursor called preproinsulin in the pancreas. The preproinsulin, made up of an A (alpha), B (beta), and C chains, and a signal peptide, is 110 amino acids long.

106. The first successful gene therapy on humans was performed in 1990 by researchers at the National Institutes of Health. The therapy treated a four-year-old child for adenosine deaminase (ADA) deficiency, a rare genetic disease in which children are born with severe immunodeficiency and are prone to repeated serious infections.

107. A person suffering from ADA deficiency can be treated by using Gene replacement therapy. Enzyme/Gene replacement therapy, involves repeated injections of the ADA enzyme.

108. In case of transgenic organisms, the genetic material from one organism is inserted in the genetic material of a completely unrelated organism. The inserted genes, called transgenes when they are inserted into the new host, may come from another organism of the same or a different species, or a completely unrelated kind of organism.

109. The human protein used for treatment of emphysema is alpha- I- antitrypsin. Alpha-1 antitrypsin (AAT) is a natural protein circulating in our blood. Its main function is to keep white blood cells from damaging normal tissues. White blood cells contain destructive substances they use to fight infections. The major known cause of emphysema is alpha-1 antitrypsin deficiency.

110. Emphysema is a respiratory disorder. Emphysema is a form of chronic (long-term) lung disease. People with emphysema have difficulty breathing from a limitation in blowing air out. There are multiple causes of emphysema, but smoking is by far the most common. It is the main type of chronic obstructive pulmonary disease (COPD).

111. *Cry*1 protoxins of Bacillus thuringiensis are insecticidal 135-kDa proteins synthesized and assembled into parasporal crystals during sporulation. Insecticidal *Cry* proteins produced by Bacillus thuringiensis are the principal active ingredients of most bacterial insecticides.

112. Protoxins are cleaved by protease in the insect gut to form active toxin, which kills the insect. The acidic environment created by the destruction of gut cells (by influx of ions) is utilized by the endospores for germination and subsequent attack.

113. RNA interference (RNAi) is a natural process that cells use to turn down, or silence, the activity of specific genes. It was discovered in 1988.

114. Proinsulin is matured into insulin by removal of the C-chain of insulin. The endopeptidase called proprotein convertase 1 plays a significant role in the removal of the C-chain in order to generate proinsulin.

115. *Trichoderma* viridae was proved as an effective biocontrol agent against two fungal pathogens, Fusarium oxysporum f. sp. adzuki and Pythium arrhenomanes, infecting soybean. During an in vitro biocontrol test, *Trichoderma* showed mycoparasitism and destructive control against the tested fungal pathogens.

116. A ribozyme is an RNA segment that has the ability to catalyze the cleavage and formation of covalent bonds in RNA strands at specific sites.

117. Terminal nucleotidyl transferase is a template-independent DNA polymerase which is capable of catalyzing the elongation of a DNA strand by the addition of nucleotides from the surrounding solution.

118. Opines are low molecular weight compounds found in plant crown gall tumors or hairy root tumors produced by parasitic bacteria of the genus Agrobacterium. Each strain of Agrobacterium induces and catabolizes a specific set of opines.

119. Functions like oncogenesis, opine synthesis, opine breakdown, differentiation of the tumor into root, shoot or root and shoot on a hormone less synthetic medium were found in T DNA. The transferred DNA (T-DNA) which is found on a large Ti (tumor inducing) plasmid is modified within the bacterium and is transferred to the plant where it becomes integrated into the plant genome. It is responsible for tumor induction on dicotyledonous plants. T-DNA leads to the formation of proteins involved with the production of auxins and cytokynines.

120. Site specific insertion of genes into Ti plasmid DNA of the agrobacterium tumefaciens tumor inducing plasmid is an approach to genetic engineering of higher plant cells.

The procedure is as follows :

A T-DNA sub fragment cloned in pBR322 is cleaved by a restriction endonuclease at a unique central site and target DNA is ligated into this site. The resulting recombinant plasmid is purified and cleaved with Eco RI; the resulting fragment bearing the kanamycin resistance marker is ligated into the unique Eco RI site of pRK290, a wide-host-range plasmid. The pRK290 recombinant plasmid is transformed into an A. tumefaciens strain containing a wild type Ti plasmid. Double recombination between the altered T-DNA fragment of the clone and its wild-type counterpart in the Ti plasmid is selected for by introduction of R751-pMG2, a plasmid incompatible with pRK290.

121. Micro propagation (tissue culture or in-vitro culture) refers to the multiplication of plants, in aseptic condition and in artificial growth medium. It facilitates long distance transport of propagation material and long term storage of clonal materials.

122. The extraction of DNA polymerase from thermophilic bacteria allowed major advances in PCR technology. Thermus aquaticus is a bacterium which lives in the hot springs at 203°F (95°C). The DNA polymerase from T. aquaticus keeps its activity at above 203°F (95°C) for many hours thereby allowing priming and extension steps to be carried out at high temperatures.

123. Transgenic microorganisms are used to improve the meat yield. Enzymes like proteases can make meat tenderer and improve aroma. They can also be used to help separate meat residues from bones.

124. Germline gene therapy involves the modification of germ cells (gametes) that will pass the change on to the next generation. With germline therapy genes could be corrected in the egg or the sperm that is being used to conceive. Because every cell descends from the fertilized egg, every cell in the offspring would possess the transplanted gene.

125. Sequence maps are very useful in understanding the nature of the gene and many functions associated with it. It shows sequences of DNA or amino acids, as well as primers, features, restriction sites and more.

126. RFLP stands for Restriction Fragment Length Polymorphism. It is a technique that is based on the observation that there are variations in the DNA sequence of restriction sites among different individuals and different species.

127. A microarray works by exploiting the ability of a given mRNA molecule to bind specifically to, or hybridize to, the DNA template from which it originated. By using an array containing many DNA samples, scientists can determine, in a single experiment, the expression levels of hundreds or thousands of genes within a cell by measuring the amount of mRNA bound to each site on the array.

128. Protein- encoding genes are identified by searching for ORFs which stand for Open Reading Frames. These are translated DNA sequences. The input sequence will be translated in 6 frames, which are +1, +2, +3, -1, -2, and -3. The + indicates the forward chain whereas - indicates the backward chain. These are used to determine the most probable gene regions in the sequence so as to be used for further analysis and to see what gene it expresses.

129. DNA markers are DNA polymorphisms that are useful for genetic engineering. It is a gene or DNA sequence with a known location on a chromosome that can be used to identify individuals or species. It can be described as a variation (which may arise due to mutation or alteration in the genomic loci) that can be observed.

130. A short tandem repeat (STR) is a type of DNA polymorphism where short sequences of DNA are repeated. STRs are usually considered "junk DNA" because they are introns and do not code for protein. The number of times a DNA sequence is repeated for a given STR is variable between different individuals and thus, STRs are often useful for forensic or genealogical studies.

131. SNPs stand for Single Nucleotide Polymorphism. SNPs are DNA sequence variations, which occur when a single base (A, C, G or T) is altered so that different individuals may have different letters at these positions.

132. RFLP analysis uses "Southern Blot Hybridization" (blot DNA samples after separating fragments with agarose gel electrophoresis) to detect specific DNA restriction fragments in genomic DNA. The Method includes, acquiring a chromosome - from suspect, saliva, hair, crime scene, etc. Then chopping up DNA with restriction endonucleases. Then acquiring the characteristic fragments. And lastly, separating the fragments with agarose electrophoresis.

133. The rate of RNA synthesis can be measured by adding a ribonucleotide (especially uridine labeled with tritium) to the cells and measuring how much radioactivity is incorporated into RNA specific for the gene of interest.

134. Given that 20 G's, 10 A's, 15 T's and 10 C's are present in one strand of the double stranded DNA. So as we know that A pairs with T and C pairs with G, so the number of G's present in the complementary strand would be 10 (as 10 C's will be converted to G in the complementary strand. Therefore the total number of G's in the double stranded DNA is 30.

Similarly the complementary strand will have 15 A's, 10 T's and 20 C's present in the sequence. So total number of A would be 25, T would be 25 and C would be 30 in the double stranded DNA.

135. In the patient with hemophilia the intronic regions that flanked each exon were sequenced and sets of facing intronic primers were designed for specific amplification of each exons. Direct sequencing of the amplified exons identified some novel missense mutations. One mutation which occurred in the patient was a point mutation wherein a lysine residue was replaced by a glutamate residue.

136. After the development of human genome project the individual differences are now detected by using SNPs (Single Nucleotide Polymorphisms). A SNP is a single base pair mutation at a specific locus, usually consisting of two alleles (where the rare allele frequency is >1%).

137. cDNA was found to contain the same nucleotide sequence as the gene that produced mRNA in the discovery of introns. After mRNA was isolated and purified, an enzyme called reverse transcriptase was used to "backtrack" and create a DNA version of the mRNA. This version is called copy DNA, or cDNA.

138. Most restriction endonucleases do not cut DNA at the same nucleotide position, but cleave the recognition sequence asymmetrically. As a result, the DNA fragments produced by cleavage with these enzymes have short, complementary single-stranded overhangs at each end.

139. As humans do not produce aromatic amino acids therefore they are not affect by glyphosphate in plant crops. Glyphosate's mode of action is to inhibit an enzyme involved in the synthesis of the aromatic amino acids: tyrosine, tryptophan and phenylalanine, therefore not affecting the humans.

140. Plant cells which receive Ti plasmid with a TMV viral gene are immune to TMV virus because TMV does not infect cells which are already infected with TMV. TMV stands for Tobacco Mosaic Virus. It is transmitted when an infected leaf rubs against a leaf of a healthy plant and not of an already diseased plant.

141. Ethylene is the plant hormone that causes fruit to ripen, also causes flowers to wilt. Therefore the gene that makes flower petals respond to ethylene by wilting is replaced with a gene insensitive to ethylene.

142. Bovine somatotropin (BST) ingested via a human's digestive system has no effect on the human. The digestive system breaks the BST down into amino acids and peptides, which remain inactive. Human growth hormone (or human somatotropin) has different amino acids than BST, which prevents BST from having any effect on humans. Milk produced by BST injected cattle is perfectly safe for human consumption. Gastric enzymes in the human stomach effectively destroy any BST from the milk.

143. Glyphosate is a non-selective systemic herbicide that is applied directly to plant foliage. The active ingredient, glyphosate, is one of the most widely used pesticides worldwide.

144. Adenoviruses are medium-sized (90-100 nm), non-enveloped, icosahedral viruses composed of a nucleocapsid and a linear double-stranded DNA (dsDNA) genome.

Parvoviruses are typically linear, non-segmented single-stranded DNA viruses, with an average genome size of 5000 nucleotides.

Retroviruses are RNA viruses that infect animal cells.

Reoviridae have a genome composed of multiple (10, 11, or 12) segments of linear double-stranded RNA (dsRNA), which are contained within virus particles that have icosahedral symmetry.

145. The first step to clone a processed mRNA into a bacterial vector is to create a cDNA molecule by using PCR. cDNA synthesis is performed in the first step using either total RNA primed with oligo(dT), random primers, or a gene-specific primer.

146. After transformation, cells are plated onto agar medium that contains selective antibiotic: only transformed cells, that acquire antibiotic resistance gene on the vector plasmid, will survive and form colonies. All the untransformed cells will die.

147. Mammals including human are made up of very small cells called mammalian cells .These differ from plant cells by not having cell wall. These are bounded by only plasma membrane so sensitive to osmotic lysis. The clones produced by these mammalian cells replicate much faster than a prokaryotic clone thereby expressing a protein product much quicker than expected. This is the only limitation of using them as cloning and expression hosts.

148. Cloning Vector provides highly efficient cloning of PCR products through TA hybridization. The vector is supplied in a linear form with a T overhang at each 3' end, which hybridizes with high specificity to the A overhang of PCR products generated by Taq and other non-proofreading DNA polymerases.

149. To clone a gene means to cut it out of its original place in the host DNA and ligate it into a suitable "vector" where multiple copies of it can be quickly reproduced. Genes are inserted in to bacteria by using a plasmid which are small pieces of circular DNA that are often found in bacteria which are self-replicating and are maintained in the cell in a stable and characteristic number of copies.

150. In whole animal cloning, researchers remove a mature somatic cell, such as a skin cell or an udder cell, from an animal that they wish to copy. Then they transfer the DNA of the donor animal's somatic cell into an egg cell, or oocyte, that has had its own DNA-containing nucleus removed.

151. Positional cloning is a technique that is used in genetic screening to identify specific areas of interest in the genome, and then determine what they do. This type of screening is sometimes referred to as reverse genetics. It can be defined as mapping a gene to a chromosomal region and then identifying and cloning a genomic copy of the gene from the region

152. Most high copy number plasmid vector used for cloning inEscherichia coli are based on Col E1 natural replicon. The replicon produces 15-20 copies per cell and does not require a plasmid-encoded function for replication for which, it uses DNA polymerase I and III, DNA-dependent RNA polymerase, and the long-living products of the bacterial genes dnaB, dnaC, dnaD, and dnaZ.

153. A bacterial artificial chromosome (BAC) is a DNA construct, based on a functional fertility plasmid (or F-plasmid), used for transforming and cloning in bacteria. While another cloning vector called PAC is produced from the bacterial P1-plasmid. The key difference between these two vectors is that BACs are always maintained at very low copy number while PAC can be maintained at a high copy number.

154. In higher eukaryotes only the genomic DNA is found to contain genes. Higher eukaryotes tend to have a larger percentage of their genes containing introns than lower eukaryotes, and the introns tend to be larger as well.

155. Yep vectors can be classified as autonomously replicating low copy-number vectors. The YEp yeast episomal plasmid vectors replicate autonomously because of the presence of a segment of the yeast 2 mm plasmid that serves as an origin of replication (2 mm ori). Therefore they do not contain an origin of replication from a yeast chromosome.

156. Yeast two hybrid technique is mainly used to detect the protein- protein interaction. In this technique, transcriptional activity is measured by using a eukaryotic transcription activator which is modular in nature and consists of 2 domains: DNA binding and activation domain. If the two proteins interact, then a reporter gene (e.g. gal1-lacZ - the beta-galactosidase gene) is transcriptionally activated and a colour reaction occurs on a specific media.

157. The baculoviral genome has a large capacity for foreign gene insert with up to 38 kb have been tried successfully. Simultaneous delivery of multiple genes to the same cell is feasible.

158. The transfer DNA (abbreviated T-DNA) is the transferred DNA of the tumor-inducing (Ti) plasmid of some species of bacteria such as Agrobacterium tumefaciens and Agrobacterium rhizogenes. By transferring the T-DNA into the plant genome, the bacterium essentially reprograms the plant cells to grow into a tumor and produce a unique food source for the bacteria.

159. The enzyme reverse transcriptase is used to synthesize a cDNA library and not a genomic library. It reverse transcribes the mRNA molecules into single stranded cDNA molecules.

160. Endonuclease enzyme is used to break the inter phosphodiester bond between the nucleic acids in a DNA molecule or RNA sequences.

161. In Cohen and Boyer's transfer of toad genes the plasmid contained a single site for the restriction enzyme EcoRI and a gene for tetracycline resistance. The restriction enzyme EcoRI was used to cut the frog DNA into small segments. Next, the frog DNA fragments were combined with the plasmid, which had also been cleaved with EcoRI.

162. A probe is used during the screening portion of the gene transfer process. Screening a library depends either on having a probe bearing part of the nucleotide sequence or an antibody or other way of recognizing the protein coded by the gene.

163. The cofactor required by the type I restriction enzymes for the methylation of DNA is S-adenosyl methionine. It can act as a methyl-donor. The coenzyme methylates DNA, RNA, proteins and small molecules.

164. A primer is a short synthetic oligonucleotide which is used in many molecular techniques from PCR to DNA sequencing. These primers are designed to have a sequence which is the reverse complement of a region of template or target DNA to which we wish the primer to anneal.

165. Topoisomerases are the key enzymes involved in the procedure of supercoiling of the DNA fragments. The phenomenon occurs when the double stranded DNA molecule is coiled upon itself after the replication process. If one of the double strands of the circular supercoiled DNA fragment is broken due to nick or cut, the double helix reverts to its normal i.e. uncoiled or relaxed state. As a result of this the plasmid DNA changes its conformation which is called as open-circular (oc) or relaxed.

166. "Gene mapping" refers to the mapping of genes to specific locations on chromosomes. It is a critical step in the understanding of genetic diseases. The ultimate goal of gene mapping is to clone genes, especially disease genes. Once a gene is cloned, we can determine its DNA sequence and study its protein product.

167. A genomic library is a set of clones that together represents the entire genome of a given organism. The number of clones that constitute a genomic library depends on

(1) the size of the genome and

(2) the insert size tolerated by the particular cloning vector system.

168. The sequences required by the bacterial expression vectors for the gene product to get expressed include:

- A Shine Dalgarno sequence- which is a ribosomal binding site in the mRNA, generally located 8 basepairs upstream of the start codon AUG.
- Transcription Termination sequences- whch are a section of genetic sequences that marks the end of gene or operon on genomic DNA for transcription.
- Sequence to promote post- translational modification of protein product- Also known as gene constructs which is necessary for the transfer or the expression of the target gene.

169. Antisense RNA is a single-stranded RNA that is complementary to a messenger RNA (mRNA) strand transcribed within a cell. Antisense RNA may be introduced into a cell to inhibit translation of a complementary mRNA by base pairing to it and physically obstructing the translation machinery.

170. The YEp vectors transformed with an efficiency of 0.5-2.0 * 104 transformants/mg input plasmid DNA, and the YRp7 plasmid produced 0.5-2.0 * 103 transformants/mg input plasmid DNA. While the YIp integrating vectors are transformed with low efficiencies, 1-10 transformants/mg. therefore the correct order of these yeast derived vectors in terms of transformation efficiency is :

Yep > YRp > YIp

171. Alkaline phosphatase enzyme is used to prevent unwanted self- ligation of vector DNA molecules in cloning procedures. As ligation requires the presence of 5′ phosphate group at the DNA site to be ligated. So this enzyme removes the phosphate group from the 5′ end of a DNA molecule and leaves a free 5′ hydroxyl group, thereby preventing self- ligation.

172. *In situ* hybridization indicates the localization of gene expression in their cellular environment. A labeled RNA or DNA probe can be used to hybridize to a known target mRNA or DNA sequence within a sample. This labeled RNA or DNA probe can then be detected by using an antibody to detect the label on the probe. The probes can therefore be used to detect expression of a gene of interest and the location of the mRNA.

173. *E. coli* DNA Ligase catalyzes the formation of phosphodiester bonds between double-stranded DNA fragments containing juxtaposed 5′-phosphate termini and 3′-hydroxyl termini in the presence of the NAD^+ cofactor. It uses nicotinamide adenine dinucleotide (NAD^+) instead of ATP as a cofactor and release AMP and nicotinamide mononucleotide (NMN) as a result of phosphodiester bond formation.

174. RAPD stands for random amplification of polymorphic DNA. It is a type of PCR reaction, but the segments of DNA that are amplified are random. The primers used in RAPD are arbitrary, short primers (8-12 nucleotides) which bind to random sites on the DNA strands as no knowledge of the DNA sequence for the targeted genome is required.

175. Those enzymes which have a VNTR segment for genetic disease analysis in RFFLP are chosen. VNTR stands for Variable Number Tandom Repeats. These are segments of DNA in which a 10- 50 bp sequence is repeated, often flanked by conserved sequence unevenly distributed in chromosomes (more frequent near ends).

176. Vitamin B7 is incorporated into dUTP as a non-radioactive label for the DNA probe. Vitamin B7 is also known as Biotin. Biotinylated probes are more frequently used in labeling the DNA.

177. Avidin is a tetrameric protein of 4 identical subunits (homotetramer) each of which can bind to biotin with a high degree of affinity and specificity. Avidin molecular weight in its tetrameric form is estimated to be between 66-69 kDa. Avidin is produced in the oviducts of birds, reptiles and amphibians and is subsequently deposited in the whites of their eggs.

178. In hierarchical shotgun approach large 150,000 base pairs go together to create chromosomes. Shotgun sequencing randomly shears genomic DNA into small pieces which are cloned into plasmids and sequenced on both strands, thus eliminating the BAC step from the HGP's approach. Once the sequences are obtained, they are aligned and assembled into finished sequence.

179. Ligases are the enzymes that seal breaks in the phosphate-sugar backbone of DNA. Thus if DNA ligase is not used, then base pairing can still occur but the sugar- phosphate backbone would not be connected/joined.

180. The DNA polymerase from Thermus aquaticus (Taq polymerase) is homologous to Escherichia coli DNA polymerase I (Pol I) and likewise has domains responsible for DNA polymerase and 5′ nuclease activities. The structures to the polymerase domains of Taq polymerase and of the Klenow fragment (KF) of Pol I are almost identical.

181. A Psi-sequence is found near the 5′ end of the RNA-genome which is necessary for packaging viral RNA into virus capsids to continue the infection of HIV in its host.

182. A heterologous protein for its expression in the milk of a transgenic animal should be under the control of the promoter of the gene which codes for beta- lactoglobulin. It is the major whey protein of cow and sheep's milk. It is found in the milk of a wide variety of species including dogs and dolphins.

183. Plasminogen activator is produced commercially from the mammalian culture. It is a protease produced in the kidney that converts plasminogen to plasmin and so initiates fibrinolysis. It is an enzymatic cascade involved in the control of fibrin degradation, matrix turnover and cell invasion.

184. Coagulation factor IX is used for the treatment of haemophiliaces. Coagulation Factor IX is an important protein in the process of hemostasis and normal blood clotting. Hemophilia B occurs because of deficiency of clotting factor IX.

185. In hybridoma technology the de novo pathway of normal cells occurs by the enzyme Di Hydro Folate Reductase (DHFR). This aminopterin is an inhibitor of DHFR enzyme. Thus, Aminopterin in turn blocks the de novo pathway. Hence, unfused myeloma cells die, as they cannot produce nucleotides by de novo or salvage pathway. Thus it is used in the production of myeloma cells.

186. Serum is added to cell culture medium as it provides growth factors like Platelet Derived Growth Factor (PDGF), Epidermal Growth Factor (EGF), and Insulin like Growth Factor (IGF) etc. Different batches of serum have different growth factors.

187. The purpose of neomycin in creating mice with knock outs in gene X is that it selects for the survival of ES cells that have incorporated the mutant gene in the place of the wild- type gene. Neomycin acts as a positive selection marker. The antibiotic neomycin is toxic to mouse cells because they do not normally contain the NeoR gene. However, when the NeoR gene is added to mouse cells, these cells can survive in the presence of neomycin.

188. The Transformed human kidney cell line, TCL-598 is used in Single chain urokinase-type plasminogen activator (scu-PA).

189. In cellular Biochemical's beta and gamma interferon's cell lines used are Chinese Hamster Ovary (CHO) cell lines. The Chinese hamster ovary (CHO) cell line producing interferon-gamma (IFN-gamma) exhibits a 2-fold increase in specific productivity when grown at 32 degrees C compared to 37 degrees C.

190. MRC 5 is a cell lines isolated from normal, non cancerous human embryonic tissue. It is a finite cell line which originated from the lungs of a human. It is diploid and has fibroblast morphology.

191. The transgene construct is prepared in a buffer solution and injected into the male pronuclei of fertilized egg. The injection of a DNA solution into the pronuclei of fertilized eggs is the most common method for making transgenic animals. Injection is done at the stage of development when the ova have two pronuclei, one from each gamete, which will later fuse to form the diploid nucleus.

192. The Salmonella typhimurium genes for serine acetyltransferase (cys E) and O-acetylserine sulphydrylase B (cys M) are isolated and characterized in order to express these as transgenes in sheep to establish a cysteine biosynthesis pathway and, thereby, to achieve an increased rate of wool growth.

193. Hybridoma cells are non anchorage dependent and they can be grown in suspension culture. Hybridoma cells which are non anchorage dependent can be grown in serum free media for mammalian culture in suspension. These cells do not require attachment for growth. All suspension cultures are derived from cells of the blood system because these cells are also suspended in plasma *in vitro*.

194. The mice which are used for immunization in the Hybridoma technology are Balb/c mice. BALB/c mice are particularly well known for the production of plasmacytomas following injection with mineral oil forming the basis for the production of monoclonal antibodies.

195. Hayflick limit is the limit upto which a population of cells can divide. And then the cell population enters the senescence phase where the cell stops dividing entirely. The cells remain alive for a time after they stop dividing, but sometime after cellular division ends, cells commit suicide.

196. Cell lines transformed under *in vitro* conditions give rise to continuous cell lines. They exhibit the property of ploidy, either aneuploid or heteroploid and there is no contact inhibition or anchorage dependence.

Finite cell lines have a limited life span and they show the property of contact inhibition, density and anchorage dependence.

197. Myeloma cells are cancerous cells which are double mutant. They are immortal and they are not capable of producing any antibody. These cells do not have Thymidine Kinase (TK) and Hypoxanthine Guanine Phospho Ribosyl Transferase (HGPRT) enzyme. Thus myeloma cells become a double mutant and become toxic to normal cells. They are fused with B- cells so that B- cells (antibody producing cells) can be cultured indefinitely.

198. Cell- cell adhesion molecules (CAMS) are calcium independent while the proteins like integrins and cadherine are calcium dependent molecules. CAMS are used by cells to attach to the surface.

199. Cultures contain a relatively uniform population of proliferating cells at low cell densities and an equally uniform, more differentiated, non-proliferating population at high cell densities.

200. Plating density is the ratio of number of cells per volume of culture medium. It is critical for various cell types.

201. Cell suspension cultures are a result of disrupting the normal relationship of animal cells with their neighboring cell wall. The tendency of animal cells in vivo to interact with one another and with the surrounding extracellular matrix is mimicked in their growth in culture. Animal cells used Cell Adhesion Molecules (CAM) like cadherine, integrins and proteoglycans to attach to the surface.

202. The generation time is the period of time required for a quantity to double in size or value. It is also called doubling time. It can also be defined as the time it takes for a parent cell to divide into two daughter cells. It can be calculated as:

$G = t \log 2 / \log Nt - \log No$

Where t -- time period

Nt -- number of cells at time t

No-- initial number of cells

203. ATCC, ECACC and NIGMS are the major animal culture collection repositories which are abbreviated forms for:

ATCC- American Type Culture Collection

ECACC- European Collection of Cell Cultures

NIGMS- A human genetic cell repository

These repositories are responsible for authenticating all the cell line submissions. ATCC uses short tandem repeat (STR) DNA fingerprinting to authenticate its cell lines.

204. The transfer of a portion of cells from the ongoing culture to the new culture vessels for the maintenance of cell lines is known as passage or sub- culture.

205. He La (Human epithelial carcinoma cell line) cell line is an immortal cell line in which He La stands for the patient's name i.e. Henrietta Lacks. Henrietta's cells were the first immortal human cells ever grown in culture. They were essential to developing the polio vaccine. The cell line was derived from cervical cancer cells taken from Henrietta Lacks, in 1951.

206. Nutrient optimization of cells in culture medium is required for proper cell growth. Components like arginine, L- glutamine, tyrosine, some salts, vitamins, glucose are added to promote nutritional uptake by the cells. Glucose acts as the primary energy source while L- glutamine acts as alternate energy source. Phenol red is also added to optimize the pH of the media. Nutrient optimization is also required as it facilitates anchorage to substrate i.e. serum and maintains a high cell density.

207. Transgenesis is the process of inserting a gene from one source into a living organism that would not normally contain it. Gene being transferred is called a transgene while the animal is called a transgenic animal.

208. During the process of fertilization, after the sperm enters the ovum, but before they fuse, the nucleus of a sperm or an egg cell is called a pro nucleus. It is a cell nucleus with a haploid set of chromosomes (23 chromosomes in humans) resulting from meiosis (germ-cell division). The male pro nucleus is the sperm nucleus after it has entered the ovum at fertilization but before fusion with the female pro nucleus. Similarly, the female pro nucleus is the nucleus of the ovum before fusion with the male pro nucleus.

209. BSA or Bovine Serum Albumin is added as a substitute to serum to nourish the cells. BSA , Carboxymethyl cellulose (CMC) or Pluronic acid are added to a serum free medium to increase the viscosity of the medium. Serum free media is specially-formulated media that has defined components in it: that is, instead of adding serum (which can vary from lot to lot, and has many less-regulated proteins, hormones, etc), the precise components are added in measured amounts by the manufacturer. These components usually include BSA, as well as other proteins, lipids, insulin, transferring and also CMC or pluronic acid.

210. YAR cell line is found in humans.

YAC-1 is a T cell lymphoma which was induced by inoculation of the Moloney leukemia virus (MLV) into a newborn A/Sn mouse. This cell line is sensitive to the cytotoxic activity of naturally occurring killer cells in mice.

Vero cells are derived from the kidney of an African green monkey, and are one of the more commonly used mammalian continuous cell lines.

Chinese hamster ovary (CHO) cells are a cell line derived from the ovary of the Chinese hamster.

211. Water for injection (WFI) can be defined as water purified by distillation for the preparation of products for parenteral use. The ingredients of cell culture media are prepared in WFI. It is the water purified by distillation or a purification process that is equivalent or superior to distillation in the removal of chemicals and micro organisms.

212. Fibronectin is known as an attachment and spreading factor. It promotes cell attachment and it is essential for anchorage dependent cell lines. Anchorage dependent cell lines need a stable or inert surface for cell growth, division, and spreading.

213. Gene knockout studies involve replacing the gene sequence with something which is not functional. It is done via recombination pathway. It is done to determine the function or role of any particular gene. It is also done to find variations in the genome which cause diseases.

214. Dediffrentiation can be defined as the regression of a specialized cell or tissue to a simpler unspecialized form by the loss of phenotypic characteristics. It can also be defined as a process by which structures or behaviors that were specialized for a specific function lose their specialization and become simplified or generalized.

215. Retrovirus mediated method is the most common method to produce transgenic animals. A retrovirus is a virus that carries its genetic material in the form of RNA rather than DNA. Retroviruses are used as vectors to transfer genetic material into the host cell, resulting in a chimera, an organism consisting of tissues or parts of diverse genetic constitution. Chimeras are inbred for as many as 20 generations until homozygous (carrying the desired transgene in every cell) transgenic offspring are born.

216. The cells which transfer the desired gene into progeny of the transfected animals are germ cells. A germ cell is any biological cell that gives rise to the gametes of an organism that reproduces sexually. In many animals, the germ cells originate near the gut of an embryo and migrate to the developing gonads.

217. Hollow fiber perfusion systems are used in case of high cell density and offers maximum surface area for the cultivation of anchorage dependent cells. They are made up of plastic capillaries. The cells are grown on the outer surface of these

micro- capillaries. These are arranged in bundles/ cartridges. The capillary system is permeable and media is perfused from the middle of the capillary. Thus media reaches each and every cell.

218. Serum is added to the culture medium for better growth of cells. It is extracted from Bovine and Equine. It is mostly extracted from the fetus or calf. It consists of proteins, growth factors, hormones, nutrients and metabolites, lipids, minerals etc.

219. Selenium (Se) is a well-known essential trace element (TE) for cell growth and development, and its positive role in biological system includes detoxification of free radicals by activating glutathione peroxidase. In cell culture, selenium in the form of selenite can help cells to detoxify the medium thus protect them from oxidative damage.

220. When animal cells divide, they exhibit a property of contact inhibition. Thus if a cell comes in contact with other cell, then it gives the signal for not dividing. Due to contact inhibition animal cells grow as a monolayer. Thus, when the entire available surface is utilized by the cells, then its culture is called as confluent culture.

221. The percentage of cells plated (seeded, inoculated) which attach to the surface of the culture vessel within a specified period of time is referred to as seeding efficiency. The conditions under which such a determination is made should always be stated.

222. The type of engineering involved in changing genes in eggs, sperms, or very early embryos is called as germline engineering. It refers to human genetic engineering used to alter genes in the first cells of the blastocyst.

223. The gene therapy is opened by Monochromosomal Transfer Techniques with the help of Mammalian artificial chromosome (MAC). MAC is a molecule of defined structure comprising combinations of various cloned or synthesized chromosomal elements that had been individually manipulated, tested and optimized as part of a controlled construction process. A MAC is constructed by using a centromere, several DNA replication origins, and two telomeric repeats. The resulting structure carries and expresses one or more selected genes (transgenes), introduced for specific purposes.

224. The method used to disrupt endogenous loci, resulting in targeted null alleles is called gene knockout. It is a genetic technique in which one of an organism's genes is made inoperative ("knocked out" of the organism). Also known as knockout organisms or simply knockouts, they are used in learning about a gene that has been sequenced, but which has an unknown or incompletely known function.

225. The enzyme that is expressed by the cells which are being continuous cultivated for several generations is the telomerase enzyme. Telomerase, also called telomere terminal transferase, is an enzyme made of protein and RNA subunits that elongates chromosomes by adding TTAGGG sequences to the end of existing chromosomes. If telomerase is activated in a cell, the cell will continue to grow and divide.

226. $NaCO_3$ acts as a buffer and is used to maintain the pH and to supply the sodium ions as sodium ions are involved in the sodium bicarbonate buffer formation.

$NaCO_3 \rightarrow Na^+ + CO^{-3}$

It is used as it is less toxic, cheaper and a natural buffer.

227. The products which do not require mammalian cells to be cultured are Atropine and Pheromones.

Atropine is a naturally occurring tropane alkaloid extracted from various plants of the family Solanaceae. It is a secondary metabolite of these plants and serves as a drug with a wide variety of effects.

A pheromone is a chemical secreted by an animal, especially an insect that influences the behavior or development of others of the same species, often functioning as an attractant of the opposite sex. Pheromones are chemicals capable of acting outside the body of the secreting individual to impact the behavior of the receiving individual

228. Some transgenic animals are produced to create specific products. Transgenic cattle have been created to produce milk that contains particular human proteins to be used to treat emphysema. Emphysema is a long- term lung disease.

229. Transgenic cows produce a particular substance that helps increase red blood cells in men. Thus a transgenic cow has RBC's.

230. The business of micro propagation is due to totipotency which can be defined as the potential of a cell to regenerate an entire organism. Every living plant cell is totipotent, i.e., each cell is capable of regenerating a whole plant. When plants are grown by tissue culture methods like organogenesis, embryogenesis etc. it is called micro propagation. It is always done with a view to marketing and commercialization.

231. The hydrocarbon secreted by plants which is used in varnishes is resin. It is secreted by plants, mainly by coniferous trees. It is a natural or synthetic compound which begins in a highly viscous state and hardens with treatment. Resin is soluble in alcohol, but not in water.

232. *In vitro* fusion of isolated protoplasts to form a hybrid cell is referred to as somatic hybridization. In other words, the process by which protoplasts of two different plant species fuse together to form hybrids is known as somatic hybridization and the hybrids so produced is known as somatic hybrids.

233. Secondary metabolites are chemicals produced by plants for which no role has yet been found in growth, photosynthesis, reproduction, or other "primary" functions. These chemicals are extremely diverse and play a role in the defense mechanism of plants against pests and pathogens. They can sometimes be used as taxonomic characters in classifying plants. These secondary metabolites include alkaloids, resins, tannins, latex etc.

234. Cytokinins are compounds with a structure resembling adenine which promote cell division, stimulate shoot proliferation and inhibit root formation. This makes cytokinins useful in culturing plant cell tissue where strong growth without root formation is desirable.

235. Terpenoids (or isoprenoids), a subclass of the prenyllipids (terpenes, prenylquinones, and sterols), represent the oldest group of small molecular products synthesized by plants and are probably the most widespread group of natural products. Terpenoids can be described as modified terpenes, where methyl groups are moved or removed, or oxygen atoms added.

Plant terpenoids are used extensively for their aromatic qualities. They play a role in traditional herbal remedies and are under investigation for antibacterial, antineoplastic, and other pharmaceutical functions.

236.
- Elements that a plant needs in large amounts to live are known as macroelements. KNO_3 is a macroelement.
- Mineral elements that are needed by plants in only trace amounts are known as microelements. KI is a microelement.
- Nicotinic acid is a form of niacin that helps in lowering your cholesterol. Not only is nicotinic acid important because it targets HDL, LDL, and triglycerides, it is also available without a prescription. Thus it is an important organic supplement.
- Plants get their carbon source from photosynthesis. Carbon source is in the form of glucose or sucrose which is stored as starch in plants and glycogen in animals.

237. Production of entire plant or organism from a single cell is called plant regeneration. Ex situ conservation of rare and endemic or threatened plants by tissue culture techniques is also called plant regeneration.

238. *Agrobacterium* mediated gene transfer is one of the most widely used vector dependent methods of gene transfer. *Agrobacterium tumefaciens* is a bacterial plant pathogen found in the soil which causes crown gall disease in plants.

The basic mechanism that works behind it is, transfer of a segment or small piece of bacterial DNA to the plant genome. This bacterial DNA segment gets incorporated in the plant genome, stabilizes there and transcribes or expresses itself. The bacterial DNA causing disease is situated on a plasmid, Ti plasmid (Tumour inducing plasmid, 200kb) instead of on a bacterial chromosome.

239. Colchicine is a toxic chemical that is often used to induce polyploidy in plants. Basically, the colchicine prevents the microtubule formation during cell division, thus the chromosomes do not pull apart like they normally do. The end result is a cell that now has double the number of chromosomes that it would normally have. If this cell divides again in the future, then the doubled number of chromosomes are passed to the offspring cells. Plants that have more than the normal two sets of chromosomes are termed "polyploidy" in general, although specific names are given to the certain chromosome numbers.

240. Haploid plant production from microspores of anthers is referred to as Androgenesis. It involves development of an egg cell with a male nucleus that has been introduced by sperm in the process of fertilization. Androgenesis is observed in certain species of animals (silkworms) and plants (tobacco, corn) in cases where the maternal nucleus is destroyed before fertilization. In androgenesis, the male gametophyte (the microspore or immature pollen) produces haploid plants by stopping the development of pollen cell into a gamete and forcing it to develop into a haploid plant.

241. Artificial culture of haploid plants is done by culturing from anther tissues and microspores. The technique of haploid production through anther pollen culture as well s ovary culture is of immense use in plant breeding employed to improve crop plants. It enables raising plants expressing traits that are otherwise recessive.

242. The other name for organic culture is Permaculture. This term was coined in 1978 by Bill Mollison. It is a contraction of "permanent culture." Permaculture is a design system for creating sustainable human environments. "Integrated farming" and "ecological engineering" are terms sometimes used to describe permaculture.

243. *Agrobacterium rhizogenes* mediated transformation efficiency was assessed under the influence of:

- **Ultra sonication of leaf segments-** In Sonication assisted *Agrobacterium tumefaciens* mediated transformation, the leaf segments were taken in a 50 ml polypropylene tube and ultrasonicated at 50MHz for 10-100 sec at 80% amplitude. The sonicated explants dipped in *Agrobacterium rhizogenes* culture incubated in a shaker at 70 rpm for 30 min in dark. The explants were blot dried using sterile filter paper and inoculated on co-cultivation medium.
- **Treatment with acetosyringone-** Acetosyringone 50 -150 µM was incorporated in the co-cultivation medium. This was filter sterilized using 0.22 m disposable sterile syringe filter added to sterilized, cooled co-cultivation medium. The sonicated explants dipped in A. rhizogenes culture incubated in a shaker at 70 rpm for 30 min in dark. The explants were blot dried using sterile filter paper and inoculated on co-cultivation medium.
- **Calcium treatment-** Five to 20 mM CaCl2 was incorporated in co-cultivation medium. The sonicated explants dipped in Agrobacterium rhizogenes culture incubated in a shaker at 120 rpm for 30 min in dark. The explants were blot dried using sterile filter paper and inoculated on co-cultivation medium.

244. Under intense (ultra) sonication enzymes or proteinscan be released from cells or subcellular organelles as a result of cell disintegration or disruption. Calcium induces in cell signaling for isolation of protoplast, specific enzymes can be used like the macerating enzymes. The virulence (vir) gene of Agrobacterium is activated by acetosyringine.

245. The selecting agent used to detect high- producing cell line is pimelic acid. It is a precursor of biotin.

246. Exogenous elicitors are considered as the primary signals in plant pathogen interactions. They are produced by microorganisms. They originate in the pathogen itself, mostly have a limited mobility within plant tissues, and evoke a response in cells in the immediate vicinity to the pathogen.

247.

- Batasins are those plant growth hormones which cause bulbs to become dormant. They are isolated from yam plants.
- Brassins aresteroid growth promoters first isolated from pollen grains of Brassica napus. Brassins have also been isolated from tea, rice and bean.
- Triacontanol is a saturated primary alcohol. It has potential importance for increasing crop yield. Its foliage spray on seedlings of maize and rice showed enhanced growth of the plants.
- In higher plants, abscisic acid and xanthoxin are two potent growth regulators. Xanthoxin has regulatory functions other than those of abscisic acid.

248. Abiotic elicitors include heavy metal ions or UV light which can induce stress responses in exposed tissues, and may provide an additional barrier to invading pathogens or alternatively, thus increasing the plant's susceptibility to infection.

249. Hempoline or hemp fuel is an environmentally friendly biodiesel made out of hemp. It's one of the most cost efficient and climate saving fuels in the world. Hemp fuel is much more efficient than other kinds of biodiesel, because the hemp plant produces more biomass than any other plant on earth.

250. Resistance to herbicides glyphosate in transgenic plants is brought about by the underproduction of EPSP synthase. **5-enolpyruvylshikimate-3-phosphate (EPSP) synthase** is an enzyme that catalyzes the chemical reaction:

Phosphoenolpyruvate + 3-phosphoshikimate ⇌ phosphate + 5-enolpyruvylshikimate-3-phosphate (EPSP)

The enzyme is a target for herbicides as these amino acids are only synthesized in plants and microorganisms. Glyphosate acts as a competitive inhibitor for phosphoenolpyruvate and is used as a broad-spectrum systemic herbicide.

251. Opines are low molecular weight compounds found in plant crown gall tumors or hairy root tumors produced by parasitic bacteria of the genus Agrobacterium. Opines are a carbon compound that the bacteria can utilize. Thus they serve as a source of carbon, nitrogen and energy for the Agrobacterium in crown gall disease.

252. Synthetic seed is produced by encapsulating somatic embryo with sodium alginate. The somatic embryos are mixed with sodium alginate (2%) and the suspension is dropped into the calcium salts solution (200mM). The principle involved is when sodium alginate dropped into the calcium salt solutions it from round firm beads due to the ion exchange between Na+ in sodium alginate and Ca2+ in calcium slat solutions and sodium alginate form calcium alginate in 20-30 minutes.

253. Meristem cultures are used for propagation and virus eradication.

The growth analysis of cells in cell suspension culture is assessed through packed cell volume (PCV).

Isolation of protoplast culture is done by incubation with 1% macerozymes for the production of protoplasts.

By the use of anther culture, homozygosity is achieved in the quickest possible way and thus a pre homozygous line is developed much faster than by using conventional plant breeding programs.

254. A gene gun or a biolistic PDS1000/He particle delivery system, originally designed for plant transformation, is a device for injecting cells with genetic information.

The Vir region on the Ti/Ri plasmid is a collection of genes whose collective function is to excise the T-DNA region of the plasmid and promote its transfer and integration into the plant genome.

The Gene Pulser apparatus is a pulse generator which uses capacitor discharge to produce controlled exponential pulses for cell electroporation.

Microinjection is normally performed under a specialized microscope with the aid of a micromanipulator which allows for small movement under high magnification

255. Protoplasts fusion is a physical phenomenon. During fusion, two or more protoplasts come in contact and adhere with one another spontaneously or in presence of fusion inducing chemicals. PEG induces protoplast aggregation and subsequent fusion.

256. The β-glucuronidase (GUS) gene reporter system is one of the most effective and employed techniques in the study of gene regulation in plant molecular biology. The GUS gene is commonly utilized as a "marker gene" for genetically engineered plants.

257. Golden Rice is an effective source of vitamin A. The β-carotene contained in Golden Rice is highly available and easily taken up into the bloodstream by the human digestive system. Golden Rice could still be a valuable complement to children's diets in many countries by contributing to the reduction of clinical and sub-clinical vitamin A deficiency-related diseases.

258. White's medium, one of the earliest plant tissue culture media, includes all the necessary nutrients and is widely used for root culture.

259. Potato extract alone or potato extract combined with components of conventional culture media is a useful media for anther culture of wheat and some other cereal plants.

260. Somaclonal variation is the variation seen in plants that have been produced by plant tissue culture. Somaclonal variation is not restricted to, but is particularly common in, plants regenerated from callus.

261. Plants developed *in vitro* culture from pollen grain are androgenic haploids. The basic principle of anther or pollen culture is the production of haploid plants exploiting the totipotency of micro spore and the occurrence of single set of chromosome (n) in the microspore. In this process, normal development and the function of pollen cell to become a male gamete is stopped and is diverted forcely to a new metabolic pathway for vegetative cell division.

262. Insects most often use sugars or polyols as cryoprotectants. Arctic frogs use glucose, but Arctic salamanders create glycerol in their livers for use as cryoprotectant.

263. Cryoprotectants operate simply by increasing the solute concentration in cells. However, in order to be biologically viable they must (1) easily penetrate cells, and (2) not be toxic to the cell.

272. The methods commonly used for gene transfer into animal cells include-Coprecipitation using Calcium Phosphate, Lipofection, Electroporation, Retroviral method and using Baculovirus.

273. To maintain the membrane potential of animal cell in culture, growth medium requires supplementation of which three ions - K^+; Na^+ and Cl^-.

274. The acyl-CoA dehydrogenase is specific for the length of the acyl chain being oxidized. Three types of the dehydrogenase exist in mitochondria; type I (EC 1.3.99.12; long chain) which oxidizes C12-C18 fatty acids, type II (EC 1.3.99.3) which oxidizes C4-C14 fatty acids, and type III (EC 1.3.99.2; butyryl dehydrogenase) which oxidizes C4 and C6 acyl-CoA substrates.

275. Continous cultures are comprised of a single cell type that can be serially propagated in culture either for a limited number of cell divisions (approximately thirty) or otherwise indefinitely. Cell lines of a finite life are usually diploid and maintain some degree of differentiation. The fact that such cell lines senesce after approximately

thirty cycles of division means it is essential to establish a system of Master and Working banks in order to maintain such lines for long periods.

276. Attached cell lines can be classified as endothelial such as BAE-1, epithelial such as HeLa, neuronal such as SH-SY5Y, or fibroblasts such as MRC-5 and their morphology reflects the area within the tissue of origin.

278. Lag Phase - at this stage the cells do not divide. During this period the cells adapt to the culture conditions and the length of this phase will depend upon the growth phase of the cell line at the time of subculture and also the seeding density.

1. Logarithmic (Log) Growth Phase – cells actively proliferate and an exponential increase in cell density arises. The cell population is considered to be the most viable at this phase, therefore it is recommended to assess cellular function at this stage. Each cell line will show different cell proliferation kinetics during the log phase and it is therefore the optimal phase for determining the population doubling time.
2. Plateau (or Stationary) Phase – cellular proliferation slows down due to the cell population becoming confluent. It is at this stage the number of cells in the active cell cycle drops to 0-10% and the cells are most susceptible to injury.
3. Decline Phase – cell death predominates in this phase and there is a reduction in the number of viable cells. Cell death is not due to the reduction in nutrient supplements but the natural path of the cellular cycle.

279. Fusion of a myeloma cell with a normal antibody-producing cell from a rat or mouse spleen yields a hybrid that proliferates into a clone called a hybridoma. Like myeloma cells, hybridoma cells are immortal. Each hybridoma produces the monoclonal antibody encoded by its B-lymphocyte partner. For this reason, researchers looked to culture techniques in order to obtain usable quantities of monoclonal antibody. Because primary cultures of normal B lymphocytes do not grow indefinitely, such cultures have limited usefulness for production of monoclonal antibody.

280. Transplant of tissue to the same person-Autograft

An allograft is a transplant of an organ or tissue between two genetically non-identical members of the same species

A transplant of organs or tissue from one species to another-xenograft

A subset of allografts in which organs or tissues are transplanted from a donor to a genetically identical recipient (such as an identical twin)-Isograft

285. Themonoclonal antibody known as OKT3 is saving organ transplants threatened with rejection and preventing bonemarrow transplants from setting off graft-versus-host disease (immune system series)

289. When human fibroblasts were cultured on a surface coated with collagen at 5.8×10^{-3} mg/cm², cell attachment and subsequent cell growth were both enhanced compared to the culture on an uncoated surface.

290. Anoikis (detachment induced cell death or suspension induced apoptosis) is the name given to a special form of cell death by apoptosis. It was observed initially after disruption of the interactions between normal epithelial cells an extracellular matrix.

292. Useful parameters include, for example, breeding value (A), which is the expected performance of offspring, and heritability (h2 = ¼ VA/VP, the ratio of additive genetic variance or variance of breeding value VA to the overall or phenotypic variance VP, but widely misunderstood). In view of the assumed complexity of the underlying gene action, involving many loci with unknown effects and interactions, much quantitative genetic analysis has, unashamedly, been at a level of the 'black box'.

293. In conventional breeding strategies, molecular markers have several short-range or immediate applications (viz. parentage determination, genetic distance estimation, determination of twin zygosity and freemartinism, and sexing of pre-implantation embryos and identification of disease carrier)

309. Due to the lack of a vascular system in the organ culture, the nutrient supply and gas exchange of the cells become limited.

310. The procedure for organ cultures has the following steps :

(*a*) The organ tissue is collected after the dissection.

(*b*) The size of the tissue is reduced to less than 1mm in thickness.

(*c*) The tissue is placed on a gas medium interface support.

(*d*) Incubation in a CO_2 incubator.

(*e*) M199 or CMRL 1066 medium is used and changed frequently.

315.

List-I	List-II
P. ECORI	3. Escherichia Coli RyB
Q. Hind II	4. Haemophilus influenzae
R. EcBII	1. E-Coli R 245
S. BgCII	2. Bacillus globigii

316.

List-I	List-II
P. Type I	2. EcoK
Q. Type II	3. HindII
R. Type III	1. EcoP1

317. Removal of the protruding ends by digestion with S1 nuclease; this enzyme digests both 3′ and 5′ protruding ends.

318.

A	B
P. BamHI	2. G/G AT CC CCTA G/G
Q. EcoRI	1. G/AA TTC C TT AA/G
R. HindIII	4. A/A GCTT TTCG A/A
S. HindII7ujm	3. G/G CC CC G/G

319. In L γ–DNA technology. Reverse transcriptase is used for producing CDNA copies of m – RNA for creation of C.DNA libraries

321. DNA ligase is termed as molecular glue

322. Short double stranded DNA molecule to convert blunt end to sticky ends is a linker

323. The sequence recognized by restriction enzymes are palindromic sequences

326. The degradative plasmids are also called dissimilation plasmids.

331. Nucleoside Enzyme Nucleotide hypoxanthine hypoxanthine/guanine phosphoribosyl transferase (HGPRT) IMP; guanine hypoxanthine/guanine phosphoribosyl transferase (HGPRT); GMP adenine adenine phosphoribosyltransferase (APRT) AMP.

332. I. Pyrosequencing uses luciferase to generate light for detection of the individual nucleotides added to the nascent DNA.

II. DNA nanoball sequencing is a type of high throughput sequencing technology used to determine the entire genomic sequence of an organism.

III. In dye-terminator sequencing, each of the four dideoxynucleotide chain terminators is labelled with fluorescent dyes, each of which emit light at different wavelengths.

333. Ct is defined as the fractional PCR cycle number at which the reporter fluorescence is greater than the threshold. The Ct is a basic principle of real time PCR and is an essential component in producing accurate and reproducible data.

339. I. Negatively charged amino acids contributing to a high isoelectric point.

II. Positively charged amino acids contributing to a low isoelectric point.

III. In 2-D SDS-PAGE, polypeptides are separated according to isoelectric point according to their molecular weight.

340. In order to make the proteins accessible to antibody detection, they are moved from within the gel onto a membrane made of nitrocellulose or polyvinylidene difluoride (PVDF).

341. The protein transfer on a suitable membrane (blotting) with protein-binding capacity (nitrocellulose, PVDF) is highly dependent on the biochemical properties of the target protein.

342. I. Genetic fingerprints are closely related between parents and child

II. The DNA fingerprinting process was developed in 1985 by the British scientist Alec Jeffreys

III. RFLP profiling takes weeks to complete

343. A mice in which specific genes have been replaced, by their disrupted counter parts through a process of homologues recombination is knockout mice.

344. Hybridization is the pairing of complementary RNA and DNA strands to give n RNA – DNA hybrid.

346. In situ hybridization is a technique used to identify a bacteria containing recombinant gene.

347. In situ hybridization is also called as Simple hybridization and Colony hybridization.

350. I. RAPD markers are dominant

II. PCR is an enzymatic reaction

III. Mismatches between the primer and the template may result in the total absence of PCR product.

351. I. RFLPs can be used determine the disease status of an individual.

II. RFLP loci measured in centimorgans.

III. RFLPs determine if a person were homozygous, heterozygous to disease.

352. I. PCR site directed mutagenesis generates a nicked, circular DNA which can undergo repair by endogenous bacterial machinery.

II. A clear disadvantage of PCR as a DNA cloning method has been the size range of the DNA sequences that can be cloned.

III. The amount of PCR product obtained in a single reaction is also much more limited than the amount that can be obtained using cell-based cloning where scale-up of the volumes of cell cultures is possible.

353. The process of the injection of DNA's directly into the cells using a micropipette is called Micro injection.

354. Cloning an entire genome in the form of library of random genomic clones is often called as Shot gun experiment.

355. Direct DNA uptake by protoplast can be stimulated by chemicals like PEG.

357. The process of introduction of a foreign DNA into a bacterial cell through a virus is called Transduction.

358. Polyethylene glycol (PEG) is a polyether compound with many applications from industrial manufacturing to medicine. It has also been known as polyethylene oxide (PEO) or polyoxyethylene (POE), depending on its molecular weight, and under the trade name Carbowax.

360. Electroporation is a dynamic phenomenon that depends on the local transmembrane voltage at each point on the cell membrane.

361. Solutions with high salt concentrations might cause an electrical discharge (known as arcing).

363. The process of infeding the divot plants with tumefaciens for gene transfer is called Agro infection.

365. I. SDSA pathway : Homologous recombination via the SDSA pathway occurs in cells that divide through mitosis.

II. The double Holliday junctions are then converted into recombination products by nicking endonucleases

III. A displacement loop (D-loop) is formed during strand invasion between the invading 3' overhang strand and the homologous chromosome.

367. The ability of bacterial cells to uptake (intake) DNAs from the medium is said to be Competence.

370. The variation may be spontaneous, i.e. already present in the original explant material, or that induced during the process of tissue culture initiation and subculture.

373. Dorenberg and knorr (1992) showed that the enzymatic permeabilization in Morinda citrifolia culture could be used which maintains the cell viability also.

374-375 :

Colchicine is one of the most effective spindle fiber inhibitors, which arrests the cells at metaphase stage of cell division.

376. Synchronization is achieved by temporarily blocking the progression of events in the cell cycle by addition of DNA synthesis inhibitors and accumulating cells in a specific stage.

Generally, 5 fluorodeoxyuridine (FudR), excess thymidine (TdR) & hydroxyl urea are used which accumulates cells at G_1/S interface stage. In this method, the cell synchrony is limited to only cell cycle.

377. The various methods employed to bring this synchronization have been listed.

(*i*) Cold treatment – Temp shocks at 4°C for few days can be used.

(*ii*) Starvation method

(*iii*) Use of inhibitors

(*iv*) Colchicines method

378. The cell cycle can be divided into interphase & mitosis (M) interphase is divided into a DNA synthesis phase (S) & preceded by a pre-synaptic phase (G_1) & followed by a post – synaptic phase (G_2).

379-280 :

Sucrose (Disaccharide) is frequently used as a carbon & energy source & leads to optimal growth rate of plant cell cultures. Both the nature & concentration of the sugar generally affects the yield of secondary product.

382. In this the growth phase is supported by a 2-stage approach the growth phase is supported on MS medium with 10^{-5} M NAA & 10^{-6} M KIN, while the production phase is optimal in MS with reduced levels of growth regulators & by omitting cause in hydrolysate from the growth medium.

384. The most successes in development of transgenic plants in monocots have been due to vectorless system. However, recently Agrobacterium mediated transformation has been reported in monocot species including most important food crops Viz. rice, maize & wheat.

385. Approximately 40 different genes conferring insect resistance have been incorporated into crops. Genes conferring insect resistance to plants have been obtained from microorganisms.

Bt gene from Bacillus thuringiensis.

ipt gene from Agrobacterium tumefaciens. Etc

386. Resistance genes of animal origin are serine proteinase inhibitors from mammals & tobacco hornworm (Manduca sexta).

387. Bt is an entomocidal bacterium that produces an insect control protein. Bt genes code for Bt toxin. Both the full lengthy truncated forms of Bt 8 – endotoxins have been introduced in plants, conferring demonstrable resistance to tobacco pests (M. Sextra), tomato pests (Heliothis virescens & Helicoverpa zea) & potato pests (phthorimaea operculella).

388. α – Amylase inhibitor. Genes for 3 α – Amylase inhibitor have been expressed in tobacco but the main emphasis has been on transforming the gene of α – Amylase inhibitor (αAl – PI) isolated from adzuki beans. This α – amylase inhibitor protein blocks the larval feeding in the midnight. The larvae secrete a gut enzyme called α – amylase that digests the starch.

390. Satellite RNAS are a class of small, single stranded RNA molecules that are dependent upon a helper virus for replication & virion packaging to cause infection elsewhere. Therefore satellite RNA depends on virus for it's replication & transmission, even though it is unrelated to viral genome.

391. Interferons induce the synthesis of additional proteins that directly lead to inhibition of virus multiplication. One of these proteins is 2′ → 5′ diego adenylate synthase. This enzyme is activated by dsRNA, the replication intermediate of RNA viruses.

392. Pathogenesis related proteins : These are low mwt proteins, which accumulate the significant levels in infected plant tissues.

393. Glucanase gene when preset with chitinase showed fungal resistance in tobacco, tomato, carrot etc.

395. The bacterial halo blight pathogen of bean, pseudomonas phaseolicola, produces a tripeptide toxin, phaseolotoxin, which causes the chlorotic halos. Phaeolotoxin inhibits the enzyme orinithine transcarbamylase.

397. HM1 gene from maize, which confers resistance to cochliobolus carbonum, has been closed by transposons tagging. This gene encodes NADPH dependent HC-toxin reductase that inactivates the fungal HC toxin.

399. Almost all the abiotic stresses such as drought, low temperature, salinity & alkalinity adversely influence growth & induce senescence, leading to cell death/reduced crop yield.

400. γ – pyrroline – 5 – carboxylate synthetase (P5CS) is the limiting enzyme in the synthesis of proline. Transgenic tobacco plants were generated P5CS enzyme & they showed enhanced expression of this enzyme & 10 – 18 fold more proline.

401. A major effort has been devoted in several laboratories to engineer herbicide resistant plants 3 approaches :

(*i*) Over production of a herbicide sensitive biochemical target

(*ii*) Structural alteration of a biochemical target resulting in reduced herbicide affinity.

(*iii*) Detoxification – degradation of the herbicide before it reaches the biochemical target inside the plant cell.

404. Improved storage / long shelf life of tomato that is suitable for processing can be developed by 2 approaches using.

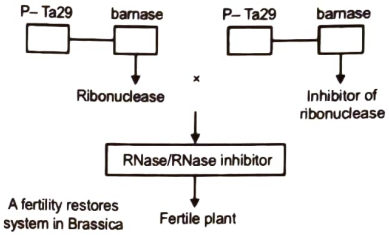
(*i*) Antisense RNA technology

(*ii*) Using gene for 1 – aminocyclopropane – 1 – carboxylic acid (ACC) deaminase, which degrades ACC to ethylene.

407. Anthocyanins are the major flower pigments in higher plants. A major target has been the development of blue pigmented flowers in species in which these colors do not naturally occur, such as rose/carnation.

Delphinidin is the anthocyanin that normally leads to blue pigmentation.

410. The female components of these plants maintain fertility. Transgenics with male sterility & fertility restoration are produced in Brassica napus.



412. The technology that terminates the viability/ fertility of seeds after a given time is known as terminator technology & the gene involved is popularly known as terminator gene. The terminator gene technology has generated for a lot of attention due to a patent No. 5,723,765 on. Control of plant Gene expression which was granted on march 3, 1998 by USA.

413. Repressible gene : A 3rd gene produces a protein called repressor protein which represses the promoter of the recombinase gene in the second gene construct. The repressor protein itself becomes inactive when it binds to a specific chemical. Ex : – Tetracycline

414-415 :

Compound	origin of gene
Application	Plant species used
Medium chain	California bay tree
Food, detergent	oilseed rape.
Fatty acids	
industrial	
Proteins	
α – trichasantin	Chinese medical Plant
Inhibition of HIV	replication
Nicotiana	benthamiana

416. Sucrose, diaccharides of linked glucose & fructose is the most preferred sweetener. Over expression of SPS holds promise for achieving the goal of increased sucrose synthesis in plants.

NUMERICAL TYPE QUESTIONS

1. The Ti plasmid is lost when the Agro bacterium is grown at 28°C. Growth above 28°C results in loss of mega plasmid that determines the virulence.

2. The range of DNA fragment which can be cloned in YAC is around 100- 3000 Kb. Therefore they are extensively used in mapping the large genomes, like the Human Genome Project.
3. To each reaction only one of the four dideoxynucleotides are added (ddATP, ddGTP, ddCTP, or ddTTP) which are the chain-terminating nucleotides, lacking a 3'-OH group required for the formation of a phosphodiester bond between two nucleotides, thus terminating DNA strand extension and resulting in various DNA fragments of varying length.
4. Whether a DNA probe is labeled by nick-translation or PCR, an important rule of the FISH protocol is that the labeled probe DNA fragments are cut to an average of 200- 500 bp before hybridization so that the background is not visible on the slide.
5. PCR leads to duplication of the original DNA molecule. So, if 10 double stranded DNA molecules are amplified by PCR, then after first cycle 20 double stranded DNA molecules can be generated. Thus after 5 cycles, 320 double stranded DNA molecules can be generated.
6. Plating efficiency ("PE") is a measure of the number of colonies originating from single cells. It can be calculated as:

   PE(%) = (Colonies formed / Cells Inoculated) × 100

   We have, colonies formed = 40

   Inoculated cell count = 200 cell/ml

   Therefore, PE = (40/200) × 100

   = 20%
7. The optimum temperature for growth of animal cells in culture is 37°C as at this temperature cells grow exponentially. The temperature 37°C is also close to the body heat of the animals in vivo and thus same temperature is required to be maintained.

   High humidity is required for animal cells to avoid evaporation of the media. Thus the humidity for animal cells is 95%.
9. In $cacl_2$ treatment, the DNA is taken up by the cells when they are subjected to heat shock for min at 42°C.
10. Herbicide tolerant transgenic crops covered 63% of the total 114.3 million hectares of the global are under transgenic transgenics in 2007. It was mainly due to herbicide tolerant soybean (51%), maize (6%), canola (5%), and cotton (1%) (Fames, 2007).

EXERCISE – II

MCQ TYPE QUESTIONS

1. Gene therapy is the therapeutic delivery of nucleic acid polymers into a patient's cells as a drug to treat disease. Avidin is a protein derived from both avians and amphibians that shows considerable affinity for biotin, a co-factor that plays a role in multiple eukaryotic biological processes.
2. The culture viability is merely a ratio of viable cells in the culture to total cells in the culture. For the purpose, the hemocytometer will only be used to obtain a viability count, not a total cell count (cells/mL). The viability will be determined by the trypan blue assay , and it can be used with the total cell count (obtained from a Coulter counter) to determine a total count of viable cells, as : Number of viable cells = (% viability)*(Total number of cells)
3. Secondary treatment is the portion of a sewage treatment sequence removing dissolved and colloidal compounds measured as BOD. Secondary treatment is traditionally applied to the liquid portion of sewage after primary treatment has removed settleable solids and floating material. Secondary treatment is typically performed by indigenous, aquatic microorganisms in a managed aerobic habitat. Bacteria and protozoa consume biodegradable soluble organic contaminants (e.g. sugars, fats, and organic short-chain carbon molecules from human waste, food waste, soaps and detergent) while reproducing to form cells of biological solids.
4. The T-DNA transfer process of Agrobacterium tumefaciens is activated by the induction of the expression of the Ti plasmid virulence loci by plant signal molecules such as acetosyringone. The overdrive proteins from acetosyringone-induced Agrobacterium cells interact with T-DNA border and overdrive sequences. The vir gene products act in trans to mobilize the T-DNA element from the bacterial Ti plasmid. The T-DNA is bounded by 25-base pair direct repeat sequences, which are the only sequences on the element essential for transfer. Thus, specific reactions must occur at the border sites to generate a transferable T-DNA copy.
5. Secondary Metabolites including Morphine, Pyrethrins, Scopolamine, Vincristine are extracted from Papaver somniferum, Tagetes erecta, Datura stramonium, Catharanthus roses respectively.
6. Certain genetic elements used in transgenic research includes Ubiquitin I promoter, Nos Transcriptional terminator, bar Selection Marker Gene, gus Reporter gene. They are obtained from Zea mays, Agrobacterium tumifaciens, Streptomyces hygroscopicus, Escherichia coli respectively.

7. Bicarbonate is often included to act as a buffer system in a conduction with the CO_2 environment (5-10%) in which the cell should be cultured. This allows the cultures to be maintained at the normally optimum pH range of 6.9 - 7.4. The CO_2 is provided by a controlled atmosphere inside an incubator or alternatively in a sealed flask.
8. CMV promoter is plant expression vector.
9. About $\frac{ddNTP_s}{dNTP_s} = \frac{1}{300}$
10. NH_4^+ product is only in anaerobic processes like waste water treatment.
11. 2nd generation crop are the value added crops i.e; increased nutritional value. Instead of insect resistance (Bt) and herbicide tolerance (HT).
12. Respective primers will be anneal to 3′ ends of the two template strands; keeping antiparallel nature of DNA duplex and complimentarity in the mind..
13. Hae III : GG | CC
    CC | GG

    BamHI : G | GATC
    CCTAG | G
14. The number of bases present in the restriction site of the restriction enzyme = 6

    We know, there are four different types of nitrogenous bases in a DNA molecule.

    So, the statistical frequency of occurrence of the restriction site would be = 4^6 = 4096
15. Serum is commonly used as a supplement to cell culture media. It supports cell growth and product formation by providing a broad spectrum of macromolecules, nutrients and vitamins as well as certain growth factors and hormones. However, serum must be filter sterilized with 0.45? filter paper before using in the media to grow mammalian cells because it contains Mycoplasma and microorganisms as contaminants.
16. The bands which are labeled as A were uncut and separated at three different regions on the agarose gel after digestion with EcoRI. It shows that the plasmid A separates is having concatemers. Plasmid B labeled as B is nicked while Plasmid C which is labeled as C has a very thick band which suggests that Plasmid C is supercoiled. Band D was the only plasmid which was the digested product of Eco RI are linear because EcoRI has already digested the plasmid.
17. Belladona plant produces a type of secondary metabolite called atropine which helps in the ocular examination procedure. Foxglove produces a secondary metabolite called digitalin that helps in the treatment of cardiac diseases. Pacific yew is another plant with medicinal use which produces taxol that is extensively used in cancer research and treatment purposes. Eucalyptus is used in the production of menthol that helps in the curing of cough and cold.
18. There are several species of mustard. Black mustard (Brassica nigra) produces small seeds that become black at maturity. They have an extremely rich and pungent flavor, stronger than yellow mustard. White mustard (Sinapis alba) produces large yellow-colored seeds with a bitter taste that is not as pungent as other varieties. Indian mustard or mustard greens (Brassica juncea) is a green vegetable whose very tasty leaves are used in the same way as spinach. The chemicals that are responsible for pungency are stored in inactive conjugate form in compartments within the seeds and are protected from enzymes. But when seeds are crushed, the pungent chemicals come in contact with the enzymes and are converted into active chemicals which give the pungent smell.
19. The length of the unique stretch of DNA sequence that can be found only once in 3 billion base pair long genome is a 16 bases long DNA sequence from which 16s rRNA is transcribed.
20. Several types of DNA library resources were sponsored by the DOE before and during the Human Genome Program (HGP). These included both prokaryotic and eukaryotic vector systems, and clone libraries representing single chromosomes. Bacterial Artificial Chromosomes (BACs) became the most broadly used resource for several reasons. The large size was a good match for capabilities of high throughput sequencing centres. A bacterial artificial chromosome (BAC) is an engineered DNA molecule used to clone DNA sequences in bacterial cells (for example, *E. coli*). BACs are often used in connection with DNA sequencing. Segments of an organism's DNA, ranging from 100,000 to about 300,000 base pairs, can be inserted into BACs. The BACs, with their inserted DNA, are then taken up by bacterial cells. As the bacterial cells grow and divide, they amplify the BAC DNA, which can then be isolated and used in sequencing DNA.
21. Natural genetic transformation is believed to be the essential mechanism for the attainment of genetic plasticity in many species of bacteria. During bacterial evolution, the ability of Bacteria and Archaea to adapt to new environments most often results from the acquisition of new genes through horizontal transfer rather than by the alteration of gene functions through numerous point mutations. Horizontal gene transfer is

defined to be the movement of genetic material between bacteria other than by descent in which information travels through the generations as the cell divides. It is most often thought of as a sexual process that requires a mechanism for the mobilization of chromosomal DNA among bacterial cells. However, because they are unable to reproduce sexually, bacterial species have acquired several mechanisms by which to exchange genetic materials. It can happen through transformation, conjugation and transduction. In bacteria, it mainly occurs by transformation and conjugation.

22. *Agrobacterium tumefaciens* is a gram-negative rod-shaped bacterium. It infects plants usually through an open wound. Once it enters its plant host, it injects a section of its DNA called the T-DNA which is derived from its Ti (tumor inducing) plasmid into its host. The T-DNA is then integrated into the plant's genome, and has two effects on the plant host. The T-DNA first directs the plant cells to make auxins and cytokinins, which causes the cells to become irregularly shaped and form a visible tumour called a gall. The T-DNA then directs the plant cell to start making opines (usually nopaline or agropine) which *A. tumefaciens* use as an energy source.

23. Glyphosate kills plants by interfering with the synthesis of the aromatic amino acids phenylalanine, tyrosine and tryptophan. It does this by inhibiting the enzyme 5-enolpyruvyl-shikimate-3-phosphate synthase (EPSPS), which catalyzes the reaction of shikimate-3-phosphate (S3P) and phosphoenolpyruvate to form 5-enolpyruvyl-shikimate-3-phosphate (ESP). Bromoxynil is a nitrile herbicide and one of its common degradation products (3, 5-dibromo-4-hydroxybenzoic acid) have been shown to undergo metabolic reductive dehalogenation by the microorganism. Sulfonylureas is used as Chlorimuron (Classic) for soybeans. Their site of action is Acetolactate synthase (ALS) enzyme. Dalapon is a synthetic herbicide and plant growth regulator. It is responsible for inhibition of lipid synthesis by acting on dehalogenase enzyme.

24. The given plasmid sequence is 5000bp long. When it is digested with Hind III, it produces 2 fragments of 1200bp and 3800bp. This means that Hind III cuts the plasmid in two different sites. So Hind III has 2 sites in plasmid. When digested with Bam HI, it produces one fragment of 5000bp. This means Bam HI has only 1 recognition site on the plasmid sequence. Eco RI cuts the fragment in two equal halves of 2500bp each. Hence Eco RI also has 2 sites of recognition on the plasmid.

25. Given, size of the genome of *E. coli*

$= 4 \times 10^3 kb = 4 \times 10^6 bp$

Size of the DNA fragment = 5000 bp

Therefore, minimum number of independent recombinant clones required to represent the DNA fragment in a genomic library

$= (4 \times 10^6)/5000 = 8 \times 10^2$

26. Given, Probability of finding the fragment in the library = 95% = 0.95

27. Both pUC and the bacterial own genome produce a faulty gene product of Lac Z gene. The *lac Z* fragment, whose synthesis can be induced by IPTG, is capable of intra-allelic complementation with a defective form of β-galactosidase enzyme encoded by host chromosome (mutation lacZDMI5)

28. DMSO is used in cell freezing media to protect cells from ice crystal induced mechanical injury.

29. The strength of ChIP assays is their ability to capture a snapshot of specific protein: DNA interactions occurring in a system and to quantitate the interactions using quantitative polymerase chain reaction (qPCR).

30. Fusion of freely isolated protoplasts from different sources with the help of fusion inducing chemical agents is known as induced fusion. It includes use of high pH or Ca++ treatment, Use of PEG (Polyethylene glycol) which induces protoplast aggregation and subsequent fusion and the use of inactivated Sendai virus for fusion which occur in four temperature-dependent stages.

Hence Colchicine is not a protoplast fusion inducing agent. Colchicine is used to prevent gout attacks in adults, and to relieve the pain of gout attacks when they occur. Colchicine is also used to treat familial Mediterranean fever (FMF; an inborn condition that causes episodes of fever, pain, and swelling of the stomach area, lungs, and joints) in adults and children 4 years of age and older.

31. Given, weight of single base pair of DNA

$= 1.1 * 10^{-21}$ grains

Length of plasmid vector= 2750bp

Purified DNA= 1ug

Thus, 0.55 picomoles of a plasmid vector of length 2750bp are contained in 1ug of purified DNA.

32. Elicitors are compounds, which activate chemical defense in plants. Various biosynthetic pathways are activated in treated plants depending on the compound used. They are compounds that when introduced into a living organism signals the activation or synthesis of another compound. Thus (R) is true but (A) is false.

33. Somatic embryogenesis can be initiated either directly without going through the callus phase from predetermined embryonic cells or indirectly through callus proliferation and differentiation into embryonic cells within the callus tissue

34. Embryonic stem cells possess the capacity to divide for long periods and retain their ability to make all cell types within the organism. These are termed pluripotent stem cells. They are derived from a fertilized embryo. Cells are obtained from an embryo in the blastula phase, when they are still only a few days old. Because they have only begun to differentiate, these cells have the capability of developing into any cell in the human body which makes them potentially important in medicine.

35. The yeast artificial chromosome (YAC) is an artificially constructed system that can undergo replication. The design of a YAC allows extremely large segments of genetic materials to be inserted. Subsequent round of replication produce many copies of the inserted sequence, is a genetic procedure known as cloning. The reason the cloning vector is called the YAC has to do with the structure of vector. YACs are shuttle vector that can be amplified in bacteria and employed for the cloning and the manipulation of large DNA inserts (up to 3mb pairs) in the yeast-Saccharomyces cerevisiae. Large DNA (>100 Kb) is ligated between two arms.

36. Tissue plasminogen activator (t-PA) is a naturally occurring protein that catalyze the conversion of the inactive proenzyme plasminogen into active serine protease plasmin. It is one of the two major endogenous mammalian enzymes with this property and is found in blood and tissues from various organs. Although it exerts a similar enzymatic action, it is structurally different from the other major endogenous enzymes, urokinase. t-PA can be used to treat some people who are having a stroke caused by a blood clot. It plays an important role in cell migration and tissue remodeling.

37. Synthetic seeds or artificial seeds are seed-like structures derived from somatic embryos under in vitro condition after encapsulation by hydrogen. These are tissue culture derived somatic embryos encased in a protective coating. Hydrated synthetic seeds are produced in those plant species where the somatic embryos are recalcitrant and sensitive to desiccation. Hydrated synthetic seeds are produced by encapsulating the somatic embryos in hydrogel capsules. The somatic embryos are mixed with sodium alginate (2 %) and the suspension is dropped into the calcium salts solution (200mM). The principle involved is when sodium alginate dropped into the calcium salt solutions it form round firm beads due to the ion exchange between Na^+ in sodium alginate and Ca^{2+} in calcium slat solutions and sodium alginate form calcium alginate in 20-30 minutes. The calcium salt used can be calcium nitrate.

38. Shoot organogenesis is one of the in vitro plant regeneration pathway. It has been widely employed in plant biotechnology for invitro micropropagation and genetic transformation, as well as in study of plant development. Organogenesis continues until the definitive characteristics of the organs or plant parts are achieved.

39. Hairy roots are produced by infecting sterile plant with a natural genetic engineer, Agrobacterim rhizogenes. Genes for auxin synthesis and sensitivity are engineered into plant cells leading to gravity- insensitive mass root production. It is very useful for products produced in roots. Aggregation and shear sensitivity are a major problem for scale up.

40. Restriction endonucleases that recognize the same sequence are isoschizomers. The first example discovered is called a prototype and all subsequent enzymes that recognize the same sequence are isoschizomers of the prototype. For example, Sph I (CGTAC/G) and Bbu I (CGTAC/G) are isoschizomers of each other. Isoschizomers are pairs of restriction enzymes specific to the same recognition sequence.

41. Phylogenetics deals with evolutionary relatedness of the species. It calculates the distances amongst the species from its origin and can track down any ancestral relationship.

42. Yeast two hybrid screening is a molecular biology technique used to discover protein-protein interaction and protein-DNA interaction by testing for physical interactions. Mass spectrometry (MS) is an analytical technique that measures the mass-to-charge ratio of charged particles.

43. Rapid amplification of cDNA ends (RACE) polymerase chain reactions (PCR) are used for independent high-throughput verification of transcriptional starting sites (TSSs) determined by genome-wide assays. 52 -RACE PCR is a well-established and widely used method to specifically amplify the 52 end of a transcript, facilitating mapping of the TSS and the approximate location of promoter elements. Conventionally, this mapping is done by cloning the 52 -RACE PCR product into a bacterial vector and sequencing a few clones by classic electrophoresis-based Sanger sequencing. Now-a-days, even 3' end transcripts are being mapped

using 3'-RACE PCR. Baculoviruses are a very diverse group of viruses with double-stranded, circular, supercoiled genomes, with sizes varying from about 80 to over 180 kb which encode between 90 and 180 genes. Baculoviruses have evolved to initiate infection in the insect midgut. Baculovirus-insect cell expression systems have the capacity to produce many recombinant proteins at high levels and they also provide significant eukaryotic protein processing capabilities.

44. Hybridoma technology is a technology of forming hybrid cell lines (called hybridomas) by fusing a specific antibody-producing B cell with a myeloma (B cell cancer) cell that is selected for its ability to grow in tissue culture and for an absence of antibody chain synthesis. The antibodies produced by the hybridoma are all of a single specificity and are therefore monoclonal antibodies Fused cells are incubated in HAT medium (hypoxanthine-aminopterin-thymidine medium) for roughly 10 to 14 days. Aminopterin blocks the pathway that allows for nucleotide synthesis. Hence, unfused myeloma cells die, as they cannot produce nucleotides by the de novo or salvage pathways because they lack HGPRT. Removal of the unfused myeloma cells is necessary because they have the potential to outgrow other cells, especially weakly established hybridomas. Unfused B cells die as they have a short life span. In this way, only the B cell-myeloma hybrids survive, since the HGPRT gene coming from the B cells is functional.

45. Let N_0 be the initial number of molecules present in the aliquot and let N be the end product of PCR amplification. Let n be the number of cycles performed by the PCR.

We, know $N = N_0 \times 2^n$ for n number of cycles since in each cycle the number of amplicons double the number of parent strands in an exponential manner.

Here, $N_0 = 100$ for a 200 µL solution (given),

$n = 10$

Therefore, $N = 100 \times 2^{10}$

$= 100 \times 1024$

$= 1.024 \times 10^5$

Hence, the required amplicons generated after 10 cycles of PCR run $= 1.024 \times 10^5$

46. From previous question, we have

Number of amplicons in 200 µL of the solution after PCR run $= 1.024 \times 10^5$

Therefore, number of amplicons in 0.1 µL solution $= 1.024 \times 10^5 \times 0.1/200 = 51.2$

Hence, the required number of amplicons present in 0.1 µL of the solution = 51.2

47. E. coli is the bacteria which are known to be naturally competent for transformation of DNA. It offers many advantageous such as:

Genetic simplicity: E. coli cells only have about 4,400 genes whereas the human genome project has determined that humans contain approximately 30,000 genes. E. coli, live their entire lifetime in a haploid state.

Growth rate: E. coli grows rapidly at a rate of one generation per twenty minutes under typical growth conditions.

Safety: E. coli is naturally found in the intestinal tracts of humans and animals where it helps provide nutrients (vitamins K and B12) to its host.

Conjugation and the genome sequence: The E. coli genome was the first to be completely sequenced. Genetic mapping in E. coli was made possible by the discovery of conjugation.

Ability to host foreign DNA: E. coli is readily transformed with plasmids and other vectors, easily undergoes transduction, and preparation of competent cells (cells that will take up foreign DNA) is not complicated.

48. Vancomycin is the antibiotic resistance marker that cannot be used in a cloning vector in gram negative bacteria. Vancomycin acts by inhibiting proper cell wall synthesis in Gram-positive bacteria. Vancomycin treats only gram positive infections (commonly a skin infection, an example of a Gram positive infection). It has little/no use against a gram negative bacteria because vancomycin acts by binding to parts of the cell wall that are present in a gram positive bacteria but not in a gram negative bacteria.

49. TDT do not require a template as it adds N-nucleotides to the variable exons during antibody gene recombination. It functions by catalysing the addition of nucleotides to the 3' terminus using 3' – overhang as the substrate. Also it can act independently on blunt or recessed 3' ends. It is expressed in immature, pre- B , pre –T lymphoid cells and uses cobalt as cofactor, also Mg/Mn^{2+} presence in-vitro. It is used in adding nucleotides labelled with radioactive isotopes in TUNEL assay (terminal deoxynucleotidyl transferase dUTP Nick End Labelling).

50. In somatic cell gene transfer, the therapeutic genes are transferred into the somatic cell or body, of a patient. It involves using a vector such as a virus to deliver therapeutic gene to the appropriate target cells. This technique is currently the basis for cloning animals (creating transgenic animals) and is used in vitro.

51. EST is a unique stretch of DNA within a coding region of a gene that is useful for identifying full-length genes and serves as a landmark for mapping. An EST is a sequence tagged site (STS) derived from cDNA library.

EST's sequence can be submiited or accessed from 3 universal databases- GenBank, DDBJ, EMBL. NCBI also has a database exclusively for EST generation are individual from a cDNA library.

52. Somatic embryogenesis is a process where a plant or embryo is derived from a single somatic cell or group of somatic cells. Somatic embryos are formed from plant cells that are not normally involved in the development of embryos, i.e. ordinary plant tissue. In plants, somatic embryogenesis comprises of embryo initiation followed by embryo production.

Embryo initiation occurs on a medium rich in auxin, which induces differentiation of localized meristematic cells. The auxin typically used is 2,4-Dichlorophenoxyacetic acid (2,4-D). thus embryo initiation is dependent on 2,4-D. While these are transferred to a medium containing low or no auxin, then embryo production does not require high concentration of 2, 4-D.

53. Fibronectin is a glycoprotein that can act as general cell adhesion molecule by anchoring cells to collagen or proteoglycan substrates. Insulin is a hormone, produced by beta cells of the pancreas, and is central to regulating carbohydrate and fat metabolism in the body Insulin is a hormone that has profound effects on metabolism. alpha-2-Macroglobulin,is a large plasma protein found in the blood. It inhibits the thrombin action at the time of couagulation by blocking trypsin enzyme. Transferrins is a iron binding glycoprotein.

54. Patatin class- I promoter is a tissue specific potato promoter which expresses the potato gene mainly in tuber.

Phaseollin is a promoter for cotyledon tissue.

Genes encoding High Molecular Weight (HMW) glutenin, a wheat seed storage protein, is expressed in developing endosperm.

Alpha amylase is a promoter for aleurone tissue.

55. T_4DNA Ligase catalyzes the formation of a phosphodiester bond between juxtaposed 5'-phosphate and 3'-hydroxyl termini in duplex DNA or RNA. The enzyme repairs single-strand nicks in duplex DNA, RNA, or DNA/RNA hybrids. It also joins DNA fragments with either cohesive or blunt termini, but has no activity on single-stranded nucleic acids. It can ligate blunt-ended DNA with much greater efficiency than E. coli DNA ligase. Unlike E. coli DNA ligase, T_4DNA ligase cannot utilize NAD and it has an absolute requirement for ATP as a cofactor.

The rates of blunt-end and cohesive-end ligation of DNA by T_4DNA ligase are increased by orders of magnitude in the presence of high concentrations of a variety of nonspecific polymers such as polyethylene glycol, Ficoll, bovine plasma albumin, or glycogen. Blunt-end ligation of small self-complementary oligodeoxyribonucleotides is also stimulated.

56. The cloning sites of the vectors are following: Lambda phage has size of 5-24kb, BACs have size of 5300kb, PACS have size of 100-300kb and Cosmids have size of 35-45kb.

57. Alpha, Beta, and Gamma are the three subunits of the G-protein. The active form is the Alpha-GTP complex. The intrinsic GTPase activity converts the bound GTP to GDP in the a - subunit, returning it to its original conformation. This results in the a - subunit diffusing from the adenylate cyclase and reassociating with the b & g subunits. Adenylate cyclase catalyses the conversion of ATP to cAMP which is then used to activate another, initially inactive, enzyme. The mutant G alpha protein with increased GTPase activity shows decreased signaling because of the group of proteins present which are specific to G alpha subunit called the regulator of G protein signaling.

58. Number of clones which have 99% probability

$= \log_{10}(1-0.99)/\log_{10}(1-5/4 * 10^5)$

$= \log_{10}(0.01)/\log_{10}(0.9999875)$

$= -4.605/-1.25 * 10^{-5}$

$= 3.5 * 10^{-5}$

59. Real time quantitative PCR allows the sensitive, specific and reproducible quantification of nucleic acids. Real time PCR allows precise quantification of specific nucleic acids in a complex mixture even if the starting amount of the material is at a very low concentration. This is accompanied by monitoring the amplification of a target sequence in real time using fluorescent technology. A common approach uses SYBR GREEN I. It binds to any double stranded DNA which could result in inaccurate data.

2-D electrophoresis is used to separate and display all gene products present. This technique separate proteins in two steps according to two independent properties: the first-dimension is isoelectric focussing (IEF), which separated proteins according to their more isoelectric points (pI); the second dimension is SDS-polyacrylamide gel electrophoresis (SDS-PAGE) which separates protein according to their molecular weights (MW). The procedure involves placing the sample

in gel with a pH gradient and applying a potential difference across it. There are two alternative methods to create pH gradient→carries ampholytes and immobilized pH gradient (IPG) gels.

Affinity chromatography is a powerful tool for the purification of specific biomolecules, including proteins. The basic principle is that a biospecific ligand is immobilized to a solid support or resin to which a solution containing the protein of interest is passed over. Ligands are often based on biological functional pairs, such as enzymes and substrates or antigens and antibodies. The specific ligand binds the protein of interest and all non-specific molecules are washed away. The protein is eluted in a specific buffer, either by pH or ionic strength shift or by competitively displacement elution. Examples of resin are→CNBr-agarose, antibody linked sepharose beads, etc.

Microarray is a multiplex lob on a chip. It is 2-D array on a solid substrate (usually a glass slide or silicon thin-film cell) that assays large amounts of biological material using high-throughput screening methods. Types of microarray include DNA microarrays (biochips are used), Protein microarrays, etc.

60. In anther culture, the anther swells and dehisces along its upper margin, lengthwise. This phenomenon helps to expose the pollen grain. Alternatively, huge amount of pollen grains can be isolated manually and can be cultured aseptically very easily. Therefore, pollen is more suitable material than egg cell for the production of haploid. The development and production of haploid plant in vitro is very important for the study of fundamental and applied aspects of genetics in the higher plants. Production of homozygous diploid by doubling the chromosome number of haploid in vitro makes a pure line in single step and such homozygous pure line is of great importance in plant breeding. Also it is easy to induce cell division in immature pollen cells in some species. By anther and pollen culture, haploids can be produced in large numbers very quickly.

61. Isoschizomers are pairs of restriction enzymes specific to the same recognition sequence. For example, Sph I (CGTAC/G) and Bbu I (CGTAC/G) are isoschizomers of each other. The first enzyme to recognize and cut a given sequence is known as the prototype, all subsequent enzymes that recognize and cut that sequence are isoschizomers. Isoschizomers are isolated from different strains of bacteria and therefore may require different reaction conditions.

62. Baculovirus expression in insect cells represents a robust method for producing recombinant glycoproteins. Baculovirus-produced proteins are currently under study as therapeutic cancer vaccines with several immunologic advantages over proteins derived from mammalian sources.

63. Cowpea Trypsin Inhibitor: The explanation for this is called co-suppression. The plant has ways of knowing that the viral coat protein should not be produced, and it has ways of eventually shutting down the protein's expression. When the virus tries to infect the plant, the production of its essential coat protein is already blocked.

64. **Statement Q:** When glyphosphate expressed in chloroplasts of transgenic plants, the insensitive enzymes produced by confer tolerance to the herbicide glyphosphate.

Statement S: Glyphosphate acts by inhibiting enolpyruvyl-shikimate-3-phosphate synthase (EPSPS), an enzyme in the pathway leading to biosynthesis of aromatic amino acids. Since transgenic plants exhibit resistance to glyphosphate, there is no inhibition of the synthesis of aromatic amino acids.

65. **P>3:** The rolA, rolB and rolC genes are plant oncogenes that are carried in plasmids of the plant pathogen Agrobacterium rhizogenes. Following agrobacterial infection, these genes are transferred into the plant genome and cause tumor formation and hairy root disease.

Q>1: Agrobacterium rhizogenes causes hairy root disease in plants. The transformed roots start producing opines which serve as carbon or nitrogen source for the infecting bacteria.

R>4: Genes in the virulence region are grouped into the operons virABCDEFG, which code for the enzymes responsible for mediating transduction of T-DNA to plant cells.

S>2: Agrobacterium rhizogenes transfers a piece of oncogenic DNA, i.e. T-DNA. The T-DNA contains onc genes, the expression of which is responsible for the tumorous character of the transformed plant cells. Three onc genes are of prime importance: cyt, aux1, aux2.

66. In cell biology, pluripotency (from the Latin plurimus, meaning very many, and potens, meaning having power) refers to a stem cell that has the potential to differentiate into any of the three germ layers: endoderm (interior stomach lining, gastrointestinal tract, the lungs), mesoderm (muscle, bone, blood, urogenital), or ectoderm (epidermal tissues and nervous system). Pluripotent stem cells can give rise to any fetal or adult cell type. However, alone they cannot develop into a fetal or adult organism because they lack the potential to contribute to extraembryonic tissue, such as the placenta.

67. • **DNA footprinting** is a method of investigating the sequence specificity of DNA-binding proteins in vitro. This technique can be used to study protein-DNA interactions both outside and within cells.

• **Yeast two-hybrid** is based on the reconstitution of a functional transcription factor (TF) when two proteins or polypeptides of interest interact.

• **VNTRs** have become essential to forensic crime investigations, via DNA fingerprinting and the CODIS database. When removed from surrounding DNA by the PCR or RFLP methods, and their size determined by gel electrophoresis or Southern blotting, they produce a pattern of bands unique to each individual. When tested with a group of independent VNTR markers, the likelihood of two unrelated individuals having the same allelic pattern is extremely improbable.

• **Serial analysis of gene expression** (SAGE) is a technique used by molecular biologists to produce a snapshot of the messenger RNA population in a sample of interest in the form of small tags that correspond to fragments of those transcripts. It has been successfully used to describe the transcriptome of other diseases and in a wide variety of organisms.

68. • **Erythropoietin (EPO)** is a hormone that prevents anemia (low blood count) by making red blood cells. Anemia causes fatigue and low energy levels. It occurs when there are not enough red blood cells to carry oxygen from the lungs to supply all the body's needs. Erythropoietin has its primary effect on red blood cells progenitors and precursors by promoting their survival through protecting these cells from apoptosis.

• **Anti-fibrin 99** is involved in animal cell separation.

• **Collagenase** enzyme helps in wound healing by assisting in blood clot formation.

• **Transferrins** are iron-binding blood plasma glycoproteins that control the level of free iron in biological fluids.

69. Since $60\% \rightarrow 692551$ cpm

So, $100\% = 1154201.67\%$ cpm

Number of moles of ^{3}H UTP

$$= \frac{1154201.67\,\text{cpm} \times \mu\text{mol}}{500 \times 2.2 \times 10^6\,\text{cpm}}$$

$$= 1050 \times 10^{-12}\,\text{mole} = 1.05\,\text{nmole}$$

There is some error in options.

70. Parthenogenesis is a form of asexual reproduction in which growth and development of embryos occur without fertilization. In plants, parthenogenesis means development of an embryo from an unfertilized egg cell, and is a component process of apomixis.

71. By clonal propogation or micropropagation thousands of plants can be produced from a single piece of plant tissue explant. They are genetically identical.

72. STATEMENT P: HAT Medium (hypoxanthine-aminopterin-thymidine medium) is a selection medium for mammalian cell culture, which relies on the combination of aminopterin, a drug that acts as a powerful folate metabolism inhibitor by inhibiting dihydrofolate reductase, with hypoxanthine and thymidine which are intermediates in DNA synthesis. Aminopterin blocks DNA de novo synthesis.

STATEMENT S: The thymidine present in the HAT medium that can be absorbed by the cells and phosphorylated by thymidine kinase (TK) into TMP. By additional phosphorylation reactions, TMP can be used to make thymidine triphosphate (TTP), one of the four nucleotide precursors that are used by DNA polymerases to create DNA. Hypoxanthine-guanine phosphoribosyltransferase (HGPRT) reacts with hypoxanthine absorbed from the medium with PRPP, liberating pyrophosphate, to produce IMP(Inosine Mono Phosphate) by a salvage pathway.

Therefore, the use of HAT medium for cell culture is a form of artificial selection for cells containing working TK and HGPRT.

73. Somatic fusion, also called protoplast fusion through plasma membrane, is a type of genetic modification in plants by which two distinct species of plants are fused together to form a new hybrid plant with the characteristics of both, a somatic hybrid. Hybrids have been produced either between the different varieties of the same species (e.g. between non-flowering potato plants and flowering potato plants) or between two different species (e.g. between wheat triticum and rye secale to produce Triticale).

74. • Dicer is an endoribonuclease in the RNase III family that cleaves double-stranded RNA (dsRNA) and pre-microRNA (miRNA) into short double-stranded RNA fragments about 20-25 base pairs long, with a two-base overhang on the 3' end. Dicer contains two RNase III domains and one PAZ domain; the distance between these two regions of the molecule is determined by the length and angle of the

connector helix and influences the length of the siRNAs it produces.

- Dicer facilitates the formation of the RNA-induced silencing complex (RISC), whose catalytic component argonaute is an endonuclease capable of degrading messenger RNA (mRNA).

75. • **Probiotics** are live bacteria that may confer a health benefit on the host. Lactic acid bacteria (LAB) and bifidobacteria are the most common types of microbes used as probiotics.

- A **xenobiotic** is a chemical which is found in an organism but which is not normally produced or expected to be present in it. The term xenobiotics is very often used in the context of pollutants such as dioxins and polychlorinated biphenyls.
- **Fructo-oligosaccharide** (FOS) and **lactulose** have been classified as prebiotics. Prebiotics are non-digestible food ingredients that stimulate the growth and/or activity of bacteria in the digestive system in ways claimed to be beneficial to health.
- **β-Lactam antibiotics** (beta-lactam antibiotics) are a broad class of antibiotics, consisting of all antibiotic agents that contains a β-lactam ring in their molecular structures.

76. In PCR, the template DNA after '*n*' cycle

$$= (\text{original DNA}) \times 2^n = 400 \times 2^{20}$$

But, this is amount in 100μl. In 0.1 μl we have,

$$= \frac{400 \times 2^{20} \times 0.1}{100} = 4.194304 \times 10^5$$

77. Cells are harvested when the cells have reached a population density which suppresses growth. For adherent cultures, we use trypsin or collagenase. Trypsin, collagenase, or pronase, usually in combination with EDTA, causes cells to detach from the growth surface, thereby aiding in harvesting of anchorage dependent animal cells from culture vessels.

78. DNA footprinting is a method of investigating the sequence specificity of DNA-binding proteins in vitro. This technique can be used to study protein-DNA interactions both outside and within cells. The regulation of transcription has been studied extensively, and yet there is still much that is not known. Transcription factors and associated proteins that bind promoters, enhancers, or silencers to drive or repress transcription are fundamental to understanding the unique regulation of individual genes within the genome. Techniques like DNA footprinting will help elucidate which proteins bind to these regions of DNA and unravel the complexities of transcriptional control.

79. Secondary metabolites often play an important role in plant defense against herbivory and other interspecies defenses. Humans use secondary metabolites as medicines, flavorings, and recreational drugs.

80. PUFAs are long-chain fatty acids containing two or more double bonds.

They provide structural & functional characteristics, and are involved in a wide range of biological components including membranes (in phospholipids). Equally important PUFAs serve as precursors for conversion into metabolites that regulate critical biological functions.Biodiesel is a non-petroleum and environmentally-friendly alternative to petroleum-based diesel fuel and not fatty acid.PUFA is fatty acid it is not used as it forms epoxide in presence of oxygen.

81. A form of asexual reproduction related to parthenogenesis is gynogenesis. Here, offspring are produced by the same mechanism as in parthenogenesis, but with the requirement that the egg merely be stimulated by the presence of sperm in order to develop. However, the sperm cell does not contribute any genetic material to the offspring. Since gynogenetic species are all female, activation of their eggs requires mating with males of a closely related species for the needed stimulus.

82.

Marker Gene	*Enzyme encoded*	*Selective agent(s)*
hpt	*Hygromycin phosphotransferase*	*Hygromycin B*
npt II	*Neomycin phosphotransferase*	*Kanamycin*
aro A	*EPSP synthase*	*Glyphosate (Roundup)*
bar	*Phosphinothricin acetyltransferase*	*Phosphinothricin (bialaphos)*

83. The various reasons attributed to the escape of the meristems by virus invasion are:

(*a*)Viruses move readily in a plant body through the vascular system which in meristems is absent,

(*b*)A high metabolite activity in the actively dividing meristematic cells does not allow virus replication and

(*c*)A high endogenous auxin level in shoot apices may inhibit virus multiplication.

84. TthPlus DNA polymerase is isolated from the Thermus thermophilus strain. TthPlus DNA polymerase is a single 92 kDa polypeptide showing a 52 -32 exonuclease activity but lacking 32 -52 exonuclease activity. It catalyzes the polymerization of nucleotides into double-stranded DNA in the presence of $MgCl_2$. Its efficiency has been shown more particularly on large DNA fragments up to 12 kb (using lambda phage DNA as a template). TthPlus DNA polymerase is also capable of catalyzing the polymerization of DNA using a RNA template in the presence of Mn^{2+}. The ability of TthPlus DNA polymerase to reverse transcribe at elevated temperatures (70 C) minimizes the problems encountered with strong secondary structures in RNA since they are unstable at higher reaction temperatures. Higher temperatures also result in increased specificity of primer hybridization and extension. In coupled RT/PCR assays, TthPlus is about 50-100 times more efficient than regular Taq DNA polymerase.

85. Gibberellins (GAs) are plant hormones that regulate growth and influence various developmental processes, including stem elongation, germination, dormancy, flowering, sex expression, enzyme induction, and leaf and fruit senescence.

86. Type I collagen, a fibrous protein abundant in connective tissues including tendon, ligament, dermis and blood vessel, is the major component and the primary determinant of tensile strength of the extracelluar matrix (ECM). It is widely used as a thin layer on tissue-culture surfaces to enhance the attachment and proliferation of a variety of cells including endothelial cells, fibroblasts, hepatocytes, epithelial cells and etc. In addition, collagen I can self-assemble into a 3-D superamolecular gel in vitro, making it an ideal biological scaffold to promote more in vivo-like cellular morphology and function.

87. Betaine improves the PCR amplification of GC-rich DNA sequences. Betaine reduces the formation of secondary structure caused by GC-rich regions and, therefore can be used to assist the amplification of GC rich sequences by PCR. It facilitates an isostabilizing effect on the differential thermal stability of AT and GC base pairs, which may result in a larger destabilizing effect on GC pairs and stabilzes AT rich sequence by binding with them and causing conformational change.

88. The efficiency of the production process depends on the rate of release of products rather than actual rate of biosynthesis. The immobilization process reduces biosynthetic capacity. The process does not allow easy removal of cell products or inhibitors as release of products/ by-products from cell into medium is hindered.

89. A lymphocyte is a type of white blood cell in the vertebrate immune system.Under the microscope, lymphocytes can be divided into large lymphocytes and small lymphocytes. Large granular lymphocytes include natural killer cells (NK cells). Small lymphocytes consist of T cells and B cells.

90. Zeatin is a plant hormone derived from the purine adenine. It is a member of the plant growth hormone family known as cytokinins. Zeatin was first discovered in immature corn kernels from the genus Zea. It promotes growth of lateral buds and stimulates cell division to produce bushier plants if sprayed on meristems. Again the Indole-3-acetic acid (IAA) is the most common, naturally-occurring, plant hormone of the auxin class. It is the best known of the auxins, and has been the subject of extensive studies by plant physiologists.

91. Required for the sandwich assay, the capture antibody binds to the antigen of interest and generates a detectable signal for quantification. This antibody is usually polyclonal in nature, and is conjugated with an enzyme (most commonly, horseradish peroxidase), so that, upon substrate addition, a colored product is formed. The density of the color can then be measured spectrophotometrically as a function of antigen concentration. The western blot (sometimes called the protein immunoblot) is a widely accepted analytical technique used to detect specific proteins in the given sample of tissue homogenate or extract. It uses gel electrophoresis to separate native proteins by 3-D structure or denatured proteins by the length of the polypeptide. The proteins are then transferred to a membrane (typically nitrocellulose or PVDF), where they are stained with antibodies specific to the target protein.

92. Lipofection (or liposome transfection) is a technique used to inject genetic material into a cell by means of liposomes, which are vesicles that can easily merge with the cell membrane since they are both made of a phospholipid bilayer. Lipofection generally uses a positively charged (cationic) lipid to form an aggregate with the negatively charged (anionic) genetic material and hence is the best mechanism to transform intact plant tissues.

93. Tissue plasminogen activator is a protein involved in the breakdown of blood clots. The importance of IFNγ in the immune system stems in part from its ability to inhibit viral replication directly, and most importantly from its immunostimulatory and immunomodulatory effects. podophyllotoxin is the pharmacological precursor for the important anticancer drug etoposide.

Polyhydroxyalkanoates or PHAs are linear polyesters produced in nature by bacterial fermentation of sugar or lipids, used in the production of biodegradable plastics.

94. Amperometric biosensors function by the production of a current generated by flux of redox electrons when a potential is applied between two electrodes.

Evanescent wave biosensors are fiber-optic sensors which detect the refractive index of the light beams.

Many enzyme catalysed reactions are exothermic, generating heat, this principle is used as a basis for measuring the rate of reaction and, hence, analyze concentration by calorimetric biosensors.

Potentiometric biosensors make use of ion-selective electrodes in order to transduce the biological reaction into an electrical signal. A recent development from ion-selective electrodes is the production of ion-selective field effect transistors (ISFETs) and their biosensor use as enzyme-linked field effect transistors.

95. Somaclonal variation is not restricted to, but is particularly common in, plants regenerated from callus. The variations can be genotypic or phenotypic, which in the latter case can be either genetic or epigenetic in origin. Typical genetic alterations are: changes in chromosome numbers (polyploidy and aneuploidy), chromosome structure (translocations, deletions, insertions and duplications) and DNA sequence (base mutations). Typical epigenetic related events are: gene amplification and gene methylation.

96. Gel filtration is a separation based on size and same is the purpose of SDS-PAGE i.e. to separate proteins according to their size, and no other physical feature.

97. Con A has varying affinities for various polypeptides and hence affinity binding will be used.

98. Number of λ-phage vectors $= \dfrac{1.0 \times 10^8}{20 \times 10^3} = 5 \times 10^3$

Number of yeast clones $= \dfrac{10^8}{250 \times 10^3} = 400$

99. λ-phage vectors

$$= \frac{\text{in}(1-p)}{\text{in}(1-a/b)} = \frac{\text{in}(0.0001)}{\text{in}\left(1-\dfrac{20\times10^3\ bp}{10^8\ bp}\right)}$$

$$= \frac{\text{in}(0.0001)}{\text{in}(1-0.0002)} = \frac{\text{in}(0.0001)}{\text{in}(0.9998)}$$

$$= \frac{-9.210}{-2.0\times10^{-4}} = 5 \times 10^4$$

$$\text{Yeast clones} = \frac{\text{in}(0.0001)}{\text{in}\left(1-\dfrac{250\times10^3}{10^8}\right)}$$

$$= \frac{-9.210}{\text{in}(1-0.0025)}$$

$$= \frac{-9.210}{\text{in}(0.9975)}$$

$$= \frac{-9.210}{-2.5\times10^{-3}}$$

$$= 4000$$

NUMERICAL TYPE QUESTIONS

1. Average number of fragments generated

$$= \left(\frac{1}{4}\times\frac{1}{4}\times\frac{1}{4}\times\frac{1}{4}\right)\times 4.3\times10^5$$

$$= \frac{4.3\times10^5}{256}$$

$$= 1.68 \times 10^3$$

2. DNA = 0.5 μg

$$\text{Dilution} = \frac{50}{200\times10000}$$

$$= 2.5 \times 10^{-5}$$

$$\text{TE} = \frac{\text{Number of colonies}}{\text{Dilution} \times \text{Amount of DNA}}$$

$$= \frac{35}{2.5\times10^{-5}\times0.5}$$

$$= 2.8 \times 10^6$$